Diagnostic Exponential Rate Attenuation in Spatial–Temporal Earthquake Models

Stability, identification and predictive evaluation of point-process models

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Diagnostic scope. The attenuation terms are statistical, isotropic point-process constructions. They do not implement Coulomb stress transfer, directional receiver-fault responses or rate-and-state friction.

AI generation and human responsibility. AI agents produced most of the code, analysis and manuscript, as detailed in Appendix B.5. The named human collaborators are responsible for deciding whether to endorse, circulate or use the work and for addressing corrections. Internal computational audits and AI review are not independent external scientific peer review. No comprehensive human verification is asserted.

Repository and correspondence. Source, data provenance and correction requests are maintained at

https://github.com/Drefetr/An_Evaluation_of_Earthquake_Models

Correspondence is through repository issues and the authors' linked profiles.

Keywords. Diagnostic inhibition; exponential rate attenuation; Hawkes and ETAS processes; fractional relaxation; identifiability; proper scoring; forecast evaluation.

Open research monograph

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Abstract

This monograph studies diagnostic exponential rate attenuation in spatial–temporal point-process earthquake models, using reconstructed New Zealand ($M \geq 5.0$), Southern California ($M \geq 4.0$) and Japanese ($M \geq 4.5$) catalogues. Against excitation baselines with jointly fitted background and triggering nuisance parameters, the reported comparisons establish no statistically detectable held-out log-score improvement from the tested diagnostic attenuation recipes. These isotropic statistical constructions do not implement physical Coulomb stress transfer, receiver-fault directionality or rate-and-state friction.

New Zealand shared attenuation loses 17.62 total held-out nats; Fold 1 contributes 16.52 nats, or 93.7%. Its fitted trend ($d = 1.790$, versus -0.252 without attenuation) is strongly collinear with depletion in event-score geometry ($\rho = 0.9341$), before extrapolation into $x_{\text{trend}} > 0$. Fold 1's block intervals exclude zero at all evaluated lengths. The second fold's interval includes zero at 90, 180 and 365 days; the third fold's interval includes zero at 90 days ($[-2.087, 0.004]$), but excludes it at 180 and 365 days. In California, immigrant-only attenuation and the magnitude-scaled ETAS attenuation extension select the nested training boundary $\hat{\eta} = 0$: exact zero held-out contrasts then follow algebraically, rather than constituting powered tests of physical absence. Raw total attenuation gives small contrasts of -0.22, -1.95 and -0.21 nats; its pooled 90-day interval is $[-6.564, -0.033]$. The magnitude-scaled ETAS procedure (EP+) gains 69.55 nats over the archived unweighted inverse-power benchmark (EP) while simultaneously changing full-plane normalization with regional leakage, training-declustered background weighting, magnitude-scaled spatial kernels and joint temporal–spatial fitting. All three spatial-scale estimates reach their 2.0 km lower bound. Both the original background-weighted offspring recipe (OC) and EP fail the continuous-statistic conditional calibration checks ($p \in [0.001, 0.012]$): their score contrasts rank conditionally misspecified procedures. Dependent calendar-block intervals and sign-flip references are descriptive sensitivity analyses, not calibrated long-memory tests.

Synthetic results depend on nuisance flexibility and event counts. Jointly fitted broad/long immigrant attenuation has mean -10.37 nats, median $+2.26$ and sample Monte Carlo standard error (MCSE) 8.56, with 10/20 wins and 16.9 test events per history on average. Broad/long total attenuation has mean -1.26 nats, median -1.26 and MCSE 2.63, with 8/20 wins and 13.25 mean test events; 98.2% of offspring proposals are rejected. Earlier large gains against a fixed-parameter null are removed or reversed when its background and temporal triggering are fitted. The narrow/short immigrant controls realize no thinning in 20/20 unit-mark and 19/20 magnitude-scaled histories. An intermediate regime retains 94.7 test events on average: the raw mean is $+0.294$ nats (MCSE 2.777; 13/20 wins and 13/20 parameter-boundary hits). A later sensitivity applying the inherited training-BIC rule gives mean $+2.495$ nats, median 0, MCSE 1.444, with 9/20 wins and 12/20 active fits. These are total test-window nats per independent synthetic history, not nats per event; neither sparse histories nor noisy moderate effects establish uniform detection power.

Component-ablation and sensitivity controls decompose the 69.55-nat EP+ gain into 29.22 nats from training-declustered background weighting, 30.03 from magnitude-scaled spatial kernels, 10.24 from joint temporal–spatial fitting and 0.06 from full-plane versus rectangle normalization. Relaxing the binding 2 km lower search bound to 0.05 km selects interior scales of 0.82–0.85 km and raises the three-fold gain to 77.06 nats with redeclustered weights (78.58 nats with the accepted weights held fixed). In New Zealand Fold 1, matched zero-trend and fixed-null-trend controls reduce the 16.52-nat deficit to 8.50 and 4.56 nats, attributing

48.5–72.4% of it to free trend extrapolation. The S8 oracle ceiling averages 18.11 nats (MCSE 4.18) against the jointly fitted null, while the fitted procedure captures 0.29 nats, an estimation loss of 17.82 nats. At 90-day blocks, the two-sided 95% minimum detectable effect is 9.24 nats in New Zealand, 48.30 in California and 15.98 in Japan (a between-window standard error for Japan). These controls sharpen the procedure-level reading; they do not turn it into a physical-absence test.

The mathematical results also have quantitative boundaries. The 195-mode weak Harris stability certificate requires a skeleton time of at least 9.8×10^{16} days because $\vartheta_{\min} = e^{-38.8}$ per day; 61 modes decay by less than 1% over the 24,394-day New Zealand horizon. Full-law active identification uses unbounded marks along zero-aftershock paths whose probabilities decay doubly exponentially as $\exp\{-a_u \kappa(q)\}$. The finite-horizon Pinsker visibility upper bound equals 1, and is therefore vacuous, for $\eta \geq 2.34 \times 10^{-3}$ on that horizon, including the positive New Zealand shared-inhibition strengths. These results establish properties of the specified statistical laws and fitted procedures; they do not identify physical stress shadows.

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Part I

Mathematical Foundations, Identifiability, and Nuisance Sensitivity

Chapter 1

Questions, Architecture, and Consolidated Findings

An earthquake changes the conditions under which later earthquakes occur. A statistical model may represent some of that change as triggering, which raises the subsequent event rate, and some as depletion, which lowers a stress-controlled contribution to that rate. Relaxation allows the influence of a past event to persist and then decay. These ingredients make a coherent modelling hypothesis, but the observed sequence of event times and magnitudes does not label which mechanism produced each event. A fitted inhibitory term can also compete with a flexible background, an alternative triggering kernel, or an imperfect observation model.

The central question of this manuscript is therefore not simply whether an inhibitory model can be fitted. It is when its event-dependent contribution can be learned, what mathematical assumptions make its dynamics legitimate, and what the available predictive comparisons actually support. The distinction matters in both directions. A weak fitted detector need not mean that two specified generating laws are intrinsically indistinguishable. Conversely, a strong detector supplied with the generating parameters need not show that an unknown-parameter procedure can achieve the same performance. Both questions must be kept separate from identifying a physical mechanism in a real catalogue.

The main empirical result is negative: in the completed observed-catalogue comparisons, none of the tested inhibition additions establishes a predictive gain over its matched excitation reference. The declared inverse-power ETAS benchmark (EP) also beats the background-weighted offspring procedure (OC) jointly in every reported California and Japan primary fold. The evidence supports this procedure-level conclusion alongside a restricted mathematical contribution; it does not falsify physical inhibition. Finite-mode stability, particular identification statements, and finite-history likelihood identities remain useful results under explicit hypotheses. Controlled experiments show substantial changes in calibrated detection when background nuisance structure is estimated. The tested inhibited recipes have not demonstrated improved predictive score in the reconstructed New Zealand and inspected California comparisons. Japan contributes transfer and competitive-benchmark evidence, including a timing reversal; it was not an LI evaluation. Neither physical inhibition nor its physical absence follows from that comparison.

The investigation developed from an earlier circulated draft. The current public manuscript is a separate restricted analysis with independently reconstructed computations; it does not reproduce the earlier draft's original calculation. The earlier draft is not redistributed, and its frozen catalogue, original fitting code, complete optimization records and editable source have not been recovered. The public implementation and selected summaries document the present reconstruction, with their availability distinguished in Appendix A.

1.1 From a material hypothesis to an observation law

A viscoelastic motivation suggests that one relaxation function might describe several responses to an event: the persistence of its stress drop, the accumulated effect of loading,

and a triggering density related to the derivative of relaxation. This is a strong modelling restriction. Demonstrating that its components are mathematically compatible is different from showing that the restriction can be distinguished from other explanations using event observations.

The analysis consequently uses two models for different purposes. The physical loading construction retains a latent stress state, a loading history and fresh event marks. Its finite-mode representation permits a Markov description and a scoped stability analysis. The diagnostic model instead specifies a deterministic background independently and asks how identifiable inhibition changes a finite observation law. In abbreviated notation it has intensity

$$\lambda_\eta(t) = B(t)\exp\{-\eta G_s(t)\} + nH_e(t), \qquad \eta \geq 0. \tag{1.1}$$

Here G_s is a marked depletion history, H_e is normalized marked triggering, and both use events strictly before t. The background B can vary with calendar time. The parameter η is a diagnostic inhibition coefficient; it is not automatically the physical stress coefficient of the loading model. The distinction is developed before any theorem or experiment in Chapter 2.

This separation makes the scope of a conclusion visible. An invariant-law theorem for an autonomous finite-mode process does not apply merely because a nonstationary diagnostic uses a similar kernel. A test of inhibition against no inhibition does not by itself test equality of two physical memory functions. A finite-window numerical approximation does not establish a stationary infinite-memory limit. These are different mathematical and observational questions rather than competing descriptions of the same result.

1.2 Four questions answered at different levels

1. **When is the proposed process well defined and stable?** The finite-mode analysis gives conditions for construction, non-explosion, an invariant law, first moments and convergence in a specified weighted cost. Heavy-tailed marks require attention to the particular moments used. Explicit no-feedback counterexamples show why finite-mode results do not justify an unrestricted exact-memory theorem.
2. **What can the observation law identify?** A stress-memory parameter disappears at zero feedback under the stated finite-nuisance or diagnostic conditions. Positive-feedback identification is possible in a restricted full-law experiment with known transforms, empty history and sufficiently rich mark support. Such a statement does not guarantee stable estimation from a finite catalogue.
3. **How does estimating nuisance structure change detection?** Controlled comparisons use the same histories, procedure-specific null reference scores and separate null queries. They distinguish strong separation of a fully specified pair from weaker performance when an uninhibited explanation may change its nuisance parameters. The comparison does not identify a unique physical cause or an avoidable fraction of algorithmic loss.
4. **Does the additional inhibited structure improve observed prediction?** A chronological comparison evaluates declared recipes using full marked-event log scores with a common fitted magnitude law. The shared-inhibition recipe loses to its noninhibited Mittag–Leffler triggering comparator in each of three test periods, while an Omori comparator scores better than both recipes. Reused periods and different kernel flexibility limit the interpretation to those exploratory recipes.

The first two questions concern dynamics and equality of probability laws. The third concerns a specified testing procedure under a declared generator. The fourth concerns the

predictive performance of fitted recipes on an observed catalogue. Evidence at one level can guide another, but cannot replace it.

1.3 The completed empirical arc

The pooled temporal failure is the beginning of the empirical question, not its endpoint. Part II examines whether magnitude history, free productivity, background coordinates or the temporal kernel explain it. Exposure orthogonalization improves numerical conditioning without materially changing the feasible optimum. Common temporal-scale estimation helps in its matched comparison, whereas the parent-class scale continuation fails its declared gate. Continuously tuned Omori excitation improves predictive score, but matched and jointly tuned inhibition still lose. These comparisons retain their own reference recipes; gains from different stages are not added into a synthetic cumulative improvement.

Part III then changes the observation representation itself. The New Zealand progression moves from latitude zones through redistribution and distance routing to projected continuous offspring geography, followed by a temporal–spatial–background factorial. Matched local inhibition does not become a predictive success merely because excitation is localized. Continuous locations improve on coarse cell-lift references under the richer background. Most of these gains repair uniform within-cell spreading; they are representation checks, not evidence of competitive spatial forecasting superiority. Kernel and background changes contain important negative contrasts.

California and Japan separate transport of an ablation from competitive performance. California retains the registered matched directions, but EP beats OC and conditional calibration exposes discrepancies in OC. Japan retains the location benefit and reverses the timing direction. Finally, immigrant-only continuous local inhibition nests OC exactly at zero strength and selects that boundary in the observed California windows. Its fixed controls include both positive-law failures and broad/long successes. The magnitude-scaled, training-weighted excitation procedure subsequently improves on the archived EP procedure, while its own attenuation extension returns to the uninhibited boundary. The completed arc therefore distinguishes excitation improvements from the failure of the tested attenuation additions to improve their matched forecasts.

The matched synthetic controls expose the role of nuisance fitting. Jointly fitting the temporal backbone in both arms gives matched broad/long immigrant and total-attenuation means of -10.3701 and -1.2607 nats, with 10/20 and 8/20 wins, respectively. The earlier apparent broad/long advantages shrink as the uninhibited background and branching ratio are freed. Narrow/short immigrant controls instead realize almost no thinning. The intermediate 20-history total-attenuation regime rejects 27.2% of proposals and retains 94.7 test events on average; its matched joint-fit mean is $+0.2941$ nats, with 13/20 wins. Thus neither fixed-parameter separation nor a positive fitted strength by itself establishes detectable inhibition.

1.4 Relation to existing work

1.4.1 Triggering and fractional relaxation

Self-exciting point processes provide a natural representation of event clustering. Classical work by Hawkes (1971) develops their second-order structure and also illustrates that second-order properties need not uniquely distinguish different process constructions. That obser-

vation is relevant to physical interpretation: a recognizable correlation pattern is not, on its own, a unique mechanism assignment.

Mittag–Leffler earthquake triggering is already developed by Davis et al. (2024a). Their construction uses a fractional relaxation function and a corresponding triggering density, with magnitude-dependent productivity. The triggering component studied here is inherited from that line of work after parameter relabelling; introducing it is not a contribution of this manuscript. The additional target is a restriction linking event-dependent background depletion and triggering memory, together with the mathematical and inferential conditions required to investigate it. The multivariate fractional model of Davis et al. (2026a) further makes mark-category probabilities history dependent. The fresh independent mark assumption used in the present restricted analysis should therefore be read as a modelling boundary, not an advance over history-dependent marks.

1.4.2 Excitation correction and spatial structure

Classical statistical stress-release models express tectonic loading minus cumulative release through

$$\lambda(t) = \exp\left\{\alpha + \beta\left(\rho t - \sum_{t_i<t} S_i\right)\right\}.$$

The initial stress is absorbed into α, and S_i is a specified magnitude-based release proxy. This tradition includes Vere-Jones (1978, 1988); Zheng and Vere-Jones (1991, 1994); linked regional stress-release models allow interactions between regions (Bebbington and Harte, 2003). The simple law above has no additive ETAS branching term. Here a decaying depletion kernel is combined with Hawkes/ETAS triggering, so the tested extension differs from classical cumulative release.

Physical stress transfer provides a separate, richer account of rate changes. Coulomb calculations produce anisotropic lobes of positive and negative ΔCFF depending on source rupture and receiver-fault geometry (King et al., 1994; Stein, 1999). Rate-and-state friction converts a stressing history into the seismicity of a population of nucleation sites (Dieterich, 1994); successive shocks can suppress previously elevated aftershock rates as well as alter background activity (Toda et al., 2005). Rupture dimensions change with magnitude, and the response applies to the stressed faults regardless of an immigrant/offspring label. The isotropic point-kernel attenuation recipes tested here omit those signed stress lobes, receiver orientations and rupture geometry. Their predictive scores evaluate these statistical recipes rather than Coulomb rate-and-state theory.

The empirical literature does not supply a single universal stress-shadow signature. Harris and Simpson (1998) reconcile regional quiescence and an apparent shadow exception using rate-and-state dependence on the pre-event fault state. Parsons (2002) relates global triggered sequences to computed stress changes and their temporal decay. Felzer and Brodsky (2006) instead interpret the spatial decay of aftershocks as evidence for dynamic triggering. For the Landers and Kobe sequences, Mallman and Zoback (2007) could not reliably resolve the predicted rate decreases and found increases in regions of both stress signs. These are distinct empirical and physical hypotheses. A positive radial attenuation coefficient cannot stand in for their signed, geometry- and state-dependent predictions.

Combining excitation with stress-related correction also predates this work. Self-correcting processes and stochastic stress-transfer models provide earlier formulations (Isham and Westcott, 1979; Borovkov and Bebbington, 2003). Petrillo et al. (2023) compare statistical mechanisms with catalogues generated by a physical spring-block model and study the interaction

of clustering and stress discharge. The relevant distinction lies in the equations: an additive correcting contribution and a positive background multiplied by a relaxing depletion factor need not have the same law or positivity properties. A claim about a shared relaxation restriction must identify that difference explicitly, rather than claim the first combination of excitation and correction.

Spatially heterogeneous backgrounds and different space–time cluster structures are established parts of earthquake point-process modelling (Ogata, 1998). The completed spatial progression in Part III addresses a targeted model-adequacy question. It asks whether localizing depletion adds predictive information beyond matched spatial background and triggering structures. The algebra in Remark 4.5 explains why a collection of locally inhibited processes generally cannot be represented by applying a single exponential of pooled depletion to the national background. It does not establish that spatial pooling caused the observed predictive loss.

Testable clustering forecasts also have a substantial history beyond the particular spatial recipes used here. Console and Murru (2001) combine spontaneous activity with rate contributions from earlier events. Helmstetter et al. (2006); Werner et al. (2011) compare smoothed-seismicity backgrounds and updated triggering forecasts, with spatial treatment that can change for larger events. Probabilistic reconstruction of direct and indirect triggering links further emphasizes that the observed sequence does not come with a unique genealogy (Marsan and Lengliné, 2008). The magnitude-scaled, training-weighted benchmark in Section 11.5.2 tests an attenuation extension against a jointly fitted excitation arm, rather than using OC as the sole competitive reference.

1.4.3 Catalogue observation and forecast evaluation

A selected magnitude threshold defines the data used in a comparison; it does not prove complete detection. Completeness estimates have uncertainty (Woessner and Wiemer, 2005), and a magnitude-dependent detection function can change the observed frequency–magnitude distribution (Mignan, 2012). Window or interaction-based declustering (Gardner and Knopoff, 1974; Reasenberg, 1985) assigns a catalogue partition under specified rules. Stochastic declustering instead weights possible background and parent assignments (Zhuang et al., 2002); neither supplies observed immigrant labels. These observation and genealogy questions determine what an apparent rate deficit can mean.

Forecast comparison and model adequacy also require different evidence. A proper logarithmic score rewards the assigned predictive law in expectation (Gneiting and Raftery, 2007); a realized advantage alone does not calibrate that law. Rhoades et al. (2011) distinguish model-consistency tests from paired forecast comparisons, while Bray and Schoenberg (2013) examine residual methods that can reveal discrepancies hidden by aggregate scores. We therefore report score contrasts and conditional simulation diagnostics separately. The common interpretation of the recorded-data target, fitted-state uncertainty and boundary selection is fixed in Section 6.1.

1.4.4 Stability, identification and nonregular testing

General nonlinear Hawkes stability theory (Brémaud and Massoulié, 1996) and coupling methods such as those of Hairer et al. (2011) provide an important context for the finite-mode argument. Their hypotheses remain part of the work: a generator drift inequality alone is not a construction, an existence theorem, or a complete contraction proof. The manuscript therefore states the state domain, mark assumptions, coupling cost and finite-mode restric-

tions alongside the result. It distinguishes a proof within the stated model from a priority claim over all earlier stability theory.

Likewise, the representation and identification results of Davis et al. (2026b) apply under their own linear-model and observation hypotheses. They do not directly settle a nonlinear stress-memory parameter that vanishes from the law at zero feedback. The present restricted identification argument addresses that singularity explicitly, while retaining its dependence on known mark transforms and unbounded mark support.

Testing at such a boundary is not justified by invoking an ordinary likelihood-ratio reference distribution. Work by Davies (1977, 1987) identifies the difficulty of a nuisance parameter present only under the alternative. Boundary asymptotics also require regularity conditions of the kind examined by Self and Liang (1987). These references motivate care; they are not a ready-made calibration theorem for the heavy-tailed inhibited model. For fitted residual diagnostics, Kling and Vetter (2024) similarly motivates accounting for estimation rather than treating an estimated compensator as known.

1.4.5 Mathematical contribution

Random-productivity spectral formulas such as those of Brémaud and Massoulié (2002) require their stated moment assumptions. The heavy-tailed regime studied here requires a separate check of those moments.

The mathematical contribution is a restricted analysis: a model-specific application of weak Harris theory, fixed-horizon coupling bounds, and self-contained calculations for stationary moments, exact-memory response, observation-law identification and finite-history inference. The approximation discussion corrects the earlier inference from local to half-line Lipschitz continuity. The triggering component is inherited, and the shared depletion/triggering relation is the particular model restriction studied. Appendix A.7 classifies the individual results and their primary-source comparisons.

1.5 Evidence and interpretation

The observed analysis uses an independently reconstructed GeoNet catalogue. It contains 913 selected events over a 24,394-day window starting in 1960 and ending before 15 October 2026. Although its count matches a count reported in an earlier manuscript, the unavailable historical event list and fitting records have not been recovered. The reconstruction is a new archived snapshot with its own selection rules, not historical reproduction. GeoNet describes its catalogue interfaces as access routes to earthquake parameters (GNS Science, n.d.); a common provider or matching event count does not make interface snapshots independent catalogues.

The historical controlled studies have a different role. They compare declared laws and procedures, using saved simulated histories and retained failure/recovery policies. When generating nuisance parameters are supplied, the known-law test detects all 150 saved alternatives and rejects 50 of 1,000 separate null queries. Estimating inhibition alone with those nuisance values supplied also detects all 150 alternatives. Estimating background level with trend supplied detects 123, whereas estimating level and trend detects 68. The earlier fuller fitted procedure detects 57 after its own recalibration. These contrasts are substantive evidence about these procedures, with conditional uncertainty and adaptive-reuse qualifications. They are not five independent replications, an additive causal decomposition, or a proof that fixing unknown background values is appropriate for the observed analysis.

The New Zealand sequence also involves adaptive holdout reuse: earlier test results informed later model classes, bounds and continuation decisions on the same 612 targets. A frozen within-stage protocol limits additional discretion but does not restore independence from prior test inspection (Dwork et al., 2015). These are exploratory retrospective comparisons, with no fresh confirmatory test set or reusable-holdout guarantee.

The predictive comparison contributes a further kind of evidence. The shared-inhibition recipe fails to improve score in each declared test period, with approximately 93.7% of its total disadvantage in the first. This supports a negative statement about the predictive return from the tested additional structure. It does not supply three independent confirmations against inhibition or identify a geological cause. The original event-only spatial group sums and archived-magnitude substitution were limited diagnostics. The subsequently completed spatial conditional-intensity comparisons include full spatial exposure; they still do not constitute a complete observation-error analysis.

1.6 How to read the manuscript

The exposition follows questions rather than the chronology of research iterations. Chapter 2 defines the models and their observation conventions. Chapters 3 and 4 develop the mathematical domain and identification boundaries. Chapter 5 derives the finite-history likelihood tools and explains what is held fixed in response and testing statements. The remaining main chapters describe the controlled design, report the synthetic and observed evidence, and explain the resulting research decision. A reader primarily interested in the empirical conclusion can read the model definitions, the design and results chapters, and the discussion before returning to the proofs.

The appendices provide an archival source map, numerical protocols, notation and contribution classifications. Operational recovery histories and file hashes are kept there or in the companion evidence. The main contribution is the connected account of mathematical validity, identification and procedure performance. The completed locality comparisons, external transfers, competitive benchmarks and continuous-inhibition controls form Part III.

Table 1.1: Levels of evidence used in this manuscript. The conclusion in the middle column is intentionally narrower than the excluded inference.

Evidence	Supported conclusion	Inference not supplied
Restricted analysis	Results under stated state, kernel and observation assumptions	General nonlinear exact-memory stability or unrestricted physical identification
Controlled histories	Calibrated procedure contrasts at the declared generating law	Universal power, avoidable-loss decomposition or real-data nuisance knowledge
Chronological scores	Performance ordering of the declared exploratory recipes	Physical exclusion, equal-complexity family ranking or untouched confirmation
Matched spatial ablations	Incremental predictive value of a specified representation on common observations	Physical offspring attribution, recovered immigrant geography or universal scale identification
External transfer and benchmarks	Regional transport of specified contrasts and ordering of complete competing procedures	A combined success when a component reverses; superiority over untested alternatives
Conditional simulation	Within-origin diagnostic discrepancies of frozen OC and EP forecasts under their declared simulated-future laws	Joint cross-origin calibration, uncertainty from refitting, or proof of adequate probabilities
Fixed-seed LI controls	Procedure behaviour under each retained generating regime	Universal power or false-positive rates from finite, regime-specific control matrices
Source and numerical checks	Specific arithmetic, approximation sensitivity and evidence bindings	Global optimization, catalogue completeness or full historical reproduction

Chapter 2

Physical Loading, Diagnostic, and Spatial–Temporal Observation Laws

The central difficulty is to separate three objects that are often conflated in stress-release modelling: the physical state that carries memory, the point process that records event occurrence, and the observation model through which events and their marks become data. The finite-mode formulation provides a transparent autonomous physical model. A diagnostic formulation specifies an independent background and separates its suppression from triggering. The distinction matters because stability, approximation and inference make different demands on the model.

Relation to stress-release and Coulomb models. The classical statistical stress-release law is $\lambda(t) = \exp\{\alpha + \beta(\rho t - \sum_{t_i<t} S_i)\}$: tectonic loading raises the hazard and cumulative release lowers it (Vere-Jones, 1978, 1988; Zheng and Vere-Jones, 1991, 1994). Linked versions transfer release between regions (Bebbington and Harte, 2003). The model below replaces persistent cumulative release by relaxation and adds an explicit branching reservoir. Its diagnostic spatial variants use isotropic point kernels. Physical Coulomb models instead resolve signed ΔCFF lobes from magnitude-dependent finite ruptures and receiver orientations (King et al., 1994; Stein, 1999); rate-and-state evolution acts on the affected fault population, including ongoing aftershocks (Dieterich, 1994; Toda et al., 2005). Immigrant-only attenuation is therefore a statistical allocation assumption, not the physical meaning of a stress shadow. Rate-and-state shadow durations depend on the fault state as well as the stress step (Harris and Simpson, 1998); indirect triggering cascades further separate a parent label from a unique physical stress pathway (Marsan and Lengliné, 2008).

2.1 The physical finite-mode model

Fix a finite number of modes $m \in \mathbb{N}$, with $j \in \{1, \dots, m\}$; every mode sum in this finite-mode model is over that set. Let $\vartheta_j > 0$ be relaxation rates and let $w_j > 0$, $\sum_{j=1}^m w_j = 1$, be fixed weights. For a constant loading $\rho \geq 0$, write

$$x_j^* = \frac{\rho}{\vartheta_j}, \qquad \bar{x} = \sum_j w_j x_j^*, \qquad \mathcal{X} = \left\{(Z,U) : Z_j \leq x_j^*,\ U_j \geq 0\right\}.$$

The physical stress and excitation reservoir are

$$X = \sum_j w_j Z_j, \qquad H = \sum_j w_j U_j. \tag{2.1}$$

Between events the coordinates relax according to

$$\dot{Z}_j = \rho - \vartheta_j Z_j, \qquad \dot{U}_j = -\vartheta_j U_j. \tag{2.2}$$

The ground event process N has compensator

$$\Lambda(t) = \int_0^t \left\{\Psi(X(s^-)) + H(s^-)\right\} \mathrm{d}s. \tag{2.3}$$

Thus the intensity is evaluated immediately before a possible event and does not include the outcome of that event.

If an event is accepted at time T_i, a fresh mark M_i is drawn from a fixed law F, conditionally independent of the pre-event history. Only after acceptance does the mark determine the jumps

$$\Delta Z_j(T_i) = -g(M_i), \qquad \Delta U_j(T_i) = \vartheta_j \kappa(M_i). \tag{2.4}$$

The functions $g, \kappa \geq 0$ may depend on different aspects of the same mark and may therefore be correlated. Their first moments are denoted by

$$m_1 = \mathbb{E}_F g(M), \qquad n = \mathbb{E}_F \kappa(M).$$

The subcritical condition $n < 1$ will control the expected branching size. Marks affect future event rates, but an accepted event cannot change the probability of its own acceptance.

The stability theorem permits a positive nondecreasing link Ψ that is Lipschitz on the reached upper-bounded stress half-line. One physically signed example is

$$\Psi(x) = e^{a+bx}, \qquad b \geq 0. \tag{2.5}$$

Consequently the loading ρ acts physically through its effect on the coordinates and hence on X; there is no separate multiplicative loading function in (2.3). Model (2.2)–(2.5), with constant ρ and fixed Ψ, is the autonomous finite-mode model used in the stability and approximation results.

2.2 Memory kernels and exact relaxation

For the finite-mode flow,

$$\Phi(t) = \sum_j w_j e^{-\vartheta_j t}, \qquad h(t) = -\Phi'(t) = \sum_j w_j \vartheta_j e^{-\vartheta_j t}. \tag{2.6}$$

Thus $\Phi(0) = 1$, Φ is completely monotone, and h is a probability density:

$$\int_0^\infty h(t)\, \mathrm{d}t = 1 - \Phi(\infty) = 1.$$

For arbitrary finite initial coordinates, the coordinate solution gives

$$X(t) = \sum_j w_j Z_j(0) e^{-\vartheta_j t} + \rho \int_0^t \Phi(u)\, \mathrm{d}u - \sum_{0<T_i\leq t} g(M_i)\Phi(t - T_i), \tag{2.7}$$

$$H(t) = \sum_j w_j U_j(0) e^{-\vartheta_j t} + \sum_{0<T_i\leq t} \kappa(M_i) h(t - T_i). \tag{2.8}$$

The first initial term reduces to $X_0\Phi(t)$ if every stress mode starts at $Z_j(0) = X_0$; an analogous reduction for excitation requires equal initial excitation coordinates. The current event is excluded when intensity is evaluated at its left limit. General finite initial coordinates need the displayed mode sums.

Formulation (2.6) is a finite exponential mixture. A non-Markovian relaxation law is obtained by replacing the finite exponential sum with

$$\Phi(t) = E_\alpha(-(t/\tau)^\alpha), \qquad 0 < \alpha < 1, \quad \tau > 0, \tag{2.9}$$

and setting $h = -\Phi'$. Here $E_\alpha(z) = \sum_{k=0}^{\infty} z^k/\Gamma(1+\alpha k)$ is the Mittag–Leffler function. For the exact kernel,

$$\Phi(t) \sim \frac{\tau^\alpha}{\Gamma(1-\alpha)} t^{-\alpha}, \qquad h(t) \sim \frac{\alpha\tau^\alpha}{\Gamma(1-\alpha)} t^{-1-\alpha}$$

as $t \to \infty$. Hence h is integrable while Φ is not. The integrated triggering density is finite, while the integrated stress relaxation of a single release is infinite. The two memory objects therefore have different long-time integrability. The exact kernel is not, for noninteger α, a finite exponential mixture; it therefore carries genuine infinite-dimensional memory.

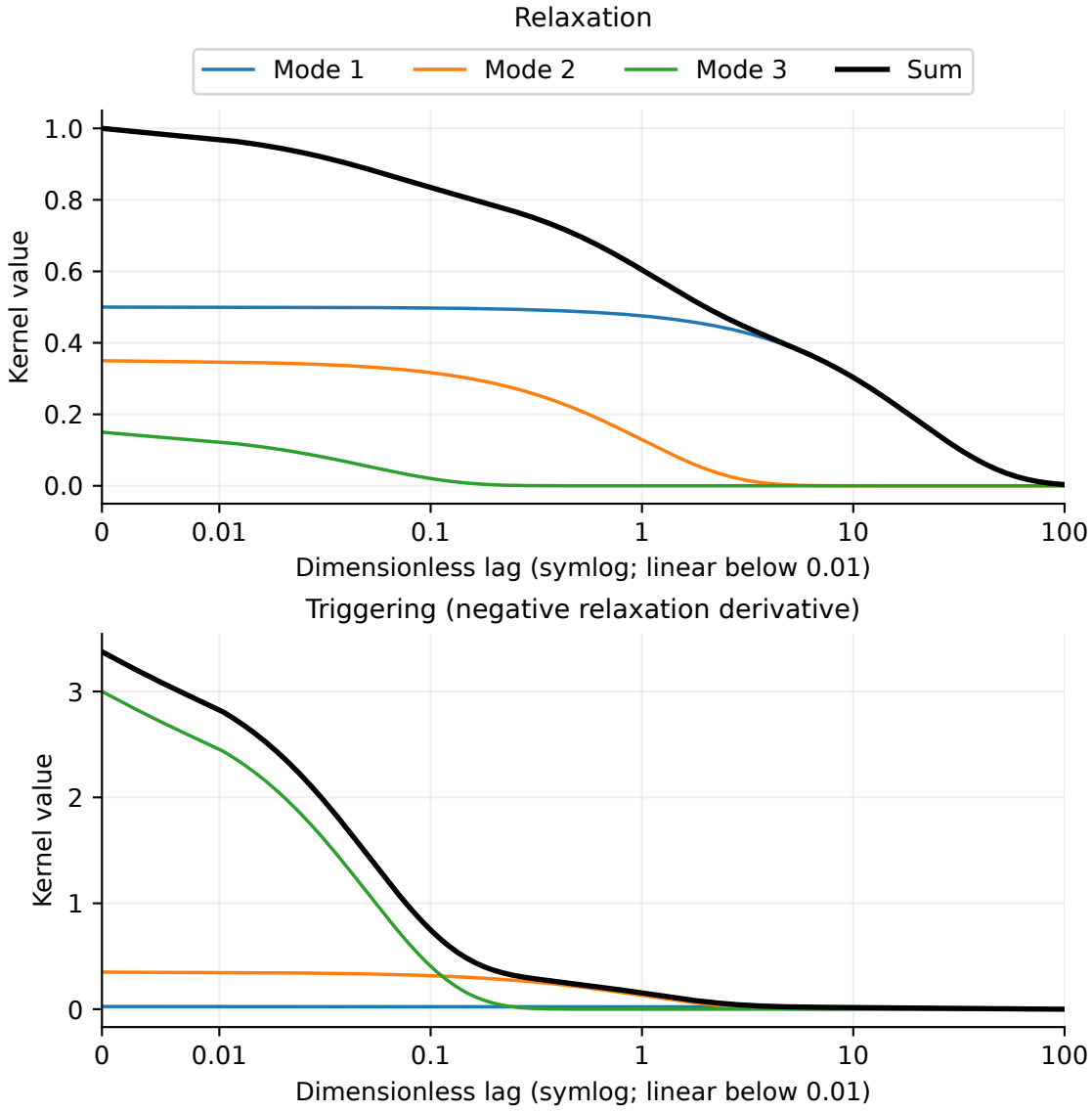


Figure 2.1: A dimensionless toy finite mixture with weights $(0.5, 0.35, 0.15)$ and rates $(0.05, 1, 20)$. The relaxation Φ and its negative derivative h weight the same modes differently. Curves are direct evaluations of the displayed finite sums; they are not fits, simulations or an exact Mittag–Leffler kernel.

In the physical model w_j is dimensionless, ϑ_j and h have inverse-time units, Φ is dimensionless, ρ is stress per unit time, and $g(M)$ is measured in stress units. The jump $\vartheta_j \kappa(M)$ has the units of U_j. With no history, the sums in (2.7)–(2.8) are empty.

2.3 The diagnostic model and observation

For diagnostic work, retain a stress relaxation kernel Φ_s and a normalized triggering density h_e, initially allowing them to vary separately. Let (T_i, Q_i) denote the accepted event times and their marks. Define the strictly prior cumulative release and excitation sums by

$$G_s(t) = \sum_{0<T_i<t} g(Q_i)\Phi_s(t - T_i), \qquad H_e(t) = \sum_{0<T_i<t} \kappa(Q_i)h_e(t - T_i), \tag{2.10}$$

and adopt the diagnostic intensity

$$\lambda_\eta(t) = B(t)\exp\{-\eta G_s(t)\} + nH_e(t). \tag{2.11}$$

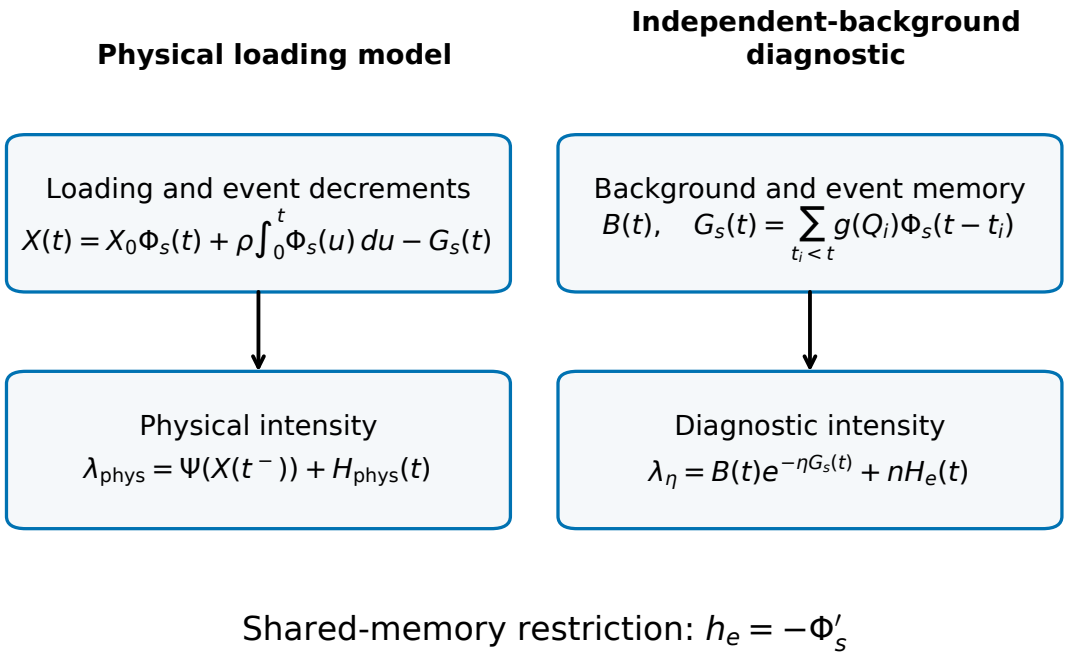


Figure 2.2: Physical loading and the independent-background diagnostic answer different questions. The left panel illustrates empty prehistory and equal initial stress coordinates $Z_j(0) = X_0$; arbitrary finite initial coordinates require the mode sum in (2.7). The physical model feeds loading through stress; the diagnostic specifies a background and suppresses its contribution by accumulated release. Triggering remains additive. The unit-productivity convention, with n outside H_e, applies to the diagnostic.

The baseline $B(t)$ is deterministic at fixed parameters and specified independently of the stress-kernel family. A useful choice is $B(t) = A\exp\{dx_{\text{trend}}(t)\}$, where $A = e^a > 0$ and $x_{\text{trend}}(t) = 2t/T_0 - 1$ for a declared fixed clock scale $T_0 > 0$ (take $T_0 = T$ for the normalized single-window formulation). It is not generated by stress loading or by the marks. The strict inequalities in (2.10) are deliberate: an event is accepted before its mark is drawn, so its own release and productivity cannot enter its intensity.

This section adopts a convention change once and for all: diagnostic productivity is scaled to have unit mean,

$$\mathbb{E}_F\kappa(Q) = 1,$$

while the coefficient n is retained *outside* H_e in (2.11). In the physical model, by contrast, $n = \mathbb{E}_F\kappa(M) < 1$ is the mean carried by the productivity marks themselves. The diagnostic normalisation removes that mean from the marks and displays it explicitly as a coefficient. This is a bookkeeping convention, not an assertion that realised productivity is one, and it is not used to reinterpret the autonomous physical stability theorem.

The physical exponential link (2.5) embeds loading through X: a constant ρ shifts the physical stress state and therefore changes $\Psi(X)$. The time-dependent baseline $B(t)$ in (2.11) is an independently specified background. It does not replace $\Psi(X)$ in the autonomous invariant-law theorem; that theorem does not cover externally varying $B(t)$.

The shared subfamily imposes the identity $h_e = -\Phi_s'$. This has substantive model content: the instantaneous excitation density is the negative derivative of the same kernel that governs stress relaxation. Consequently the decay of past stress and the contribution of a mark to the future compensator come from one memory structure, including its normalisation. Choices of Φ_s and h_e that do not satisfy this identity define a separate-memory model. Normalization alone does not impose their equality.

A ground process records only the event times T_i, whereas a marked process records the pairs (T_i, Q_i). The ground law is a projection of the marked law. The event-time contribution to a joint marked likelihood, evaluated using observed past marks, is a different object from the marginal likelihood of times alone. Event times alone generally do not identify g and κ separately, nor their within-mark dependence, without an additional mark model. Conversely, the marked likelihood includes both ground acceptance probabilities and mark densities. Since marks are fresh after acceptance, past marks enter (2.11) through the strict sums (2.10), while a newly drawn mark first affects the future. This observational distinction is essential: marginalizing to times alone and evaluating a joint marked likelihood answer different questions.

The likelihoods studied below condition on a declared starting history. Empty prehistory is a model assumption rather than a statement that no earlier earthquakes occurred. Likewise, a fresh common mark law does not establish completeness or homogeneous measurement of the observed catalogue. Unobserved parents, omitted regions and uncertain magnitudes can change the relationship between the mathematical history and its recorded counterpart. The subsequent identification statements concern the declared complete marked process; their transfer to an earthquake catalogue requires an observation model.

2.4 Master model and parameter dictionary

The same letter can occur in a mathematical proof and in an archived program field. The following dictionary fixes the scientific meaning across the empirical chapters; explicitly local theoretical uses retain their stated assumptions. In particular, X_{phys} denotes latent physical stress when a distinction is needed, whereas $x_{\mathrm{trend}}(t) = 2t/T_0 - 1$ is an affine clock, not a stress measurement. The earlier physical equations use X for the former. A likelihood is written ℓ_{score} when it could be confused with the LI range ℓ; the auxiliary frozen response slope is a separate local theoretical quantity.

Table 2.1: Unified parameter meanings. Bounds and estimated/supplied status are procedure-specific and remain in the experimental chapters.

Symbol	Meaning
a, d	Log immigrant/background amplitude and affine-clock trend. Physical exponential-link slope b is a different parameter.
$\widetilde{a}, \widetilde{d}$	Exposure-orthogonalized background coordinates; a feasible reparameterization need not alter the represented intensity.
n	Diagnostic excitation amplitude with unit-mean productivity; in the autonomous physical theorem it instead denotes the mean of the productivity mark.
η	Nonnegative attenuation strength of the stated diagnostic history feature.
α, τ	Mittag–Leffler fractional order and temporal scale; the implemented finite mixture is explicitly distinguished from the exact kernel.
c, p	Omori time offset and power, with normalized density k_O below. The magnitude-class cut is a separately declared threshold, not this time offset.
r, β	Productivity exponent and exponential magnitude-excess rate. A training-only rate estimate and a freely fitted productivity exponent are different nuisance decisions.
ω	Redistribution share in the discrete spatial routing continuation.
σ	Excitation spatial scale in the declared projected-domain density; a fitted/grid-selected model scale is not a measured interaction distance.
ℓ, ψ	LI attenuation range in kilometres and memory in days. Both disappear from the law at exact $\eta = 0$.
h, δ	Magnitude-class association and parent-class fertility contrast. These do not denote the triggering density h_e or a score difference Δ.

The labels FC through OL refer to the final smooth-background comparisons unless a factorial background is explicitly named. The earlier factorial also crosses cell and KDE-enriched backgrounds. EP is an excitation model; it is neither the no-trigger N0 recipe nor a constant-Poisson optimization floor. Comparisons of whole procedures must not be described as single-parameter inhibition ablations.

2.5 Marked history and spatial observation laws

For a magnitude excess q under the working exponential law, the normalized productivity is

$$\kappa_r(q) = (1 - r)e^{\beta r q}, \qquad 0 \le r < 1.$$

Its unit mean is a property of that working law. The free-productivity experiment estimates r within its declared continuous box; it is not a grid permitting $r \ge 1$. The class-history extension replaces the fresh marginal mark allocation by a predictable class probability and a conditional within-class density. Its mark score includes that class probability once. Chapter 9 gives the Perron normalization and explains why association can affect a joint fit without directly altering a fixed-history ground intensity after effective-scale matching.

Table 2.2: Procedure names used in Parts III–IV. Names identify archived fitted recipes rather than new universal model families.

Arm	Temporal law	Spatial/benchmark role
FC	Fixed-scale ML	Continuous reference-weighted offspring density
FL	Fixed-scale ML	Cell-lift offspring counterpart
MC	Free common-scale ML	Continuous offspring density
OC	Omori	Continuous reference-weighted offspring density
OL	Omori	Cell-lift offspring counterpart
EP	Omori ETAS adaptation	Separately fitted unweighted inverse-power radial offspring law
EP+	Omori	Jointly fitted magnitude-scaled unweighted offspring and training-declustered background (Section 11.5.2)
N0	No offspring excitation	Separately refitted smooth immigrant background
LI	Matched OC excitation	Unrenormalized immigrant-only local attenuation

On a fixed projected domain D, a noninhibited continuous model has

$$\lambda(t,z) = e^{a+dx_{\mathrm{trend}}(t)} b_{\mathrm{bg}}(z) + nH(t,z), \tag{2.12}$$

$$H(t,z) = \sum_{t_j<t} \kappa_j k_\theta(t-t_j) s_\sigma(z \mid z_j),$$

$$k_O(u) = (p-1)c^{p-1}(u+c)^{-p}, \qquad u>0, \quad c>0, \quad p>1,$$

$$s_\sigma(z \mid z_j) = \frac{b_{\mathrm{ref}}(z)(1+\|z-z_j\|^2/\sigma^2)^{-2}}{\int_D b_{\mathrm{ref}}(v)(1+\|v-z_j\|^2/\sigma^2)^{-2}\,\mathrm{d}v}. \tag{2.13}$$

Both spatial densities integrate to one over D; the temporal density integrates to one over all positive lags. Exposure uses the actual finite time window. The cell-lift counterpart allocates target-cell mass using its declared centre-based rule and spreads that mass uniformly inside the cell, retaining actual parent coordinates. EP normalizes a separately fitted radial law directly against projected area without b_{ref}. Thus a location contrast always specifies its measure and reference.

Continuous local inhibition changes only the immigrant contribution:

$$\lambda_{\mathrm{LI}}(t,z) = e^{a+dx_{\mathrm{trend}}(t)} b_{\mathrm{bg}}(z) e^{-\eta G(t,z)} + nH_{\mathrm{OC}}(t,z), \tag{2.14}$$

$$G(t,z) = \sum_{t_j<t} e^{-(t-t_j)/\psi}(1+\|z-z_j\|^2/\ell^2)^{-2}.$$

The stress multiplier is one for every eligible recorded parent. The attenuation field is not a normalized spatial density; renormalizing the attenuated immigrant term would change the model. The observed California and synthetic-control clocks have distinct declared T_0, given in Chapter 13. Exact $\eta = 0$ restores OC at matching shared coordinates and makes every ℓ, ψ pair observationally equivalent. This boundary is the spatial counterpart of the identifiability limitation in Chapter 4.

The common isotropic spatial scale is a restriction of these candidate laws. It omits magnitude-dependent rupture extent and directional fault geometry; magnitude scaling of

attenuation amplitude does not supply either feature. Space–time ETAS formulations already allow magnitude-dependent spatial spread (Ogata, 1998), and forecasting procedures can use the early aftershock cloud to represent the spatial extent of larger sources (Helmstetter et al., 2006; Werner et al., 2011). Multiplication by the piecewise-constant reference density also introduces cell-boundary jumps in the OC offspring density. The spatial ablations compare consequences of this construction, rather than identifying a physical aftershock or stress-transfer law.

The observation law is separate from the latent occurrence law. Magnitude-dependent detection can alter the recorded magnitude distribution (Mignan, 2012), and short-term incompleteness can depend on the high rate following a mainshock (Hainzl, 2016). The present likelihood models the selected recorded stream directly; it estimates neither a detection function nor a magnitude-error distribution. Immigrant-only attenuation cannot reduce the offspring component directly. Multiplying both components by an attenuation field supplies a different occurrence model, not an estimated observation mechanism. Section 6.1 fixes the resulting interpretation throughout the empirical chapters.

2.6 Canonical proper-score semantics

The logarithmic score is proper when competing predictive laws are evaluated on the same observation event and reference measure (Gneiting and Raftery, 2007). Its expectation concerns the predictive distribution; a realized score ranking does not establish calibration or a unique physical mechanism.

Let $R(t) = \int_D \lambda(t, z)\, \mathrm{d}z$ and let $\mathcal{W}$ be the scored window. All quantities below use the same fitted state and the same predictable observed history:

$$\begin{aligned} P &= \sum_{i \in \mathcal{W}} \log \lambda(t_i, z_i), & Q &= \sum_{i \in \mathcal{W}} \log R(t_i), \\ E &= \int_{\mathcal{W}} \int_D \lambda(t, z)\, \mathrm{d}z\, \mathrm{d}t, & C &= -E, \\ M &= \sum_{i \in \mathcal{W}} \log f(m_i \mid \mathcal{H}_{t_i}, t_i, z_i). \end{aligned} \tag{2.15}$$

Thus

$$\boxed{\begin{aligned} \text{Timing} &= Q + C = Q - E, \\ \text{Conditional location} &= P - Q, \\ \ell_{\text{score}} &= P + C + M = \text{Timing} + \text{Conditional location} + M. \end{aligned}} \tag{2.16}$$

For temporal laws $P = Q$ and the conditional-location term is zero. For class-history laws the conditional class probability belongs to M, avoiding double counting. E is nonnegative exposure, with expected-count dimension; it contributes negatively to a log score. A candidate-minus-reference difference obeys $\Delta C = -\Delta E$ and $\Delta \ell_{\text{score}} = \Delta P + \Delta C + \Delta M$. Positive differences favour the candidate. There is no allocation of national exposure to event-only geographical groups unless an explicit spatial intensity supplies it.

The pinned `validation/authorities/score_field_semantics_v2.json` is the documentary authority for this translation. It changes neither source fields nor scalar values. The repeated score check in Chapter 6 fixes signs before any table is interpreted.

Table 2.3: Master score-semantics crosswalk. Raw keys remain unchanged in the saved authorities; their scientific meaning is translated here.

Saved context/key	Canonical meaning
Early temporal or LI pooled raw `E`	Event point-logsum contrast ΔP (temporal ΔQ), not positive exposure
Early temporal or LI pooled raw `C`	Signed exposure contrast $\Delta C = -\Delta E$
Sequence `timing_compensator`	Difference ΔE in nonnegative exposure; it can have either sign and is subtracted
Benchmark `integrated_intensity`	E in arm rows, ΔE in contrast rows
JMA `positive_C` / `signed_negative_C`	Positive exposure / its negative contribution
Marginal mark score	M, included once; it cancels only when the compared conditional mark laws agree

Chapter 3

Finite-Mode Stability and Exact-Memory Obstructions

The finite-mode model admits a comparatively complete stability theory under autonomous loading and integrable marks. The theory combines a branching domination, a Lyapunov function, a synchronous coupling and a small-set estimate. Kernel approximation is then controlled on a fixed horizon. Finally, exact memory reveals a sharp obstruction without feedback and a resolvent response for an associated linear system; neither statement supplies a general nonlinear stationary theory.

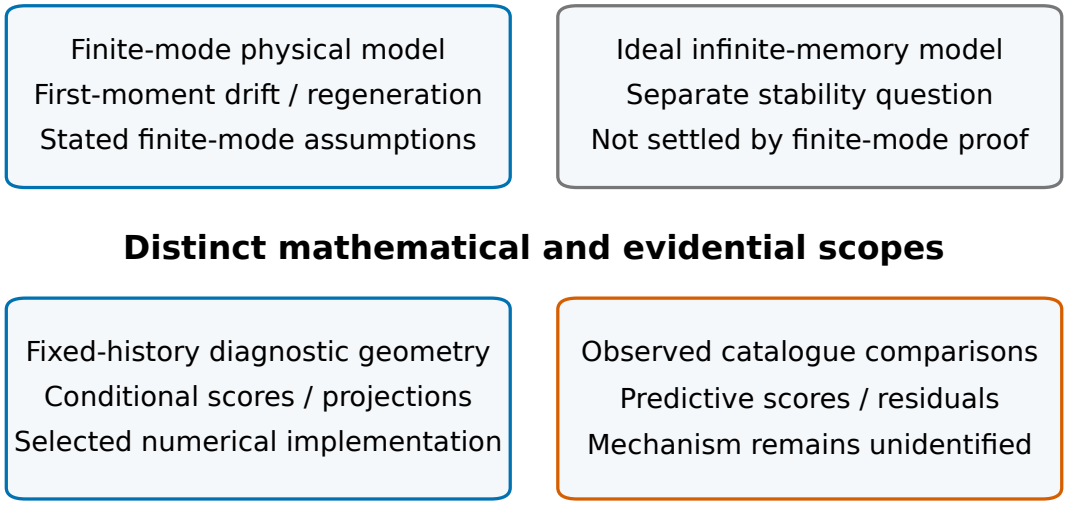


Figure 3.1: The positive results have different assumption sets and limiting regimes. Finite-mode invariant laws, fixed-horizon approximation and the auxiliary response do not combine into a nonlinear exact-memory ergodicity theorem.

Initialization and theorem scope. The autonomous finite-mode result concerns the stated finite-dimensional state and its admissible initial states. It must be distinguished from the later finite-horizon kernel comparison, which fixes a common finite initial stress and no prehistory, and from the marked-resolvent identity, which assumes scalar initial stress $X_0\Phi(t)$, zero initial excitation and no additional prehistory. An arbitrary initialized excitation state contributes an additional homogeneous term; it is not covered by silently reusing that scalar resolvent display. None of these statements proves convergence of invariant laws as the number of modes increases. The physical stress coordinate in this chapter is the theoretical state, rather than the observed diagnostic's affine time coordinate.

3.1 Autonomous finite-mode stability

Fix the finite index set $j \in \{1, \dots, m\}$, $m < \infty$, from Section 2.1, and return to the physical convention $n = \mathbb{E}_F \kappa(M)$. Suppose $m_1 < \infty$, $n < 1$, and $\Psi : (-\infty, \bar{x}] \to (0, \infty)$ is nondecreasing and Lipschitz on the whole half-line. Write $\bar{\Psi} = \Psi(\bar{x})$. Put

$$y_j = x_j^* - Z_j, \qquad Y = \sum_j w_j y_j, \qquad W = \sum_j \frac{w_j U_j}{\vartheta_j}, \qquad V = 1 + Y + cW,$$

where

$$c > \frac{m_1}{1-n}. \tag{3.1}$$

The following theorem concerns constant ρ and fixed Ψ; it is not a theorem about a time-varying diagnostic baseline.

Theorem 3.1 (Finite-mode stability)**.** *From every finite state in $\mathcal{X}$, the autonomous finite-mode model has a pathwise unique, non-explosive, cadlag strong Markov construction and a Feller semigroup P_t. There is a unique invariant probability π satisfying $\pi(V) < \infty$. For suitable positive ϵ, β, C, r, with*

$$\|z - z'\| = \sum_j w_j (|Z_j - Z_j'| + |U_j - U_j'|), \qquad d_\epsilon(z, z') = 1 \wedge \frac{\|z - z'\|}{\epsilon},$$

and

$$\widetilde{d}(z, z') = \sqrt{d_\epsilon(z, z')\{1 + \beta V(z) + \beta V(z')\}},$$

one has

$$W_{\widetilde{d}}(P_t(z, \cdot), \pi) \le C e^{-rt} \sqrt{1 + V(z)}, \qquad t \ge 0. \tag{3.2}$$

In particular, $P_t(z, \cdot)$ converges weakly to π.

Here W_c is the infimum of the expected cost c over couplings. The weighted cost $\widetilde{d}$ need not satisfy a triangle inequality; the conclusion is not a total-variation mixing statement.

The constants generally depend on the rates, weights, m_1, n, the Lipschitz constant of Ψ and $\bar{\Psi}$. No uniformity is claimed as $\vartheta_{\min} = \min_j \vartheta_j$ tends to zero; the contractive time and the selected radius may deteriorate in that limit.

Numerical scale of the certificate. For the implemented 195-mode mixture, the smallest rate is $\vartheta_{\min} = e^{-38.8}\,\text{day}^{-1}$. The proof below requires $t_0 \ge \log(4)/\vartheta_{\min}$, already about $9.8284 \times 10^{+16}$ days. Thus its sufficient drift and coupling bounds give no useful mixing-time guarantee on the observed horizon. Of these modes, 61 decay by less than 1% over 24,394 days, but their combined normalized relaxation weight is only 0.00042934. This weight describes their contribution to Φ, not a bound on arbitrary initialized states or accumulated stress. Neither the mode count nor the conservative proof certificate estimates the actual process mixing rate. The scalar calculation and saved-fit protocol comparison are reproducible through `experiments/methodological_audit/stability.py`.

Proof. Construction and nonexplosion. Use a marked Poisson embedding with thinning by the predictable intensity $\Psi(X) + H$. The process is dominated by a marked Hawkes process with immigration $\bar{\Psi}$, kernel

$$h_d(t) = \sum_j w_j \vartheta_j e^{-\vartheta_j t},$$

and initial excitation $\sum_j w_j U_j(0) e^{-\vartheta_j t}$. The latter integrates to $W(z)$, and each cluster has expected total size $1/(1-n)$. For every nonnegative mark weight χ with finite mean,

$$\mathbb{E}_z \sum_{T_i \le T} \chi(M_i) \le \frac{\mathbb{E}_F \chi(M)}{1-n} \{\bar{\Psi} T + W(z)\}. \tag{3.3}$$

A generation is created before its marks are drawn, so (3.3) remains valid when χ and κ are correlated. Finite expected counts imply nonexplosion; the deterministic flow and fresh Poisson increments yield the strong Markov property.

Lyapunov estimate. Since the flow gives $\dot{Y} = -\sum_j w_j \vartheta_j y_j$ and $\dot{W} = -H$, while a jump adds (g, κ), the extended generator satisfies

$$\mathcal{A}V = -\sum_j w_j \vartheta_j y_j - \delta H + (m_1 + cn)\Psi(X), \qquad \delta = c(1-n) - m_1 > 0.$$

Because $\Psi(X) \le \bar{\Psi}$ on $\mathcal{X}$, direct comparison with the positive relaxation terms gives

$$\mathcal{A}V \le -\kappa_V V + D, \tag{3.4}$$

where

$$K_0 = (m_1 + cn)\bar{\Psi}, \qquad \kappa_V = \vartheta_{\min} \min\{1, \delta/c\}, \qquad D = \kappa_V + K_0.$$

The weighted jump bound (3.3) makes the stopped compensated jump integral integrable. Applying the product formula to $e^{\kappa_V t} V(z_t)$ and then Fatou's lemma gives

$$P_t V(z) \le e^{-\kappa_V t} V(z) + \frac{D}{\kappa_V}(1 - e^{-\kappa_V t}). \tag{3.5}$$

Coupling and smallness. Couple two copies synchronously until their first unmatched event σ. Identical jumps cancel before σ, so

$$\|z_t - z'_t\| \le e^{-\vartheta_{\min} t} \|z - z'\|$$

on $\{\sigma > t\}$. Bounding the compensator of the difference of their rates by $L\|z_t - z'_t\|$, with $L = \max\{L_\Psi, 1\}$, gives

$$\mathbb{P}(\sigma \le t) \le \frac{L}{\vartheta_{\min}} \|z - z'\|.$$

Thus, whenever $d_\epsilon(z, z') < 1$,

$$W_{d_\epsilon}(P_t z, P_t z') \le \left(e^{-\vartheta_{\min} t} + \frac{L\epsilon}{\vartheta_{\min}} \right) d_\epsilon(z, z'). \tag{3.6}$$

First fix $R = 4D/\kappa_V$ from the Lyapunov bound. Because the mode set is finite, $\{V \le R\}$ is compact and has finite diameter D_R. Next choose $\epsilon > 0$ so that $L\epsilon/\vartheta_{\min} \le 1/4$, and then choose t_0 so that

$$e^{-\vartheta_{\min} t_0} \le 1/4, \qquad e^{-\vartheta_{\min} t_0} D_R \le \epsilon/2.$$

Thus (3.6) has coefficient at most $1/2$ for every $t \ge t_0$. On the fixed sublevel set, event-free flow decreases both Y and W, and the event rate is at most $\Lambda_R = \bar{\Psi} + \vartheta_{\max} R/c$. Use the same marked Poisson random measure Π for both copies, with intensity $\mathrm{d}s\ \mathrm{d}u\, F(\mathrm{d}m)$ and the usual acceptance threshold. The event

$$\Pi([0, t] \times [0, \Lambda_R] \times \mathsf{M}) = 0$$

has probability $e^{-\Lambda_R t}$, since F has total mass one. On this event both copies follow their event-free flows and their rates remain at most Λ_R, so neither jumps. Thus the synchronous coupling gives probability at least $e^{-\Lambda_R t}$ that neither copy jumps. Once $e^{-\vartheta_{\min} t} D_R \le \epsilon/2$,

$$\sup_{V(z),V(z')\le R} W_{d_\epsilon}(P_t z, P_t z') \le 1 - \tfrac{1}{2} e^{-\Lambda_R t} < 1.$$

The same coupling shows that the state at a fixed time converges in probability as the initial state varies continuously. Expectations of bounded continuous functions therefore converge, establishing the Feller property. Taking $t = t_0$ verifies contraction and smallness at the same time; the choices were made in the order R, ϵ, t_0.

Weighted contraction and continuous time. The deduction has four steps.

1. The function $d_\epsilon = 1 \wedge (\|z - z'\|/\epsilon)$ is a bounded complete metric on the closed finite-dimensional state space $\mathcal{X}$. Equation (3.5) supplies the Lyapunov condition (4.1) of Hairer et al. (2011), with $C_V = 1$ and $K_V = D/\kappa_V$. Equation (3.6) gives contraction with coefficient $1/2$ at t_0 when $d_\epsilon < 1$, and the preceding display gives d_ϵ-smallness of $\{V \le 4K_V\}$ at the same time. These are their Definitions 4.7 and 4.4, respectively. Increasing t_0 if necessary preserves both estimates and ensures the drift reduction required in the proof of their Theorem 4.8.
2. The weighted-cost construction in that theorem gives $\beta > 0$, $\widetilde{\alpha} \in (0,1)$ and a skeleton time (denoted again by t_0) for which

$$W_{\widetilde{d}}(P_{t_0}\mu, P_{t_0}\nu) \le \widetilde{\alpha} W_{\widetilde{d}}(\mu,\nu), \qquad \widetilde{d}(z,z') = \sqrt{d_\epsilon(z,z')(1 + \beta V(z) + \beta V(z'))}.$$

   The skeleton time need not equal the first time at which the unweighted hypotheses hold. Tight time-averaged laws, compact sublevels, the Feller property and (3.5) give an invariant law π; the theorem gives uniqueness. Iteration and invariance give

$$W_{\widetilde{d}}(P_{nt_0} z, \pi) \le \widetilde{\alpha}^n W_{\widetilde{d}}(\delta_z, \pi).$$

3. For $0 \le s < t_0$, use the same synchronous coupling. When $d_\epsilon(z,z') < 1$, (3.6) bounds its expected d_ϵ-cost by $(1 + L\epsilon/\vartheta_{\min}) d_\epsilon(z,z') \le (5/4) d_\epsilon(z,z')$. When $d_\epsilon = 1$ this also holds since the cost is at most one. Moreover, (3.5) gives $P_s V(z) + P_s V(z') \le V(z) + V(z') + 2D/\kappa_V$ uniformly in s. Cauchy–Schwarz on this same coupling therefore gives

$$\begin{aligned} W_{\widetilde{d}}(P_s z, P_s z') &\le \sqrt{\tfrac{5}{4} d_\epsilon(z,z')(1 + \beta V(z) + \beta V(z') + 2\beta D/\kappa_V)} \\ &\le C_0 \widetilde{d}(z,z'), \qquad C_0 = \sqrt{\tfrac{5}{4}(1 + 2\beta D/\kappa_V)}. \end{aligned}$$

   Integrating these measurable coupling kernels against any coupling of μ, ν and taking the infimum proves $W_{\widetilde{d}}(P_s\mu, P_s\nu) \le C_0 W_{\widetilde{d}}(\mu,\nu)$.
4. Write $t = nt_0 + s$, $0 \le s < t_0$. Then $P_t = P_s P_{nt_0}$ and $P_s \pi = \pi$. The invariant moment estimate proved below gives $\pi(V) \le D/\kappa_V$; hence Jensen's inequality gives

$$W_{\widetilde{d}}(\delta_z, \pi) \le \sqrt{1 + \beta V(z) + \beta D/\kappa_V}.$$

   Consequently

$$W_{\widetilde{d}}(P_t z, \pi) \le C_0 \widetilde{\alpha}^n \sqrt{1 + \beta V(z) + \beta D/\kappa_V} \le C e^{-rt} \sqrt{1 + V(z)}, \qquad r = -\frac{\log \widetilde{\alpha}}{t_0},$$

   where $\widetilde{\alpha}^n \le \widetilde{\alpha}^{-1} e^{-rt}$. This is (3.2); it uses only first mark moments.

To see that every invariant probability has the same finite moment without subtracting potentially infinite integrals, suppose $PV \le aV + b$ with $a < 1$, and put $f_K(v) = \min(v, K)$. If ν is invariant, concavity gives

$$\nu\{f_K(V) - f_K(aV + b)\} \le 0.$$

For $K \ge b/(1-a)$, the integrand is at least $[(1-a)V - b]\mathbf{1}_{\{V \le K\}}$. Hence $\nu(V\mathbf{1}_{\{V\le K\}}) \le b/(1-a)$. Monotone convergence and (3.5) yield $\nu(V) \le D/\kappa_V$. The weak Harris theorem supplies uniqueness. □

Under the invariant law, let $\Lambda = \pi(\Psi(X) + H) > 0$. Since $\pi(V) < \infty$, every U_j and $|Z_j|$ is integrable, and $\Lambda < \infty$. The mild coordinate equations and fresh-mark compensation therefore give, for $t > 0$,

$$\pi(U_j) = e^{-\vartheta_j t}\pi(U_j) + n\Lambda(1 - e^{-\vartheta_j t}),$$
$$\pi(Z_j) = e^{-\vartheta_j t}\pi(Z_j) + \frac{\rho - m_1\Lambda}{\vartheta_j}(1 - e^{-\vartheta_j t}).$$

All expectations are finite before cancellation: the expected marked jump sums are bounded by their first mark moments times Λt. Cancelling $1 - e^{-\vartheta_j t} > 0$, then using $H = \sum_j w_j U_j$, gives

$$\pi(U_j) = n\Lambda, \qquad \pi(Z_j) = \frac{\rho - m_1\Lambda}{\vartheta_j}, \qquad \Lambda = \frac{\pi(\Psi(X))}{1-n},$$

and hence

$$\pi(X) = (\rho - m_1\Lambda)\sum_j \frac{w_j}{\vartheta_j}.$$

Under $\mathbb{P}_\pi$, intensity ergodicity and the compensated counting martingale give $N(t)/t \to \Lambda$, while the fresh-mark strong law gives $t^{-1}\sum_{T_i \le t} g(M_i) \to m_1\Lambda$.

Proposition 3.2 (Infinite stationary second moments)**.** *If $\mathbb{E}_F g(M)^2 = \infty$, then $\pi(Y^2) = \infty$ and consequently $\pi(X^2) = \infty$. If $\mathbb{E}_F \kappa(M)^2 = \infty$, then $\pi(W^2) = \infty$ and consequently $\pi(H^2) = \infty$.*

Proof. Let $\Phi_d(t) = \sum_j w_j e^{-\vartheta_j t} > 0$. The mild equations for the nonnegative depletion modes y_j and the integrated excitation modes give, on any stationary interval of length t,

$$Y(0)^2 \ge \Phi_d(t)^2 \sum_{-t < T_i \le 0} g(M_i)^2, \qquad W(0)^2 \ge \Phi_d(t)^2 \sum_{-t < T_i \le 0} \kappa(M_i)^2.$$

Replace a squared mark by $g_K^2 = (g \wedge K)^2$. Fresh-mark compensation, applied before each mark is revealed, gives

$$\mathbb{E}_\pi \sum_{-t < T_i \le 0} g_K(M_i)^2 = \mathbb{E}_F g_K(M)^2 \int_{-t}^{0} \mathbb{E}_\pi \lambda_s \, \mathrm{d}s = t\Lambda \mathbb{E}_F g_K(M)^2.$$

Thus $\pi(Y^2) \ge \Phi_d(t)^2 t\Lambda \mathbb{E}_F g_K(M)^2$. Letting $K \uparrow \infty$ proves the first implication. The productivity argument is identical. Since $X = \bar{x} - Y$, infinite $\pi(Y^2)$ implies infinite $\pi(X^2)$. Since W and H are comparable through the finitely many positive rates, infinite $\pi(W^2)$ implies infinite $\pi(H^2)$. □

The argument concerns the corresponding jump transform: it asserts stress divergence when the stress-mark second moment diverges, and excitation divergence when the productivity second moment diverges. Finite first-moment stability and infinite stationary variance coexist. A quadratic theorem under stronger finite-moment assumptions is not contradicted; those stronger assumptions cannot be inferred for an untruncated heavy-tailed fitted law.

3.2 Fixed-horizon kernel approximation

Let Φ and Φ_d be nonnegative decreasing absolutely continuous kernels, normalised to one at zero and tending to zero at infinity, so that $h = -\Phi'$ and $h_d = -\Phi_d'$ are probability densities. Use the same scalar initial term $X_0\Phi(t)$ (and $X_0\Phi_d(t)$), zero initial excitation, constant loading, no prehistory, fresh marks, and the first-moment assumptions above. In a finite mixture this scalar initial term means equal initial stress coordinates. These are not stationary initializations drawn from π_d. Let Ψ be positive, nondecreasing and locally Lipschitz on $\mathbb{R}$. For fixed $T < \infty$, put

$$B = A_T = |X_0| + \rho T, \qquad K_T = \frac{\Psi(B)T}{1-n},$$

$$\Delta_\Phi = \sup_{[0,T]} |\Phi_d - \Phi|, \qquad \Delta_h = \int_0^T |h_d - h|.$$

Proposition 3.3 (Quantitative half-line approximation)**.** *If Ψ is Lipschitz on $(-\infty, B]$ with constant L_B, then, under the same Poisson-measure coupling,*

$$\mathbb{P}(\textit{marked catalogues differ on } [0,T]) \le \min\{1,\ L_B T(A_T + m_1 K_T)\Delta_\Phi + nK_T\Delta_h\}. \tag{3.7}$$

For $\Psi(x) = e^{a+bx}$, $b \ge 0$, one may take $L_B = be^{a+bB}$.

Proof. All stresses remain at most B. Until the two coupled processes first separate,

$$|X_d(s) - X(s)| \le \Delta_\Phi \left(A_T + \sum_{T_i < s} g(M_i) \right),$$

and the integrated excitation discrepancy is bounded by $\Delta_h \sum_{T_i \le T} \kappa(M_i)$. Common pre-separation events form a subset of the reference events, which are dominated by the Hawkes process with immigration $\Psi(B)$. Its expected event count is at most K_T, while the corresponding weighted expectations are at most $m_1 K_T$ and nK_T. Bounding the compensated difference of the two event intensities by the stress discrepancy and the excitation discrepancy and integrating its expectation gives (3.7). □

Local Lipschitz continuity does not support the same linear error bound. It nevertheless gives qualitative convergence through a separate truncation.

Proposition 3.4 (Qualitative local approximation)**.** *For $R > 0$, let L_R be a Lipschitz constant of Ψ on $[-|X_0| - R, B]$. Then*

$$\mathbb{P}(\textit{catalogues differ on } [0,T]) \le \min\left\{1, \frac{m_1 K_T}{R} + L_R T(A_T + R)\Delta_\Phi + nK_T\Delta_h\right\}. \tag{3.8}$$

Proof. On common prehistories whose cumulative release is at most R, both stresses lie in the compact interval on which L_R is valid, and the half-line calculation applies with R in place of the full released mass. The complement has probability at most $m_1 K_T / R$ by Markov's inequality. Fix R, let the two kernel errors tend to zero, and then let $R \to \infty$. □

Thus half-line Lipschitz continuity gives a quantitative error rate, whereas local Lipschitz continuity gives only qualitative fixed-horizon convergence. A finite half-line constant cannot be inferred from local Lipschitz continuity. The constants depend on T through B, K_T and the relevant Lipschitz constant, and can deteriorate rapidly as the horizon increases. Neither bound is uniform in T, and neither exchanges the kernel-approximation and long-time limits.

3.3 Exact-memory obstruction and response

Exact memory neither prevents all stationary constructions nor yields a general nonlinear stationary theorem. The first point is visible already in a zero-history model without feedback.

Proposition 3.5 (No-feedback obstruction)**.** *Let $0 < \alpha < 1$, $\tau > 0$, and $\Phi(t) = E_\alpha(-(t/\tau)^\alpha)$. Consider the zero-history model with $X_0 = 0$, constant immigration rate $\mu > 0$, no triggering, and loading $\rho = \mu \mathbb{E}S$. Let $N(\mathrm{d}v, \mathrm{d}S)$ be Poisson with intensity $\mu \, \mathrm{d}v \, F_S(\mathrm{d}S)$, let $\widetilde{N}$ be its compensated measure, and define*

$$X(t) = -\int_0^t \Phi(t-v) S \, \widetilde{N}(\mathrm{d}v, \mathrm{d}S).$$

If $S = 1$, no sequence $t_k \to \infty$ has a tight subsequence of stress laws for $\alpha \le 1/2$. If $P(S > s) = s^{-q}$ for $s \ge 1$ and $1 < q < 2$, the corresponding obstruction occurs when $\alpha q \le 1$. For $S = 1$, $\alpha > 1/2$ admits a stationary two-sided compensated moving average in L^2, and the zero-history one-time laws converge. For Pareto marks, $\alpha q > 1$ admits such a construction in L^p for every $1/\alpha < p < q$.

Proof. The finite-interval Poisson exponential formula gives

$$\left| \mathbb{E} e^{iuX(t)} \right| = \exp\left\{ -\mu \int_0^t \mathbb{E}_F\{1 - \cos(uS\Phi(v))\} \, \mathrm{d}v \right\}. \tag{3.9}$$

For unit marks, the integrand is asymptotic to $u^2\Phi(v)^2/2$ for fixed $u \neq 0$. For Pareto marks and $0 < x \le 1$, restricting S to $[1/x, 2/x]$ yields

$$\mathbb{E}_F\{1 - \cos(xS)\} \ge (1 - \cos 1)(1 - 2^{-q}) x^q.$$

Since $\Phi(v) \sim \tau^\alpha v^{-\alpha}/\Gamma(1-\alpha)$, the modulus (3.9) tends to zero at every nonzero frequency in the stated regimes. Tightness would give a weak subsequential limit; by the continuity theorem its characteristic function would be continuous at zero, but the limit would equal zero at every $u \neq 0$ and one at $u = 0$, a contradiction.

Above the threshold, the two-sided compensated integrals converge in L^2 by the isometry for unit marks. For Pareto marks, choose $1/\alpha < p < q$. Then $\mathbb{E}S^p < \infty$ and $\int_0^\infty \Phi(v)^p \, \mathrm{d}v < \infty$, so the compensated Poisson L^p estimate gives the two-sided construction. □

This is a no-feedback statement. It does not assert nonstationarity of the stress-sensitive nonlinear model, and a stationary compensated construction is not the subtraction of two separately divergent, uncompensated past integrals.

Even the stationary unit-mark construction for $\alpha > 1/2$ has unbounded upper support. At a fixed stationary time split its stress as

$$X_\infty = -\int_0^L \Phi(u) \, \widetilde{N}(\mathrm{d}u) + R_L, \qquad \mathbb{E}R_L^2 = \mu \int_L^\infty \Phi(u)^2 \, \mathrm{d}u \longrightarrow 0,$$

where the recent and remote Poisson restrictions are independent. For any K, choose L large enough that $\mu \int_0^L \Phi(u) \, \mathrm{d}u > K + 1$ and $\mathbb{E}R_L^2 < 1$. No recent event has probability $e^{-\mu L} > 0$, and $\Pr(R_L > -1) > 0$ by the second-moment bound. Their intersection gives $X_\infty > K$. Hence $\operatorname{ess\,sup} X_\infty = \infty$: finite variance does not provide a deterministic upper stress cap. For $b > 0$, e^{a+bX_∞} is unbounded above, so the finite-mode half-line argument cannot be transferred to this law. Introducing that stress-dependent rate changes the no-feedback process and requires separate analysis; unbounded support alone proves neither nonexistence of a nonlinear invariant law nor divergence of the mean exponential rate.

For an auxiliary excitation-inclusive system, a positive stress-to-rate slope changes the response through an integrable resolvent with a heavy tail. Let $m_1 > 0$, $0 \le n < 1$ and $\ell > 0$, and consider

$$x = -m_1 \Phi * y, \qquad y = \ell x + n h * y + \xi, \qquad h = -\Phi', \tag{3.10}$$

with zero initial perturbation and additive event-flux forcing ξ. Set

$$\gamma = m_1 \ell, \qquad \beta_{\mathrm{rel}} = 1 - \alpha, \qquad a_{\mathrm{rel}} = (1-n)\tau^{-\alpha}.$$

Proposition 3.6 (Excitation-inclusive response)**.** *The causal solution of* (3.10) *is* $x = -m_1 R_\gamma * \xi$*, where*

$$\widehat{R}_\gamma(s) = \frac{1}{s + a_{\mathrm{rel}} s^{\beta_{\mathrm{rel}}} + \gamma}. \tag{3.11}$$

For $\gamma > 0$*,* R_γ *is completely monotone,* $R_\gamma(0+) = 1$*,* $\int_0^\infty R_\gamma(t)\,\mathrm{d}t = 1/\gamma$*, and*

$$R_\gamma(t) \sim \frac{a_{\mathrm{rel}} \beta_{\mathrm{rel}}}{\gamma^2 \Gamma(1-\beta_{\mathrm{rel}})} t^{-1-\beta_{\mathrm{rel}}}. \tag{3.12}$$

In particular, $\int_0^\infty t R_\gamma(t)\,\mathrm{d}t = \infty$*.*

Proof. Eliminating y in the Laplace domain gives

$$\widehat{\Phi}(s) = \frac{s^{\alpha-1}}{s^\alpha + \tau^{-\alpha}}, \qquad \widehat{h}(s) = \frac{\tau^{-\alpha}}{s^\alpha + \tau^{-\alpha}}.$$

Substitution into (3.10) yields

$$\widehat{x}(s) = -\frac{m_1 s^{\alpha-1} \widehat{\xi}(s)}{s^\alpha + \gamma s^{\alpha-1} + (1-n)\tau^{-\alpha}} = -m_1 \widehat{R}_\gamma(s) \widehat{\xi}(s),$$

which is (3.11). The reciprocal of the nonnegative complete Bernstein function $s + a_{\mathrm{rel}} s^{\beta_{\mathrm{rel}}} + \gamma$ is a Stieltjes function, hence R_γ is completely monotone (Meerschaert and Toaldo, 2019, Section 2.1). Its spectral density is

$$k_\gamma(r) = \frac{a_{\mathrm{rel}} \sin(\pi\beta_{\mathrm{rel}}) r^{\beta_{\mathrm{rel}}}}{\pi\{[\gamma - r + a_{\mathrm{rel}} r^{\beta_{\mathrm{rel}}} \cos(\pi\beta_{\mathrm{rel}})]^2 + a_{\mathrm{rel}}^2 r^{2\beta_{\mathrm{rel}}} \sin^2(\pi\beta_{\mathrm{rel}})\}},$$

so $R_\gamma(t) = \int_0^\infty e^{-rt} k_\gamma(r)\,\mathrm{d}r$. The limits of (3.11) at infinity and zero give $R_\gamma(0+) = 1$ and $\int R_\gamma = 1/\gamma$. Near $r = 0$,

$$k_\gamma(r) \sim \frac{a_{\mathrm{rel}} \sin(\pi\beta_{\mathrm{rel}})}{\pi\gamma^2} r^{\beta_{\mathrm{rel}}}.$$

Rescaling the spectral integral and using the gamma reflection formula gives (3.12). The tail exponent $-1-\beta_{\mathrm{rel}}$ lies between -2 and -1, so the first temporal moment diverges. □

The weak-feedback crossover is equally explicit. Put $t_\gamma = (a_{\mathrm{rel}}/\gamma)^{1/\beta_{\mathrm{rel}}}$. As $\gamma \downarrow 0$, the probability densities $\gamma t_\gamma R_\gamma(t_\gamma u)$ on $(0,\infty)$ converge weakly to the law with Laplace transform $(1+z^{\beta_{\mathrm{rel}}})^{-1}$, because

$$\int_0^\infty e^{-zu} \gamma t_\gamma R_\gamma(t_\gamma u)\,\mathrm{d}u = \frac{1}{1 + z^{\beta_{\mathrm{rel}}} + z/(\gamma t_\gamma)} \longrightarrow \frac{1}{1+z^{\beta_{\mathrm{rel}}}}.$$

At $\gamma = 0$, $R_0(t) = E_\alpha(-a_{\mathrm{rel}} t^\alpha)$ is not integrable. The crossover is separated from the early-time scale $a_{\mathrm{rel}}^{-1/\alpha}$ only when $\gamma \ll a_{\mathrm{rel}}^{1/\alpha}$. This is a statement about an auxiliary impulse response, not a nonlinear mixing time.

An abstract signed-response equation illustrates the limit of this resolvent method. Let $\psi : \mathbb{R} \to \mathbb{R}$, $\psi(0) = 0$, and consider

$$y = \psi(x) + nh * y + \xi, \qquad \mathcal{E}(x) = m_1\psi(x) - \gamma x.$$

If $\mathcal{E}$ is globally Lipschitz with constant $L < \gamma$, the map

$$x = -R_\gamma * (m_1\xi + \mathcal{E}(x))$$

has contraction constant at most $L\|R_\gamma\|_1 = L/\gamma < 1$ on bounded causal functions, and $\|x\|_\infty \le m_1\|\xi\|_\infty/(\gamma - L)$. This nonempty abstract class includes the signed linear response $\psi(x) = \gamma x/m_1$. For jointly measurable stationary forcing on $\mathbb{R}$ with an almost surely uniform deterministic bound, the analogous two-sided causal resolvent map has a stationary fixed point by shift-equivariant Picard iteration. This assertion concerns the integrable-resolvent equation, not absolute convergence of a two-sided $\Phi * y$.

It is *not* a global contraction criterion for the physical positive link. If $\psi(x) = \Psi(x + x_*) - \Psi(x_*)$ on its lower-unbounded domain and $\Psi > 0$, then $\psi(x) \ge -\Psi(x_*)$ and $\mathcal{E}(0) = 0$. For negative x,

$$\mathcal{E}(x) \ge \gamma|x| - m_1\Psi(x_*),$$

so any global Lipschitz constant of $\mathcal{E}$ is at least γ. No differentiability assumption is needed for this obstruction. In particular, every positive nondecreasing physical link is excluded, not only the exponential example. A contraction restricted to a compact stress band would additionally require that the map preserve that band; no such invariant-band or exit control is proved here.

The obstruction and resolvent calculation therefore delimit the demonstrated scope. Neither supplies a generic stationary theorem for the nonlinear stress-sensitive exact-memory process.

3.3.1 What the auxiliary response omits

The actual marked process has two noise sources rather than a single prescribed event-flux input. In the physical productivity convention, define compensated marked measures by

$$\mathrm{d}M_g = \mathrm{d}N_g - m_1\lambda\,\mathrm{d}t, \qquad \mathrm{d}M_\kappa = \mathrm{d}N_\kappa - n\lambda\,\mathrm{d}t,$$

where N_g and N_κ accumulate the corresponding marks. These martingales are generally correlated: both jump at the same events, and the two transforms of a fresh mark need not be independent.

In the following exact identity, $\gamma > 0$ is an auxiliary parameter added and subtracted algebraically; it is not an established lower physical restoring slope. Set $Q_\gamma = R_\gamma - nR_\gamma * h$, and let $X^+(t)$ denote post-event stress, including releases at time t. For scalar initial stress $X_0\Phi(t)$, zero initial excitation and no additional prehistory, finite-horizon compensation and the same Laplace-resolvent algebra give the exact identity

$$\begin{aligned} X^+(t) = X_0Q_\gamma(t) &+ \rho\int_0^t Q_\gamma(u)\,\mathrm{d}u \\ &+ [R_\gamma * (\gamma X^+ - m_1\Psi(X^+_-))](t) \\ &- \int_{(0,t]} Q_\gamma(t-u)\,\mathrm{d}M_g(u) \\ &- m_1\int_{(0,t]} (R_\gamma * h)(t-u)\,\mathrm{d}M_\kappa(u). \end{aligned} \tag{3.13}$$

Here $X^+_-(u) = X^+(u^-)$, and all assertions are on finite horizons under compensator integrability. At an event time, the intensity uses the left limit; the displayed post-event identity

must be left-limited consistently if it is used to evaluate that intensity. The deterministic kernels in the two noise terms are bounded and integrable, but this does not by itself control the nonlinear residual in the second line. In particular, the stress can leave any band on which a positive lower slope is available. Establishing tightness or a uniform occupation bound outside that band remains a separate task. The identity explains the relevance of the auxiliary response while retaining the nonlinear term and both correlated mark martingales.

Compensation and cancellation. An event-free path has $\mathrm{d}N_g = \mathrm{d}N_\kappa = 0$, but not $\mathrm{d}M_g = \mathrm{d}M_\kappa = 0$. Instead,

$$\mathrm{d}M_g = -m_1\lambda\,\mathrm{d}t, \qquad \mathrm{d}M_\kappa = -n\lambda\,\mathrm{d}t.$$

The compensated integrals must therefore be retained even when no event occurs. The causal kernels satisfy

$$Q_\gamma + nR_\gamma * h = R_\gamma, \qquad Q_\gamma + \gamma R_\gamma * \Phi = \Phi.$$

For the stated initialization, put $f = X_0\Phi + \rho(1 * \Phi)$, where $(1 * \Phi)(t) = \int_0^t \Phi(u)\,\mathrm{d}u$. In the following ordinary time convolutions write X for X^+, retaining X_- in the predictable intensity. The original equations give

$$X = f - \Phi * \mathrm{d}N_g, \qquad \lambda = \Psi(X_-) + h * \mathrm{d}N_\kappa.$$

Substitution of both compensated measures into the displayed identity, followed by the first kernel relation, reduces its right-hand side to

$$X_0Q_\gamma + \rho(1 * Q_\gamma) - Q_\gamma * \mathrm{d}N_g + \gamma R_\gamma * X.$$

The second kernel relation makes the first three terms $X - \gamma R_\gamma * X$, proving the cancellation. These are finite-horizon causal convolution identities under the stated integrability conditions; no infinite-history transform of a nonlinear sample path or stationary exact-memory limit is asserted.

The archival E04 worked example and expanded derivations remain in the public repository at `paper/supplements/explanatory_supplement.pdf`; compact standalone release bundles omit this supplement.

A uniform invariant-mean bound cannot be inferred from fixed-mode stability. If $\sup_d |\mathbb{E}_{\pi_d} X_d| < \infty$, $m_1 > 0$ and $A_d = \int_0^\infty \Phi_d(t)\,\mathrm{d}t \to \infty$, stationary balance gives the following conditional implication; it establishes neither invariant-law convergence nor an exchange of limits:

$$\mathbb{E}_{\pi_d} X_d = (\rho - m_1\Lambda_d)A_d \quad \Longrightarrow \quad m_1\Lambda_d \longrightarrow \rho.$$

Chapter 4

Structural Identifiability, Boundary Singularity, and Regional Nonclosure

An identifiability question asks whether distinct generative structures produce the same observed law. A visibility question asks how strongly those laws separate on a finite experiment. Exact nonidentification prevents recovery of the unidentified quantity from the observed law, whereas distinct laws can still be difficult to distinguish on a finite horizon. The distinction is essential here: stress memory disappears from the rate at zero feedback, yet under stronger assumptions it becomes identifiable when inhibition is active.

4.1 Physical loading and the diagnostic family

Let Q_i denote marks, let Φ_s be a nonnegative normalized relaxation kernel, and put

$$X(t) = X_0\Phi_s(t) + \rho\int_0^t \Phi_s(u)\,\mathrm{d}u - G_s(t), \qquad G_s(t) = \sum_{T_i<t} g(Q_i)\Phi_s(t-T_i).$$

The physical rate family is

$$\lambda_{\mathrm{phys}}(t) = \exp\{a + bX(t)\} + nH_e(t), \qquad H_e(t) = \sum_{T_i<t} \kappa(Q_i)h_e(t-T_i). \tag{4.1}$$

Here κ is a unit trigger amplitude, $\mathbb{E}\kappa(Q) = 1$, and n is outside H_e. Marks are generated from the common fresh-mark law specified for the experiment. The diagnostic experiment instead uses

$$\lambda_\eta(t) = B(t)e^{-\eta G_s(t)} + nH_e(t), \qquad B(t) = Ae^{d x_{\mathrm{trend}}(t)}, \qquad \eta \geq 0, \tag{4.2}$$

where $A = e^a > 0$ and $x_{\mathrm{trend}}(t) = 2t/T_0 - 1$ uses the fixed declared clock scale $T_0 > 0$. This clock is distinct from physical stress $X(t)$. The same event and mark parents supply G_s and H_e. Thus nH_e has the same unit interpretation as in (4.1).

The independent background $B(t)$ in the diagnostic family is not a reparameterization of the physical loading law. In the diagnostic model, $B(t) = Ae^{d x_{\mathrm{trend}}(t)}$ is an explicitly fitted background trajectory; in the physical model, the stress response is constrained by the relaxation kernel and by the latent initial stress. This distinction is part of the inferential target, not merely a matter of notation. It is also the reason a fitted background trend cannot by itself establish earthquake-induced inhibition. Productivity is displayed in the unit convention, with its branching coefficient separate.

4.2 Exact nonidentification at zero feedback

Remark 4.1 (Stress-kernel nonidentification at zero feedback). In the diagnostic family, $\exp(-\eta G_s)|_{\eta=0} = 1$, so the intensity $B + nH_e$ contains no stress kernel. With background,

triggering and fresh mark law fixed, every normalized stress kernel therefore gives the same observed marked law. In the physical family the identical conclusion follows from $bX = 0$ at $b = 0$ with finite X_0, ρ. A shared kernel can still be estimated through triggering; this does not identify the latent stress response.

The singular closure $b \downarrow 0$ with $p = bX_0$ or $q = b\rho$ fixed is a different model: its log-background is $a + p\Phi_s + q\int_0^t \Phi_s - rG_s$, $r = b$, with a kernel-shaped deterministic trend at $r = 0$. Nonzero p or q requires divergent X_0 or ρ.

4.3 Restricted active identification

Nonidentification in the physical family does not mean that active inhibition is never identifiable. A deliberately restricted marked experiment can remove the zero-feedback ambiguity by exposing the nonlinear intervention term. The following statement is an idealized full-law result, not a finite-sample guarantee.

Proposition 4.2 (Restricted active identification)**.** *Consider the diagnostic family (4.2) with empty initial history, a positive continuous deterministic background B, known positive continuous g, and nonnegative η. Let the known transform $\kappa > 0$ be continuous with $\mathbb{E}\kappa(Q) = 1$ and $\kappa(q) \to \infty$, and let the common fresh mark law have a positive density on $[0, \infty)$. Let h_e be locally integrable at zero, continuous and positive for $u > 0$, normalized to integrate to one, and let Φ_s be nonnegative continuous with $\Phi_s(0) = 1$. Assume all models are nonexplosive with finite mean event counts. Equality of two observed marked laws identifies $B(t)$ on $[0, T]$, the product $nh_e(u)$ for observed lags $0 < u < T$, and η; if $\eta > 0$, Φ_s is identical on $[0, T]$. For the scaled finite positive exponential-mixture family with known weights and relative rates, and $n > 0$, the mixture scale and n are also identified within that restricted family.*

Proof. Before the first event $G_s = H_e = 0$, so the hazard is $B(t)$. The distribution of the first waiting time identifies $\int_0^t B(u)\,\mathrm{d}u$, and continuity identifies B itself. Marked-law equality therefore permits comparison along a history containing one event at s with mark q and no further event before t. Local integrability of h_e gives a positive no-further-event survival on each finite interval. For almost every $s < t$ and $q \geq 0$, equality of the two hazards gives

$$B(t)\exp\{-\eta g(q)\Phi_s(t-s)\} + n\kappa(q)h_e(t-s) = \widetilde{B}(t)\exp\{-\widetilde{\eta} g(q)\widetilde{\Phi}_s(t-s)\} + \widetilde{n}\kappa(q)\widetilde{h}_e(t-s).$$

Continuity extends this almost-everywhere identity to every finite q and positive observed lag. After $B = \widetilde{B}$, divide by $\kappa(q)$ and let $q \to \infty$. The first term is bounded by $B(t)$, while $\kappa(q) \to \infty$; hence the normalized triggering term is identified as $nh_e(u) = \widetilde{n}\widetilde{h}_e(u)$. Subtract this common term. The remaining background is positive on the observation window, so taking logarithms shows that $\eta g(q)\Phi_s(u) = \widetilde{\eta} g(q)\widetilde{\Phi}_s(u)$. Dividing by the known positive $g(q)$, and then letting $u \downarrow 0$ with $\Phi_s(0) = 1$, identifies η. If $\eta > 0$, division by η identifies $\Phi_s(u)$ throughout the observed window.

For the finite mixture $h_e(u) = \sum_j w_j(c_j\tau_e)^{-1}\exp\{-u/(c_j\tau_e)\}$, with known positive c_j, w_j and unknown $\tau_e > 0$, the ratio $-h_e'(0)/h_e(0)$ identifies the common scale τ_e when the identified product nh_e is positive. Its value at zero then identifies n. □

Several limitations are part of the result. If $\eta = 0$, the stress kernel is unidentified; if $n = 0$, the triggering scales are unidentified, even though the background may be known. The proof uses arbitrarily large marks across the full probability law. Bounded marks do not inherit this positive-identification proof: the limiting operation $q \to \infty$ is unavailable, so a separate argument would be needed. Shared triggering may identify a trigger scale when $\eta = 0$, but

it does not validate equality of the latent physical loading laws; the exact nonidentification argument remains applicable to the latent target. Unknown state-dependent mark laws, unmatched histories, random prehistory, and kernel behavior beyond the observation window are outside this restricted statement. Near-zero η, structural identification can coexist with very weak finite-horizon separation.

Frequency of the identifying histories. The positive support used in the proof does not imply that a finite catalogue contains useful instances of these histories. For a first event at s with mark q, the conditional probability of no further event before $s+u$ is

$$S_q(s,u) = \exp\Big\{-\int_0^u B(s+v)e^{-\eta g(q)\Phi_s(v)}\,\mathrm{d}v - n\kappa(q)\int_0^u h_e(v)\,\mathrm{d}v\Big\}.$$

It is positive for every finite q, but for $n>0$ and fixed $u>0$, $S_q(s,u)\le \exp\{-a_u\kappa(q)\}$, where $a_u = n\int_0^u h_e(v)\,\mathrm{d}v > 0$. With exponential productivity this is exponential decay in an exponentially growing function of the mark. The full-law argument remains valid because each finite-mark history has positive support before taking the limit. It supplies no lower bound on the frequency of informative histories, estimation rate, or finite-catalogue uncertainty. Constant productivity, including the productivity-matched candidate in Section 9.16, does not satisfy the unbounded-productivity assumption; this proof does not settle its identification.

4.4 Instantaneous and finite-horizon visibility

To quantify separation without invoking asymptotics, compare the diagnostic intensity λ_η with the noninhibited intensity $\lambda_0 = B + nH_e$ on the same realized history. Define

$$p_B(t) = \frac{B(t)}{B(t)+nH_e(t)}, \qquad \delta_\eta(t) = \frac{\lambda_0(t)-\lambda_\eta(t)}{\lambda_0(t)} = p_B(t)\left(1-e^{-\eta G_s(t)}\right).$$

The identity is an instantaneous comparison at fixed history. It does not simulate the different event history that inhibition would have generated, and therefore it is not a bound on the long-run causal effect of removing immigrant arrivals. Those immigrants would alter later descendants. Likewise, in a stationary uninhibited Hawkes model, $1-n$ can be interpreted as an average immigrant fraction; that is not a pointwise identity for p_B in the varying-background inhibited family. A further masking mechanism can arise in marks: if $\kappa(Q)$ grows faster with mark size than $g(Q)$, then at a fixed positive lag the product $p_B G_s$ can shrink even though the stress drop grows. Recovery time, background level, overlapping events, and kernel tails therefore matter as much as a single extreme mark.

The pointwise inequality $0 \le \delta_\eta \le p_B$ is sharp as $\eta G_s \to \infty$ with p_B held fixed. It is useful for score geometry but is not itself a finite-horizon testing bound. The following entropy result supplies one.

Lemma 4.3 (A weighted-history moment bound)**.** *On $[0,T]$, start with empty history and use the same parents in the nonnegative sums G_s, H_e, with fresh marks, $\kappa>0$ almost surely, $h_e(u)>0$ for $0<u\le T$, and finite $\mathbb{E}N(T)$. Suppose*

$$C_T = \int_0^T \frac{\Phi_s(u)^2}{h_e(u)}\,\mathrm{d}u < \infty, \qquad m = \mathbb{E}\left[\frac{g(Q)^2}{\kappa(Q)}\right] < \infty.$$

With $G_s^2/H_e = 0$ on empty histories,

$$\mathbb{E}\int_0^T \frac{G_s(t)^2}{H_e(t)}\,\mathrm{d}t \le C_T m\,\mathbb{E}N(T). \tag{4.3}$$

Proof. Weighted Cauchy–Schwarz gives

$$\frac{G_s(t)^2}{H_e(t)} \le \sum_{T_i<t} \frac{g(Q_i)^2}{\kappa(Q_i)} \frac{\Phi_s(t-T_i)^2}{h_e(t-T_i)}.$$

Integrating each nonnegative summand over the remaining window and applying Tonelli bounds the left side of (4.3) by $C_T \mathbb{E} \sum_{T_i<T} g(Q_i)^2/\kappa(Q_i)$. Fresh-mark compensation makes the last expectation $m\mathbb{E}N(T)$, even though each mark can affect later event counts. This proves the bound. □

Proposition 4.4 (Finite-horizon visibility bound)**.** *Work on $[0,T]$ with empty history, the same bounded nonnegative deterministic background, normalized nonnegative trigger kernel, stress kernel, almost surely finite nonnegative mark transforms and fresh mark law under every $\eta \ge 0$. Assume $\mathbb{E}\kappa = 1$ and $0 \le n < 1$. For the resulting diagnostic laws,*

$$D_{\mathrm{KL}}(P_\eta \| P_0) \le \mathbb{E}_\eta \int_0^T \frac{B(t)^2}{B(t) + nH_e(t)} \min\{\eta^2 G_s(t)^2, 1\} \, \mathrm{d}t \tag{4.4}$$

$$\le \mathcal{B}, \qquad \mathcal{B} = \int_0^T B(t) \, \mathrm{d}t. \tag{4.5}$$

If also $n > 0$, $B \le B_{\max}$, $\kappa > 0$ almost surely, $h_e(u) > 0$ for $0 < u \le T$, $C_T = \int_0^T \Phi_s(u)^2/h_e(u) \, \mathrm{d}u < \infty$, and $m = \mathbb{E}[g(Q)^2/\kappa(Q)] < \infty$, then

$$D_{\mathrm{KL}}(P_\eta \| P_0) \le \eta^2 K_T, \qquad K_T = \frac{B_{\max}^2 C_T m \mathcal{B}}{n(1-n)}.$$

Consequently every possibly randomized test $\varphi \in [0,1]$ satisfies

$$\mathbb{E}_\eta \varphi \le \mathbb{E}_0 \varphi + \min\{1, \eta\sqrt{K_T/2}\}.$$

Proof. The common mark measure cancels from the likelihood ratio. Ratios in integrated expressions are set to zero where both intensities vanish; such times have no event contribution. On a fixed finite path,

$$L_\eta = \exp\left\{\int_0^T (\lambda_0 - \lambda_\eta) \, \mathrm{d}t\right\} \prod_i \frac{\lambda_\eta(T_i)}{\lambda_0(T_i)} \le e^{\mathcal{B}}.$$

The same bound holds after localization, so the likelihood local martingale is uniformly integrable. Subcritical Hawkes domination gives nonexplosion and $\mathbb{E}_\eta N(T) \le \mathcal{B}/(1-n)$. These facts justify the changed law. The entropy identity is well-defined: $\lambda_\eta \log(\lambda_0/\lambda_\eta) \le \lambda_0 - \lambda_\eta \le B$ bounds the event-log expectation, and the integrated difference is also bounded by $\mathcal{B}$. Substituting $\lambda_\eta = \lambda_0(1-\delta_\eta)$ gives

$$D_{\mathrm{KL}}(P_\eta \| P_0) = \mathbb{E}_\eta \int_0^T \lambda_0 \{(1-\delta_\eta)\log(1-\delta_\eta) + \delta_\eta\} \, \mathrm{d}t.$$

For $0 \le \delta \le 1$, the bracket is at most δ^2, because their difference is $(1-\delta)(-\delta - \log(1-\delta)) \ge 0$. Combining this with $1 - e^{-\eta G_s} \le \min\{\eta G_s, 1\}$ proves the first displayed bound. Since $B^2/(B+nH_e) \le B$, it is at most $\mathcal{B}$.

For the quadratic bound, $B^2 G_s^2/(B+nH_e) \le B_{\max}^2 G_s^2/(nH_e)$. Lemma 4.3 and the count bound therefore give

$$\mathbb{E}_\eta \int_0^T \frac{B^2 G_s^2}{B + nH_e} \, \mathrm{d}t \le \frac{B_{\max}^2 C_T m}{n} \mathbb{E}_\eta N(T) \le K_T.$$

Pinsker's inequality then bounds total variation by the square root of half the entropy, which bounds the difference in expectations of any $[0,1]$-valued test. □

This is an upper bound on distinguishability with nuisances fixed; fitting cannot create information. It is not a necessary detection threshold and does not convert a feasibility calculation into a power theorem. It also does not make structural identification uniform near $\eta = 0$. The next chapter studies the likelihood and score geometry that connect these visibility statements to calibrated inference.

These constants are horizon-dependent. For the exact shared Mittag–Leffler pair, (2.9) gives

$$\frac{\Phi_s(u)^2}{h_e(u)} \sim \frac{\tau^\alpha}{\alpha\Gamma(1-\alpha)} u^{1-\alpha}, \qquad C_T \sim \frac{\tau^\alpha}{\alpha(2-\alpha)\Gamma(1-\alpha)} T^{2-\alpha}.$$

With a fixed positive constant background and fixed $m > 0$, K_T consequently grows as $T^{3-\alpha}$.

Numerical visibility on the catalogue horizon. At $T = 24{,}394$ days, $\alpha = 0.5$ and $\tau = 1$ day, the asymptotic expression is $C_T \simeq 2.8660801 \times 10^6\,\text{days}^2$. Direct quadrature of the implemented 195-mode pair gives

$$C_T^{(195)} = \int_0^T \frac{\Phi_{195}(u)^2}{h_{195}(u)}\,\mathrm{d}u = 2.8662509 \times 10^6\,\text{days}^2.$$

For the frozen New Zealand reference background $B(t) = \exp\{-4.16210847 + 0.23211266(2t/T - 1)\}$ and $n = 0.50177516$, $\mathcal{B} = 383.34958$ and $B_{\max} = 0.01964375\,\text{day}^{-1}$. Under $Q \sim \text{Exp}(\beta)$ with $\beta = 2.84940176$, $r = 0.7$, $g(Q) = e^{\gamma(Q-1)}$, $\gamma = 0.75 \log 10$, and $\kappa(Q) = (1-r)e^{r\beta Q}$,

$$m = \frac{e^{-2\gamma}\beta}{(1-r)\{\beta(1+r) - 2\gamma\}} = 0.21606515, \quad K_T = 3.6644466 \times 10^5, \quad \sqrt{K_T/2} = 428.0448.$$

Thus the Pinsker upper bound is exactly one, and completely vacuous, whenever $\eta \geq \sqrt{2/K_T} \simeq 0.00233620$, including every fitted $\eta \in [0.10, 1.65]$. For a constant positive background, a non-vacuous upper bound requires $\eta = O(T^{-(3-\alpha)/2})$ with a sufficiently small constant; a bound tending to zero requires $\eta = o(T^{-(3-\alpha)/2})$. These are sufficient upper-bound scales, not necessary detection thresholds. The quadrature, mark moment and background integration are checked by `scripts/check_mathematics.py`.

4.5 What spatial pooling preserves

Remark 4.5 (A conditional failure of closure under pooling). Let $B_k \geq 0$, $B_+ = \sum_k B_k > 0$, $p_k = B_k/B_+$, and $\eta > 0$. Jensen's inequality for the strictly convex function $x \mapsto e^{-\eta x}$ gives

$$J_{\text{local}} = B_+ \sum_k p_k e^{-\eta G_k} \geq B_+ e^{-\eta \sum_k p_k G_k},$$

with equality only when the G_k on positive-background regions are equal. Its effective depletion is

$$G_{\text{eff}} = -\eta^{-1} \log\left(\sum_k p_k e^{-\eta G_k}\right). \tag{4.6}$$

It depends on regional allocation, not only on $\sum_k G_k$. For example, with two equal backgrounds B and $g > 0$, the allocations $(0, g)$ and $(g/2, g/2)$ have the same total depletion but local aggregates differ by $B(1 - e^{-\eta g/2})^2 > 0$. Thus no function of the total depletion alone represents every local aggregate. This algebra does not identify the cause of any observed score difference.

Chapter 5

Finite-History Likelihood Identities and Response Tools

The marked likelihood provides the common bridge from structural and visibility questions to fitted inference. It records event times, pre-event intensities, the complete observation window, and the mark factors under the same mark model. Likelihood geometry can then be separated into three questions: whether a score direction has finite second moment, whether amplitude profiling gives a well-defined finite-history optimization, and how the rejection probability of a calibrated rule responds when the generating inhibition changes.

5.1 The full marked likelihood and a score bound

For an observed marked history $(T_i, Q_i)_{i=1}^{N(T)}$, write $G_i = G_s(T_i-)$ and $H_i = H_e(T_i-)$, so the event at T_i is excluded from both. With the common mark density p_Q,

$$\ell(\vartheta, \eta) = \sum_{i=1}^{N(T)} \log\left\{B(T_i)e^{-\eta G_i} + nH_i\right\} - \int_0^T \left\{B(t)e^{-\eta G_s(t)} + nH_e(t)\right\} \mathrm{d}t + \sum_{i=1}^{N(T)} \log p_Q(Q_i). \tag{5.1}$$

Every event term uses its pre-event stress and trigger history, the time integral covers the full horizon, and no first event is deleted. The mark sum belongs to the full likelihood. The first two terms form its event-time contribution, evaluated on the observed marked history; they are not the marginal time-only likelihood. The mark sum cancels in a contrast only when both candidates evaluate the same marks under the same mark law and support; it is constant in ground-rate coordinates when that mark law is held fixed. Ground-rate equality without equality of mark laws is insufficient for equality of marked likelihoods. The likelihood is conditioned on the declared starting history, rather than on the observed count.

For example, let a magnitude excess $Q \geq 0$ have exponential density $\beta e^{-\beta Q}$, and let

$$\kappa(Q) = (1-r)e^{r\beta Q}, \qquad 0 \leq r < 1.$$

Then $\mathbb{E}\kappa = 1$, and the raw triggering amplitude is $n(1-r)e^{r\beta Q}$. Changing β while holding r fixed changes both the future mark distribution and the raw productivity exponent. To preserve a raw law Ce^{cQ} on the observed history under a changed β', one must instead use

$$r' = c/\beta', \qquad n' = C/(1-r'),$$

subject to $\beta' > c$ and the branching constraints. This can preserve the event-time likelihood contribution evaluated on the observed marked history; it does not preserve the full marked law. A mark-only estimate from a training prefix is also not automatically the joint marked-process maximum-likelihood estimate, since the ground-rate term can depend on β through productivity.

For fixed nuisance parameters, the inhibition score is

$$U_\eta = \ell_\eta = \int_0^T -\frac{B(t)e^{-\eta G_s(t)}G_s(t)}{B(t)e^{-\eta G_s(t)} + nH_e(t)} \left\{\mathrm{d}N(t) - \lambda_\eta(t)\,\mathrm{d}t\right\}. \tag{5.2}$$

At $\eta = 0$, this is the score for the declared boundary direction, with the ordinary Hawkes law as reference. The next bound shows that raw heavy-tailed marks need not imply an infinite variance for this particular intensity-weighted score.

Proposition 5.1 (Restricted finite score variance)**.** *On $[0,T]$, with empty initial history and $\eta \geq 0$, let $0 < B(t) \leq B_{\max} < \infty$, $0 < n < 1$, let $\kappa > 0$ almost surely with $\mathbb{E}\kappa = 1$, and let h_e be normalized with $h_e(u) > 0$ for $0 < u \leq T$. Assume the same parents contribute to G_s and H_e, with almost surely finite nonnegative mark transforms and nonnegative Φ_s, $C_T = \int_0^T \Phi_s(u)^2/h_e(u)\,\mathrm{d}u < \infty$, and $m = \mathbb{E}[g(Q)^2/\kappa(Q)] < \infty$. Then the score (5.2) is square integrable and its predictable variance is*

$$\mathbb{E}_\eta U_\eta^2 = \mathbb{E}_\eta \int_0^T \frac{(Be^{-\eta G_s} G_s)^2}{\lambda_\eta}\,\mathrm{d}t \leq \frac{B_{\max}^2}{n} C_T\, \mathbb{E}_\eta N(T) m < \infty.$$

Proof. The normalized subcritical Hawkes envelope gives $\mathbb{E}_\eta N(T) \leq \mathcal{B}/(1-n) < \infty$, where $\mathcal{B} = \int_0^T B(t)\,\mathrm{d}t$. Since $\lambda_\eta \geq nH_e$ and $Be^{-\eta G_s} \leq B_{\max}$, Lemma 4.3 bounds the expected bracket by

$$\mathbb{E}_\eta \int_0^T \frac{(Be^{-\eta G_s} G_s)^2}{\lambda_\eta}\,\mathrm{d}t \leq \frac{B_{\max}^2}{n} \mathbb{E}_\eta \int_0^T \frac{G_s^2}{H_e}\,\mathrm{d}t \leq \frac{B_{\max}^2}{n} C_T m \mathbb{E}_\eta N(T).$$

The score is therefore an L^2 compensated-counting martingale with the stated variance. More generally the same argument works for $n > 0$ if a finite mean count is supplied separately; subcriticality is the sufficient count control used here. □

Raw $\mathbb{E}g^2$ or $\mathbb{E}\kappa^2$ may be infinite while the intensity-weighted ratio has finite moment. Positive n and matched triggering parents supply the required comparison history; the bound is not uniform as $n \downarrow 0$ or as the kernel-ratio integral grows.

Example for the declared temporal, positive-productivity mark law. For $Q \sim \mathrm{Exp}(\beta)$, $g(Q) = e^{\gamma(Q-1)}$ and $\kappa(Q) = (1-r)e^{r\beta Q}$, where $\gamma = \frac{3}{4}\log 10$ is the local stress-mark exponent, distinct from the auxiliary response gain in Chapter 3. The magnitude-six reference factor is included in g; the outer branching coefficient n is not included in κ. Then

$$\mathbb{E}\left[\frac{g(Q)^2}{\kappa(Q)}\right] = \frac{e^{-2\gamma}}{1-r}\frac{\beta}{\beta(1+r)-2\gamma}, \qquad \beta(1+r) > 2\gamma.$$

At the declared synthetic values $\beta = 2.849401764894274$ and $r = 0.7$, the denominator is approximately 1.390105, giving $m \approx 0.216065$. Although both first moments are finite, $2\gamma > \beta$ and $2r\beta > \beta$, so $\mathbb{E}[g(Q)^2] = \mathbb{E}[\kappa(Q)^2] = \infty$. The finite weighted moment verifies this mark condition in Proposition 5.1, not all its other hypotheses by itself. The positive-triggering, matched-parent, finite-horizon and kernel-ratio conditions remain essential to that bound; the bound is not uniform near the simultaneous zero boundary, and no joint Fisher-information, central-limit or Wilks conclusion follows.

Which fitted directions inherit the mark bound. The preceding calculation is specific to its declared productivity ratio. In S02, the matched M2 candidate sets $r = 0$ but retains $g(Q) = e^{\gamma(Q-1)}$; hence its ratio moment is $\mathbb{E}g^2 = \infty$ at the declared mark rate. In the spatial LI+ controls of Section 13.10, the transform is instead $g_+(Q) = e^{\gamma Q}$, with $\beta = 2$ and $r = 0.7$. Its ratio moment is infinite because $\beta(1+r) < 2\gamma$. Original unit-mark LI does not share this stress-mark obstruction. These statements concern ideal exponential mark laws. The

executable calculation at `experiments/methodological_audit/moments.py` records the transforms, moment inequalities and source identities.

Failure of a sufficient upper bound is not by itself a divergence proof. The following argument supplies one for the relevant unstopped null laws with bounded kernels.

Proposition 5.2 (A positive-triggering null obstruction)**.** *Under the uninhibited temporal law, suppose the history starts empty, B is positive continuous and bounded on the finite horizon, $0 < n < 1$, marks are fresh and independent with $\mathbb{E}\kappa = 1$ and $\mathbb{E}g < \infty$, the normalized trigger kernel is bounded, and the nonnegative stress kernel is continuous with $\Phi_s(0) = 1$. If $\mathbb{E}[g^2/(1+\kappa)] = \infty$, the immigrant-only null score has infinite second moment.*

Proof. Condition on the first event time $S = s$ in an early interval of positive probability and its mark $Q = q$, allowing all later immigrants and descendants. On a later interval of positive length, the first parent's contribution gives $G_s(t)^2 \geq c\, g(q)^2$. The branching construction and fresh marks bound the expected total productivity of that parent's cluster by $\kappa(q)/(1-n)$; future immigrant clusters contribute a finite bound independent of q. Boundedness of the trigger kernel therefore gives $\mathbb{E}[H_e(t) \mid S = s, Q = q] \leq C(1+\kappa(q))$. Positivity of B and Jensen's inequality yield, on this interval,

$$\mathbb{E}\left[\frac{B(t)^2 G_s(t)^2}{B(t) + nH_e(t)}\middle| S = s, Q = q\right] \geq c' \frac{g(q)^2}{1+\kappa(q)}.$$

Integration over the first mark and that interval makes the expected bracket infinite. Meanwhile, $\mathbb{E}g < \infty$, the finite mean count and bounded stress kernel imply $\mathbb{E}\int_0^T BG_s \, \mathrm{d}t < \infty$, so the compensated score is an integrable martingale. A square-integrable terminal score would imply finite expected bracket, a contradiction. □

The same argument applies on a finite spatial domain when the trigger density is bounded and the background and first-parent stress kernel have positive lower bounds on a time–space region of positive measure. The ideal spatial Omori null with positive temporal offset satisfies these conditions, as does the finite-mixture M2 candidate under its own fresh-mark null. For the exponential productivity above, $1+\kappa$ and κ have equivalent tails, so the spatial LI+ ratio divergence gives the claimed null obstruction. For M2, $\kappa = 1$. This proof does not condition on absence of descendants: at positive triggering that condition would tilt the first mark's law. Nor does it identify the exact Mittag–Leffler or persistent-mark S02 generators with the finite-mixture candidate.

At the simultaneous boundary $n = \eta = 0$, the conclusion can fail. Take a constant Poisson background $B > 0$, marks $Q \sim \mathrm{Exp}(\beta)$, and $g(Q) = e^{\gamma(Q-1)}$ with $\gamma < \beta \leq 2\gamma$. The first stress-mark moment is finite while the second is infinite. For one exponential relaxation mode of rate $k > 0$, the resulting positive shot noise has a stationary version of mean $B\mathbb{E}g/k$; copies with the same future points and marks have differences that decay exactly as e^{-kt}. Thus a first-moment stable state process is compatible with the following score irregularity. On the positive-probability event of exactly one event at $S < T$, the pre-event stress is zero and the score contribution has the form

$$U_0 = g(Q) \int_S^T B(t)\Phi_s(t-S) \, \mathrm{d}t.$$

The fresh mark is independent of that event configuration, and the integral is strictly positive on a positive-probability set of S. Hence the score has infinite second moment. At positive η, even with $n = 0$,

$$\frac{(G_s B e^{-\eta G_s})^2}{\lambda_\eta} \leq BG_s^2 e^{-\eta G_s} \leq \frac{4B}{e^2\eta^2},$$

so integrable B controls the score bracket without a raw second mark moment. This regularization is not uniform as $\eta \downarrow 0$ and does not establish null-boundary asymptotic theory.

Total attenuation and positive generating inhibition. For the separate family $\lambda_\eta = (B + nH_e)e^{-\eta G_s}$, the fixed-nuisance score and its null expected predictable bracket are

$$U_{\eta,\mathrm{total}} = -\int_0^T G_s\,(\mathrm{d}N - \lambda_\eta\,\mathrm{d}t), \qquad I_{\mathrm{total}}(0) = \mathbb{E}_0 \int_0^T G_s^2(B + nH_e)\,\mathrm{d}t.$$

The immigrant-only ratio bound does not apply. A first parent's background contribution requires $\mathbb{E}g^2 < \infty$, and its same-parent stress–trigger contribution, where the kernels overlap and $n > 0$, requires $\mathbb{E}[g^2\kappa] < \infty$. For $g(Q) = Ce^{\gamma Q}$ and the exponential productivity above, these conditions respectively require $2\gamma < \beta$ and $2\gamma + r\beta < \beta$. Both fail for the spatial LI+ law. Thus its expected null bracket diverges. These diagonal conditions are necessary obstructions, not a sufficient theorem for the cross-parent terms. Even absolute first-moment control involves $\mathbb{E}[g\kappa]$, requiring $\gamma + r\beta < \beta$; without an integrability argument, an infinite expected bracket should not be equated with an ordinary terminal-score variance.

Positive generating inhibition is different. The immigrant-only positive-η bound above holds regardless of the raw second mark moment. For total attenuation,

$$G_s^2\lambda_\eta \le \frac{4}{e^2\eta^2}(B + nH_e), \qquad \eta > 0.$$

The fresh-mark subcritical Hawkes envelope gives a finite expectation of the integrated right-hand side. Both families therefore have finite score brackets at each fixed positive generating η under these assumptions; the bounds are not uniform at the null.

A finite horizon alone does not truncate the ideal marks or counts. The implemented generators additionally use finite uniform grids, representable marks and explicit resource/numerical stopping rules. Their finite saved histories and any law conditioned on successful completion are distinct from the ideal unstopped law used above. Null moment irregularity neither proves that a positive-inhibition generating score is irregular nor establishes the cause of held-out losses, selected parameter values or optimizer behavior.

Finite score variance alone is not a central limit theorem. It also does not establish differentiability in quadratic mean for the full searched model or a regular information matrix at the boundary. This distinction is visible in the pathwise negative Hessian at the null:

$$-\ell_{\eta\eta}(\vartheta, 0) = A_T^{(2)} - C_T^{(2)}, \qquad A_T^{(2)} = \int_0^T BG_s^2\,\mathrm{d}t, \qquad C_T^{(2)} = \sum_i \frac{B(T_i)G_i^2 nH_i}{\lambda_0(T_i)^2}.$$

For example, suppose B is continuous and positive, Φ_s is continuous with $\Phi_s(0) = 1$, and $\mathbb{E}g(Q)^2 = \infty$. Let S be the first event time. Its null density on $(0, T)$ is $B(s)\exp\{-\int_0^s B(v)\,\mathrm{d}v\}$, and its fresh mark is independent of S. The contribution of that first parent gives

$$A_T^{(2)} \ge g(Q_1)^2 \int_S^T B(t)\Phi_s(t - S)^2\,\mathrm{d}t \quad \text{on } \{S < T\}.$$

The integral is strictly positive, so Tonelli yields $\mathbb{E}_0 A_T^{(2)} = \infty$. This applies to the displayed exponential-mark example whenever $2\gamma \ge \beta$. Under Proposition 5.1's additional hypotheses, define the finite null score bracket mean

$$I_{\eta\eta}^{(0)} = \mathbb{E}_0 \int_0^T \frac{B^2 G_s^2}{\lambda_0}\,\mathrm{d}t < \infty.$$

Fresh-event compensation and the nonnegative integrand identity

$$BG_s^2 = \frac{BG_s^2 nH_e}{\lambda_0} + \frac{B^2 G_s^2}{\lambda_0}$$

then imply $\mathbb{E}_0 C_T^{(2)} = \infty$ as well. Thus taking the expectations of the two Hessian components separately produces an undefined $\infty - \infty$, not a justified information identity. This fact alone does not determine integrability of their pathwise difference; cancellation would need its own argument. Finite score variance cannot be used as permission for that termwise interchange.

5.2 Nuisance projection and local information

Suppose the relevant scores are square integrable. Define the ensemble inner product

$$\langle f, h \rangle_0 = \mathbb{E}_0 \int_0^T \lambda_0(t) f(t) h(t) \, \mathrm{d}t.$$

For a nuisance parameter ζ, the score integrand is $f_\zeta = \partial_\zeta \lambda_0 / \lambda_0$. If the nuisance scores span a closed subspace, the residual after projecting f_η onto that subspace is the efficient-score integrand. In finite nonsingular notation,

$$I_{\eta|\zeta} = I_{\eta\eta} - I_{\eta\zeta} I_{\zeta\zeta}^{-1} I_{\zeta\eta} \le I_{\eta\eta}.$$

When nuisance scores are redundant, a generalized inverse and its range condition replace the ordinary inverse.

This projection gives a quantitative meaning to masking by fitted background and triggering. Slowly varying depletion can resemble a background trend, while post-event depletion can resemble a triggering scale or productivity. It also shows why effective information is an ensemble object: it requires integrability and a reference law, not merely the algebra of one fixed path. At $\eta = 0$, the information contributed by an independently parameterized stress-kernel direction is zero, because $\partial_\theta \lambda = -\eta B e^{-\eta G_s} \partial_\theta G_s$. An $O(\eta^2)$ information statement away from zero requires a justified uniform moment bound for that derivative; the zero first derivative alone is not enough.

A related calculation can be made on one fixed reference path without the ensemble expectation. For $B = \exp(a + dx_{\mathrm{trend}})$, the four null log-intensity directions are

$$s_a = \frac{B}{\lambda_0}, \quad s_d = \frac{x_{\mathrm{trend}} B}{\lambda_0}, \quad s_n = \frac{H_e}{\lambda_0}, \quad s_\eta = -\frac{BG_s}{\lambda_0}.$$

The realized Gram matrix $J_{ij} = \int_0^T \lambda_0 s_i s_j \, \mathrm{d}t$ is positive semidefinite. Its residual inhibition norm after projecting on (s_a, s_d, s_n) is the minimum of $\int \lambda_0 (s_\eta - b_a s_a - b_d s_d - b_n s_n)^2 \, \mathrm{d}t$. If $G_s = c_0 + c_1 x_{\mathrm{trend}} + c_2 H_e / B$, that residual is exactly zero. This algebra explains how depletion can resemble a fitted level, trend and triggering component.

On two frozen reference paths, the archived residual/raw squared-norm ratios were approximately 10.82% and 3.26%. These are two realized-path projections, rather than estimates of ensemble Fisher information. The paths and their calibrations differ; their ordering does not establish that greater exposure reduces expected information. Additional score directions can only reduce the residual for the same path and inner product, but no power decomposition follows from that projection fact.

5.3 Conditional amplitude profiling

Fix one observed marked history and fix the trend coefficient d and inhibition η. Put

$$q_i = \exp\{dx_{\text{trend}}(T_i) - \eta G_i\}, \qquad c_i = nH_i, \qquad J = \int_0^T \exp\{dx_{\text{trend}}(t) - \eta G(t)\}\ \mathrm{d}t,$$

and let $A = \exp(a)$ with $A \in [L, U]$, where $L = e^{-12}$ and $U = 1$. The negative event-time contribution to the joint marked log-likelihood, omitting the common mark constant and writing $K = \int_0^T H_e(t)\ \mathrm{d}t$, so that nK is the full-window triggering compensator, is

$$F(A; d, \eta) = AJ + nK - \sum_{i=1}^{N} \log(Aq_i + c_i). \tag{5.3}$$

The full horizon and all pre-event intensities are retained. The mark factors in (5.1) are constant in these fitted coordinates under the common mark model.

Proposition 5.3 (Conditional amplitude profile)**.** *Assume a finite history and finite event features and parameters. Let every $c_i \geq 0$, let K be finite, and require $0 < J < \infty$. The specified finite-mode observation window satisfies these conditions when x_{trend} is bounded and all recorded marks are finite. Then (5.3) has a unique minimizer on $[L, U]$:*

1. *if $F_A(L) \geq 0$, the minimizer is L;*
2. *otherwise, if $F_A(U) \leq 0$, the minimizer is U;*
3. *otherwise, the unique interior root of the continuous strictly increasing F_A is the minimizer.*

If $N = 0$, the unique minimizer is L.

Proof. Ordinary finite-sum differentiation gives

$$F_A = J - \sum_{i=1}^{N} \frac{q_i}{Aq_i + c_i}, \qquad F_{AA} = \sum_{i=1}^{N} \left(\frac{q_i}{Aq_i + c_i}\right)^2.$$

For $N \geq 1$, $q_i > 0$, so $F_{AA} > 0$ and F_A is strictly increasing and continuous on the compact interval. If its endpoint values place zero inside the interval, it has one root, and the sign change from negative to positive identifies that root as the unique minimizer; endpoint cases place the minimizer at the corresponding boundary. If $N = 0$, then $F(A) = AJ + nK$ is strictly increasing because $J > 0$, so the lower endpoint is the unique minimizer. □

The amplitude result is a finite-history optimization fact, not a global fit certificate. Convexity in A does not make the profiled surface $V(d, \eta) = F(A_*(d, \eta); d, \eta)$ convex in (d, η). A finite grid or local search establishes only its declared numerical search outcome.

For $a = \log A$, set $p_i = Aq_i/(Aq_i + c_i)$. Then

$$g(a) = F_a = AJ - \sum_i p_i, \qquad g'(a) = AJ - \sum_i p_i(1 - p_i),$$

and g need not be monotone even though F_A is increasing in A. A bracketing method for a can rely on continuity and endpoint signs, not on monotonicity of g. At an interior root the exact balance is

$$AJ = \sum_i p_i.$$

The curvature identity in log amplitude is

$$F_{aa} = \sum_i p_i^2 + R, \qquad R = AJ - \sum_i p_i.$$

Thus at the balanced root, the negative log-likelihood satisfies $F_{aa} = \sum_i p_i^2$, while the log-likelihood $\ell = -F$ satisfies $\ell_{aa} = -\sum_i p_i^2$. At an approximate root, the balance residual R must be added to $\sum_i p_i^2$; it is not exact curvature at that numerical point. In a region where the optimizing amplitude is interior, profile differentiation also gives

$$V_d = AJ_x - \sum_i x_{\text{trend}}(T_i)p_i, \qquad V_\eta = -AJ_G + \sum_i G_i p_i,$$

where J_x and J_G integrate $x_{\text{trend}}(t)$ and $G(t)$ against the background weight. Here $J_x = \int_0^T x_{\text{trend}}(t)e^{dx_{\text{trend}}(t)-\eta G(t)}\,\mathrm{d}t$ and $J_G = \int_0^T G(t)e^{dx_{\text{trend}}(t)-\eta G(t)}\,\mathrm{d}t$. At fixed d, implicit differentiation in an interior-amplitude region gives

$$\frac{\partial a_*}{\partial \eta} = \frac{AJ_G - \sum_i G_i p_i(1-p_i)}{\sum_i p_i^2}.$$

The numerator has no universal sign. An amplitude curve that also refits trend has a different total derivative. The identity $AJ = \sum_i p_i$ equates integrated fitted background with event-wise background mixture weights; these weights are not observed physical parent labels or a stationary balance law. With empty prehistory and positive background, the first event has $c_1 = 0$, so $p_1 = 1$. No initial quiet interval or first event can be discarded from this argument.

These are local envelope and implicit formulas. Active endpoint transitions require one-sided treatment, and none of them implies that every starting point reaches the global profile minimum.

Stable evaluation also matters for the conditional proof. In logarithmic coordinates compute $\log \lambda_i$ by a log-sum-exp of $a + \log q_i$ and $\log c_i$. When $c_i = 0$, the mathematically positive background gives $p_i = 1$, even if exponentiating $\log q_i$ underflows. A numerical zero is not evidence that the exact objective is flat. The same amplitude algebra holds with positive quadrature weights, but agreement between quadrature orders at selected points is a numerical sensitivity check rather than a rigorous global integration bound.

5.4 Boundary hypotheses and calibration

Feedback detection and testing equality of memories are different tasks. The first compares $\eta = 0$ with an active alternative; the second compares shared and separate stress/trigger families while permitting inhibition. At zero feedback, an independently varied stress-kernel parameter disappears from the observed law. Together with the one-sided constraint $\eta \geq 0$, this produces a nonregular null: there is a nuisance parameter present only under the alternative. A fixed-kernel boundary formula cannot automatically calibrate a maximization over stress kernels and fitted nuisances. The general difficulty is discussed by Davies (1977); the regularity conditions of boundary likelihood theory, including those in Self and Liang (1987), need their own verification here.

A calibration recipe must match the complete statistic actually computed. Under a fitted null, this means simulating new marked catalogues, refitting the allowed nuisance parameters under both candidates, repeating the declared kernel search and using the same failure and selection rules. A reference distribution for one nuisance block cannot be substituted for another. Such a plug-in procedure is a proposed finite-sample method; the present score bounds do not prove its uniform validity over a composite null or near weak identification. Non-rejection of shared memory cannot confirm that restriction without evidence that meaningful separate-memory alternatives are distinguishable.

The distinction between supplied and estimated nuisances is equally substantive. A likelihood ratio between two fully specified laws addresses a simple testing problem. A composite

null can change its background and other permitted parameters, admitting alternative explanations that the simple null excludes. Superior calibrated detection with supplied generating values therefore does not measure purely avoidable fitting inefficiency. It combines different statistical questions with different procedures; neither an arbitrary optimizer improvement nor fixing real-data nuisance values without justification is guaranteed to recover the supplied-law performance.

5.5 Response of a calibrated fixed-rule test

Calibrated inference starts with a complete-data decision rule rather than with a particular fitted parameter value. Let $I(\mathcal{H}) \in \{0,1\}$ be a fixed bounded measurable rejection indicator defined on every admissible marked history. The reference bank, tuning and model-selection procedure, critical threshold, treatment of empty histories, and any auxiliary randomization law are fixed independently of the true generating inhibition. The rule may estimate nuisances or select a model from the data, because these operations are part of the same measurable mapping from history to $\{0,1\}$; what is held fixed is the rule as the generating η varies. Define

$$p(\eta) = \mathbb{E}_\eta I(\mathcal{H}).$$

Proposition 5.4 (Positive-inhibition response of a fixed rule)**.** *Fix empty history, a common fresh mark law with finite nonnegative transforms, $\mathbb{E}\kappa = 1$, $0 \le n < 1$, nonnegative kernels bounded on the finite window with normalized h_e, and a deterministic nonnegative background with $\mathcal{B} = \int_0^T B(t)\,\mathrm{d}t < \infty$. Hold all these generating nuisances fixed as η varies. Then $p(\eta)$ is continuously differentiable for $\eta > 0$, and*

$$p'(\eta) = \mathbb{E}_\eta[I(\mathcal{H})S_\eta] = \mathrm{Cov}_\eta(I(\mathcal{H}), S_\eta).$$

Moreover,

$$|p'(\eta)| \le \frac{2\sqrt{\mathcal{B}\,p(\eta)(1-p(\eta))}}{e\eta} \le \frac{\sqrt{\mathcal{B}}}{e\eta}.$$

If in addition the fixed generating ingredients satisfy Proposition 5.1, put

$$K_T = \frac{B_{\max}^2 C_T m \mathcal{B}}{n(1-n)}.$$

Then the stronger bound

$$|p'(\eta)| \le \sqrt{K_T p(\eta)(1-p(\eta))} \le \frac{\sqrt{K_T}}{2}, \qquad \eta > 0,$$

is uniform in positive η with those ingredients fixed. It does not assert existence of $p'(0)$.

Proof. Let P_0 be the noninhibited Hawkes law. Its mean count is at most $\mathcal{B}/(1-n)$, and it dominates each inhibited construction. On a finite path the likelihood ratio is

$$L_\eta = \exp\left\{\int_0^T B(1-e^{-\eta G_s})\,\mathrm{d}t\right\} \prod_i \frac{Be^{-\eta G_i} + nH_i}{B + nH_i} \le e^{\mathcal{B}}.$$

At actual event times the reference intensity is positive almost surely; where both intensities vanish away from events, set their likelihood ratio to one. The stopped density martingale is uniformly bounded and hence defines the changed law. Its score $S_\eta = \partial_\eta \log L_\eta = U_\eta$ satisfies

$$\mathbb{E}_\eta S_\eta = 0, \qquad \mathbb{E}_\eta S_\eta^2 \le \frac{4\mathcal{B}}{e^2\eta^2},$$

by the positive-inhibition bracket bound, without a raw stress-mark second moment.

For clarity, write

$$\mathcal{B} = \int_0^T B(t)\,\mathrm{d}t, \qquad J_\eta = \int_0^T B(t)(1 - e^{-\eta G_s(t)})\,\mathrm{d}t,$$
$$\varrho_i(\eta) = \frac{B_i e^{-\eta G_i} + c_i}{B_i + c_i}, \quad c_i = nH_i \geq 0,$$

where $B_i = B(T_i)$ and G_i, H_i are the pre-event features. Then $L_\eta = e^{J_\eta} \prod_i \varrho_i(\eta)$, with $0 \leq \varrho_i \leq 1$ and $0 \leq J_\eta \leq \mathcal{B}$. At actual reference-law events $B_i + c_i > 0$ almost surely. On a compact positive interval $\eta \in [\eta_-, \eta_+]$, direct differentiation gives

$$|\varrho_i'(\eta)| = \frac{B_i}{B_i + c_i} G_i e^{-\eta G_i} \leq \frac{1}{e\eta_-}, \qquad |J_\eta'| \leq \frac{\mathcal{B}}{e\eta_-}.$$

The product rule, without division by any ϱ_i, therefore yields

$$|\partial_\eta L_\eta| \leq e^{J_\eta} \left(|J_\eta'| \prod_i \varrho_i + \sum_i |\varrho_i'| \prod_{j \neq i} \varrho_j \right)$$
$$\leq \frac{e^{\mathcal{B}}(\mathcal{B} + N(T))}{e\eta_-}.$$

The same bound applies to $|\partial_\eta(L_\eta I)|$ for the fixed bounded indicator. It is integrable under the reference law because $\mathbb{E}_0 N(T) \leq \mathcal{B}/(1-n)$. The bound is for the derivative of L_η, not its logarithm. The finite-path derivative is continuous in η. Dominated convergence gives $p'(\eta) = \mathbb{E}_0[I \partial_\eta L_\eta] = \mathbb{E}_\eta[I S_\eta]$. No derivative of the data-dependent fitted outputs is taken: I is the same mapping on every history. The mean-zero score gives the covariance identity. Cauchy–Schwarz with $\mathrm{Var}(I) = p(1-p)$ gives the general bounds. Under the additional weighted-moment hypotheses, Proposition 5.1 and $\mathbb{E}_\eta N(T) \leq \mathcal{B}/(1-n)$ instead give $\mathbb{E}_\eta S_\eta^2 \leq K_T$; the same covariance inequality proves the uniform bound. □

The covariance has no proved sign. Stronger generating inhibition need not increase detection probability for a particular calibrated rule. The general positive-η estimate gives local differentiability and moment control but is not uniform as $\eta \downarrow 0$. The stronger bound needs positive triggering and the finite weighted-history moment; it bounds the positive-parameter slopes but supplies neither monotonic power nor null-boundary asymptotic calibration. Unknown, failed, capped, or otherwise unresolved computational outcomes remain unknown; they must not be recorded as nonrejections. Conditioning on completed cases estimates a different functional from the unconditional $p(\eta)$ defined by the fixed rule.

The proposition also does not compare a simple hypothesis with a known generating law to a composite hypothesis whose nuisances are fitted. Those are different operating characteristics, with different permitted null explanations and decision rules. An oracle that knows the nuisance values and a composite rule that estimates them are not the same test function, so their performance gap is not by itself pure fitting waste.

5.6 Scope and unresolved inference

Taken together, the results separate four questions. Exact marked-law equality shows that zero feedback leaves a separately varied stress kernel unidentified; the restricted active-inhibition experiment can identify the inhibition parameter and kernel in an idealized full-law sense; the

restricted score can remain square integrable even when raw mark moments diverge; and a calibrated fixed rule has a local covariance response at positive inhibition. Amplitude profiling adds a unique conditional optimization in A, while leaving the remaining trend–inhibition surface without a global certificate.

These statements do not imply a Wilks law, a general Fisher-information theorem, likelihood-ratio asymptotics at $\eta = 0$, uniform validity near the boundary, a necessary detection threshold, or a finite-sample power result. They also provide no new numerical evidence and do not replace a specified calibration procedure by a general asymptotic claim. The unresolved inference is therefore explicit: structural identification in a restricted family is not practical detectability; finite score variance is not a distributional approximation; conditional amplitude uniqueness is not global identifiability; and a signed response formula for a fixed rule is not a theorem that stronger inhibition creates a more powerful test.

Chapter 6

Experimental Design, Sequential Conditional Scoring, and Audit Protocols

This chapter defines the finite-history diagnostic experiment, its complete generating laws, the separate oracle and fitted comparisons, and the reference-calibration rule. The observed-catalogue analyses use the same predictable-history and scoring conventions.

6.1 Common interpretation of the diagnostic experiment

The empirical target is the selected recorded event stream and its stated magnitude and location measure. The fitted occurrence intensity has no estimated detection function, missing-parent filter or magnitude-error law. Completeness is a property to estimate from an observation system, with uncertainty (Woessner and Wiemer, 2005; Mignan, 2012); selecting a threshold does not establish it. The synthetic laws, in contrast, supply their own complete histories and mark distributions. Their recovery results concern those laws.

The attenuation fields use nonnegative radial point kernels. Reference geography and finite-domain truncation can make the resulting full intensity anisotropic, but no fitted arm calculates signed Coulomb stress on receiver faults or propagates a physical rate-and-state population. Physical stress transfer depends on source geometry, receiver orientation and fault state (Dieterich, 1994; King et al., 1994; Harris and Simpson, 1998). The comparisons estimate the incremental predictive value of the stated exponential rate attenuation after the excitation and background representation is specified.

The paired calendar-block intervals condition on the saved fitted states. They summarize variation of the observed score contrasts across retained dependent blocks. The conditional simulation reference likewise holds each origin's fitted parameters, nuisance choices and pre-history fixed. These are different experiments: neither refits the full procedure over repeated catalogues or propagates catalogue revision and measurement uncertainty. Their interval labels and simulation ranks do not assert nominal calibration after adaptive model selection. Model comparison and conditional consistency are reported separately, following the distinction emphasized by Rhoades et al. (2011); Bray and Schoenberg (2013).

A selected $\widehat{\eta} = 0$ means that the stated training optimization returns the uninhibited boundary. It is a result of that fitted procedure, not a measurement of a physical attenuation strength or a calibrated power estimate. Power is a rejection probability under a declared alternative-generating law and testing rule; the synthetic reference-calibrated comparisons supply those experiments separately.

6.2 The diagnostic law and its history

For a history $\mathcal{H}_t$, let

$$\lambda_\eta(t) = \exp\left\{a + d\left(\frac{2t}{T} - 1\right)\right\} \exp\{-\eta G_s(t)\} + nH_e(t), \tag{6.1}$$

where

$$B(t) = \exp\left\{a + d\left(\frac{2t}{T} - 1\right)\right\}.$$

Here a controls the background level, d the linear trend in transformed time, n the additive triggering scale, and η the diagnostic inhibition of marked stress memory. The feature $G_s(t)$ is a nonnegative marked stress-memory feature, while $H_e(t)$ is a normalized marked triggering feature. The interpretation of η is conditional on the represented triggering process and on the specified background: it is an attenuation of the first term in (6.1), not a declaration that the physical process is governed by an equivalent coefficient.

Both G_s and H_e are functions of the strictly earlier history. If event i has time t_i and mark m_i, the predictable intensity used for that event is $\lambda_\eta(t_i-)$. The mark m_i affects later stress memory and later triggering but cannot affect the intensity immediately before its own occurrence. This timing convention prevents circular use of the event being evaluated and is essential to the likelihood interpretation below.

The numerical kernels are positive normalized finite spectral mixtures. They represent the stress and triggering responses with a finite number of modes rather than by evaluating the exact Mittag–Leffler kernel. In the shared implementation, the kernel scale parameters are $\alpha = 0.5$ and $\tau = 1$, and the normalized mixture comprises 195 modes. Although the modes may approximate a long-memory response, the finite mixture is itself the implemented model. For magnitude excess $m_i = M_i - M_0$, the implemented features are

$$G_s(t) = \sum_{t_i<t} e^{\gamma(m_i-1)}\Phi_s(t-t_i), \qquad \gamma = \tfrac{3}{4}\log 10, \tag{6.2}$$

$$H_e(t) = \sum_{t_i<t} (1-r)e^{\beta r m_i} h_e(t-t_i). \tag{6.3}$$

The relaxation kernel has the form $\Phi_s(u) = \sum_q w_q e^{-\rho_q u}$, with positive weights summing to one. The corresponding triggering density is $h_e(u) = \sum_q v_q \zeta_q e^{-\zeta_q u}$, with positive weights summing to one and positive rates. Thus $\Phi_s(0) = 1$ and $\int_0^\infty h_e(u)\,\mathrm{d}u = 1$. Sharing the relaxation parameters means using a common survival kernel and its negative derivative as the triggering density, rather than setting the two functions equal. The stress-mark normalization is the declared magnitude-six reference when $M_0 = 5$. These formulas are explicit in public restricted-history implementation (`S10`) and public observed implementation (`S17`). Conclusions therefore apply to the finite mixture that was used, not automatically to every exact infinite-mode limit.

Finite spectral recipe. For $0 < \alpha < 1$ and the dimensionless rate variable u, the continuous representation (Mainardi, 2014, Eqs. (2.12)–(2.13)) is

$$E_\alpha(-t^\alpha) = \int_0^\infty e^{-ut} K_\alpha(u)\,\mathrm{d}u, \qquad K_\alpha(u) = \frac{\sin(\pi\alpha)}{\pi} \frac{u^{\alpha-1}}{u^{2\alpha} + 2u^\alpha \cos(\pi\alpha) + 1}.$$

Writing $u = e^v$ gives the log-rate density

$$K_\alpha(e^v)e^v = \frac{\sin(\pi\alpha)}{2\pi\{\cosh(\alpha v) + \cos(\pi\alpha)\}}.$$

For the scaled relaxation $E_\alpha(-(t/\tau)^\alpha)$, the rates are e^v/τ. The following finite rule applies trapezoidal weights on the stated log-rate grid, halves its endpoints, truncates and renormalizes. The continuous representation and this numerical discretization are distinct; the citation does not provide a universal likelihood- or inference-error bound. The generating relaxation uses log rates $v_j = 0.4j$, $j = -97, \ldots, 97$, and rates $\rho_j = e^{v_j}/\tau$ in days^{-1}, with $\alpha = 0.5$ and $\tau = 1$ day. Before normalization its weights are

$$\widetilde{w}_j = \frac{0.4 \sin(\pi\alpha)}{2\pi\{\cosh(\alpha v_j) + \cos(\pi\alpha)\}},$$

with both endpoint weights halved, then $w_j = \widetilde{w}_j / \sum_k \widetilde{w}_k$. More generally the implemented endpoint index is

$$J = \left\lceil \frac{1}{s} \max\left\{ \frac{\log 4}{\alpha}, \frac{\log[4\sin(\pi\alpha)/(\pi\alpha\varepsilon)]}{\alpha} \right\} \right\rceil, \qquad s = 0.4, \quad \varepsilon = 10^{-8}.$$

Thus the 195-mode count specifies the generating case, while candidate $\alpha = 0.7$ uses 137 modes with the same spectral rule. The machine-readable methods supplement supplies the rates and normalized weights. The finite triggering density remains $h = -\Phi'$; its finite value at zero does not reproduce the exact fractional singularity.

Marks are generated with excess density

$$f_\beta(m) = \beta e^{-\beta m}, \qquad m > 0,$$

with $\beta = 2.849401764894274$. The normalized productivity factor is

$$(1 - r)e^{\beta r m}, \qquad r = 0.7.$$

This normalization gives unit mean productivity at the reference scale and keeps the productivity multiplier positive. The exponential excess itself has finite moments. It is the exponential stress and productivity transforms that have finite first but infinite second moments at the declared parameters. That feature motivates bounded-indicator simulation checks rather than variance formulas or finite-information shortcuts based only on the raw marks.

6.3 Generating histories and evaluating both laws

The main generating experiment has duration $T = 24394$ days, begins with an empty prehistory, and produces random event counts. Counts are not conditioned on a target total; in particular, the design does not condition on a 913-event count. The generating nuisance parameters include

$$a = -4.162108473051707, \qquad d = 0.23211265877958467, \qquad n = 0.5017751599706763,$$

and the main diagnostic value is

$$\eta = 0.4885010300470942.$$

The null experiment sets $\eta = 0$ while retaining the generating nuisance values in the null law. This is a comparison between a marked process with stress inhibition and a marked process without that inhibition, not a comparison against a law with triggering or marks removed.

The null reference bank is generated at the known simulation nuisance vector, not at an estimated null fit from each query. Thus even the procedures that estimate nuisance parameters are calibrated here against an oracle-supplied generating law. This controlled comparison isolates specified fitting procedures; it is not an implementable plug-in bootstrap for an unknown composite null, nor a test with demonstrated size uniformly over nuisance values.

Simulation and event clock. The simulator starts both modal states at zero. At each proposal it samples an exponential wait with envelope $e^{a+|d|}+\sum_q w_q E_q$, where E_q is the current excitation state, advances and decays both modal states, and accepts with probability equal to the new predictable intensity divided by that envelope. Nonnegative inhibition bounds the background by $e^{a+|d|}$, and excitation decays between events. Only after acceptance is a fresh exponential excess mark drawn; its productivity and stress increments then update future states. Proposals reaching the horizon add no event. The accepted clock retains the legacy compensated binary64 projection and the exact Decimal sum of sampled binary64 waits at precision 1600, together with residual coordinates and positive relative event gaps. Projected ties are retained without jitter or resampling; paired-clock likelihood evaluation uses those gaps. Unrepresentable waits, mark-transform overflow and resource caps remain explicit unresolved cases, with restart state retained.

For observed event times t_i and marks m_i, the marked point-process log likelihood is

$$\ell(\vartheta, \eta; \mathcal{H}_T) = \sum_{i:t_i \le T} \log \lambda_\eta(t_i-) - \int_0^T \lambda_\eta(t)\, \mathrm{d}t + \sum_{i:t_i \le T} \log f_\beta(m_i). \tag{6.4}$$

Evaluation uses the full predictable intensity and the full integral, including quiet exposure before, between, and after events and the terminal interval up to T. It is not a count-conditional likelihood on the event times alone. When the same mark density is held fixed for both laws, the mark terms in (6.4) cancel in their difference. They are present in both likelihoods; their cancellation is an algebraic consequence of the common mark law, not an omission of marks from the model.

Two comparison designs are distinguished. An oracle signed log-likelihood ratio compares two fully supplied laws. A fitted generalized statistic instead permits separate null and alternative optimizations over the nuisance parameters allowed by each procedure. The oracle therefore has additional information: it knows nuisance values that an estimation procedure must recover. This changes a simple-law comparison into a composite-law task. A contrast such as 100% detection for the known-law procedure versus roughly 38% for a fitted procedure is not, by itself, evidence of avoidable numerical inefficiency, nor does it establish the power of a hypothetical oracle for the composite problem. Estimating nuisance changes both the test statistic and its null distribution. Each procedure consequently requires its own reference calibration. In symbols, the oracle uses the signed contrast

$$L_{10} = \ell(\vartheta_\star, \eta_\star; \mathcal{H}_T) - \ell(\vartheta_\star, 0; \mathcal{H}_T), \tag{6.5}$$

whereas a fitted procedure uses its selected maxima to form

$$D = 2\left\{\widehat{\ell}_1 - \widehat{\ell}_0\right\}. \tag{6.6}$$

The hats denote the declared finite fitting recipe, rather than a proof that a global maximum has been found. The oracle contrast need not be nonnegative; a fitted statistic with an exact eligible null floor is nonnegative. Their numerical scales and critical values cannot be exchanged.

6.4 Reference calibration and inferential scope

For each historical matched-law method described here, the null calibration uses 999 reference null history identifiers and a separate bank of 1,000 null query histories. The same reference identities are reused across methods, but each method fits and scores its own statistic on those histories. The 150 stronger alternative histories are also held in common across procedures.

Reference histories are not scored as queries. For a query statistic D_q, the reported rank probability is

$$p = \frac{1 + \sum_{j=1}^{999} \mathbf{1}\{D_j \geq D_q\}}{1000}, \qquad \text{reject if } p \leq 0.05. \tag{6.7}$$

All ties count through the indicator, with no random tie breaking or exclusion. The add-one term and the denominator of 1,000 make the query effectively one member of a rank set containing 999 calibration references.

The reported summaries condition on the realized reference banks and generating law. For a procedure fixed in advance whose reference and query histories are exchangeable under the null, the rank rule has the usual finite-sample conservativeness of a Monte Carlo rank test. That ideal argument is not retrospective proof for a procedure selected and reused during the research process. Adaptive selection followed by scoring can alter both the statistic and the comparison being calibrated. The reported Clopper–Pearson intervals are nominal exact binomial summaries under fixed-bank, fixed-procedure and independent-query assumptions. They describe Monte Carlo uncertainty in query outcomes, rather than conditioning away those outcomes. They do not resample the reference bank, the generator, or the adaptive procedure-selection process. Similar realized null rejection rates therefore do not prove equal test size uniformly over a composite null.

The null value $\eta = 0$ lies on the boundary of the permitted nonnegative parameter space. The usual interior-parameter Wilks argument does not apply directly. This study uses empirical matched-null calibration rather than asserting a chi-squared reference law. The inferential calculations are based on the separately calibrated procedures and their realized reference distributions. Because each rejection decision is a bounded indicator, conditional Monte Carlo standard errors remain meaningful even when individual marked histories are heavy tailed. This does not imply that a finite joint Fisher information exists for the fitted model.

6.5 Numerical controls and completed evidence

The evaluation retains the full event clock, including extreme histories, rather than replacing generated histories with summaries. Required optimizer starts must resolve; an unresolved calculation stopped at a cap is recorded as unknown and is never converted into a non-rejection. Selected-point likelihood and intensity comparisons were checked across integration orders and backends, and direct likelihood contrasts were used where possible to reduce cancellation between nearly equal large integrals. Numerical envelopes serve as sensitivity controls for these calculations. They are not a universal quadrature bound and do not establish global maxima for an optimization.

The completed historical oracle, supplied-nuisance, estimated-background and level-only evaluations each retain all 2,149 declared histories. No failed or extreme history is removed to obtain these denominators. Reporting policies, inclusion rules, and provenance for the numerical controls are recorded in the appendix. The completed-history evaluation and its reference-calibration conventions are documented in archival `S09` (recovered archive); the matched synthetic construction and likelihood accounting are recorded in archival `S11` (recovered archive), and the final reporting structure is recorded in archival `S16` (recovered archive).

Procedure-specific estimation. The completed supplied-trigger studies hold $n, r, \beta, \alpha_e, \tau_e$ and the stress kernel fixed. The eta-only procedure uses starts 0 and 0.2 over $\eta \in [0, 3]$. The level-and-trend procedure runs two null optimizations in $(a, d) \in [-12, 0] \times [-2, 2]$ and selects the better null fit. It then runs two alternative optimizations from that one fit, initialized at

$\eta = 0$ and $\eta = 0.2$. Thus level-and-trend has four prescribed optimizations; the two procedures together have six. Both use L-BFGS-B with maxiter = 100, maxls = 40, ftol = 10^{-10} and gtol = 2×10^{-6}. Failure alone enables the same-start CPU retries (line-search limits 40 then 200), followed by bounded Powell; the supplied-trigger procedures never free n. Every prescribed start must resolve, the exact matching null remains eligible, and selected order 16 values are checked independently at CPU order 32 to tolerance 10^{-4}. The distinct level-only procedure supplies d and profiles $a \in [-12, 0]$ for each η. It checks 17 equally spaced η-values in $[0, 3]$, the saved supplied-background alternative and zero, refines strict local mesh minima, and includes an 11-point profile. Its saved eligible floors and all mandatory solver returns are retained. Complete bounds, starts, stopping and recovery rules for this procedure, the historical fuller fit and the observed predictive fits are given separately in the methods supplement; none is a claim of global optimization.

Numerical integration and stable evaluation. The positive-gap history geometry retains initial and terminal quiet exposure. Within each event gap it inserts the offsets shorter than the gap from 23 geometric values spanning 10^{-6} to 10^{5} days and their 22 adjacent geometric means, then applies Gauss–Legendre order 16 on each subpanel. Selected controls use independent CPU order 32; the oracle refinement can require order 64. The triggering compensator uses the finite-mixture CDF, evaluated with $-\operatorname{expm1}(-\rho_q u)$. Event log intensity uses a log-sum-exp evaluation, including an explicit zero-triggering branch. Each event reads the decayed pre-event state before adding its mark. The historical fuller search and observed search have separate parameter and recovery contracts in the companion specification.

Companion procedure specification. The portable methods supplement and machine-readable settings in `supplements/` distinguish the generating kernel, the historical fuller calibration, the supplied-trigger procedures, the level-only profile and the observed predictive search. They record estimated and supplied coordinates, bounds, starts, quadrature, recovery and unresolved availability limits at their respective accepted versions. The metadata arrays are deterministic evaluations of the documented mixture rule, not new event histories or refits.

The public detection summary and the availability guide provide public entry points. The original primary history bank and the complete executed known-law likelihood-and-rank replay are now supplied under `reproduction/synthetic/` and `reproduction/oracle/`. The replay starts from event histories and reproduces the saved oracle decisions. The original fitted-procedure records and their source reports are supplied under `reproduction/historical_fits/`.

6.6 Sequential conditional prediction and regional cohorts

The observed comparisons use chronological training prefixes and later scored windows. Parameters, fitted mark-rate recipes and spatial nuisance choices are estimated from training information and frozen for each test window. During that window, strictly earlier observed test events enter the history. This is sequential conditional scoring: it differs from issuing a distribution for a complete future catalogue at a training cutoff. Equal timestamps are handled by the source-specific strict-history/batch convention; an event never predicts itself.

New Zealand uses the reconstructed GeoNet snapshot and its common historical targets. The baseline and primary middle stages retain the original chronological windows; constant-background and post-1980 sensitivities are separate procedures, not extra primary replications. California uses the repaired preferred-record schema with an explicit warmup interval.

Its higher-magnitude sensitivity selects a different target set. Japan uses complete reported-primary magnitude records from the declared annual archives and an explicit missing-primary exclusion rule. Exact cohort sizes, time windows, warmup and training-prefix counts are retained in Chapters 8 and 11. They must not be pooled into a single denominator or repeatedly counted as fresh evidence.

Training-only spatial density selection prevents direct use of future test locations in nuisance fitting. It does not identify immigrant geography: the training density describes observed events, including clustering. Leave-one-out bandwidth selection and subsequent full-training plug-in estimation are also different operations. Shared nuisance scales in a factorial comparison mean that the result is conditional on the shared rule, not an independent full optimization of every family.

6.7 Score sign, denominator and common-space checklist

Every comparison is candidate minus a named reference on common observations and a common dominating measure. The master definitions in Section 2.6 give

$$\Delta\ell_{\text{score}} = \Delta P - \Delta E + \Delta M = \Delta Q + \Delta C + \Delta(P - Q) + \Delta M.$$

The event contribution is ΔP, never canonical E. Positive integrated exposure is subtracted; a raw archived key called `C` can denote a signed contribution or positive exposure, so the source-field crosswalk must be consulted. Quiet periods, all scored spatial cells and the terminal censor remain in the compensator. Spatial densities measured per projected square kilometre cannot be compared directly with label-only scores without a declared common-space lift.

Pooled nats per target are the sum of the window score differences divided by the sum of their test counts. An unweighted average of window-normalized differences is generally a different quantity. Fold signs, component signs and pooled signs answer different questions. The class-history mark likelihood is scored once; it is not silently assumed to cancel when the model changes the conditional class law. Neither a positive event-only contrast nor a larger location term establishes a positive complete joint score.

6.8 Three separate calibration questions

The historical nuisance ladder calibrates a procedure-specific test statistic against a matched reference bank. A separate observed in-sample exercise calibrates the no-feedback and common-memory restrictions under their fitted null laws, as reported in Chapter 8; those conditional restriction tests are distinct from both synthetic detection and later predictive scoring. California conditional simulation instead starts from each original frozen OC state and generates complete branching futures for diagnostic coordinates. Its insertion-rank intervals, ties and audit scope are given in Chapter 12. That exercise does not calibrate EP and does not supply a composite boundary test for inhibition. The original LI fixed-seed histories are engineering/regime controls. The expanded matrix estimates regime-specific score and selection frequencies; its BIC-style gate is not a calibrated boundary test.

6.9 Interpretation shared by the empirical comparisons

The observational score tables compare declared fitted procedures on the inspected chronological windows. Their contrasts are descriptive; no confidence interval or familywise significance

claim is attached to them. Reuse of targets and overlapping training prefixes prevents treating successive model studies as independent replications. Finite multistart convergence and saved-state score checks do not certify a global optimum. These conventions apply throughout the empirical chapters; local qualifications identify additional restrictions of a particular comparison. Synthetic win intervals and sample standard errors have the separate definitions stated with their generating regimes.

The regional laws forecast the recorded catalogue under its declared selection rules. They do not include a time-varying detection process. A lower recorded rate after a mainshock cannot therefore be assigned to physical inhibition by these comparisons. Nor does a poor score for one attenuation recipe rule out other spatial or observation laws.

6.10 Custody and review protocol for this edition

The public package supplies the recovered inputs, original executed fit and score records, and separately identified reproduction runs. Historical authorities are preserved unchanged. Withdrawn results remain verifiable as historical records and are excluded from current manuscript-display counts. The claim ledger records the authority path, JSON pointer or textual locator, SHA-256 identity, transformation, score component, comparator and denominator, and derived presentation values retain their rounding rule. Source identity, arithmetic reconciliation, numerical controls, sampled or full-path computational replay, and external scientific review remain distinct forms of evidence.

The appendix retains the numerical rosters supporting the reported comparisons. A completed process exit is not proof of scientific correctness, and a successful multistart fit is not a global-optimization certificate. Negative contrasts, boundary selections, endpoint choices, failed continuation gates and retained limitations remain integral parts of the reported findings. The reproduction guide identifies the original executions and subsequent replays for each study; source and publication checks have separate records.

Chapter 7

Controlled Nuisance Sensitivity and Reference Calibration

This chapter reports what the matched synthetic studies establish about detecting marked stress sensitivity when parts of the background and triggering process are estimated. The historical comparisons below concern their stated law class and finite numerical implementation; the recovered original fit records support the procedures reported here. The common alternative bank contains 150 stronger histories, and each procedure has 1,000 null queries with 999 separately identified null references. Success is a calibrated rejection on an alternative history; the null rejection rate counts rejections among uninhibited query histories. Figure 7.1 summarizes the detection rates and their conditional binomial intervals.

For the first four procedures, the triggering process, mark exponent β, and both kernels are supplied. The first row supplies both laws completely. The second estimates η while supplying the remaining nuisance values. The third estimates η and background level while supplying the trend, and the fourth estimates η, background level and trend. The final row is the fuller historical fitted procedure, which additionally estimates or searches the triggering, mark, and kernel components under the earlier recipe. It is therefore not a purely nested extension of the fourth row, and its count should not be interpreted as an independent replication.

The known-laws (fixed η) procedure detects every one of the 150 stronger alternatives. This demonstrates that, for the evaluated generating law and numerical kernels, the stronger marked sensitivity is distinguishable when the comparison laws are supplied. It is not a statement of intrinsic inability to distinguish arbitrary physical mechanisms, nor a proof that all mechanisms would be easy to distinguish in observed data. The η-estimated, nuisances-supplied procedure also detects all 150. There is no observed loss from estimating that parameter alone on this bank; this does not prove equality of population detection probabilities. Estimating the background level leaves 123 detections, while estimating both level and trend leaves 68. These named-procedure differences are not an additive attribution of the gap. Moving from 150 to 123 and from 123 to 68 describes the tested configurations; it does not decompose the loss into independently estimable physical or numerical contributions.

The realized null rates lie between 4.4% and 5.0%. In row order, the conditional 95% null intervals are 3.73–6.54%, 3.65–6.43%, 3.39–6.09%, 3.21–5.86%, and 3.73–6.54%. These

Table 7.1: Historical detection on the common stronger alternative bank and null rejection on 1,000 null queries. Intervals are nominal 95% Clopper–Pearson intervals conditional on the fixed bank and law.

Procedure	Detections	95% interval	Null rejections
Known laws (fixed η)	150/150	97.57–100%	50/1000
η estimated; nuisances supplied	150/150	97.57–100%	49/1000
Level estimated; trend supplied	123/150	74.90–87.79%	46/1000
Level and trend estimated	68/150	37.20–53.66%	44/1000
Historical fuller fitted procedure	57/150	30.21–46.28%	50/1000

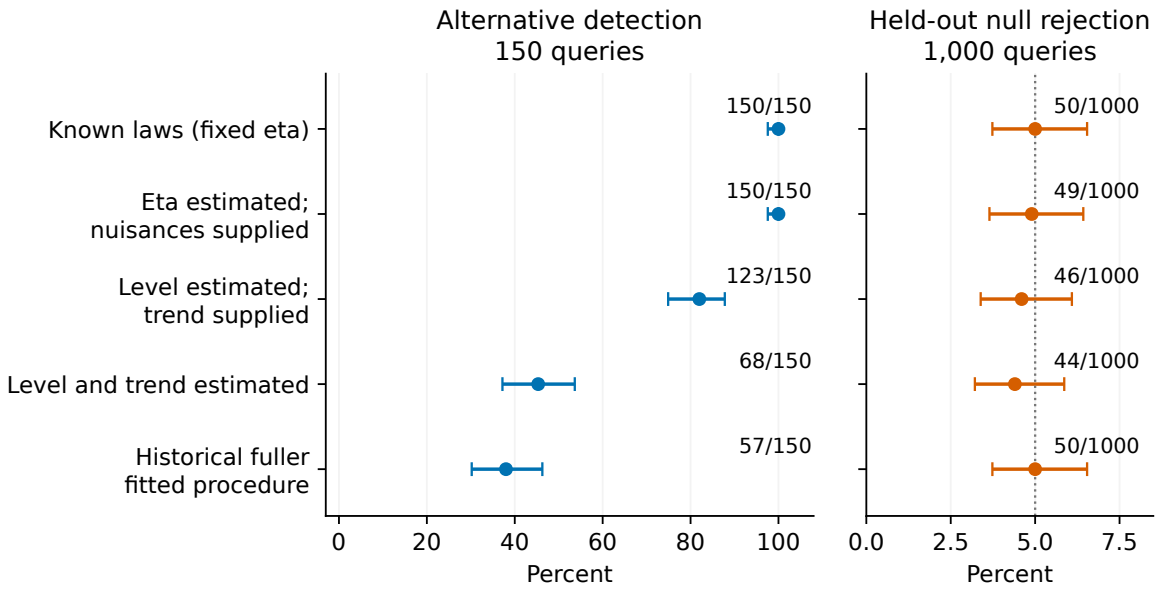


Figure 7.1: Historical matched-null detection on the same 150 stronger alternatives. Error bars are nominal conditional 95% binomial intervals. Each procedure has its own 999 reference statistics and 1,000 separate null queries; adaptive selection and reference-bank uncertainty are omitted.

similar observed rates are useful calibration summaries, but they do not establish equal test size uniformly over the composite null. The intervals are conditional on the realized reference bank and law, and the procedures were selected through research reuse.

7.1 Sensitivity to the diagnostic value

The value of η changes the histories as well as the fixed-history intensity separation. In a paired comparison using the same 150 fresh pairs, the initial value $\eta = 0.2839974288868791$ produced 49/150 detections for the fuller fitted procedure, whereas the adjusted value $\eta = 0.4885010300470942$ produced 57/150. The paired gain is therefore 8 detected histories, or 5.33 percentage points, with a descriptive Monte Carlo standard error of 4.71 percentage points. A separate 25-pair planning sample gave 6/25 and 11/25 under the two values; it is a planning exercise and is not pooled with the 150-pair comparison.

The descriptive paired Monte Carlo standard error uses the sample variance of paired decision differences. For N_{pair} pairs, let $D_i \in \{-1, 0, 1\}$ be the first decision minus the second, with positive and negative discordance counts b, c. Then

$$\overline{D} = \frac{b - c}{N_{\text{pair}}}, \qquad \widehat{\text{MCSE}} = \sqrt{\frac{b + c - N_{\text{pair}}\overline{D}^2}{N_{\text{pair}}(N_{\text{pair}} - 1)}}.$$

This is the sample standard deviation of D_i divided by $\sqrt{N_{\text{pair}}}$. Taking the adjusted-value decision first gives $b = 29, c = 21$ and $N_{\text{pair}} = 150$, hence 4.7096 percentage points, rounding to 4.71. Using empirical-variance divisor N_{pair} instead gives a slightly different value from the same outcomes. The stated fixed-bank and adaptive-reuse qualifications still apply; no new test is added.

The mean achieved fixed-history total intensity suppression was 14.159% for the initial value and 19.360% for the adjusted value. For a realized nonempty history and fixed generating

nuisance, the event-mean functional is

$$S(\eta; \mathcal{H}_T) = \frac{100}{N_T} \sum_{i=1}^{N_T} \frac{B(t_i)\{1 - e^{-\eta G_s(t_i)}\}}{B(t_i) + nH_e(t_i)}. \tag{7.1}$$

Both intensities in each comparison use the same strictly earlier history. The reported mean then gives equal weight to each catalogue's event mean, rather than pooling every event across catalogues. This is the functional implemented in archival `S07` (recovered archive). It is an event-mean description of separation on a common history, not a controlled reduction in earthquake count. Increasing η also changes the synthetic histories that are subsequently realized. Counts remain random, and no monotonicity of realized counts is guaranteed. An extreme retained pair, with 449 events under one member and 6,553 under the other, illustrates why suppression of a fixed-history rate cannot be reported as a promise about realized totals.

An earlier archive used a 1,999-reference rank denominator of 2,000. The later recalibration using 999 references and 1,000 separate null queries produced the same count of 57 detections only by happenstance. It is a consistency observation, not an independent replication, and the two reference constructions should not be combined. The public paired summary retains aggregate pairing results; the historical study is reported in archival `S08` (recovered archive). The completed-history provenance for later recalibration is recorded in archival `S09` (recovered archive) and the final results in archival `S16` (recovered archive).

7.2 Paired comparison of level and trend

The level-and-trend comparison is more informative when the two procedures are paired on the same 150 histories rather than read only from marginal detection counts. Figure 7.2 crosses their decisions. Both procedures detect on 62 histories, the level-only procedure detects on 61 where the combined procedure does not, the combined procedure detects on 6 where level-only does not, and neither detects on 21. The marginal counts are therefore 123 and 68, and the difference is $55/150 = 36.67$ percentage points, with a descriptive paired Monte Carlo standard error of 4.58 percentage points, using the same sample-variance convention with $b = 61, c = 6$ and $N_{\text{pair}} = 150$ (4.5776 before rounding).

There is no pathwise dominance. The level-only procedure detects on 61 histories where the combined procedure does not, while the reverse occurs on 6. The 55-history marginal gap is real for this realized paired bank, but the paired arrangement shows that it is not produced by a uniform improvement. Among the 999 null reference scores for the level-only procedure, 604 were exactly zero. Ties and the add-one rank rule are therefore material in that calibration. Different null thresholds can alter the comparison and must be considered when interpreting the paired decisions.

A ten-history raw-statistic pilot contained six null and four alternative histories. On that small pilot, level-only contraction exceeded trend-only contraction on four alternatives. This is a descriptive diagnostic of local score behavior, not a calibrated power comparison. That historical experiment does not contain a fully calibrated trend-only row, so the level-only discordance cannot be assigned uniquely to trend. The absence of that row does not imply that trend-only would perform best, nor does it automatically identify trend-only calibration as the best next experiment. A fair decision among procedures would require its own reference bank and calibration under the same reporting rules.

All 1,150 decisions in the reported historical detection-and-null evaluation resolved under the empirical numerical envelopes. That is a completeness statement for this realized study, not a rigorous universal bound on numerical integration or optimization. The selected-ten

pilot is documented in archival S15 (recovered archive). The paired detection table and matched-null results are documented in archival S16 (recovered archive).

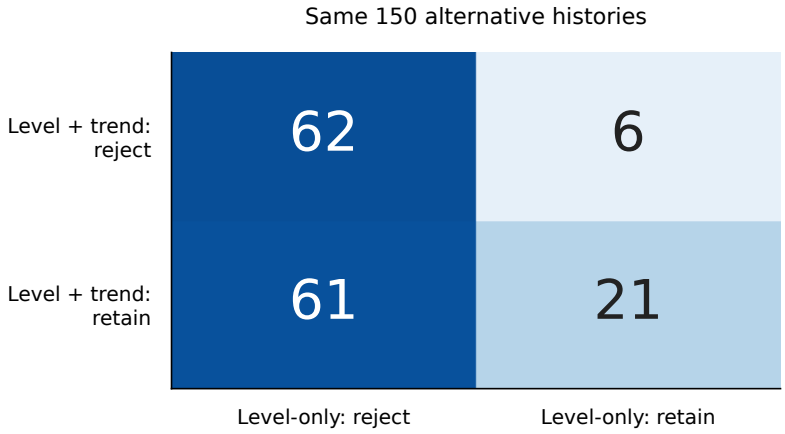


Figure 7.2: Paired decisions for background level alone versus level and trend estimated. The 61 versus six discordant histories yield a 36.67 percentage-point net difference, but neither procedure dominates on every history.

7.3 Diagnostics and limits of explanation

A separate audit examined ten selected difficult completed histories using a larger iteration limit and a Powell search. The largest phase improvement in negative log likelihood was $1.94995664 \times 10^{-8}$, and the largest direct generalized-likelihood-ratio change was $3.89681745 \times 10^{-8}$, below the declared diagnostic screens of 0.001 for phase negative log likelihood and 0.1 for the direct statistic. These histories were selected because they were difficult, so the audit is neither representative of the full bank nor a global optimization guarantee. Within the audited set, however, it weakens an explanation of the fitted-law gap based on inadequate optimizer stopping. It does not eliminate the possibility of optimizer limitations on other histories.

Two pre-existing fixed-history score projections onto a, d, and n retain 10.82% and 3.26% of weighted η squared-score norms. These percentages show local overlap between the directions in which the background parameters and the diagnostic parameter move the score. They are not ensemble Fisher information, not fractions of statistical power, and not an explanation of the observed 38% fitted detection rate.

Saved likelihood accounting provides one further descriptive check: estimating background level and trend improved the selected null likelihood more than the alternative likelihood on all 150 histories, with median null negative-log-likelihood gain 28.665 versus 0.862 for the alternatives. This records how the supplied comparison accounted for background in these specific histories. It is not a measurement of physical loading. The optimizer audit is reported in archival S14 (recovered archive); the score geometry is reported in archival S02 (recovered archive); and the saved likelihood accounting is documented in archival S11 (recovered archive).

Taken together, the matched studies locate a substantial change in calibrated detection when background parameters are estimated: under the stated synthetic law, the stronger alternatives are distinguishable with supplied laws and with η alone estimated, while fitting background components changes the calibrated detection outcome. The experiments cannot separate the unavoidable difficulty of a composite statistical problem from recoverable loss in

a particular fitted procedure. They are controlled matched-law tests rather than empirical physical discovery, and the reported boundaries follow from that design. The scoped contribution is a reproducible, numerically explicit assessment of where the marked-sensitivity signal is retained or attenuated by estimation, together with the calibration limits needed to read it.

The public detection summary and the availability guide provide public entry points. Raw histories, role/seed/clock mappings and the complete executed known-law replay are supplied in `reproduction/synthetic/` and `reproduction/oracle/`. The original fitted-procedure records and supporting analyses are supplied with their input identities. The separate F continuation has been withdrawn from this edition because its preserved original restricted-fit states could not be recovered; its source, results and withdrawal record remain in the historical archive.

Part II

Temporal, Mark-History, and Orthogonalized Extensions in New Zealand

Chapter 8

Reconstructed New Zealand Baseline and Temporal Diagnostic

8.1 Scope and provenance of the archived catalogue

The three test windows contain 203, 155 and 254 events, totalling 612. These same targets are inspected again in the mark-history, productivity, orthogonalization, temporal-kernel and spatial studies of Chapters 9 and 10. Later choices were informed by earlier results. Training-only fitting within a frozen stage therefore coexists with sequential adaptive reuse of the evaluation archive; the programme does not provide independent confirmatory holdouts (Dwork et al., 2015).

Execution provenance. The recovered-input edition reran the frozen diagnostic fitting recipe from the exact reconstructed catalogue; Section 8.7 reports a separately executed autonomous fit. Earlier restriction calibrations below retain their original records and were not recomputed during this edition.

The empirical analysis begins with an archived catalogue rather than with the unavailable inputs to the original manuscript. The primary archive is a GeoNet WFS snapshot downloaded on 15 October 2026. Events were selected with magnitude $M \geq 5$ and depth no greater than 40 km, within latitude $[-48.5, -33.5]$ and longitude $[165, 180]$. The depth sentinel value -9 was excluded, while other negative depths were retained. The resulting file contains 913 selected events in a 24,394-day observation window, from 1960-01-01 inclusive to 2026-10-15 exclusive. The event count is recorded, together with the filter definition and catalogue metadata, in the archived research report and processed metadata (archival `S25` (recovered archive); public metadata extract (`S24`)).

The reappearance of the number 913 does not establish identity with the event list analysed in the original work. The original frozen data, source configuration, and executable code are absent, so neither event identity nor processing history can be recovered. A second archived FDSN interface from the same provider returned 914 selections, consisting of 913 events common to the primary snapshot and the additional event 2025p672513. These are interface selections from one national provider, not independent physical catalogues. The interfaces share the same provider and much of their processing history; agreement between them cannot serve as independent validation of completeness or magnitude accuracy.

This distinction governs the interpretation of every later comparison. The reconstruction is an independent analysis applied to a documented archive snapshot, not a numerical replay of the historical analysis. The fitted catalogue slope for a continuous excess-exponential magnitude model is 2.849401764894274. For comparison with the earlier manuscript's half-bin convention, define

$$\widehat{\beta}_{\mathrm{adj}} = \frac{1}{\overline{M - M_0} + 0.05}, \qquad M_0 = 5.$$

Applied to the archived selection, this gives 2.494071218330174, consistent in rounded form with the reported value near 2.494. This reconciles an estimator convention; mixed precision

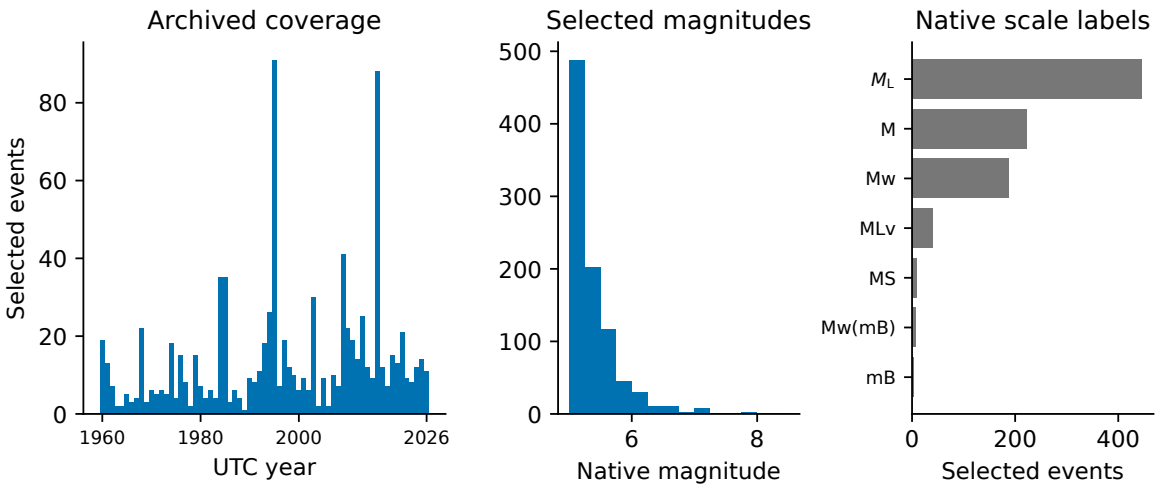


Figure 8.1: Descriptive summaries of the 913-event archived selection: annual counts, magnitudes and native magnitude labels. The final calendar year is partial. These plots describe this selection and do not establish catalogue completeness or homogeneous observation through time.

and native magnitude scales do not establish a common half-bin adjustment as a generally validated correction for this catalogue. The primary continuous estimate, raw magnitudes, productivity normalization and fitted-law records remain unchanged. It does not demonstrate that the historical event sample or likelihood calculation has been reproduced. Completeness above a time-varying threshold, the relation between native magnitude scales, missing parents of smaller or deeper events, activity before 1960, and the adequacy of national pooling therefore remain explicit assumptions (archival S27 (recovered archive)).

A magnitude threshold specifies the selected observations; it does not establish complete detection or a homogeneous magnitude scale. Completeness estimates have uncertainty (Woessner and Wiemer, 2005), and a detection function can alter the recorded frequency–magnitude distribution (Mignan, 2012). Changes in that observation law can resemble changes in occurrence rate or productivity. The common measurement and uncertainty interpretation is stated in Section 6.1.

8.2 A restricted diagnostic intensity

The observed comparison uses the temporal diagnostic defined in the experimental design:

$$\lambda_\eta(t) = B(t)e^{-\eta G_s(t)} + nH_e(t), \qquad B(t) = e^{a+d(2t/T_0-1)}, \quad \eta \geq 0. \tag{8.1}$$

Here the subscript s denotes stress memory and e denotes excitation. Neither denotes a spatial coordinate. Both features depend only on strictly earlier events. The implemented Mittag–Leffler (ML) kernels are finite positive spectral mixtures. A shared-memory recipe supplies the same kernel parameters to the stress and triggering features, although those features enter different parts of the intensity. The background is specified independently of the stress history. Thus this family diagnoses a restricted form of inhibition; it does not reproduce the original physical loading construction.

The five observed candidate families were constant Poisson, loglinear Poisson, Omori triggering, ML triggering, and ML triggering with shared inhibition. The principal contrasts compare the last three families. Model labels use "ML" for Mittag–Leffler. The native local-magnitude label in Figure 8.1 is displayed as M_L; the original archive label is retained in the

data. All five families and their search rules are recorded in public search protocol (operational adaptation) (`S18`).

The temporal Omori comparator uses the normalized triggering density

$$h_{\mathrm{O}}(u; c, p) = \frac{p-1}{c}\left(1+\frac{u}{c}\right)^{-p} = (p-1)c^{p-1}(u+c)^{-p}, \qquad u \geq 0, \quad c > 0, \quad p > 1.$$

The lag u is measured in days. Its integral on $[0, \infty)$ is one. The scale c is measured in days and is stored as `KernelSpec.tau`; the triggering multiplier n remains separate. This specifies the temporal comparator used here, not a complete space–time ETAS construction.

The split is chronological. The absolute trend coordinate is $x = 2t/T_0 - 1$, where $T_0 = 24394$ days. Candidate fits use the first 0.50, 0.65, and 0.80 fractions of the archive horizon as training prefixes, comprising 301, 504, and 659 events. Tests use the following disjoint intervals 0.50–0.65, 0.65–0.80, and 0.80–1.00, containing 203, 155, and 254 events and 612 test events in total. These intervals correspond approximately to 1993–2003, 2003–2013, and 2013–2026, but the fractions and archive day indices are the operative definitions.

All retained history before each origin, including earlier test history, is available to the candidate intensity. Each fit uses only its training prefix. The exponential mark-rate estimate is obtained from training marks alone and is common across candidate families; it is a plug-in estimate rather than a joint marked-intensity maximum-likelihood estimate. The reported score is the full marked-event score: the sum of event log intensities and log mark densities minus the integrated intensity over quiet exposure, with the final censor handled explicitly in the declared protocol. The uninhibited fit is retained as an exact eligible point in the inhibited search, so nonnegative training gain is partly built into candidate selection. This fact limits any inference from training improvement alone. The stored training gains for the shared diagnostic are 2.731009, 0.302486, and 0.143866, but those numbers are not treated as evidence of predictive success.

8.2.1 Saved parameters at the predictive origins

Table 8.1 displays the saved selected records for the three triggering families at the three predictive origins, not new fits. The selected-parameter companion retains all fifteen records for the five families, full precision, source hashes and field pointers. The constant Poisson model fixes $d = n = \eta = 0$; loglinear Poisson fixes $n = \eta = 0$. The table's two uninhibited triggering families fix $\eta = 0$.

All six selected Mittag–Leffler kernels use $\alpha = 0.5$, the lower searched grid endpoint, and $\tau = 1$ day, an interior grid value. The common selected productivity ratio is $r = 0.4$ at the first origin and $r = 0.7$ at the later origins; the former is the lower searched ratio endpoint. The Omori coordinates c, p were optimized continuously with a retained coarse-grid floor; they are not merely grid winners. These endpoint descriptions are newly checked arithmetic against the frozen search grids. Historical boundary flags, tolerance-based near-boundary classifications and standard errors were not saved. The selected continuous coordinates are strictly inside their recorded bounds under exact comparison. No uncertainty estimates are inferred.

The rates β_{train} are estimated from training marks alone, held common across families, and frozen during each test. Inactive serialization fields are not fitted coordinates: the saved p field of a Mittag–Leffler kernel and α field of an Omori kernel have no role in their respective formulas.

At the three predictive origins, the selected shared-inhibition recipe's unattenuated background intercept is approximately 9.60, 1.57 and 1.34 times that of the noninhibited Mittag–

Table 8.1: Saved selected parameters at three predictive origins. The background is $\exp(a + dz(t) - \eta G_s(t))$ per day, with $z(t) = 2t/24394 - 1$. Values are rounded for display; zeros in the uninhibited families are fixed.

Origin	Family	a	d	n	η
50%	Omori	−4.712517	−0.457071	0.685640	0.000000
50%	Mittag–Leffler	−4.458572	−0.252150	0.455824	0.000000
50%	Shared inhibition	−2.196743	1.789757	0.481700	1.646714
65%	Omori	−4.291554	0.048036	0.570076	0.000000
65%	Mittag–Leffler	−4.188810	0.147441	0.497996	0.000000
65%	Shared inhibition	−3.841844	0.492682	0.503490	0.211710
80%	Omori	−4.394252	−0.196162	0.561216	0.000000
80%	Mittag–Leffler	−4.282575	−0.056651	0.470292	0.000000
80%	Shared inhibition	−4.103408	0.127638	0.473071	0.105254

Origin	β_{train}	r	Omori c (days)	Omori p
50%	3.101622	0.4	0.012995	1.113368
65%	3.149488	0.7	0.020798	1.198604
80%	3.021907	0.7	0.006972	1.155845

Leffler recipe. In the first training prefix, the trend coordinate lies in $[-1, 0]$. Of the 195 relaxation modes, 61 decay by less than 1% over the 24,394-day horizon, but they contain only 0.043% of the normalized mixture weight. The full stress history has jumps and decay; it is not monotone. Its training event-time projection $G_s = c_G + k_G x + R_G$, with $c_G = 2.7507$ and $k_G = 2.2417$, gives the exact identity

$$\log B_\eta(t) = (a - \eta c_G) + (d - \eta k_G)x(t) - \eta R_G(t).$$

The affine part of depletion can therefore exchange with intercept and trend. The projection explains 14.9% of the training pre-event stress variance; the residual remains part of the intensity. With event-score weights $(B_\eta/\lambda)^2$, the intercept-adjusted trend–depletion correlation is 0.9341, quantifying the local confounding of these directions. The first inhibited fit combines $d = 1.789757$, compared with -0.252150 for uninhibited ML, with the 9.60-fold larger intercept. Extrapolation to $x > 0$ changes that balance: the training projection's residual has positive mean on the test events. The held-out accounting below gives the observed direction: the inhibited integrated intensity is lower than ML's, and its negative event log-intensity contrast outweighs the compensator saving. The largest subinterval loss occurs in 1995–1998. The executable projection and archived-intensity checks are in `experiments/methodological_audit/trend_depletion.py`.

8.3 Earlier calibrated restrictions

The earlier reconstructed analysis tested two restrictions: no feedback and a common rather than separate memory scale. Its observed statistics were 0.7036472 and 0.0829952, respectively. The completed fitted-law calibration retained 1,999 null cases per restriction, giving add-one ranks of 0.365 and 0.263. Neither restriction was rejected by that declared five-percent procedure. These are conditional ranks, not probabilities that the hypotheses are true. In particular, non-rejection does not demonstrate equal physical memories, and the stress kernel's disappearance at zero feedback limits what such a restriction can establish. The calibration

authority is archival S01 (recovered archive); its interpretation is summarized in archival S27 (recovered archive).

The later chronological comparison asks a different question: do the declared inhibited recipes predict subsequent observations better than serious triggering alternatives? A negative predictive result need not contradict a non-rejecting in-sample restriction test. Their targets, fitting rules and calibration differ.

8.4 Held-out score comparison

Table 8.2 reports score differences relative to the ML triggering-only benchmark. Negative values favour the benchmark; positive values favour the named alternative.

Table 8.2: Held-out marked-process score differences relative to the ML triggering-only benchmark. Tests are disjoint chronological intervals in the declared archive protocol. Values are differences, not log-likelihoods for separate datasets. Paired calendar-block intervals for shared inhibition follow below.

Contrast	Test 1	Test 2	Test 3	Total
Shared inhibition − ML	−16.517557	−0.075277	−1.030025	−17.622859
Omori triggering-only − ML	+3.106623	+2.837708	+6.513790	+12.458121

The shared diagnostic loses in all three folds. Its total disadvantage is driven overwhelmingly by the first block: approximately 93.7% of the total deficit occurs there. The second-block difference is nearly zero, but that is not strong independent evidence of parity because the training histories overlap and later folding choices were inspected during development. The calculations are exploratory and were not designed as confirmatory or as an independent sequence of repeated tests (archival S19 (recovered archive)).

The continuous Omori recipe improves on the finite ML kernel grid in every fold. This result cannot be converted into a universal law ranking. The Omori refinement and the finite spectral ML candidate have unequal flexibility, and the contrast therefore favours the declared Omori recipe under its fitting convention. It is not an isolated estimate of the effect of η, nor a demonstration that every continuously refined triggering model dominates every grid-based alternative. Residual diagnostics were descriptive, varied over the record, and did not certify any winner as an adequate physical model.

The catalogue, frozen fitting recipe and executable score reconstruction are supplied in `reproduction/observed/`.

8.4.1 Paired calendar-block score uncertainty

For each test fold, contiguous 90-day calendar blocks retain both the event log-intensity contrast and the corresponding integrated-rate contrast, including empty blocks and terminal exposure. We resample these paired block scores 49,999 times with replacement to obtain percentile 95% intervals for the total contrast. A two-sided paired sign-flip reference independently reverses each block contrast. Training estimates remain fixed in these conditional comparisons. Pooled totals resample separately within each disjoint fold before summing. We also repeat the calculation with 180- and 365-day blocks.

Calendar splitting does not make these blocks independent or strictly exchangeable. In the New Zealand model, 61 of the 195 finite-mixture modes decay by less than 1% over 24,394 days, with minimum rate $e^{-38.8}$ day^{-1}; their aggregate weight is small, but the state is carried across block boundaries. Triggered sequence clustering and fitted nonstationary rates provide further

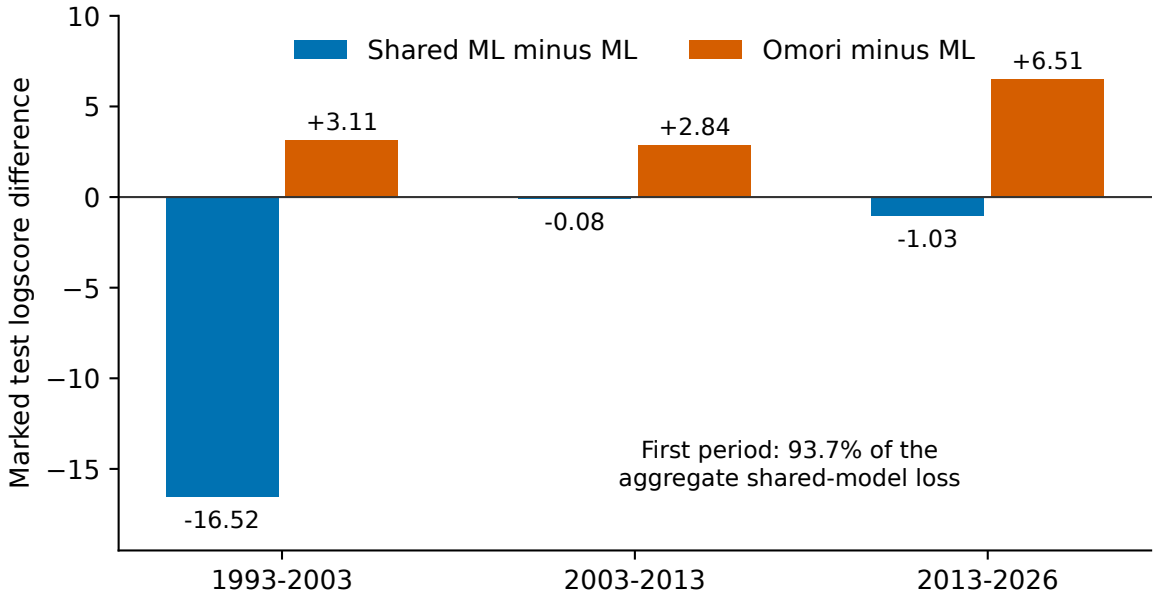


Figure 8.2: Chronological test-score contrasts for the declared recipes. The first test block contributes about 93.7% of the aggregate shared-inhibition disadvantage. Training prefixes overlap and the periods were reused in exploratory development; the three periods are not independent confirmatory tests.

dependence across months and years. An independently reversed sign for each observed block is therefore a reference calculation under an additional symmetry/exchangeability assumption, not a justified randomization of independent earthquake sequences.

Small nominal sign-flip values and wide block intervals address different features of the same contrasts. For example, Fold 1 has nominal $p = 0.00104$ and interval $[-25.526, -7.459]$ nats, while its strongest subinterval loss is concentrated in 1995–1998. The sign-flip reference counts directional extremeness under independent reversals; the bootstrap expresses magnitude variability under its chosen block resampling. Neither operation establishes long-memory calibration. Prioritize the agreement or disagreement of the 90-, 180- and 365-day interval summaries and the fold-level directions over a nominal 90-day p-value. The intervals also remain conditional sensitivity summaries: their nominal coverage is not proved under the catalogue's full dependence and fitted-parameter uncertainty. Section 6.1 states the common inferential scope.

Fold	Contrast (nats)	Paired 95% interval	Nominal sign-flip p
1	-16.518	$[-25.526, -7.459]$	0.00104
2	-0.075	$[-1.320, 1.215]$	0.91248
3	-1.030	$[-2.087, 0.004]$	0.05766

Table 8.3: Shared inhibition minus the uninhibited Mittag–Leffler model: paired calendar-block uncertainty. Negative contrasts favour the uninhibited model.

Fold 1 contributes 93.7% of the total deficit and its intervals exclude zero at all three block lengths. Fold 2's intervals contain zero throughout: its 180- and 365-day intervals are $[-1.109, 1.014]$ and $[-1.511, 1.324]$. Fold 3 is sensitive to the dependence scale: the 90-day interval includes zero, whereas the 180- and 365-day intervals are $[-2.185, -0.091]$ and $[-2.149, -0.225]$; the annual-block nominal sign-flip value is $p = 0.015$. This change in zero inclusion is part of the block-size sensitivity, not a validated long-memory rejection.

8.5 Score accounting and temporal localization

For a fold, write the score difference as

$$\Delta\ell = \Delta Q + \Delta C, \qquad \Delta Q = \sum_i \log \frac{\lambda_{\mathrm{sh},i}}{\lambda_{\mathrm{ML},i}}, \qquad \Delta C = \Lambda_{\mathrm{ML}} - \Lambda_{\mathrm{sh}}, \tag{8.2}$$

For this temporal model $P = Q$, $E = \int \lambda(t)\, dt > 0$, and $C = -E$. Thus $\Delta C = E_{\mathrm{ML}} - E_{\mathrm{sh}}$. The historical event-sum symbol E has been replaced here by ΔQ. The event sum uses pre-event intensities at the realized times, and Λ denotes the corresponding integrated intensity. This separates event-term disagreement from realized compensator disagreement. In the first test block,

$$\Delta Q = -33.715205, \qquad \Delta C = 17.197647, \qquad \Delta\ell = -16.517557.$$

The integrated intensities are 115.871595 under the shared diagnostic and 133.069242 under the ML benchmark for 203 events. These are not conventional expected counts from a fixed fitted Poisson model: the compensator is evaluated along an updating estimated history. Its realized value is not automatically the future expected count conditional only on information at the test origin. No naive Poisson count probability value is attached to their difference.

Four equal-duration bins within the first test interval sharpen the accounting without allocating events to bins by any counterfactual rule. Each bin uses the complete preceding history and integrates over its own realized path. Approximate calendar labels are displayed only for orientation; the archive-day boundaries define the calculation.

Table 8.4: Descriptive score accounting within four equal-duration subintervals of the first test block. Each row uses full preceding history and realized integration within that subinterval. Confidence intervals are not reported.

Approximate interval	ΔQ	ΔC	$\Delta\ell$
1993-05-24 to 1995-11-24	−2.891244	2.464270	−0.426975
1995-11-24 to 1998-05-27	−16.875378	7.143815	−9.731563
1998-05-27 to 2000-11-27	−8.696131	4.642115	−4.054016
2000-11-27 to 2003-05-31	−5.252451	2.947447	−2.305004

The deficit is localized in time rather than uniformly spread across the first block. The strongest subinterval loss occurs between approximately late 1995 and mid-1998. This localization does not identify an earthquake sequence, a geological province, or a causal mechanism. It records that the aggregate score is sensitive to when an updating history and its compensator disagree (archival `S20` (recovered archive)).

A spatial and mark audit found that all five occupied coarse cells had negative event-term contributions for the shared-minus-benchmark contrast. The same direction appeared for native ML and M_{w} magnitude conventions and for all declared magnitude and depth groups. The shared specification contains one regional compensator rather than cell-specific or group-specific exposure. Accordingly, these sums are diagnostics of event-term sign under the fitted family, not local likelihoods, local rates, or fractions of the full score. Unequal cell areas, time-location confounding, sparse cells, and reuse of events across overlapping marginal classifications prevent such a local interpretation. Grouping by the current event's mark is descriptive: that mark affects subsequent intensity and does not enter its own pre-event intensity. Native magnitude-label composition changes most strongly in the last block, whereas the dominant loss is in the first, weakening a simple explanation based only on a modern label transition. Completeness and legacy magnitude-scale concerns remain.

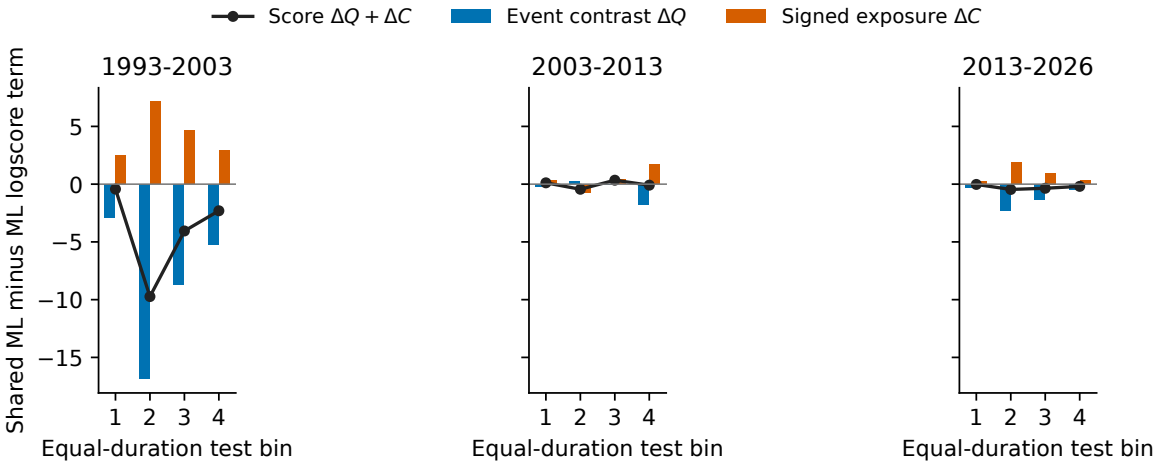


Figure 8.3: Event and compensator contributions in four equal-duration bins in each of the three test periods. The event log-intensity contrast is ΔQ and the signed exposure contrast is ΔC. Each bin is integrated with complete preceding history. In the first period a favorable compensator contribution partly offsets the negative event term; the full score is $\Delta Q + \Delta C$.

8.5.1 Trend–depletion controls in the first New Zealand fold

The first fold accounts for 16.518 nats of the 17.623 nat shared-attenuation deficit. A matched zero-trend fit reduces the first-fold loss to 8.504; freezing the uninhibited trend reduces it further to 4.556. Relative to the free fit, the zero-trend and fixed-null controls therefore remove 48.5% and 72.4% of the deficit, leaving a detrended residual of 8.504 nats. The positive fitted trend, not depletion alone, accounts for most of the original first-fold loss. The clamped-clock counterfactual is not a refit and produces a 24.791 nat loss; it is reported only to show that removing positive extrapolation without re-estimating the intercept is not an admissible control.

The middle fold's zero-trend fit selects the zero boundary and is tied at displayed precision, while the late fold retains a small positive strength. These two folds do not support the same attribution as the first fold. The autonomous zero-trend comparison in Table 8.6 remains a separate fitted procedure with a different nuisance treatment.

Fold	Control	a	d	n	η	$\Delta\ell_{\mathrm{tr}}$	ΔQ	ΔC	$\Delta\ell$
F0	Free d	-2.197	1.790	0.482	1.647	2.731	-33.715	17.198	-16.518
F0	Fixed null d	-4.329	-0.252	0.461	0.178	0.299	-12.971	8.415	-4.556
F0	Zero d	-4.064	0.000	0.464	0.357	1.225	-26.025	17.521	-8.504
F1	Free d	-3.842	0.493	0.503	0.212	0.302	-1.786	1.711	-0.075
F1	Fixed null d	-4.152	0.147	0.499	0.034	0.048	-2.033	2.709	0.675
F1	Zero d	-4.241	0.000	0.499	0.000	0.000	-0.000	0.000	0.000
F2	Free d	-4.103	0.128	0.473	0.105	0.144	-4.431	3.401	-1.030
F2	Fixed null d	-4.262	-0.057	0.471	0.016	0.022	-3.397	2.271	-1.126
F2	Zero d	-4.213	0.000	0.472	0.043	0.162	-9.213	6.286	-2.927

Table 8.5: New Zealand trend–depletion controls. Free d reproduces the archived fits. Fixed null d freezes the uninhibited `ml_trigger` trend while fitting (a, n, η); the zero-trend row fits (a, n, η) with a zero linear trend and the archived productivity ratio and kernels. Training and held-out entries are nats; held-out $\Delta\ell = \Delta Q + \Delta C$ and the common mark cancels.

8.6 Archive sensitivity

A replacement audit matched all 913 event identifiers and found 241 numerical magnitude differences, the largest absolute difference being 0.164143665. Many differences were only in small decimal places. The full history was rescored using alternate matched magnitudes while primary clocks, locations, depths, the fitted β, and the recipe definitions were fixed. The additional event in the 914-event interface selection was excluded, and no parameters were refitted or physically converted between magnitude scales.

Under this replacement, the total shared score changed as follows:

$$-17.622859365 \longrightarrow -17.617975447, \qquad \text{change } +0.004883918.$$

No fold changed sign. The Omori total changed from 12.458121 to 12.441651, shown to six decimal places. Thus the tested archive value discrepancy does not explain the recipe failure. Equally, this bounded audit does not eliminate every observational uncertainty in magnitudes, depths, locations, completeness, or event identity.

The observed evidence in this chapter is therefore selective and conditional. The held-out scores are unfavourable to the implemented shared-inhibition recipes, the source-magnitude audit does not reverse that pattern, and the strongest loss is temporally localized. These findings constrain the current diagnostic families. They do not establish the absence of physical shared memory in the Earth system.

The public predictive summary and the availability guide provide public entry points. The exact selected catalogue and complete historical and newly executed fitting outputs are supplied in `reproduction/nzca/` and `reproduction/observed/`. The command `scripts/reproduce_observed.py` rebuilds the fitted results from the catalogue and compares their scientific fields.

8.7 Fitting an autonomous finite-mode subfamily

This section reports a separately executed fit of the raw constant-background intensity family already used in Chapter 9 (Tables B.8 and B.10). It makes the mapping to the autonomous theorem explicit; it does not introduce a distinct intensity family. Set constant loading $\rho = 0$, link $\Psi(X) = \exp(a + X)$, and initial modes $Z_j(0) = U_j(0) = 0$. Retain the diagnostic recipe's fixed finite mixture and mark scalings. With physical stress release ηg_d and productivity $n\kappa_d$, the mode equations give

$$G(t) = \sum_{T_i<t} g_d(Q_i)\Phi(t - T_i), \qquad H(t) = \sum_{T_i<t} \kappa_d(Q_i)h(t - T_i),$$
$$\lambda(t) = \exp\{a - \eta G(t)\} + nH(t), \quad h = -\Phi'.$$

Here $g_d(Q) = \exp\{\gamma(Q-1)\}$ and $\kappa_d(Q) = (1-r)\exp(\beta r Q)$ use the existing fixed diagnostic scalings. Under the posited independent exponential mark law, $\mathbb{E}[\eta g_d] = \eta e^{-\gamma}\beta/(\beta-\gamma)$ when $\eta > 0$ and $\beta > \gamma$, while $\mathbb{E}[n\kappa_d] = n < 1$. At $\eta = 0$ stress release is identically zero. Since $X \le 0$, the link is bounded and Lipschitz on the reached half-line. These are the relevant first-moment conditions of Theorem 3.1; zero initialization is not a stationary draw.

For general constant loading, event-free equilibrium initialization $Z_j(0) = \rho/\vartheta_j$ instead gives

$$X(t) = \rho \sum_j \frac{w_j}{\vartheta_j} - \eta G(t), \qquad a_{\text{eff}} = a + \rho \sum_j \frac{w_j}{\vartheta_j}.$$

Loading and intercept are therefore confounded; zero loading is a convention, not an estimate. Equal non-equilibrium initial modes contribute $X_0\Phi(t) + \rho \int_0^t \Phi(u)\ \mathrm{d}u$, generally not an affine trend. Neither convention is stationary event-process initialization. Section 3.1 quantifies the certificate's practical limits.

Each training prefix fixes the common estimate $\hat{\beta} = N_{\text{train}}/\sum Q_i$. The null fits (a, n) at $\eta = 0$; the inhibited arm fits (a, n, η) and retains the exact fitted null as a candidate. Selection uses training likelihood only. All prescribed starts completed, every returned vector and solver diagnostic is retained, and the training and held-out quadrature checks and mark-moment gates passed. Earlier parents and the final event-free tail are included. The frozen protocol, executable fit and complete outputs are supplied under `experiments/autonomous/` and `validation/live_results/`.

The earlier raw constant-background procedure used the conditional-within-bin training estimate of the magnitude rate. The present estimate changes the productivity weights as well as the common mark score. The prescribed starts, solver tolerances, recovery settings and candidate accounting also differ. Consequently these are different fitted procedures within the same intensity family, and their score difference is not an isolated effect of changing β. The earlier aggregate remains in Table B.10; the supplied protocol comparison records both recipes without replacing either result.

Fold	Null score	Inhibited score	Gain
Early	-291.5675	-301.5859	-10.0184
Middle	-170.2334	-170.2334	0.0000
Late	-523.5999	-526.5270	-2.9272
Pooled	—	—	-12.9456

Table 8.6: Held-out marked log scores and inhibited-minus-null gains for the autonomous finite-mode subfamily, in nats. The common mark contribution cancels from each gain; this is an occurrence-score comparison. The middle fold selects $\hat{\eta} = 0$: its displayed tie contains only a small shared-coordinate optimization difference. These contrasts are descriptive; confidence intervals are not reported.

The early and late fits select positive inhibition and lose held-out score; the middle fit selects zero inhibition. The pooled gain is -12.9456 nats. This exploratory comparison fits the autonomous theorem's model class on reused chronological observations. Mathematical stability of the autonomous model does not imply a predictive advantage on finite catalogues, and the fixed mark and kernel settings preclude full joint nuisance estimation.

Chapter 9

Magnitude History, Productivity Sensitivity, Exposure Orthogonalization, and Kernel Refinements

The historical observed comparison is retained in Chapter 8. This chapter consolidates the separately frozen magnitude-history and productivity studies, followed by exposure orthogonalization, parent-class temporal scales and continuously tuned Omori triggering. Each comparison uses its own matched reference. The same historical targets recur across sensitivity settings; they do not become independent replications.

Execution provenance. The magnitude-history, productivity, orthogonalization and temporal-kernel results here retain their original fit and audit records. The later selected-state replay recomputes the declared likelihoods from recovered histories or cached numerical features and reconciles the score tables and sampled profile-support statements. It does not rerun the original optimizer searches. The separate frozen diagnostic replay is reported in Chapter 8.

This reuse is sequential and adaptive at programme level: inspected results helped motivate later model families, parameter domains and reporting contrasts. A protocol frozen before one stage constrains that stage's execution but does not restore an unused holdout after earlier inspection. The resulting scores are exploratory comparisons on a shared archive, subject to the adaptive-data-analysis distinction discussed by Dwork et al. (2015).

For the temporal models, $P = Q$, timing is $Q+C$, and the joint marked score is $P+C+M$, with $E > 0$ integrated exposure and $C = -E$. Reported differences are candidate minus the named reference. The source reports' event field (sometimes called E historically) is displayed here as ΔQ; their compensator contrast is ΔC. The magnitude excess $q = M_{\text{magnitude}} - 5$ must not be confused with the score component M. The class cutoff denoted c in the class-history equations is $c_{\text{class}} = 0.5$ magnitude units, whereas c in the Omori sections is a temporal scale in days. These are separate local namespaces. The free-productivity coordinate is continuous on $[0, 0.95]$; the seven-node grid below is a descriptive profile, not its optimization domain.

9.1 The earlier magnitude-history misspecification pilot

The run `magnitude_history_20261023T205716Z` precedes the observed class-history forecast investigation. It asks whether changing magnitude dependence alters a preserved inhibitory fitting procedure's preference under a designed, entirely excitatory generating family. It is neither a fitted comparison of the richer excitatory family nor a forecast experiment.

For magnitude excess q with exponential rate b and bin cutoff a_{bin}, let $p_0 = 1 - e^{-ba_{\text{bin}}}$ and $p_1 = e^{-ba_{\text{bin}}}$. The bins are $[0, a_{\text{bin}})$ and $[a_{\text{bin}}, \infty)$. Let $P_{ij}(h) = (1-h)p_i + h\mathbf{1}_{i=j}$. With

common normalized waiting density k and no magnitude productivity in the generator,

$$\lambda_i(t) = \mu p_i + n_g \sum_{k:t_k<t} P_{iJ_k}(h)k(t-t_k), \qquad \Lambda(t) = \mu + n_g \sum_{k:t_k<t} k(t-t_k).$$

Generate the untyped timed branching forest first. Since the columns of P sum to one and $Pp = p$, assigning immigrant classes from p and child classes from the parent's column preserves each node's marginal class law by induction. Fixed-window censoring and chronological ordering depend only on the untyped forest, so this remains an exact finite-window statement conditional on the entire timed forest and its event count. Mixing the exponential law restricted to each bin recovers exactly the same exponential one-event magnitude law. At $h = 0$ the marks are iid; positive h changes labelled-history dependence without changing total timing or one-event marginals. Class counts at positive h need not be multinomial, and conditioning on label-selected subsets does not preserve this argument. In particular, with class histories S_i and $S = \sum_i S_i$,

$$\frac{\lambda_i}{\Lambda} = p_i + \frac{n_g h}{\Lambda}(S_i - p_i S).$$

Each absolute class intensity is nonnegative and receives nonnegative event increments even though its conditional class probability can decrease.

The same timed forest is labelled under all three persistence scenarios using fixed per-node random inputs and separate time/mark streams. The independent replicate is the forest, not each scenario catalogue or each event. The old candidate uses magnitude-productivity exponent $r = 0.7$, whereas the generator has no magnitude productivity ($r = 0$, or $\gamma = 0$ in its parameterization). Its finite-mixture temporal representation also differs from exact Mittag-Leffler generation. The candidate therefore remains misspecified even at $h = 0$; the mismatches are common across scenarios. Supplied-background settings do not become true generator parameters. Positive nested fit gains therefore cannot be interpreted as calibrated false-positive rates.

9.1.1 Generator settings and fitted statistic

The generator is a designed two-class MDFHP-form excitatory family with common Mittag-Leffler shape 0.5 and scale 1/day, total immigrant rate $\exp(-4.162108473051707)$, offspring mean 0.5 and empty prehistory over 24,394 days. Magnitude excess has marginal exponential rate 2.849401764894274, with bins $[0, 0.5)$ and $[0.5, \infty)$. The $\gamma = 0$ productivity boundary and unbounded final bin are explicit extensions of the published fitted specification. Selection was frozen as direct GLR > 0.001 and selected $\eta >$ `1e-6`. No Wilks reference law, calibrated rejection cutoff, power ranking or physical-mechanism interpretation was used.

For scenario h and planned independent skeleton s, let $G(s, h)$ be the procedure's resolved direct likelihood-ratio statistic and $\eta(s, h)$ its selected inhibition parameter. The frozen descriptive preference indicator equals 1 when G exceeds 0.001 and η exceeds `1e-6`. These thresholds define an operational preference; they are not calibrated significance cutoffs. The primary contrasts compare the preference indicator and continuous statistic at positive h with the matched baseline at $h = 0$ on the same skeleton.

Every scenario's planned denominator is retained. Resolved-only frequencies estimate a resolution-conditional descriptive quantity, with its denominator stated. If k cases select inhibition and u cases are unknown out of N planned cases, the all-planned selection frequency is bounded by $[k/N, (k+u)/N]$. These are missing-outcome accounting bounds, not confidence intervals. Continuous-statistic summaries based on resolved cases remain conditional summaries; missing continuous values are not replaced by zero.

The frozen persistence values are $h = 0$, 0.5 and 0.9. The actual pilot kernel has Mittag-Leffler shape 0.5 and inverse-time scale $c = 1\,\mathrm{day}^{-1}$, corresponding to the time scale $1/c = 1$ day. This pilot c is distinct from the Omori time scale used later in this chapter. The independent compiled core supports only shape 0.5 and 1; support for another shape is not claimed. Here b is the continuous mark rate and n_g is the generator branching mean. They are distinct from similarly named shape/productivity parameters in the preserved comparison engine.

The frozen primary procedure estimates background level and trend together with the inhibition alternative, while retaining the staged procedure's supplied triggering, stress and mark-rate settings. Its prescribed starts comprise two null fits followed by two alternative starts from the selected null. A secondary supplied-background procedure uses two inhibition starts. The supplied-background label does not mean its nuisance settings are true generating parameters.

This source-bound six-start staged comparison is narrower than the historical full kernel/productivity searches, the separate later nuisance continuation or the observed-catalogue fitting recipe. Results must identify this exact procedure. They cannot be substituted into historical counts, reference banks or claims about the larger searches. Solver starts, explicit $\eta = 0$ eligibility, subset floors, failure-only recovery and original numerical controls remain part of the procedure. Local optimization is not a global optimum certificate.

The $h = 0$ scenario is the experiment's independence baseline, not an in-model null-size experiment for the old procedure. The generator has $\gamma = 0$ magnitude productivity, while the old candidate intensities depend on prior continuous marks through their productivity and stress transforms. Exact Mittag-Leffler generation and the old finite-mixture representation also remain distinct. These mismatches are common to the persistence scenarios; the paired persistence contrast isolates the change in magnitude history within this deliberately specified stress family, without establishing correct specification at $h = 0$.

The primary comparison uses the same realized continuous marks and exposure for its excitatory and inhibitory candidates. A common mark-density term can cancel within that paired comparison, but marks still enter subsequent candidate intensities. Dropping a common density term does not justify discarding the marks. No comparison of raw continuous densities with discrete bin masses is made, and no held-out forecast-score claim follows from these fitted-statistic preferences.

9.1.2 Pairing, numerical exposure and reproducibility limits

The plan contains 20 independent untyped skeletons, each labelled under all three persistence scenarios. There are 60 scenario catalogues but 20 independent replicate blocks. Pairing shares the untyped skeleton and fixed per-node random inputs, while each scenario generates its own conditional labels and within-bin magnitudes. One fixed iid mark sequence must not be reused as the observed marks in every scenario.

Separate time and mark pseudorandom streams, their exact implementation, the compiler and runtime are pinned. Equal seeds in another language or standard-library implementation are not asserted to reproduce the same streams. Uncertainty for persistence contrasts resamples skeleton blocks. Resolved-only paired contrasts state the number of paired-resolved skeletons; all-planned bounds additionally retain unknown preference indicators. Pilot intervals are exploratory Monte Carlo summaries, not confirmatory coverage certification.

The synthetic window is fixed independently of the first and last events. The likelihood retains initial quiet exposure, every positive event gap and the final no-event interval. Saved two-part timestamps and positive gaps define ordering, including cases where a lossy binary64

projection ties. The generator reports a precision failure when its declared parent-child wait-error gate is exceeded; it does not redraw a wait, jitter an event or replace a magnitude to force completion. Resource caps and unresolved mandatory solver states stay explicit unknown outcomes.

The computational route combines an independently implemented C++ generator/numerical core, an R interface and the preserved Python comparison machinery with its declared CUDA/FP64 primary route and existing failure-only recovery. It is not a full rewrite of the historical engine. Independent mathematical/spectral-quadrature, saved-clock and waiting-law controls have separate evidence and declared finite scopes.

An additional reference bridge ran the original released ml.m in GNU Octave. A private byte-preserving copy changed only the C++ dispatch switch to select its original language implementation; the copyright notice and original clone were retained. Fixed-input survival and kernel calculations were compared against the new independent core. This is executable original-source validation without MATLAB, not reproduction of the original MATLAB optimizer, all fitted examples or every numerical regime.

The exact author preprint is separately pinned from the version of record. The external study's exact published supplement remains unavailable, and its released Japan CSV contains no event rows. Those external inputs are distinct from this project's recovered regional catalogue snapshots and selected historical simulation banks, now supplied under `reproduction/`. Provider terms and source identities are recorded with the recovered inputs; the project licence does not replace third-party terms.

$\gamma = 0$ is a boundary/limiting case of the released positive-productivity fitting transform. The unbounded final magnitude bin extends the article's finite upper magnitude support. The accurate model label is a designed MDFHP-form excitatory stress family with matched exponential marginal marks, not an exact recovered author fit or the entire MDFHP family.

No correctly specified richer excitatory model is fitted in this bounded pilot. A changed inhibitory preference would demonstrate a vulnerability or response of the selected old procedure under this generator; it would not establish that an estimated MDFHP practically absorbs the effect in observed data. Conversely, a small or unresolved contrast would be limited to this family, sample budget and fitting procedure; it would not rule out other magnitude-history mechanisms.

There is no fresh null calibration, size-matched power comparison, observed-catalogue evaluation, causal intervention or physical inhibition identification. Pointwise checks cannot establish uniform performance over nuisance parameters. No combined inhibitory MDFHP, general exact-memory stability theorem, exhaustive novelty claim or external-publication conclusion is asserted. Final claims require terminal saved outputs, complete failure accounting and independent review.

9.2 Historical pilot results and interpretation

The validated machinery is feasible, and the full bounded pilot completed: 20 independent timed forests, three magnitude-dependence scenarios, 60 catalogues and 120 resolved method outcomes. No planned outcome is missing.

Under the frozen descriptive rule, the primary fitted-background comparison preferred positive inhibition in 15/20 independent-mark controls, 17/20 moderate-dependence histories and 18/20 strong-dependence histories. However, both primary paired GLR-difference intervals span zero. This pilot does not establish a clear increase in likelihood evidence specifically caused by magnitude history, or explain previous observed-catalogue evidence.

A nested, unpenalized alternative can improve fit under an excitatory generator. The rule below resolves a positive fit preference near numerical tolerance; these frequencies are not calibrated false-positive rates and do not demonstrate failure of any advertised calibrated inhibition test.

9.2.1 Complete outcomes

All 20 cases per scenario resolved, so resolution-conditional and all-planned frequencies coincide here. Wilson intervals describe Monte Carlo selection frequencies under this rule. They do not calibrate the test. Missing-outcome bounds are retained in the machine-readable summary and are not confidence intervals.

Table 9.1: All six descriptive pilot frequency cells. Wilson endpoints are frequency fractions across independent forests; they do not calibrate the inhibitory test.

Procedure	h	Selected/planned	Unknown	Wilson lower	Wilson upper	Median GLR
Supplied	0.0	6/20	0	0.145477	0.518973	0.000000
Supplied	0.5	6/20	0	0.145477	0.518973	0.000000
Supplied	0.9	4/20	0	0.080658	0.416017	0.000000
Fitted (primary)	0.0	15/20	0	0.531299	0.888138	1.205812
Fitted (primary)	0.5	17/20	0	0.639581	0.947631	1.053431
Fitted (primary)	0.9	18/20	0	0.698966	0.972134	1.052331

9.2.2 Paired changes from the independent-mark control

Each contrast uses all 20 paired forests and the predeclared 20,000-resample skeleton bootstrap, seed 2026102499. These are exploratory percentile summaries, not confirmatory coverage guarantees. A zero or zero-touching interval is not proof of equivalence.

The primary selection frequency rises, but its median GLR does not increase and its paired mean GLR changes are small relative to their uncertainty. The secondary procedure moves in the opposite direction at strong persistence. These observations do not support a simple, procedure-independent claim that stronger magnitude history yields stronger inhibitory evidence.

The post-execution class-manipulation diagnostic is recorded separately in methods/pilot_input_diagnostics.json: observed parent-child same-class fractions were approximately 0.6304, 0.8207 and 0.9656, versus design expectations 0.6346, 0.8173 and 0.9635. This is a descriptive implementation diagnostic, not a new primary test; no event-iid uncertainty is asserted.

The pilot implementation, validation and resources are retained in Appendix B.7.1.

Table 9.2: All four paired changes from the independent-mark control. Selection changes are frequency fractions (multiply by 100 for percentage points); endpoints are exploratory paired-forest bootstrap summaries, not confirmatory coverage.

Procedure	h	Selection diff.	Lower	Upper	GLR diff.	Lower	Upper
Supplied	0.5	0.000000	0.000000	0.000000	-0.010306	-0.039395	0.018140
Supplied	0.9	-0.100000	-0.250000	0.000000	-0.040430	-0.073392	-0.012346
Fitted (primary)	0.5	0.100000	0.000000	0.250000	-0.162155	-0.846990	0.354902
Fitted (primary)	0.9	0.150000	0.000000	0.300000	0.030885	-1.114732	1.029795

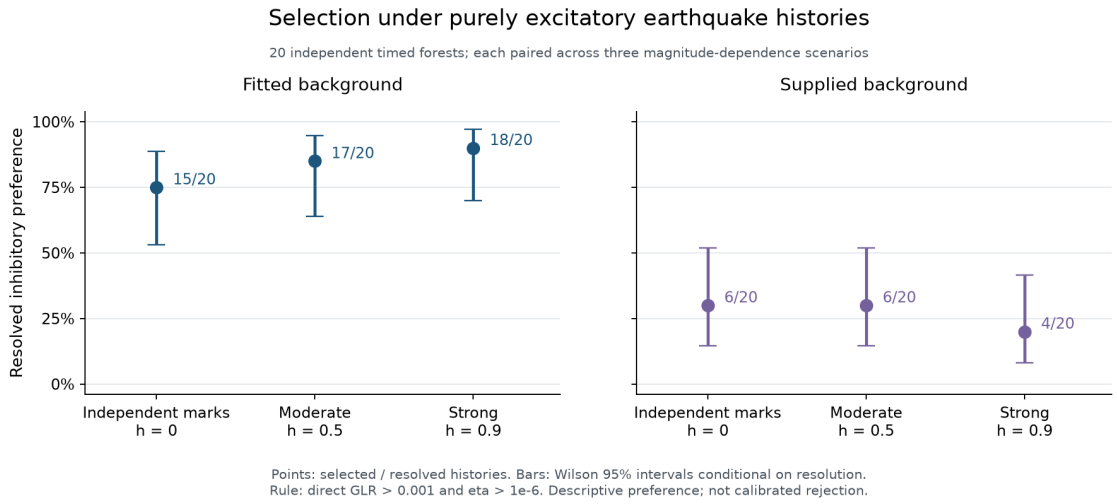


Figure 9.1: Saved final magnitude-history pilot selection frequencies. Forests are paired across scenarios. These descriptive Monte Carlo intervals do not calibrate a rejection test or establish a magnitude-history mechanism.

9.2.3 Interpretation

The feasibility and bounded descriptive pilot gates pass. The machinery can generate the intended history-dependent marks and apply the selected existing comparison reproducibly. The completed observation is that this narrow procedure can show positive inhibitory fit preference under a purely excitatory designed family, including its independent-mark control.

Whether accounting for magnitude history explains previous observed evidence remains unresolved. No correctly specified richer excitatory model was fitted in this cycle; known-law scoring controls are separate from fitting. There was no fresh null calibration, size-matched power comparison, observed-catalogue evaluation or held-out forecast comparison.

The next scientific stage should first define the claim and fit the competing excitatory family under a reviewed protocol, while separating common productivity misspecification from the history contrast. Any advertised rejection test then needs fresh calibration and independent evaluation banks reproducing nuisance fitting and selection. An observed-data comparison would require matched scoring targets/exposure and separate catalogue/completeness gates. No additional campaign was launched here.

The pilot reproducibility artifacts are retained in Appendix B.7.3.

The pilot evidence map is retained in Appendix B.7.4.

9.3 Purpose and relationship to the earlier releases

The original GitHub research manuscript separates mathematical legitimacy, identification of an observation law and the predictive value of a fitted procedure (Chapter 8; Appendix B). This chapter adds a different empirical axis: whether magnitude-class history or a more flexible magnitude-productivity law improves the declared predictive comparisons. It reports the two completed follow-ups, their numerical evidence and their stopping decisions, preserving the earlier scientific records.

The distinction between releases is substantive. The manuscript's observed reconstruction selected productivity ratios $r = 0.4, 0.7, 0.7$ at its three origins. The new matched reference candidates use $r = 0.7$ throughout and a different, common training-only magnitude-rate

Table 9.3: Scientific lineage and recipe differences. These studies are related releases, not interchangeable fits or pooled calibration results.

Study	Question/target	Productivity and magnitude treatment
Original manuscript	Restricted stability/identification, controlled nuisance comparisons, and exploratory observed prediction	Observed recipe selected $r = 0.4, 0.7, 0.7$ at the three origins; training plug-in magnitude rates 3.101622, 3.149488, 3.021907.
Class-history follow-up	Does a restricted class-history extension improve matched forecasts?	Fixed $r = 0.7$; shared training-only conditional-bin magnitude estimator; source fertility δ and association h.
Productivity follow-up	Does a training-estimated continuous slope improve matched forecasts?	Same conditional-bin magnitude estimator; fixed $r = 0.7$ versus estimated $r \in [0, 0.95]$ in E0 and I0; class-history continuation gated off.

estimator (Table 9.3). They are separately specified follow-up experiments, not replacements for the manuscript's historical score table. Both use the same hash-pinned GeoNet selection as our observed reconstruction. That archive identity does not establish identity with the unavailable earlier analysis discussed in the manuscript's catalogue-provenance section.

9.3.1 External framework and attribution

Self- and mutually exciting point processes provide a natural setting for event-triggered clustering (Hawkes, 1971); earthquake triggering models include the ETAS tradition (Ogata, 1988). Davis, Baeumer and Wang's fractional earthquake model supplies context for Mittag-Leffler triggering with magnitude productivity (Davis et al., 2024a). Their multivariate fractional construction represents magnitude ranges as subprocesses and permits history-dependent magnitude proportions (Davis et al., 2026a). We inspected its model definitions in the separately identified author preprint, Section 2, equations (2)–(4) (Davis et al., 2024b); those locators do not assert final-journal pagination. For positive linear marked Hawkes models, partitioned multivariate constructions provide an approximating parametric description under stated assumptions (Davis et al., 2026b). No stationarity or identification theorem from that setting is transferred to our nonlinear inhibitory comparator. These sources motivate the question and model language; they do not supply the empirical findings reported here.

Our low-dimensional two-class shared-kernel family simplifies some class and kernel structure while changing mark support and nuisance treatment. The consulted MDFHP preprint (Davis et al., 2024b, Section 2) uses truncated within-bin magnitude densities, including an upper magnitude of 10. Our working exponential excess law is unbounded. Consequently, our complete joint model is not a strict submodel of that specification. The present comparisons neither reproduce nor refute the authors' Japan and Middle America fitted results. Their results and software must be credited in their own scope, and the locally derived summation identity below is a result for the family actually implemented.

9.4 Restricted class history and productivity sensitivity

The class-history candidate is a restricted two-class, shared-kernel construction motivated by multivariate representations of marked excitation (Davis et al., 2026a,b). The equations below specify the implemented follow-up family; they are not the full fitted model or a reproduction

of the empirical comparisons in those papers. It simplifies class and temporal-kernel structure while changing the mark support, and is not a strict submodel of the authors' bounded-mark family. Shared target kernels and the low-dimensional association parameter are choices of this experiment. Notation is local: our β denotes the magnitude rate, rather than a source paper's fractional-kernel order.

9.4.1 Class association and the conditional ground intensity

Write $q = M - 5$, let the class cut be $c = 0.5$, and define $\mathcal{C}_0 = [0, c)$, $\mathcal{C}_1 = [c, \infty)$. Write $p_0 = 1 - e^{-\beta c}$, $p_1 = e^{-\beta c}$ for the background class probabilities under the working excess-exponential law. The common $\beta > 0$ is estimated using training marks only and held fixed in each forecast window. Let $C_k \in \{0, 1\}$ be the class label of event k. For source class j, define the strictly past history

$$H_j(t) = \sum_{k:\, t_k < t,\ C_k = j} \kappa_{0.7}(q_k) g(t - t_k), \qquad R_0(\delta) = e^{-\delta/2}, \quad R_1(\delta) = e^{\delta/2},$$

where g is the retained unit-integral finite positive exponential mixture and κ_r is defined in equation (9.3). Write $\mu(t) = \exp\{a + dx_{\text{trend}}(t)\}$ for the background, with the affine observation coordinate specified in Section 9.5. The implemented target-class intensity is

$$\begin{aligned} \lambda_i(t) &= p_i \mu(t) + \frac{n}{\rho(h, \delta)} \sum_{j=0}^{1} P_{ij}(h) R_j(\delta) H_j(t), \\ P_{ij}(h) &= (1 - h) p_i + h\, \mathbf{1}_{\{i=j\}}. \end{aligned} \tag{9.1}$$

Here n is the class-history candidate's scalar branching parameter and $\rho(h, \delta)$ is the normalizing Perron root specified below. The frozen parameter box is

$$-0.95 \min(p_0/p_1, p_1/p_0) \le h \le 0.95, \qquad -1 \le \delta \le 1, \qquad 0 \le n \le 0.98.$$

The factor 0.95 retains a margin inside the mathematical nonnegativity limit for P; the Perron root $\rho(h, \delta)$ is strictly positive. For the working within-class magnitude laws, let

$$m_j = E[\kappa_{0.7}(Q) \mid Q \in \mathcal{C}_j], \qquad m_0 = \frac{1 - e^{-0.3\beta c}}{p_0}, \quad m_1 = e^{0.7\beta c}.$$

The raw mean offspring matrix and its normalization are

$$A_{ij}(h, \delta) = P_{ij}(h) R_j(\delta) m_j, \qquad \rho(h, \delta) = \rho_{\text{PF}}(A), \qquad K = \frac{n}{\rho} A,$$

where ρ_{PF} denotes the Perron root. Thus K is the class-history candidate's mean offspring matrix, with Perron radius n. Association h and fertility contrast δ play different roles. Signed association can describe class anti-persistence while all excitation coefficients remain nonnegative. At $n = 0$, $\lambda_i = p_i \mu$; trigger association and fertility are unidentified.

For every source class, $\sum_i P_{ij}(h) = 1$. Thus, conditional on a fixed observed marked history,

$$\Lambda(t) = \sum_i \lambda_i(t) = \mu(t) + s\{R_0(\delta) H_0(t) + R_1(\delta) H_1(t)\}, \qquad s = \frac{n}{\rho(h, \delta)}. \tag{9.2}$$

The direct redistribution across target classes cancels from the ground intensity. The conditional timing log likelihood is

$$\ell_{\text{time}} = \sum_k \log \Lambda(t_k) - \int_{\mathcal{W}} \Lambda(t)\, dt.$$

For these classes, the conditional within-class magnitude density is

$$f_i(q) = \frac{\beta e^{-\beta q}}{p_i} \mathbf{1}_{\{q \in \mathcal{C}_i\}}.$$

The mark density at an observed event of class i is $\{\lambda_i(t)/\Lambda(t)\} f_i(q)$, giving mark log density $\log(\lambda_i/\Lambda) + \log\beta - \beta q - \log p_i$. This gives the saved decomposition $\ell_{\text{joint}} = \ell_{\text{time}} + \ell_{\text{mark}}$.

Equation (9.2) has several qualifications. First, holding n fixed does not remove h, because ρ may depend on h. Second, matching an effective scale at $h = 0$ requires

$$n_0 = n\frac{\rho(0,\delta)}{\rho(h,\delta)}.$$

This mapping preserves timing intensities only if the mapped parameter is admissible, in particular $0 \le n_0 \le 0.98$; $n = 0$ gives the trivial no-trigger case. The frozen bound on n, and any relevant stability constraint, can make feasible effective-scale ranges depend on h. We therefore do not assert that all constrained timing families are identical. Third, joint timing-and-mark estimation can shift s, δ and background parameters, producing indirect changes in timing forecasts. Finally, the identity conditions on the actually observed marked past. When future marks themselves are generated by the model, their class composition can depend on h, and so can a multistep or unconditional timing law.

At $h = 0$, $\lambda_i(t) = p_i\Lambda(t)$; the conditional magnitude law is history-free even when δ remains fitted. Consequently, we denote this association-zero ablation $E1_{h=0}$; it is the appropriate primary ablation for the association parameter within this restricted family. It is not the same restriction as setting both h and δ to zero.

9.4.2 Continuous productivity and dynamic evaluation

The second follow-up replaces fixed $r = 0.7$ by a training-estimated $r \in [0, 0.95]$ in

$$\kappa_r(q) = (1-r)e^{\beta r q}, \qquad \partial_r \kappa_r(q) = \kappa_r(q)\left(\beta q - \frac{1}{1-r}\right). \tag{9.3}$$

For $Q \sim \text{Exp}(\beta)$, $E[\kappa_r(Q)] = 1$ when $r < 1$. This is a normalization under the working law, not an empirical statement about the historical magnitude catalogue.

The geometry records $D_{ik} = g(t_i - t_k)\mathbf{1}_{\{t_k < t_i\}}$ and each parent's full-window exposure J_k, independently of r. Every C++ value and gradient evaluation recomputes

$$H_r(t_i) = \sum_k D_{ik}\kappa_r(q_k), \quad K(r) = \sum_k J_k \kappa_r(q_k)$$

and their r-derivatives. In the excitatory candidate, and in the inhibitory candidate whose supplied stress feature is independent of r,

$$\partial_r \text{NLL} = n\left\{K'(r) - \sum_i \frac{\partial_r H_r(t_i)}{\Lambda(t_i)}\right\}. \tag{9.4}$$

The common training magnitude law gives exactly the same mark score to the fixed- and free-r stage-one candidates. Their score differences therefore concern timing. At $n = 0$, the timing score is independent of r, its derivative vanishes and r is unidentified. Complete quiet exposure, pretraining parents, strictly past event histories and the final censor are retained.

For the same unbounded exponential working law and $m > 0$,

$$E[\kappa_r(Q)^m] = \frac{(1-r)^m}{1-mr} \quad \text{when } mr < 1;$$

the moment diverges for $mr \geq 1$. In particular, productivity variance is infinite at $r \geq 0.5$, including the baseline $r = 0.7$. Finite saved-catalogue likelihoods and numerical checks do not supply variance-based confidence or power conclusions.

9.4.3 Synthetic normalization and information

The engineering controls are execution witnesses. For a multitype mean offspring matrix K, fixing its Perron radius does not fix the expected first-generation offspring count of an immigrant with class law p: the latter is $\mathbf{1}^\top Kp$. The class-active history control changed this quantity and the resulting offspring/event composition despite using the same Perron branching parameter as its control. Its offspring dilution must therefore be retained when interpreting successful fits or forecast gains. Neither these controls nor the small fixed-r and free-r engineering examples constitute matched-power, bias, coverage or reliable-recovery studies.

9.5 Frozen exploratory protocols and score targets

The primary observed selection contains 913 GeoNet earthquakes with $M \geq 5$, in the archived 24,394-day interval beginning 1960-01-01. The provider is the GeoNet earthquake catalogue (GNS Science, n.d.). Both follow-ups use the exact selected snapshot rather than a new provider download (Appendix B.7). Their training origins are $0.50T, 0.65T, 0.80T$, with forecast endpoints $0.65T, 0.80T, T$, where $T = 24{,}394$ days. Initial training counts are 301, 504 and 659; the consecutive held-out counts are 203, 155 and 254. The frozen history rule retains all available strictly earlier parents, including events before the training-likelihood start. In the post-1980 sensitivity that start is day 7,305. Matching saved training-count differences imply 164 earlier selected events. Their retention follows the protocol rule and is documented by historical audit records; this editorial aggregate check does not newly verify parent geometry. The constant-background sensitivity fixes the background trend at zero.

The ground candidates are

$$\Lambda_{E0}(t) = e^{a+dx_{\mathrm{trend}}(t)} + nH_r(t), \qquad \Lambda_{I0}(t) = e^{a+dx_{\mathrm{trend}}(t)-\eta G(t)} + nH_r(t), \quad x_{\mathrm{trend}}(t) = 2t/T - 1.$$

Here G is the retained stress feature, H_r is productivity-weighted triggering, and the finite positive exponential mixture and stress-law settings are fixed within each declared protocol. The $E1$ class candidate is given by equation (9.1); $E1_{h=0}$ estimates fertility while fixing association at zero. Continuous free-r estimation is performed in E0 and I0, with the class-history extension conditional on the declared continuation gate.

The retained temporal kernel is a normalized finite positive exponential approximation to Mittag-Leffler memory. In particular, its finite value at zero is not the exact fractional kernel's short-lag singularity. Earlier closed-form or reference-kernel checks and the later predictive mixture representation are distinct numerical objects.

At a forecast event, the information set contains the actually observed marked past only. The ground intensity uses a left limit, and all initial, inter-event and terminal quiet exposure enters the compensator. The common conditional-within-bin exponential magnitude-rate estimate is computed from training marks at cut $c = 0.5$, then frozen for all matched candidates in that window. The joint score is a density log score with timing and mark components, not a comparison of a bin probability mass to an unrelated continuous density. The use of proper scores motivates evaluating complete predictive distributions (Gneiting and Raftery, 2007); it does not make previously inspected windows independent validation data.

Table 9.4: Descriptive history-study forecast gains in nats per event. Positive values favor the first model. Post-1980 changes the training start while retaining earlier parents. The $h = 0$ ablation was required in primary only. All settings reuse the same 612 targets. Confidence intervals are not reported.

Setting	Contrast	Timing	Mark	Joint	Winning windows
Primary	$E1 - E0$	-0.003092	-0.005590	-0.008682	1/3
Primary	$E1 - I0$	+0.026624	-0.005590	+0.021034	2/3
Primary	$E1 - E1_{h=0}$	+0.000242	-0.005590	-0.005348	1/3
Constant background	$E1 - E0$	-0.005894	-0.005602	-0.011496	1/3
Constant background	$E1 - I0$	+0.015408	-0.005602	+0.009806	3/3
Post-1980	$E1 - E0$	-0.006146	-0.002550	-0.008696	1/3
Post-1980	$E1 - I0$	+0.025900	-0.002550	+0.023350	2/3

The historical class-history fits required eight starts for every procedure. The productivity follow-up likewise required eight starts per main procedure, with seven fixed-r profile nodes and four starts per node for primary free-r models. Nested fixed-parameter candidates and same-r excitation-to-inhibition floors are admitted using training scores. All profile candidates enter the training pool before forecasts are evaluated. The selected parameter vector minimizes the prescribed training objective; descriptive profile forecasts cannot replace it. The saved near-training-equivalent band and its forecast-spread diagnostic assess the retained candidate pool, not global optimizer uncertainty.

For clarity, held-out scores intentionally decide whether the next prespecified scientific stage is warranted. They do not choose parameter values, optimization starts or an adaptively refined profile grid. Failure and unknown states remain explicit. The follow-up comparisons have no new null calibration, prospective holdout or size-matched power interpretation.

9.6 Saved predictive results

Both follow-ups used the same pinned 913-event GeoNet selection and three chronological prediction windows containing 203, 155 and 254 events. The three settings reuse those 612 distinct targets; they do not create 1,836 independent observations or nine independent trials. Parameters are fitted using each training prefix and then held fixed. Genuinely observed past events update the forecast history, with future events excluded. The reported quantities are event-weighted log-score differences over the complete forecast exposure, including quiet periods.

9.6.1 The class-history extension

Saved source records report nine observed cases, 30 fitting procedures and all 240 required starts for the completed history study. They record a passed independent saved-output audit, with no unknown procedure or forecast-ambiguity flag. The supplied audit/start evidence excerpt retains these completion records. The present editorial reconciliation checks saved records and aggregates rather than repeating that audit. These are completed local searches, not global-optimum certificates (Appendix B.7).

All nine constant-background fits prescribe $d = 0$, and the three primary $E1_{h=0}$ ablations prescribe $h = 0$. The middle-window $I0$ fit also reaches the estimated-coordinate boundary $\eta = 0$. In post-1980 fold 0, $I0$ reaches $d = 2$ and $E1$ reaches $\delta = -1$. These boundary-dependent local fits do not establish interior identification of the corresponding coordinates.

Table 9.5: Descriptive free-productivity forecast gains, in nats per event. The mark-score differences are exactly zero, so the joint differences concern timing (up to floating-point summation). Positive values favor the first model. Confidence intervals are not reported.

Setting	E0 free - fixed	I0 free - fixed	E0 free - I0 free	E0 positive windows
Primary	-0.008782	-0.009105	+0.030039	1/3
Constant background	-0.010447	-0.008689	+0.019544	1/3
Post-1980	-0.060929	-0.054066	+0.025183	0/3

The full $E1$ candidate beat $I0$ but lost to $E0$ in every aggregate setting. The primary joint loss is -5.313603 total nats over 612 targets. Its primary association-specific gain relative to $E1_{h=0}$ was also negative: a small timing difference of 0.000242 nats/event was outweighed by a mark-score loss of 0.005590. Joint estimation can alter fitted effective scale and fertility, so the nonzero timing difference does not contradict the conditional ground-intensity identity.

The direct baseline contrast $E0 - I0$, obtained as $(E1 - I0) - (E1 - E0)$, was positive in all three settings: 0.029716, 0.021302 and 0.032046 nats/event, respectively. Beating the inhibitory candidate therefore does not establish an added benefit from class history. The frozen criteria required at least 0.01 nats/event over both baselines and at least two shared winning windows in primary, with the declared sensitivity requirements. Those conditions failed, yielding `stop_no_robust_joint_gain`.

9.6.2 Continuous-productivity sensitivity

The productivity study's saved source records report nine cases, 36 main procedures, 42 primary profile procedures and all 456 mandatory starts. They record an independent computational auditor that reconstructed geometry from the raw local catalogue and checked 1,116 train/fine candidate evaluations, 288 near-candidate forecasts and 42 descriptive profile forecasts. The supplied audit/start evidence excerpt preserves the relevant saved counts and completion records. The editorial reconciliation checks these saved records and aggregates; it does not repeat the raw-data audit (Appendix B.7).

The primary $E0$ estimates were $r = 0.471908$, 0.605853, 0.610446. Across both families and all settings, 16 of 18 estimates were below 0.7; the two middle-window post-1980 estimates were above it. All 18 estimated r values were interior. Other estimated coordinates reached constraints: $I0_{\text{fixed}}$ and $I0_{\text{free}}$ both reached $\eta = 0$ in the middle constant-background window and $d = 2$ in post-1980 fold 0. Thus the productivity study has four such estimated-boundary fits, distinct from the history-study cases above. The 12 prescribed constant-background $d = 0$ coordinates and 18 fixed $r = 0.7$ coordinates are restrictions rather than estimated-boundary findings. These results do not establish interior identification of every coordinate. Both nested free-r families improved training fit but lost forecast score in every aggregate setting. The primary free-r excitation loss is -5.374505 total nats (-0.008782 per event). Training improvement is partly guaranteed by admitting the exact fixed-r floor, and is not evidence of a correct physical slope.

The primary fixed-r profiles used the frozen grid 0, 0.2, 0.35, 0.5, 0.7, 0.85, 0.95. For both families in the first window, sampled nodes 0.35 and 0.5 lay within two nats of the selected optimum. In the other four family/window profiles, no sampled node lay within that band of the off-grid optimum. These are sparse descriptive support sets, not confidence intervals or evidence of precise identification. The maximum saved forecast spread among near-

Table 9.6: Primary cross-run gains over each study's matched $E0$ baseline, in total nats. Training gains are partly ensured by nested candidate admission. The shared pattern is descriptive and uses the same historical folds, rather than independent validation. Confidence intervals are not reported.

			Step fertility		Free r	
Fold	N_{train}	N_{test}	Training	Test	Training	Test
0	301	203	+7.260	-5.209	+8.622	-7.121
1	504	155	+2.254	+5.006	+2.296	+8.167
2	659	254	+4.491	-1.838	+5.228	-6.420

training-equivalent candidates was approximately 9.04×10^{-5} nats/event, below the frozen 0.01 diagnostic threshold. It describes the saved candidate pool, not global uncertainty.

The stage-one continuation gate required primary $E0$ gain of at least 0.01 nats/event with at least two positive windows, plus positive aggregate gains and at least two positive windows in both sensitivities, complete audit and no forecast ambiguity. It failed while the resource reserve remained available. The checkpoint was `stop_no_robust_productivity_gain`; the free-r class-history models $E1_r$ and $E1_{h=0,r}$ were not run. These planned labels would allow r to vary in the full class-history candidate and its association-zero restriction, respectively. Held-out scores intentionally governed this prespecified continuation decision. They did not select parameters, admit profile candidates, refine the grid or choose among fits.

9.6.3 A descriptive cross-run comparison

The history-free step-fertility restriction $E1_{h=0}$ and continuous $E0_{\text{free}}$ are two different extensions of their matched fixed-r $E0$ baselines. In the same primary windows, both favored lower relative productivity for large magnitudes during training, and their forecast differences had the same signs (Table 9.6). The fixed-r baseline forecasts agree across the two implementations to approximately 5.57×10^{-5} nats or better over all nine cases, consistent with small local-optimizer coordinate differences.

This correspondence is consistent with both adjustments responding to weaker apparent training magnitude-productivity scaling. It does not establish exact equivalence of the fitted features, changing physical productivity through time, or a specific underprediction of large-event aftershocks as the cause of the losses. Background, branching and compensator terms also change. Because the free-r association/fertility stage was not executed, the experiments do not determine whether δ would shrink when jointly fitted with r.

9.7 Interpretation, limitations and disposition

The original manuscript, the F nuisance companion and the present follow-ups address related but distinct questions. The original observed recipe selected $r = 0.4$ at its first origin and $r = 0.7$ later, with a different training magnitude-rate estimator. The new comparisons fix $r = 0.7$ in their reference fits and use a shared training-only conditional-within-bin estimator. Their matched contrasts are valid within those separately frozen protocols; they are not corrected entries in the original paper's predictive table (Chapter 8; Appendix B).

The class-history construction is deliberately restricted. The consulted MDFHP preprint permits class-specific temporal and magnitude-productivity parameters and bounded class mark distributions, with upper magnitude 10 (Davis et al., 2024b, Section 2). Our common-kernel, low-dimensional $P(h)$ and source-fertility contrast, together with an unbounded excess-

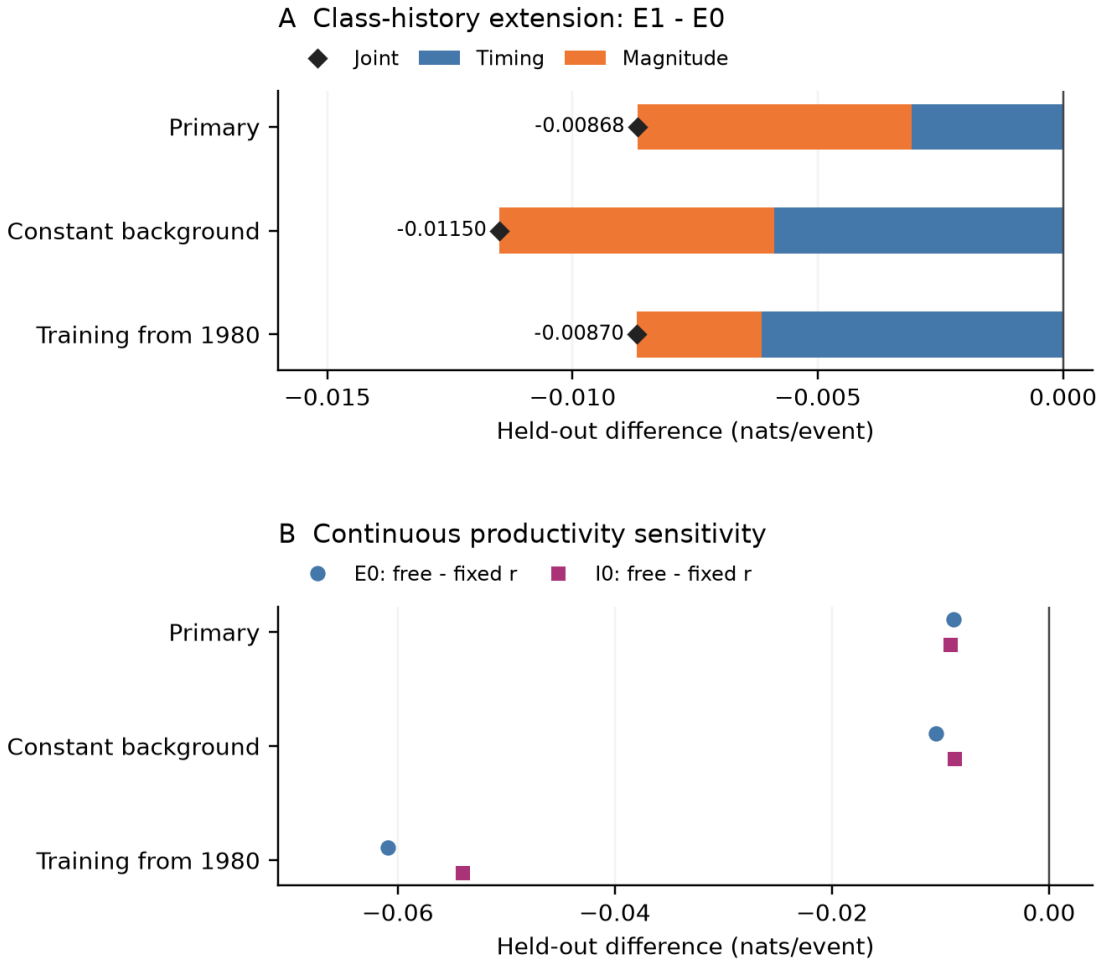


Figure 9.2: Saved aggregate forecast differences. Panel A decomposes the class-history contrast into timing and magnitude terms; diamonds show their joint difference. Panel B compares free with fixed productivity in each ground family. Positive values favor the extension. Each setting reuses the same 612 targets, the panel scales differ, and no uncertainty intervals are claimed. Source values are the accompanying reconciled gain tables.

exponential working law, are different modelling choices. The conditional summed-intensity identity depends on those choices. It does not refute the broader class-history framework, reproduce the authors' Japan or Middle America fits, or transfer identification and stationarity theorems to a nonlinear inhibitory family.

All forecast windows had been inspected in earlier research. Training prefixes overlap and the sensitivity settings reuse the same 612 targets. The results are retrospective exploratory procedure comparisons, not prospective confirmation or independent replications. Three windows do not support treating clustered events as independent draws for standard errors or significance claims. Native magnitude rounding, mixed magnitude conventions, time-varying detection, missing smaller or deeper parents, pre-1960 activity and national spatial pooling remain observation/model assumptions. They were not resolved by these extensions.

The engineering controls and complete numerical audits make the implementation inspectable, but do not establish global optima, precise identification, confidence coverage or power. Positive control examples show that the fitted pipeline can return gains on particular generating histories. Equal Perron branching radius did not equate immigrant-weighted offspring production in the class-active control; different offspring composition and event counts limit any information-matched comparison. Sparse profiles and near-training-equivalent candidate spreads retain their descriptive meanings.

The negative forecast results do not establish that physical inhibition is absent. Nor do lower fitted slopes establish that $r = 0.7$ is physically wrong, or show that the earlier negative fertility contrasts were caused by that fixed value. The cross-run pattern motivates a possible compensation interpretation, while joint estimation of the fertility/association extension with free r remains unperformed. The persistence of $E0$'s forecast advantage over $I0$ is specific to the tested recipes and windows.

Both investigations are therefore retained at their respective negative checkpoints. The fixed-r $E0$ candidate remains the predictive reference for these follow-up comparisons. Broken-slope, saturation, spatial-locality and independent-data extensions were outside their executed scope. A future study would require a separately justified question, information set and protocol, rather than further tuning of these already examined forecast windows.

9.8 Evidence and reproducibility scope

The companion records exact protocol, source, score-table and review identities. The deterministic editorial reconciliation recomputes display contrasts and counts from saved aggregate/model records using an explicit hash-bound input list. It does not regenerate scientific histories, refit models, recalibrate tests or repeat the raw-catalogue auditors.

The original history audit and the later source-reconstructed productivity audit have different recorded scopes. The latter evaluated all saved candidates using independently reconstructed geometry rather than merely checking cache hashes and scoring from production arrays. Their passed receipts are evidence about the executed checks, not certification of every mathematical or scientific assertion.

The public companion contains the reviewed source and derived score and parameter display packet, source/result identities and accepted figures. The exact public manifest identifies its membership. The reconstructed observed catalogue is now supplied with its raw-to-cohort replay under `reproduction/nzca/`. That historical study's complete geometry-cache and fitted-state replay remains distinct from that cohort reconstruction. External article PDFs, private literature inputs and provider conversations remain excluded; frozen absolute paths require an explicit relocation mapping. Availability of a compact saved-result packet does

not imply availability of every historical dependency. The accompanying historical-execution excerpt retains mandatory-start completion slots and the relevant saved auditor counts and geometry-check flags, with selectors and original source hashes. It reports prior checks; no raw-source reconstruction was repeated for that historical study. A separate boundary inventory distinguishes prescribed coordinates from estimated constraint hits.

Likelihood and analytic-gradient evaluation run in C++; Python handles geometry, optimization orchestration, artifact analysis and audits. MATLAB was unnecessary for these two predictive follow-ups. An earlier reference-kernel bridge and inspected upstream source are attributed separately from the independently implemented repeated likelihood. Existing contributor and AI-assistance disclosures are retained; computational-agent review is distinct from independent external human peer review.

9.8.1 Data, source and software attribution

The GeoNet catalogue DOI identifies a continuously updated data resource; the analyses use the frozen archived selection, whose processed SHA-256 is

feacccb7581d1c4c33f88874cd6b198c41639da6930a7fc35e46f910a594fb3e.

We acknowledge GeoNet and its programme sponsors for the data underlying the studies. Native magnitude scales and the original selected values are retained. Neither source citations nor matching archive hashes validate homogeneity or catalogue completeness.

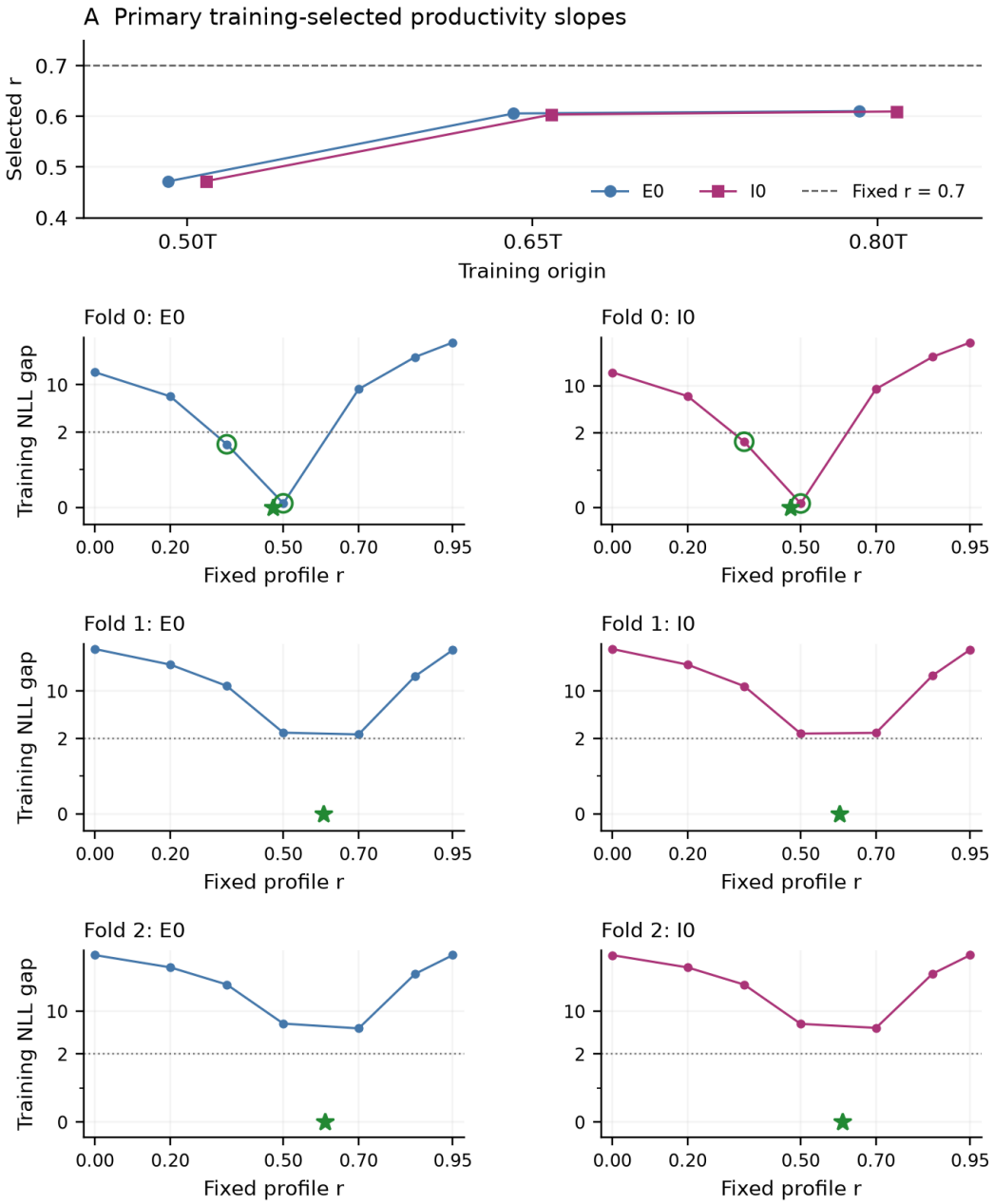
The Otago implementation was inspected at the separately identified *MDFHPCode* source commit (davlo199, 2024). An earlier feasibility reference-kernel bridge is distinct from the independently implemented finite-mixture objective used in the two predictive follow-ups. Its source context includes Garrappa's numerical Mittag-Leffler work (Garrappa, 2015). We do not redistribute the upstream general contour implementation, reference-paper PDFs, upstream binaries or private literature packet. Citation of a paper and compliance with a code file's licence are separate responsibilities; no claim of a full MATLAB optimizer reproduction follows from the reference check.

The recorded pilot environment used Python 3.13.4, NumPy 2.2.6 and SciPy 1.15.3. NumPy, SciPy and Matplotlib are credited for their actual array, optimization and display roles (Harris et al., 2020; Virtanen et al., 2020; Hunter, 2007). Repeated likelihood and analytic-gradient work was compiled C++; changing the compiler or rebuilding a library creates a new binary identity. The editorial build uses Tectonic 0.17.0, and its source, tool and figure hashes are recorded separately from the scientific execution records. Font and package resources were not snapshotted as a byte-identical historical environment.

9.8.2 Data and numerical specification

The companion data provide complete case/model score and parameter tables, the count derivation for the 164 earlier selected parents, and the aggregate values used here. Kernel and stress constants, solver settings and profile policies are supplied in `scientific_settings.json`. For the Mittag-Leffler representation, the serialized $p = 1.3$ field is inactive rather than an additional fitted kernel shape. Computational provenance is documented in `docs/result_provenance.md`.

The complete magnitude-extension fold accounting is retained in Appendix B.7.5.



B Stars: selected off-grid optima. Circles: sampled nodes within 2 nats.
Profile axes are linear near zero and logarithmic above 2 nats.
Sparse sampled support is descriptive, not a confidence interval.

Figure 9.3: Primary training-selected productivity ratios and the seven-node descriptive profiles. Stars show selected off-grid optima; circles identify sampled points within two nats. The profile vertical axes are linear near zero and logarithmic above two nats. Empty sampled support in four panels does not define an empty confidence interval. All values are saved training results; profile forecasts did not select parameters.

9.9 Exposure orthogonalization and the Stage A coarse-kernel comparison

Let $w_l > 0$ be the normalized training exposure quadrature weights and write $\bar{x} = \sum_l w_l x_{\mathrm{trend}}(t_l)$, $\bar{G} = \sum_l w_l G_l$. Provided $V_x = \sum_l w_l (x_{\mathrm{trend}}(t_l) - \bar{x})^2 > 0$, define

$$\gamma_1 = \frac{\sum_l w_l (x_{\mathrm{trend}}(t_l) - \bar{x})(G_l - \bar{G})}{V_x}, \qquad \gamma_0 = \bar{G} - \gamma_1 \bar{x}, \qquad R = G - \gamma_0 - \gamma_1 x_{\mathrm{trend}}.$$

Then $\sum_l w_l R_l = \sum_l w_l x_{\mathrm{trend}}(t_l) R_l = 0$ algebraically. The coefficients are estimated from training exposure only and frozen for prediction and order-32 checking. Refitting them on the verification grid would change the model. Degenerate exposure variance invalidates the two-coordinate projection; zero residual variance reports redundancy rather than identification. The unrestricted map is exact:

$$\widetilde{a} = a - \eta\gamma_0, \quad \widetilde{d} = d - \eta\gamma_1, \qquad a = \widetilde{a} + \eta\gamma_0, \quad d = \widetilde{d} + \eta\gamma_1.$$

Thus $a + d x_{\mathrm{trend}} - \eta G = \widetilde{a} + \widetilde{d} x_{\mathrm{trend}} - \eta R$ at every time. Equal numerical boxes in these coordinates describe different constrained families; mapped floors are admissible only when all destination restrictions hold, without clipping. An η-only ridge is invariant under this map. Raw $d = 0$ differs from residual $\widetilde{d} = 0$. The residual can be negative, so the detrended background may locally exceed its nominal undepleted level. Exposure orthogonality is not likelihood-score orthogonality: event scores contain B/λ, information involves B^2/λ, and projection onto $(1, x_{\mathrm{trend}})$ does not remove dependence on the excitation-amplitude score.

For $B = \exp(\widetilde{a} + \widetilde{d} x_{\mathrm{trend}} - \eta R)$ and $\lambda = B + nH$, let J be complete productivity-weighted trigger exposure. The timing negative log likelihood and gradient are

$$L = \sum_l q_l B_l + nJ - \sum_i \log \lambda_i,$$

$$L_{\widetilde{a}} = \sum_l q_l B_l - \sum_i B_i/\lambda_i, \qquad L_{\widetilde{d}} = \sum_l q_l x_{\mathrm{trend}}(t_l) B_l - \sum_i x_{\mathrm{trend}}(T_i) B_i/\lambda_i,$$

$$L_n = J - \sum_i H_i/\lambda_i, \qquad L_\eta = -\sum_l q_l R_l B_l + \sum_i R_i B_i/\lambda_i.$$

The raw formulas replace R by G. At $n = 0$, L_n remains well defined; at $\eta = 0$, values and shared-coordinate gradients must nest using a common integration branch or an explicitly controlled numerical seam.

Stage A selected Omori triggering by training excitation-only fits on the nine-point grid $c \in \{0.1, 1, 10\}$ days and $p \in \{1.1, 1.3, 1.7\}$. The selected kernel was frozen for its excitation/inhibition pair. The stress survival law and $r = 0.7$ remained fixed. This is a restricted coarse-grid procedure, distinct from the historical continuously refined Omori fit and the later continuous Extension O.

9.10 Stage A saved evidence and interpretation

All planned procedures are complete and have independent computational audits. A negative predictive result is a completed Stage A result.

9.10.1 Aggregate held-out comparisons

The variants reuse the same targets. Mark-score differences cancel; timing is the event plus compensator contribution. In primary, exposure orthogonalization changes the pooled ML

inhibition contrast by only −0.000045575 nats, from −18.186415 to −18.186460 nats. The raw interior optimum maps feasibly into the orthogonal box; better Gram conditioning therefore does not change the represented optimum materially.

9.10.2 Interpretation limits

Orthogonalization is an invertible unrestricted coordinate transformation. Identical boxes in raw and transformed coordinates define different constrained families, including intercept constraints. The projection removes exposure drift; it does not establish likelihood-score orthogonality. Constant-background restrictions have different meanings in the two coordinates.

Omori parameters were selected by training E0 on the fixed nine-point grid and shared within each excitation/inhibition pair. The ML stress survival law and r=.7 remained fixed. This is not exact replication of Iteration27 continuous kernel/productivity selection.

Historical Fold0 raw inhibition loss around -16.99 nats is contextual. Compare against the newly recomputed raw baseline and retain differences due to nuisance/optimizer estimation. Selected parameters, mapped feasibility, bound flags, training-only coefficients and Gram diagnostics are saved in results/analysis.

These comparisons use the inspected chronological windows and the common interpretation in Section 6.9. Stage A preceded the completed Stage B and C studies reported later in this volume.

9.10.3 Audit

The independent computational auditor reconstructed local raw rows, nuisance estimates, kernels, quadrature and projection; checked every saved candidate and solver return, exact floors, all planned start slots and aggregate score entries. Internal optimizer line-search iterates were not all serialized.

9.10.4 Fold0 reconciliation

Historical rawI-minus-E0: -16.988778741 nats; recomputed rawI-minus-E0: -16.988735123; difference: +0.000043618. Orthogonal ML inhibition minus E0: -16.988768914; orthogonal-minus-raw improvement: -0.000033791 nats. Matched Omori inhibition minus its E0: -16.404033164; Omori E0 minus ML E0: -3.210666185; combined orthogonal Omori minus ML E0: -19.614699350 nats. These are arithmetic contrasts of separately trained constrained procedures, not a causal decomposition.

The stage A complete fold score accounting is retained in Appendix B.7.6.

9.10.5 Stage A complete aggregate score accounting

Comparison keys are local to this table.

AG1 `I0_raw_ml-E0_ml`
AG2 `I0_orth_ml-E0_ml`
AG3 `I0_orth_ml-I0_raw_ml`
AG4 `E0_omori-E0_ml`
AG5 `I0_orth_omori-E0_omori`
AG6 `I0_orth_omori-I0_orth_ml`
AG7 `(I0_orth_omori-E0_omori)-(I0_orth_ml-E0_ml)`

The stage A aggregate sensitivity accounting is retained in Appendix B.7.7.

Table 9.7: Stage A complete aggregate score accounting; primary. Total nats unless stated otherwise. Positive favors the candidate. The same 612 targets enter each aggregate; positive-fold counts are out of three. Descriptive contrasts without confidence intervals.

Key	ΔQ	ΔC	Timing	ΔM	Joint	Joint/event
AG1	-40.854975	22.668560	-18.186415	0.000000	-18.186415	-0.029716
AG2	-40.855042	22.668582	-18.186460	0.000000	-18.186460	-0.029716
AG3	-0.000068	0.000022	-0.000046	0.000000	-0.000046	-0.000000
AG4	-38.017200	-0.154092	-38.171292	0.000000	-38.171292	-0.062371
AG5	-36.772664	20.496091	-16.276573	0.000000	-16.276573	-0.026596
AG6	-33.934822	-2.326583	-36.261405	0.000000	-36.261405	-0.059251
AG7	4.082379	-2.172491	1.909887	0.000000	1.909887	0.003121

The stage A training exposure geometry is retained in Appendix B.7.8.

9.11 Stage B: common and parent-class temporal scales

For a pinned positive mixture with $\sum_m w_m = 1$, use $s = \log \tau$ and $a_m = \lambda_m e^{-s}$ to write

$$k_s(u) = \sum_m w_m a_m e^{-a_m u}, \qquad S_s(u) = \sum_m w_m e^{-a_m u}.$$

Every scale preserves unit mass; the parent's observed class selects its entire kernel. The prospective range was $\tau \in [0.001, 20]$ days, including subhour scales. This is no cutoff on the retained parent history. At $y = a_m u$,

$$\partial_s k_s(u) = \sum_m w_m a_m e^{-y}(y - 1).$$

For a parent's finite-window lag interval $[L, U]$,

$$J_k(s) = \sum_m w_m e^{-a_m L}\{-\operatorname{expm1}[-a_m(U - L)]\},$$
$$\partial_s J_k = \sum_m w_m\{a_m L e^{-a_m L} - a_m U e^{-a_m U}\}.$$

Productivity weights multiply these expressions; all earlier parents and quiet exposure remain. Lags, rather than edited cached kernel values, define dynamic evaluation. The fixed-scale, common-free and class-free models respectively fix both scales to one day, estimate a common scale, and estimate two parent scales. All use $h = \delta = 0$, fixed $r = 0.7$ and the common training magnitude law. The common gradient is the sum of class gradients on the diagonal; at $n = 0$ all scale derivatives vanish and the scales are unidentified.

The optional association family preserves the original Perron normalization:

$$\lambda_i = p_i B + \frac{n}{\rho(h)} \sum_j P_{ij}(h) H_j, \qquad \Lambda = B + \frac{n}{\rho(h)}(H_0 + H_1).$$

Unit temporal mass makes ρ independent of the scales. Column stochasticity still cancels target redistribution even with different parent kernels. At fixed effective amplitude $n/\rho(h)$, timing is invariant to h wherever the map remains feasible; conditional marks can change.

Joint refitting and box constraints may indirectly change timing. No target-class temporal family was introduced to evade this identity.

For $\Gamma = n/\rho$, $T_i = \sum_j P_{ij} H_j$ and $e_i = p_i B + \Gamma T_i$, the optional joint negative log likelihood is

$$L = \int B + \Gamma(J_0 + J_1) - \sum_i \log e_i - \sum_i \{\log f_\beta(q_i) - \log p_i\}.$$

Its scale gradient is $L_{s_j} = \Gamma J'_j - \sum_i \Gamma P_{ij} H'_j / e_i$ and

$$L_h = -\Gamma \frac{\rho'}{\rho}(J_0 + J_1) - \sum_i \frac{\Gamma\{\sum_j P'_{ij} H_j - (\rho'/\rho) T_i\}}{e_i}, \quad P'_{ij} = \mathbf{1}_{i=j} - p_i.$$

The nuisance β, p_i are training-frozen. At $n = 0$, association and scale gradients vanish. All prescribed starts and exact nested floors must resolve; a finite candidate cannot replace an incomplete start. Near-training-equivalent candidates within 4×10^{-4} NLL receive full forecast checks with spread strictly below 0.01 nats/event.

9.12 Stage B audited results, historical bridge and failed association gate

Allowing one common Mittag-Leffler timescale improves pooled event-timing forecasts in all three retained windows. Additional parent-class timescales give a small positive aggregate gain, but improve only one window against the common-scale model. The predeclared association gate therefore fails, and no observed h model was fitted. This is a completed gated investigation, not a negative result for the unexecuted h branch.

9.12.1 Audited comparisons

All nine observed fit procedures and 72 prescribed starts completed. The independent raw-row audit reconciled 90 candidates and 72 solver returns, with no failed finite solver returns. The two fixed-seed engineering histories also completed eight procedures/64 starts, including the optional h engine. Near-optimal forecast stability and source/input integrity gates passed.

The 612 targets occur in disjoint chronological windows containing 203, 155 and 254 events; their training prefixes overlap. All three observed models use fixed r=0.7, the same training-only beta per case, h=0 and the same magnitude density. Mark differences cancel up to floating-point roundoff, so timing and joint gains coincide.

Here the historical E field is the ground event-log-intensity contrast, denoted ΔQ in this volume and C is the baseline-minus-candidate compensator, so $\Delta Q + \Delta C$ equals timing. Timing plus the conditional-mark contrast equals joint score. These are event-weighted sums divided by 612, rather than unweighted averages of fold rates.

The common-scale gain is consistent across these windows, although Fold 0's gain is only 0.077011 nats. The class-specific aggregate is positive, not a net loss. Its inconsistency against the stronger common-scale comparator is the reason the next branch stops. Winning two windows against the fixed tau=1 baseline is not the declared gate.

9.12.2 Scale and selection limits

All selected parameters are interior to their prescribed bounds. The allowed tau range was 0.001 to 20 days, including subhour values, but none of the observed fitted scales is subhour.

There is no lower-bound truncation explanation for these selected estimates and no bound expansion was performed.

Class 0 means $5.0 \leq M < 5.5$ and class 1 means $M \geq 5.5$. The kernels attach to the parent's class. The fitted ordering changes: class 0 is faster in Fold 0 but slower in Folds 1 and 2. These fits do not establish a stable physical magnitude ordering of aftershock timescales. Tau is a normalized-kernel scale, not a finite aftershock-duration cutoff, and is not interchangeable with Omori c.

Training NLL improves under exact nested floors. The additional class-specific training gains against the common model are approximately 0.012055, 1.867655 and 5.248592 nats, yet two held-out windows lose. Neither training improvement nor passing near-optimal forecast checks certifies physical scale identification or a global optimizer.

9.12.3 Optional association branch

The prospective rule required class-specific h=0 timing to exceed common-scale timing by more than 0.0001 total nats in at least two windows, after complete fitting, independent computational audit and forecast stability. Only Fold 1 qualifies. The audited gate is false: zero of the conditional three observed h procedures/24 starts were executed. The h code, derivatives and synthetic engineering were validated before observing this decision; engineering validation is not observed evidence for or against association.

The optional mathematical family preserves the legacy Perron normalization. With parent-class kernels, summing the column-stochastic transition matrix still gives ground intensity $B+(n/\rho(h))(H0+H1)$. Association h adds no new direct temporal shape conditional on observed parent history; its scalar n/ρ effect is already an excitation-amplitude coordinate. At fixed effective amplitude, timing is invariant to h, while conditional mark probabilities can change. No target-class kernels or post-gate model family were introduced.

9.12.4 Historical Omori reference

The immutable historical Omori total is -990.170063 joint nats. The new fixed-scale, common-scale and class-scale procedures have raw joint totals of approximately -984.684158, -977.300484 and -976.292651. Their raw differences from historical Omori are +5.485905, +12.869579 and +13.877412 nats, respectively.

These are descriptive cross-procedure comparisons, not matched kernel-only gains. Historical selected productivity ratios are [0.4,0.7,0.7], whereas B fixes 0.7 throughout; only Fold 0 has that productivity mismatch. All three historical beta estimates use a different global-MLE convention. Beta also enters productivity weights, so this difference affects timing as well as magnitude scores. Historical Omori also used its own continuous shape/scale selection.

For common free tau, the raw historical difference splits into +10.389214 timing and +2.480366 magnitude nats. Magnitude terms therefore cannot be cancelled across those recipes. The matched B free-scale gain remains +7.383674 nats; the larger historical difference cannot be attributed solely to freeing tau. No prior Omori fit was rerun or overwritten, and no new matched Omori model was fitted.

9.12.5 Delimited conclusion

The supported result is a modest, consistent timing gain from a common free ML scale in this bounded pooled procedure. Parent-class scales add a smaller positive aggregate but fail the

predeclared across-window continuation criterion. Optional h remains untested on observed data.

This retains the original regional B design despite earlier spatial results; it is not a combined spatial-temporal validation or an independently replicated catalogue. All windows were previously examined. The common inferential and optimization boundaries are stated in Section 6.9.

The stage B complete fold score accounting is retained in Appendix B.7.9.

9.12.6 Stage B complete aggregate score accounting

Comparison keys are local to this table.

BG1 `E0_tau_free-E0_tau1`
BG2 `E1_tau_class-E0_tau_free`
BG3 `E1_tau_class-E0_tau1`

Table 9.8: Stage B complete aggregate score accounting; primary. Total nats unless stated otherwise. Positive favors the candidate. The same 612 targets enter each aggregate; positive-fold counts are out of three. Descriptive contrasts without confidence intervals.

Key	ΔQ	ΔC	Timing	ΔM	Joint	Joint/event	Positive folds
BG1	4.405481	2.978194	7.383674	0.000000	7.383674	0.012065	3
BG2	5.297977	-4.290144	1.007833	0.000000	1.007833	0.001647	1
BG3	9.703458	-1.311951	8.391507	0.000000	8.391507	0.013712	2

The stage B selected temporal scales are retained in Appendix B.7.10.

9.13 Extension O: continuously tuned, kernel-matched inhibition

The Omori–Utsu temporal law is a standard aftershock-decay model (Utsu et al., 1995); the comparisons below concern its declared fitted implementation and bounds. For $u \geq 0$, use the normalized Omori density and survival

$$k(u) = \frac{p-1}{c}(1+u/c)^{-p}, \quad S(u) = (1+u/c)^{1-p}, \qquad c > 0,\ p > 1.$$

With $s = \log c$, $A(u) = \log(1+u/c)$ and $v = u/(c+u)$,

$$\partial_s \log k = -1 + pv, \quad \partial_p \log k = (p-1)^{-1} - A, \quad \partial_s S = (p-1)vS, \quad \partial_p S = -AS.$$

Parent exposure is $J = S(L) - S(U)$ and its derivatives use the same endpoint differences; stable evaluation uses $S(L)\{-\operatorname{expm1}[(1-p)(A(U) - A(L))]\}$. No lag truncation is used. The prospectively fixed bounds are $c \in [10^{-4}, 20]$ days and $p \in [1.01, 2.5]$. These numerical domains do not identify physical duration or justify an outcome-dependent expansion. For $z \in \{s, p\}$ the additional timing gradient is

$$L_z = n \sum_k \kappa_k \partial_z J_k - \sum_i \frac{n \partial_z H_i}{\lambda_i}.$$

The stress feature remains the original unit-scale ML survival history and has no c, p derivative. Both kernel gradients vanish at $n = 0$.

The primary excitation fit selects (c, p) by training NLL and shares those values exactly with raw and orthogonal matched inhibition. The jointly tuned orthogonal inhibitory fit is secondary. Exact Poisson/nested floors and the sealed coarse Stage A excitation point are reevaluated in the new objective; old scores are neither overwritten nor substituted. Every procedure uses its prescribed starts, with training-only selection and the saved forecast-stability gate. The same coordinate map and feasibility requirements from Stage A apply.

For the raw inhibited engineering law, the valid thinning envelope is the undepleted background plus excitation; current inhibited intensity is not a monotone envelope because stress may recover between events. Independent time and mark streams, rejection records, caps and unreplaced failed histories are part of engineering eligibility. A matched training-selected kernel need not equal the generating kernel; no invalid truth-containment claim follows.

9.14 Extension O audited results and sensitivity interpretation

Continuously training-tuned Omori excitation improved held-out timing over the fixed ML baseline in all three primary windows. Stress inhibition still reduced aggregate held-out scores under the same E-selected Omori kernels. The primary matched orthogonal contrast lost 14.137174 nats across 612 targets, and the secondary jointly tuned inhibitory model lost 11.803237 nats. This is a negative result for the implemented bounded stress-depletion procedures, alongside a positive result for temporal-kernel tuning.

All 45 planned procedures and 360 prescribed starts completed; the independent raw-row audit reconciled 456 candidates and 360 solver returns and found stable near-optimal forecasts. Model selection used training NLL only. Magnitude scores are identical within each case, so every displayed timing difference is also the joint difference.

The matched orthogonal model won only Fold 1 by 0.175158 nats and lost Folds 0 and 2. The jointly tuned inhibitory model lost all three primary windows. Its 2.333937-nat aggregate improvement over matched orthogonal inhibition is secondary; it does not turn the inhibitory versus excitatory comparison positive.

For E_omori minus E_ml, the event-log-intensity gain was 17.242145 nats and the compensator contribution was -5.036642 nats, giving +12.205503. For matched orthogonal inhibition minus E_omori, the event contribution was -37.002313 and the compensator contribution +22.865140, leaving -14.137174. Lower exposure could not offset lost event intensity.

The training-selected primary excitation kernels are reported in Table B.20 in Appendix B.7.13.

These c and p estimates are interior to the prospectively fixed bounds. The short selected c values explain why the earlier coarse grid with minimum c=0.1 day did not adequately search this continuous family. This is a scale parameter of a heavy-tailed kernel, not an estimated physical recovery duration. Stress itself still uses the original fixed ML survival kernel; only triggering was continuously tuned.

Orthogonalization did not rescue the primary loss. In Fold 0 raw inhibition reached its raw d=2 upper boundary. Its selected point maps feasibly into orthogonal coordinates, while the orthogonal optimum maps back to raw d=2.114439 and lies outside the raw box. The orthogonal fit improves training NLL by only 0.005858 nats but loses a further 0.361471 test nats. In Folds 1 and 2 the selected points map feasibly both ways and their forecasts agree up to numerical optimization differences. Thus the small primary difference between raw and

orthogonal procedures is explained by a changed bounded parameter domain, rather than a new unrestricted model space. Exposure orthogonality is not likelihood-score orthogonality.

The constant-background sensitivity fixes raw d=0 or detrended d_tilde=0 respectively. These are different restrictions; all selected mapped raw/orth constant-case points violate the destination constant restriction. Their contrast cannot be interpreted as pure reparameterization. The post-1980 variant changes the training window and its projection/nuisance estimates, while retaining full prior event history.

Omori excitation wins all three windows in each sensitivity variant. Every inhibitory aggregate remains negative. Constant-background detrended inhibition improves over raw inhibition by 5.449474 nats, but still loses 8.796918 nats against its matched excitatory comparator. The same test events are reused across variants and must not be pooled as 1,836 independent targets.

Within this bounded, regionally pooled fixed-productivity family, inhibitory failure therefore persists after replacing the inferior fixed ML trigger with continuously tuned Omori excitation. It cannot be attributed solely to that fixed-trigger choice or solely to the raw trend boundary. This does not establish that physical stress inhibition is absent, and it does not test continuous spatial stress or a joint spatial/temporal extension. The historical Iteration 27 Omori result uses a different nuisance recipe, including the Fold 0 productivity ratio and training magnitude estimator; the present gain is a new internally matched comparison, not a numerical reproduction of that historical score.

The catalogue and windows were previously inspected, and three chronological tests provide limited robustness evidence. The common inferential and optimization boundaries in Section 6.9 apply. S was designed and frozen before these outcomes; its launch is governed by correctness and resource admission, not this score direction.

Exact per-window scores, parameters, boundary flags and both-direction mapping diagnostics are in summary.json. The independent raw-row reconciliation is in observed_saved_audit_v1.json.

The extension O complete fold score accounting is retained in Appendix B.7.11.

9.14.1 Extension O complete aggregate score accounting

Comparison keys are local to this table.

OG1 `E_omori-E_ml`
OG2 `Iorth_matched-E_omori`
OG3 `Iraw_matched-E_omori`
OG4 `Iorth_joint-E_omori`
OG5 `Iorth_joint-Iorth_matched`
OG6 `Iorth_matched-Iraw_matched`

The extension O aggregate sensitivity accounting is retained in Appendix B.7.12.

Table 9.9: Extension O complete aggregate score accounting; primary. Total nats unless stated otherwise. Positive favors the candidate. The same 612 targets enter each aggregate; positive-fold counts count joint gains exceeding 10^{-4} nats out of three. Positive values at or below that threshold are treated as numerical ties. Descriptive contrasts without confidence intervals.

Key	ΔQ	ΔC	Timing	ΔM	Joint	Joint/event	Positive folds
OG1	17.242145	-5.036642	12.205503	0.000000	12.205503	0.019944	3
OG2	-37.002313	22.865140	-14.137174	0.000000	-14.137174	-0.023100	1

Key	ΔQ	ΔC	Timing	ΔM	Joint	Joint/event	Positive folds
OG3	-36.491541	22.715835	-13.775706	0.000000	-13.775706	-0.022509	1
OG4	-27.179752	15.376515	-11.803237	0.000000	-11.803237	-0.019286	0
OG5	9.822562	-7.488625	2.333937	0.000000	2.333937	0.003814	1
OG6	-0.510772	0.149304	-0.361468	0.000000	-0.361468	-0.000591	0

The extension O selected continuous kernels are retained in Appendix B.7.13.

9.15 Engineering witnesses and constrained-map diagnostics

9.15.1 Control normalization and saved forecast contrasts

These are particular fixed engineering realizations. Equal Perron radii do not equalize immigrant-weighted first-generation production, event counts or statistical information. In particular the class-active laws can dilute the offspring contribution. All controls remain visible, including losses, rather than retaining only successful examples. These tables support implementation inspection, not calibrated power, bias, coverage or reliable recovery.

The engineering normalization, forecast, profile and map witnesses are retained in Appendix B.7.14.

9.15.2 Coarse-grid loss and the later continuous-kernel gain

The coarse Stage A excitation-only Omori procedure lost to its matched ML reference in every aggregate setting: primary -38.171292, post-1980 -15.005799 and constant-background -43.433403 nats. Every excitation-selected scale in Table B.25 is at the lower coarse-grid endpoint, $c = 0.1$ day. In contrast, continuous Extension O gained 12.205503 primary nats over its matched ML reference, with all three windows positive. This is a materially different outcome from the coarse-grid loss. The continuous primary selected scales lie below the earlier grid, but these results compare bounded fitted procedures; they do not identify a universal kernel ranking.

The stage A selected coarse-grid kernels are retained in Appendix B.7.15.

9.15.3 Engineering-clock and evaluation scope

Stage B's two engineering laws fixed a common scale of 0.02 day or parent scales $(0.005, 0.5)$ day, with log immigration $a = \log(0.4)$, $d = 0$, $n = 0.2$, $\beta = 2.85$, $r = 0.7$, a 500-day horizon and training ending at day 350. The horizon was fixed before generation to retain expected immigration while reducing binary64 clock spacing relative to the earlier 5000-day engineering design. It did not rescale observed data or change the absolute scale bounds. In the uninhibited constant-background controls, inversion of the exact integrated hazard needs no acceptance uniform because envelope and actual intensity agree. Independent time and mark streams, retained sampled versus actual binary64 elapsed times and fail-without-jitter clock rules remain essential. Known synthetic nuisance values and the observed training-only nuisance adapter have distinct verification roles. The optional association engine was exercised on both controls before the observed gate; this supplies no observed evidence about the unexecuted association branch.

The common-scale and parent-scale engineering laws are deliberately fixed examples, not estimates of a physical ordering. Saved numerical derivatives, exact nested floors, left limits, complete parent exposure and direct-history truth checks delimit implementation validity. A repaired control or audit interface does not authorize replacing failed histories or promoting an unresolved production outcome. No new engineering computation accompanied that historical consolidation.

9.16 Completed matched magnitude-history experiment

The experiment uses 100 paired forest units and 300 catalogues across persistence settings $h \in \{0, 0.5, 0.9\}$. The strictly subcritical generator has productivity ratio $r = 0$ and reproduction mean $n = 0.5$. Each catalogue is fitted by M1, the misspecified $r = 0.7$ candidate, and M2, the productivity-matched $r = 0$ candidate. Both hold $n = 0.5$ and use the same inherited kernels and eight-start null/alternative procedure, fitting the background level, trend and stress coefficient. All 600 procedure–catalogue cases are complete; no seed was redrawn.

M1 is a structural replication arm, not an exact numerical replication of the earlier pilot: the common reproduction mean and strengthened starts are disclosed. M2 matches productivity at $h = 0$, but the exact Mittag–Leffler generator and finite 195-mode candidate still differ numerically. It is therefore not described as a literally exact full null law.

The M2 productivity change does not remove magnitude-scaled stress: the implemented transform remains $g(Q) = e^{\gamma(Q-1)}$, with γ as in Chapter 5, while $\kappa = 1$. Its ratio moment is therefore $\mathbb{E}g^2 = \infty$ under the ideal fresh exponential mark law at the declared rate. Proposition 5.2 describes this candidate's own finite-mixture null, not the misspecified score under the exact generator. At positive persistence the offspring magnitude-class probability depends on the parent's class, so the fresh-mark hypotheses require a separate conditional-mark argument. These regularity distinctions do not determine the finite selection counts below.

Table 9.10: Completed selection counts and descriptive empirical calibration. Selection uses GLR $>$ 0.001 and $\hat{\eta} > 10^{-6}$. All entries have 100 resolved catalogues; intervals are 95% Wilson intervals for selection proportions.

Procedure	h	Selected	Wilson interval	Empirical $h = 0$ threshold	Above threshold
M1	0.0	79/100	[0.700, 0.858]	7.665955	4/100
M1	0.5	85/100	[0.767, 0.907]	7.665955	6/100
M1	0.9	85/100	[0.767, 0.907]	7.665955	6/100
M2	0.0	75/100	[0.657, 0.825]	4.629883	4/100
M2	0.5	75/100	[0.657, 0.825]	4.629883	4/100
M2	0.9	69/100	[0.594, 0.772]	4.629883	4/100

The empirical threshold is the higher empirical 95th percentile of the same 100 $h = 0$ GLRs within each procedure. The four strict exceedances among those same controls are a consequence of this in-sample construction and ties, not independent evidence of 5% Type I error control. Selection by the small fixed GLR cutoff and exceedance of this descriptive threshold are distinct summaries.

These paired results retain all forest units. Neither the large fixed-cutoff selection proportions nor the descriptive self-calibrated thresholds establish a generally valid test for stress inhibition. There is no independent calibration sample in this threshold comparison.

Table 9.11: Paired GLR changes from $h = 0$ on the same forest units. MC SE is a forest-unit sample standard error.

Procedure	h	Mean change	Median change	MC SE
M1	0.5	-0.009000	0.000000	0.146627
M1	0.9	0.010831	0.046652	0.416236
M2	0.5	-0.106278	0.000000	0.098682
M2	0.9	-0.140055	0.000000	0.118097

Table 9.12: Within-forest M2 minus M1 GLR contrasts. Negative values favour a larger fitted GLR under the misspecified M1 procedure, not better predictive calibration.

h	Mean GLR contrast	MC SE
0.0	-1.132859	0.425666
0.5	-1.230138	0.423813
0.9	-1.283745	0.211306

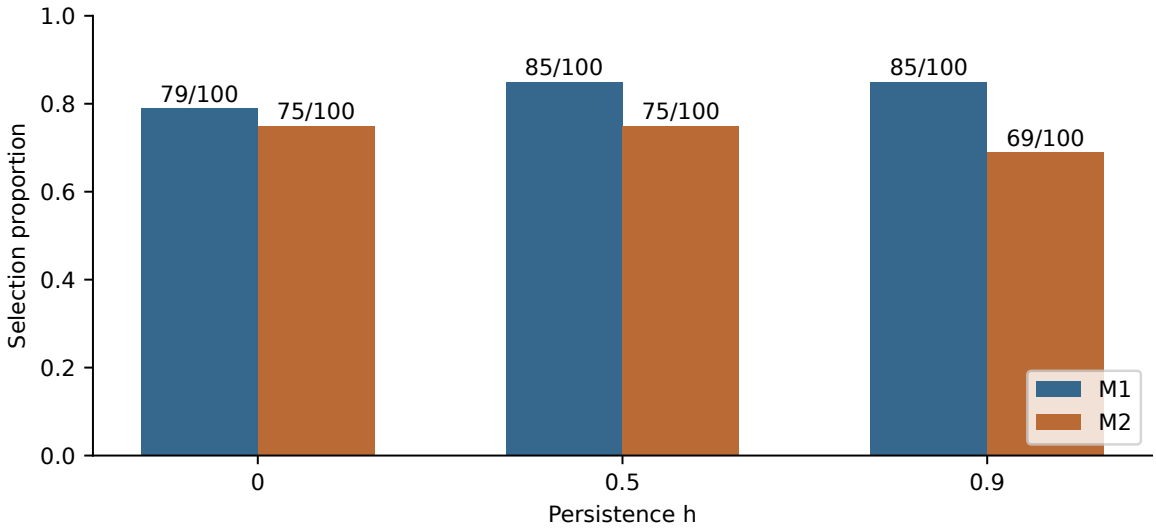


Figure 9.4: Completed selection proportions under M1 and M2, each based on 100 paired forest units. The display is descriptive; saved custody checks and sampled raw validation passed.

Part III

Spatial Structure, Regional Transfer, Competitive Benchmarks, and Continuous Local Inhibition

Chapter 10

Discrete Routing, Continuous Offspring Geography, and Background Nuisance in New Zealand

In the tables, $I_{\mathrm{d,g}}$ denotes distance excitation with global stress (the saved identifier `I_distance_globalG`). A stacked mantissa and power of ten form one scientific-notation value; their full displayed precision is retained.

10.1 From regional aggregation to matched spatial forecasts

The successive studies below turn the locality question into matched predictive comparisons. They progress from three labelled zones (C1), through a conserved cross-zone mixing law (W), finite-cell distance allocation (D), and continuous-area offspring density (S), to a joint temporal–spatial/background factorial. These stages reuse the historical windows; their gains cannot be concatenated into an independent replication count. The regional algebra motivates the completed comparisons without identifying a geological mechanism.

The completed spatial programme compared, on the same region-labelled observations, three deliberately matched candidates: no inhibition, pooled inhibition across regions, and local inhibition. For a given regional history, the motivating schematic intensities are

$$\lambda_k^{(0)} = B_k + H_k, \tag{10.1}$$

$$\lambda_k^{(\mathrm{pool})} = B_k \exp\left(-\eta \sum_j G_j\right) + H_k, \tag{10.2}$$

$$\lambda_k^{(\mathrm{local})} = B_k \exp(-\eta G_k) + H_k, \tag{10.3}$$

where H_k includes the trigger amplitude and must be defined identically across alternatives. A fair comparison matches spatial backgrounds, triggering families with a serious Omori baseline, mark and history rules, observation rules, and fitting effort. With region label r_i for event i, every candidate uses the same joint score

$$\ell = \sum_i \log \lambda_{r_i}(t_i-) - \sum_k \int_0^T \lambda_k(t)\, \mathrm{d}t + \sum_i \log f(m_i). \tag{10.4}$$

The integrals include quiet regions and quiet intervals. Omitting them would reward or penalize models according to event allocation without accounting for exposure. Alternatively, candidates must be marginalized to the same label-discarded observations; a label-aware likelihood and a label-discarded likelihood cannot be ranked as if they used the same data event.

The subsequent stages declare their own background and geometry recipes. A later replay may differ at solver precision from an older sealed result; it does not replace that result.

Throughout this chapter, an "Event" column is the point-event logsum contribution and "Compensator" is the signed contribution $C = -E$. For discrete zones/cells, the point intensity is with respect to counting measure; continuous comparisons below first put every arm on the same projected-area measure.

10.2 C1: region-labelled local and global controls

C1 supports localizing excitation for time-zone prediction, but does not support adding local stress inhibition to that localized excitation model. All three folds, four models, 12 case fits and 96 prescribed starts completed. The independent computational audit passed, reconciling 114 candidates and 96 solver returns; saved near-optimal forecasts passed the declared stability gate. The catalogue and historical authorities were preserved.

10.2.1 Separate timing from zone prediction

On the 612 held-out targets, local excitation beats global excitation by 349.786446 joint nats (+0.571546481 nats/event), positive in all three folds. Of this gain, 347.958257 nats comes from conditional-zone prediction and only 1.828190 nats from regional timing. Joint gains are +105.104546, +119.000978 and +125.680922 nats; timing differences are -2.083237, +1.797562 and +2.113864. The strong joint result therefore concerns predicting which of three fixed latitude bands receives the next event. It is not evidence of a comparably large improvement in event timing or of continuous-location forecast skill.

At the same parameters, local E0 and global E0 have exactly the same regional ground intensity and compensator. Their conditional-zone factors differ, so selecting each by full time-zone training NLL produces different fitted parameters and modestly different timing forecasts. The observed timing difference does not contradict the fixed-parameter identity.

Local inhibition also beats global inhibition strongly: +364.573096 joint nats (+0.595707674 nats/event), including +17.936860 regional-timing and +346.636237 conditional-zone nats. Both joint and timing differences are positive in all three folds. This comparison includes both localized triggering and localized stress, and the poor globally inhibited baseline. It does not establish the value of inhibition beyond a matched local excitation model.

10.2.2 The matched inhibition result remains negative

Adding local inhibition to local excitation loses 3.399761 joint nats (-0.005555164 nats/event): -2.077741 regional timing and -1.322020 conditional zone. Fold differences are -0.207862, -3.191896 and approximately zero (-0.000003) nats. The last difference is below the 0.0001-nat score-reconciliation tolerance and the selected eta is exactly zero; it should be described as a numerical tie rather than a substantive third forecast loss.

The local inhibitory fit lowers training NLL by approximately 0.112071, 1.061892 and zero nats in Folds 0-2, respectively, yet none yields a substantive held-out gain. These are modest training improvements of a nested bounded fitting procedure, not evidence for a nonzero physical depletion parameter.

Global inhibition loses 18.186411 nats (-0.029716357 nats/event). The difference of matched inhibition contrasts is therefore +14.786650 nats (+0.024161193 nats/event): localization makes the inhibitory penalty much less severe, while the local incremental inhibition contrast

remains negative. This descriptive difference is positive in Folds 0 and 2, negative in Fold 1. It is not a causal difference-in-differences design.

10.2.3 Fold 0 and proper exposure accounting

Fold 0's global inhibitory contrast is -16.988735 nats, decomposed into -33.904743 event and +16.916008 compensator nats. The matched local inhibitory contrast is -0.207862 nats: -3.000762 event and +2.792900 compensator nats. Regional timing improves from a -16.988735 global inhibitory penalty to a -0.831697 local inhibitory penalty; the local conditional-zone contribution is +0.623835 nats.

The proper Fold 0 local inhibitory score differences by zone are +0.289381 south, -0.141682 central and -0.355561 north. Their sum gives -0.207862. Across all three folds, the matched local inhibitory differences are negative in each zone: -1.199738, -1.134201 and -1.065822 nats. These are full zone contrasts with their own valid integrated exposures, not event-only geographical summaries.

The large local-versus-global joint gains cannot be reduced to a positive event term everywhere. For example, Fold 0 local E0 has a negative central-zone event contrast (-2.324219 nats), offset by a +17.184730 compensator contribution, giving a +14.860511 central-zone total. Quiet exposure and all-zone compensators are essential to the result.

10.2.4 Parameters and historical reconciliation

The selected global eta values are 1.451113, 0.230849 and 0.116715; local eta values are 0.374699, 0.517222 and 0. Local eta is smaller in Folds 0 and 2 but larger in Fold 1. There is no uniform shrinkage result. Local and global stress histories also differ in scale and composition, so eta alone is not a directly comparable measure of physical stress strength. No selected coordinate hits a restrictive box boundary except the local eta=0 boundary in Fold 2.

The fresh global inhibitory contrasts reproduce Stage A's global raw contrasts to numerical precision: Fold 0 shifts by about 1e-11 nats, Fold 1 by +0.000013869 and Fold 2 by -0.000009878. Fixed-parameter global reduction was independently verified; these tiny separately optimized fit shifts are not failures of that identity. Historical Iteration28's -16.517557 Fold 0 contrast belongs to its own fitted/nuisance recipe and remains a separate authority.

The old five-degree cell crosswalk regroups event log-intensity ratios only. C1 defines three zone intensities, without a within-zone cell probability model; consequently cell compensators and cell full likelihoods are undefined. Proper zone compensators remain separate. Historical cell event penalties are retained with their source hashes and are not spliced into new denominators.

10.2.5 Delimited conclusion

The useful result is strong retrospective categorical-zone prediction from localized triggering, with a small mixed-sign timing benefit for local E0. Localizing the whole inhibitory procedure largely removes the catastrophic global inhibition penalty, but the matched local inhibitory extension still adds no held-out value. This supports retaining $E0_{\text{local}}$ as the better of these declared local procedures; it does not demonstrate physical stress inhibition or identify a causal spatial mechanism.

The result applies to three predeclared latitude bands, omega=0, fixed r=0.7 and the retained ML law on one previously examined regional catalogue. Training-only pi and beta were frozen for test; observed history updated predictably. No event-iid confidence interval, Wilks boundary test, global-optimum certificate, new partition, kernel search, C2 or other model stage follows from this review.

Evidence: results/analysis/summary.json; methods/observed_saved_audit.json; results/analysis/historical_reconciliation.json; results/analysis/cell_event_crosswalk.json; design/spatial_zone_counts.json; protocol_observed_v1.json. This interpretation introduces no numerical refit or pinned-source change.

10.2.6 C1: complete audited comparisons

All three folds, four models (12 case fits), and 96 prescribed starts are complete. The same 612 held-out targets are scored once per model. The interpretation in Section 6.9 applies.

Comparison	Regional timing	Conditional zone	Magnitude	Joint nats	Nats/event	Positive folds
$E0_{local}$-$E0_{global}$	1.82819	347.958	0	349.786	0.571546	3/3
$I0_{local}$-$I0_{global}$	17.9369	346.636	0	364.573	0.595708	3/3
$I0_{local}$-$E0_{local}$	-2.07774	-1.32202	0	-3.39976	-0.00555516	0/3
$I0_{global}$-$E0_{global}$	-18.1864	-5.11591×10^{-13}	0	-18.1864	-0.0297164	0/3
($I0_{local}$-$E0_{local}$)-($I0_{global}$-$E0_{global}$)	16.1087	-1.32202	0	14.7866	0.0241612	2/3

The complete C1 per-fold contrasts and exact zone decompositions are retained in Appendix B.8.1.

10.2.7 Scope

K=3 fixed latitude zones, raw local stress, fixed ML excitation and r=.7, omega=0. Background proportions and magnitude nuisance use training only. Local triggering and local inhibition are compared with matched global/local excitation controls. Joint gains can come from zone prediction without improved regional timing; the two components must not be conflated.

The C1 model does not define within-zone cell probabilities, so cell event crosswalks have no invented cell compensators. Full compensator differences are reported per zone. Historical Iteration28 cell event penalties and recipe differences are retained separately in historical_reconciliation.json.

All comparisons are retrospective on previously examined chronological windows. The common inferential and optimization boundaries in Section 6.9 apply. No follow-up model/kernel/productivity search or other stage was run.

10.3 W: conserved cross-zone redistribution

W supports a small training-estimated global redistribution component in the otherwise local excitation model for zone prediction. It does not support an incremental inhibitory extension

or a direct timing benefit from the mixing weight. All 27 case fits and 216 prescribed starts completed; the independent computational audit passed, checking 279 candidates and 216 solver returns. Reported differences use 612 targets and common magnitude scores cancel.

10.3.1 Excitation mixing improves the joint score through zone prediction

Free-omega excitation improves on purely local excitation by 10.553372 joint nats (+0.017244071 nats/event), positive in all three held-out windows: +2.293398, +8.176718 and +0.083256 nats. Its conditional-zone gain is +13.789492 nats, while regional timing worsens by 3.236121 nats. Timing differences are +0.865688, -0.561462 and -3.540347 nats. The joint improvement should therefore be described as a categorical-zone forecast improvement with a net timing cost.

Selected excitation ω values are 0.046125, 0.045172 and 0.044339. These are weights on a globally redistributed trigger component; they are not literal cross-zone offspring probabilities. The globally redistributed component can also return to the parent's own zone, so a parent's actual cross-zone allocation is $\omega(1 - \pi_{z_j})$, where z_j is the parent's zone. No physical transfer rate is identified by these fitted routing weights.

The free model's training NLL gains against omega0 are approximately 2.163997, 4.867366 and 13.271381 nats. Exact completed endpoint floors preserve the nested training comparison, and all prescribed starts and near-optimal forecast checks passed. Three similar estimates from overlapping training prefixes do not provide an independent replication or confidence interval for a physical mixing parameter.

Free excitation beats global excitation by 360.339778 joint nats (+0.588790487 nats/event), with +361.747747 conditional-zone nats and -1.407969 timing nats. This chiefly retains the large spatial prediction benefit already seen in C1, with an additional modest gain over the strictly local endpoint.

10.3.2 Omega has no direct timing effect

For either fixed stress choice and the same remaining parameters,

$$\sum_z \lambda_z = \sum_z \pi_z e^{a+dx_{\mathrm{trend}}(t)-\eta G_z^{\mathrm{stress}}} + nH_{\mathrm{global}}. \tag{10.5}$$

Omega cancels because excitation allocation is column-stochastic. This holds for E and both local/global-stress I families. Ground likelihood and total compensator are omega-invariant at fixed (a,d,n,eta). The observed timing changes arise indirectly because full time-zone training refits those other coordinates. They do not overturn the shared-kernel timing-invariance result or establish direct improvement in event timing from cross-zone redistribution.

10.3.3 Matched free inhibition remains negative

With omega free in both matched models, local-stress inhibition loses 3.637297 joint nats (-0.005943296 nats/event) against excitation: -2.292422 timing and -1.344875 conditional-zone nats. Fold differences are +0.087297, -3.232617 and -0.491977 nats. Even the small Fold 0 joint gain combines worse timing (-0.704286) with better zone prediction (+0.791583). Regional timing is worse in all three windows.

Global-stress inhibition loses 12.887727 joint nats (-0.021058377 nats/event): -11.593479 timing and -1.294248 zone nats. Fold differences are -11.191948, +0.639379 and -2.335157 nats. Most of its aggregate loss remains in Fold 0. Local-stress inhibition outperforms global-stress inhibition in the free-omega comparison by 9.250429 nats, positive in two windows, but neither improves on matched excitation overall.

The selected free local-stress eta values are 0.281470, 0.579390 and 0.059749, with omega near 0.044-0.045. Free global-stress eta values are 0.910443, 0.133890 and 0.162321; their omega values are 0.068905, 0.048547 and 0.046997. These are fitted constrained prediction coordinates under different stress histories, not directly comparable physical stress strengths or a causal explanation of the losses.

10.3.4 The factorial endpoints prevent a blanket localization claim

At fixed local excitation (omega0), global-stress inhibition loses only 1.005026 aggregate nats, whereas local-stress inhibition loses 3.399764 nats. Thus local stress is not uniformly the better stress choice. The global-stress eta is 0.068165 in Fold 0 and exactly zero in Folds 1 and 2; its tiny positive score differences in those latter windows are numerical ties, not meaningful forecast wins.

At fixed global excitation (omega1), local-stress eta is exactly zero in all three windows. Its aggregate score difference from excitation is -0.000031725 nats, below the 0.0001-nat saved-score reconciliation tolerance, and should be treated as a nested numerical tie. Global-stress inhibition at that same excitation endpoint loses 18.186431 nats, including -16.988727 in Fold 0.

The factorial result is an interaction among excitation routing, stress choice and joint nuisance refitting. It does not justify a universal statement that localizing stress rescues inhibition. Much of the earlier local-versus-global improvement is attributable to excitation routing and conditional-zone forecasting. The raw positive-window counters in the audit retain arithmetic signs; eta0 differences of order1e-5 or1e-6 are not scientific evidence for an inhibitory gain.

10.3.5 Proper exposure and conclusion

All score contrasts include complete zone compensators. For example, Fold 0 free-local-inhibition versus free excitation has an event contribution of -2.002970 nats and a compensator contribution of +2.090266, giving +0.087297. Event penalties alone would give the wrong sign. No cell-within-zone score is present in this three-zone W stage; its full score is timing plus conditional zone plus common magnitude.

The delimited result favors free-omega excitation for categorical-zone forecasting among the declared W procedures, while the incremental local/global inhibitory alternatives remain negative overall. There is no direct omega timing effect, physical stress-transfer claim, event-iid confidence interval, Wilks boundary inference or global-optimum certificate.

D's model support, smoothing, distance grid, synthetic laws and selection rules were frozen before W outcomes. This interpretation changes none of them. Its historical admission depended on engineering correctness and protected resource/audit reserves; a positive W result was not an admission condition. The same held-out events were reused in D, rather than forming a fresh independent validation catalogue.

Evidence: results/W/analysis/summary.json; methods/W_observed_saved_audit.json; protocol_W_observed.json; reports/W_decision_report.md. No new fit, source modification or extension was performed for this interpretation.

10.3.6 W: complete audited comparisons

All declared procedures and starts resolved. Source and audit bind this report to the frozen protocol. The interpretation in Section 6.9 applies.

Comparison	Regional timing	Conditional zone	Within-zone cell	Joint nats	Nats/event	Positive folds
E_{wfree}-E_{w0}	-3.23612	13.7895	0	10.5534	0.0172441	3/3
E_{wfree}-E_{w1}	-1.40797	361.748	0	360.34	0.58879	3/3
$Ilocal_{wfree}$-E_{wfree}	-2.29242	-1.34487	0	-3.6373	-0.0059433	1/3
$Iglobal_{wfree}$-E_{wfree}	-11.5935	-1.29425	0	-12.8877	-0.0210584	1/3
$Ilocal_{w0}$-E_{w0}	-2.07775	-1.32201	0	-3.39976	-0.00555517	1/3
$Iglobal_{w0}$-E_{w0}	-0.528304	-0.476721	0	-1.00503	-0.0016422	2/3
$Ilocal_{w1}$-E_{w1}	-3.17254×10^{-5}	-1.7053×10^{-13}	0	-3.17254×10^{-5}	-5.18389×10^{-8}	0/3
$Iglobal_{w1}$-E_{w1}	-18.1864	-1.7053×10^{-13}	0	-18.1864	-0.0297164	0/3
$Ilocal_{wfree}$-$Iglobal_{wfree}$	9.30106	-0.0506271	0	9.25043	0.0151151	2/3

The complete W window-level event/compensator contrasts and target-region exposures are retained in Appendix B.8.2.

10.3.7 Routing interpretation

Magnitude-score differences cancel because the mark forecast is shared. W estimates cross-zone redistribution and independently varies stress localization. Column-stochastic mixing preserves total triggering at fixed remaining parameters; a timing difference after joint refitting is indirect. The same previously inspected targets recur in D and the later density comparisons, so completed stage protocols do not make these independent replications.

10.4 D: finite-distance cell allocation

The fixed-cell distance excitation procedure improves categorical location forecasts on this catalogue, but not regional timing against its primary C 1-cell comparator. Adding either tested inhibitory extension remains negative in aggregate. All 33 D procedures (21 main fits and 12 scale selectors), 264 starts and the saved-output artifact audit completed; the audit checked 312 candidates and 264 solver returns.

The primary $E_{distance}$-E_{bridge} gain is 74.843795 nats (+0.122293783 nats/event), positive in all three windows: +3.445007, +33.268354 and +38.130434. It decomposes into -3.259471 regional timing, -5.704315 conditional-zone and +83.807581 conditional-within-zone cell nats. The improvement concerns which of nine fixed cells receives an event. It is not a timing or continuous-location density gain. E_{bridge} is the strict omega 0 C 1 procedure refined onto the same cell space; it is not W's improved free-omega reference.

Training E selected sigma=150,150,50 km from the frozen four-point grid. The third choice reaches the grid's lower boundary. Cross-grid near-optimal forecast spreads were approximately 0.000002113, 0.000011607 and 0.000002524 nats/event, below the declared 0.01 gate. This establishes saved forecast stability under that criterion, not identification of a physical distance scale or an optimum beyond the grid. The global infinite-scale model remains a separate baseline, not a fifth selected point.

At the same background/productivity parameters, normalized excitation columns preserve the ground intensity for every E scale. The observed E timing differences after full-cell joint training are indirect consequences of refitting other parameters. Distance inhibition can additionally change timing through its unnormalized stress similarity, which is distinct from the normalized triggering probability. Its stress scale was tied to the E-selected scale; no separate stress-range search occurred.

$I_{distance}$-$E_{distance}$ is -2.275081 nats (-0.003717452 nats/event), comprising -0.516136 timing, -6.109160 zone and +4.350215 within-zone cell nats. Two windows gain (+0.452200 and +0.013479), but the middle window loses 2.740760 nats and determines the negative aggregate. Positive-window counts do not replace the event-weighted total. The global-stress alternative loses 2.990767 nats; local distance stress improves on that alternative by 0.715686 nats but neither beats matched excitation overall.

The local distance eta estimate equals its upper bound 3 in ALL THREE folds. Its training NLL gains against E are approximately 2.025365,5.832549 and 1.033420 nats, yet aggregate held-out performance is negative. These are bound-limited results for the declared procedure. They do not establish an unconstrained eta optimum, absence of physical stress effects, or performance under a larger bound or different independently selected stress range. No bound expansion or refit was performed.

All nine cells remain supported by predeclared hierarchical half-count smoothing, which preserves the original training zone proportions. This includes the held-out event in a previously unoccupied training cell. Proper compensators cover every target cell, including quiet/unoccupied cells. Unlike the old C 1 event-only cell crosswalk, D explicitly supplies cell intensities, so its cell P+C contrasts are defined. The new C 1-cell bridge is a declared q allocation, not a retroactive alteration of historical C 1 scores.

The separately labeled secondary comparison refines fixed W E_{wfree} to this same cell space without refitting. $E_{distance}$ gains 64.290424 nats (+0.105049712 nats/event), positive in all three windows; its timing difference is -0.023350, zone difference -19.493808 and within-zone cell gain +83.807581 nats. This supports an additional categorical-within-zone benefit even against the improved W reference, while providing no net timing improvement. It remains adaptive reporting, planned after W and before opening D outcomes, rather than an original primary contrast or independent validation.

Scope: fixed ML/r=.7, nine categorical cells, the predeclared four-scale grid and eta bound, one previously examined catalogue and reused chronological targets. No continuous spatial-density validation, physical causality, event-iid interval, Wilks boundary claim, global optimum certificate or further search follows. Evidence is in the D analysis/audit tables and the separately audited secondary common-space outputs.

10.4.1 D: complete audited comparisons

All declared procedures and starts resolved. Source and audit bind this report to the frozen protocol. The interpretation in Section 6.9 applies.

Comparison	Regional timing	Conditional zone	Within-zone cell	Joint nats	Nats/event	Positive folds
E_{bridge}-E_{global}	1.82817	347.958	1.42109×10^{-14}	349.786	0.571546	3/3
I_{bridge}-I_{global}	17.9368	346.636	-2.13163×10^{-13}	364.573	0.595708	3/3
I_{bridge}-E_{bridge}	-2.07775	-1.32201	2.84217×10^{-14}	-3.39976	-0.00555517	1/3
$E_{distance}$-E_{bridge}	-3.25947	-5.70432	83.8076	74.8438	0.122294	3/3
$E_{distance}$-E_{global}	-1.4313	342.254	83.8076	424.63	0.69384	3/3
$I_{distance}$-$E_{distance}$	-0.516136	-6.10916	4.35022	-2.27508	-0.00371745	2/3
$I_{d,g}$-$E_{distance}$	-2.85336	0.0832518	-0.220656	-2.99077	-0.00488687	1/3
$I_{distance}$-$I_{d,g}$	2.33723	-6.19241	4.57087	0.715686	0.00116942	2/3

The complete D window-level contrasts and target-cell exposures are retained in Appendix B.8.3.

10.4.2 Common-observation reconciliation

Secondary/adaptive reporting after W, planned before opening D outcomes; not an original primary contrast or a new fit. These contrasts are descriptive; no confidence intervals are supplied.

The reference is the fixed audited W E_{wfree} forecast, refined to nine cells with the already frozen training q(c|z). Timing and conditional-zone scores remain unchanged; the within-zone cell term is sum log q and each cell compensator is q times its W zone compensator. All seven D main models are compared without refitting or reselection.

D candidate minus refined W reference	Timing	Zone	Within-zone cell	Magnitude	Joint nats	Nats/event	Positive folds
E_{global}	1.40795	-361.748	1.42109×10^{-14}	0	-360.34	-0.588791	0/3
I_{global}	-16.7785	-361.748	0	0	-378.526	-0.618507	0/3
E_{bridge}	3.23612	-13.7895	1.42109×10^{-14}	0	-10.5534	-0.0172441	0/3
I_{bridge}	1.15837	-15.1115	0	0	-13.9531	-0.0227992	0/3
$E_{distance}$	-0.02335	-19.4938	83.8076	0	64.2904	0.10505	3/3
$I_{distance}$	-0.539486	-25.603	88.1578	0	62.0153	0.101332	3/3
$I_{d,g}$	-2.87671	-19.4106	83.5869	0	61.2997	0.100163	2/3

The complete window arithmetic for the secondary common-space comparison, together with its reconciliation scope, is retained in Appendix B.8.4.

10.4.3 Shared limits of the W and D comparisons

W and D completed 60 observed procedures, 480 mandatory starts and 591 audited observed candidates; their fixed-seed engineering schedules completed 40 procedures and 320 starts. The secondary common-space comparison retained all 21 per-window contrasts and seven aggregates without refitting. Its refinement preserves the W timing and zone scores, adds the fixed conditional-cell allocation, and supplies corresponding cell compensators. The distance excitation gains against refined W are +1.151609, +25.091636 and +38.047178 nats by window.

Its aggregate timing difference is small in score units but is not a formal equivalence result. The two distance inhibitory models also beat refined W by +62.015343 and +61.299657 nats, while both remain below matched distance excitation. Beating a coarser reference therefore does not establish incremental inhibition. The scientific scope is coarse spatial allocation under declared grids and bounds. Normalized routing does not directly change regional intensity at fixed parameters, selected ranges do not identify physical interaction distances, and no event-iid interval, global optimum or causal mechanism follows. Detailed contrasts, parameters and exposures are retained in Appendix B.8.

10.5 S: continuous area density

Continuous-area triggering substantially improved held-out point-location density forecasts relative to the projected cell-center lift, but did not improve regional event timing. Across the same 612 targets, E_{cont} minus E_{lift} gained 1,458.803884 joint nats (+2.383666477 nats/event), while losing 10.470041 timing nats. Most of the joint gain came from resolving locations within cells. Neither the zone-stress nor global-stress inhibitory extension supplied a meaningful incremental forecast gain.

This is principally a check of spatial discretization: a continuous density is compared with uniform spreading of cell mass, not with an equally resolved competitive spatial model. The +1,452.412080 within-cell nats account for almost all of the joint gain. The score improvement is real for these declared forecast distributions and common units, but its size does not establish a new physical mechanism or superiority over a strong continuous benchmark. The earlier national-routing reference is likewise a coarse spatial ablation.

All 42 planned procedures and 336 prescribed starts completed. The two selected E models per fold are aliases of existing selector results, not extra fits. The independent computational audit replayed 384 candidates and 336 solver returns, reconstructed geometry and positive rectangle integrals independently, and found stable candidate forecasts. The score contrasts below are descriptive, without confidence intervals.

Contrast	Fold 0 (203)	Fold 1 (155)	Fold 2 (254)	Total joint nats	Nats/event
E_{cont}-E_{lift}	+549.866921	+457.463557	+451.473406	+1458.803884	+2.383666477
E_{cont}-E_oldD_ lift	+549.307187	+462.729938	+451.046777	+1463.083902	+2.390659970
E_{cont}-E_{global}	+657.856745	+614.999257	+614.858124	+1887.714125	+3.084500205
I_cont_ zone-E_{cont}	+0.000032	-0.211419	-0.186526	-0.397913	-0.000650184
I_cont_ globalG-E_{cont}	+0.000032	-0.000016	-0.000004	+0.000013	+0.000000021
E_{lift}-E_oldD_lift	-0.559734	+5.266381	-0.426630	+4.280018	+0.006993493

The continuous model wins joint point scores in all three windows against each excitatory comparator. Against E_{lift}, its timing differences are -3.163215, +0.705482 and -8.012308 nats; it improves timing in only Fold 1. The aggregate joint decomposition is:

Component, E_{cont} minus E_{lift}	Nats
Regional timing	-10.470041
Conditional zone	-4.471988
Conditional cell within zone	+21.333833

Continued on next page

Component, E_{cont} minus E_{lift}	Nats
Within-cell area logdensity	+1452.412080
Magnitude	0.000000
Joint	+1458.803884

The corresponding point-event contribution is +1448.297321 nats and compensator contribution +10.506563, which reconcile to the joint gain because the common magnitude term cancels. This is evidence of improved conditional location prediction, predominantly inside the original coarse cells. It must not be reported as a 1,458.8-nat improvement in event timing. Normalized spatial triggering leaves the regional ground intensity unchanged at fixed nuisance parameters; the smaller timing differences arise through refitting those nuisance parameters to joint point likelihood.

Both families use actual unsnapped parent coordinates and the same equal-area projection, background density, six-point scale grid and square-kilometre area measure. The continuous model evaluates actual target points and analytically integrates each parent kernel over the rectangular domain. E_{lift} allocates parent mass at projected target-cell centers and spreads that mass uniformly within a cell. Its within-cell density is therefore fixed by cell area, whereas E_{cont} predicts subcell locations. The decomposition is proper: the within-cell term is an area logdensity, not a discrete probability.

Fold	E_{cont} selected sigma (km)	E_{lift} selected sigma (km)	Zone-I eta	Global-I eta
0	5	150	0.000000000	0.000000000
1	15	150	0.068954189	0.000000000
2	15	5	0.021703682	0.000000000

The continuous scale is at the 5-km lower grid boundary in Fold 0 and at 15 km in Folds 1 and 2. The data and this finite grid do not identify a universal physical interaction length or support extrapolation below 5 km. Each family selects its own scale using training likelihood; E_{cont} versus E_{lift} is a comparison of complete selection procedures, rather than a fixed-scale-only geometric effect. The small near-optimal forecast spreads certify local forecast stability under the prescribed candidate pool, not unique identification of sigma.

Zone inhibition selects eta=0 in Fold 0 and small positive eta in Folds 1 and 2. Its aggregate change is -0.397913 joint nats (-0.000650184 per event), including -0.115282 timing nats. Fold 0 differs from E only by approximately 0.000032 nats at eta=0, a solver-scale tie; the other windows lose 0.211419 and 0.186526 nats. Global inhibition selects eta=0 in every fold. Its aggregate +0.000013 nats is below the 0.0001-nat reconciliation tolerance and is a numerical tie, not evidence of benefit. Both inhibition procedures use the same E_{cont}-selected sigma without reselection. Stress remains a raw three-zone or global temporal field, not a continuously resolved spatial stress field.

The sealed old C $E_{distance}$ and E_{global} references were not refitted. They were lifted to point densities by dividing cell intensity by the exact projected cell area; their categorical joint totals subtract the same sum of test log-cell-areas. The independently checked lift identity puts every displayed contrast on the same km^{-2} measure. E_{cont} gains 1,463.083902 nats over the old distance lift. The new E_{lift} gains only 4.280018 aggregate nats over that reference, winning one window. That latter contrast changes projection/target-center geometry, scale-grid range and fresh fitting, so it is not a pure metric effect. Neither comparison directly subtracts categorical and density likelihoods.

This is retrospective evidence on a previously inspected catalogue and fixed finite domain. Isotropy is Euclidean in the declared cylindrical equal-area projection, not geodesic equivalence. The background remains nine-cell piecewise density, temporal excitation is fixed unit-scale ML with r=0.7, and no joint O/S model was fit. There are no event-iid intervals, Wilks boundary claims, physical-causality conclusions or universal bandwidth claims. The appropriate scientific conclusion is a strong location-density improvement for this continuous triggering procedure, with no accompanying timing improvement and no meaningful incremental inhibition gain.

Exact score and boundary tables are in summary.json, with the terminal saved-output artifact audit in observed_saved_audit_v3.json.

10.6 Joint temporal–spatial and background comparison

A matched comparison of twelve earthquake forecasting models finds positive predictive evidence from continuous event locations and, within that representation, Omori temporal decay. The experiment reuses 612 historical targets across three chronological windows. A training-only kernel-density background reduces but does not remove the continuous-location advantage over a cell approximation: gains remain 2.326-2.369 nats per event across three temporal families. With that stronger background and continuous locations, Omori improves on fixed-scale Mittag-Leffler (ML) triggering by 15.474 joint-score nats, including 10.368 timing nats, with both components positive in every window. The fully combined procedure improves aggregate timing and location scores over its current matched coarse comparator; location prediction accounts for most of the gain. Negative temporal contrasts and losing 1-2 km sensitivity fits limit the interpretation. These are retrospective fitted-procedure comparisons, not independent confirmation, identification of physical triggering, or evidence for inhibition.

10.7 Relation to the temporal and mark comparisons

Chapter 9 reports negative exploratory results from shared-kernel magnitude-class history and free productivity estimation. This chapter asks whether temporal decay and finer spatial representation improve forecasts after a more flexible background geography is supplied. These comparisons retain the earlier matched results and restricted theoretical conclusions. The withdrawn background-nuisance continuation remains a separate historical study.

The original discussion proposed that a spatial conditional intensity should be examined with common observations, proper compensators and serious temporal benchmarks. Subsequent research explored those directions. This chapter reports the jointly fitted temporal-spatial/background experiment, rather than attempting to concatenate gains from separate historical runs. Its key contribution is a controlled comparison showing that the location improvement persists under a stronger background, while the temporal benefit depends on the spatial representation and comparator.

The Hawkes/ETAS and fractional-triggering literature already cited in the manuscript bibliography supplies the modelling context. No novelty or reproduction claim for those model families is made here.

The authority is the sealed run temporal_spatial_background_20261024T211225Z, source commit 764f7e74576d71a3bd224990aa00491a533c13de. The original sealed manuscript and addenda remain historical source authorities for this unified account. The decision report and complete audited summary provide the full comparison roster.

10.8 Matched design and scoring

The local GeoNet snapshot contains 913 events of magnitude at least 5.0. Training prefixes contain 301, 504 and 659 events; their disjoint test windows contain 203, 155 and 254 targets. Times are days since 1960-01-01 UTC, with horizon 24,394 days. All earlier observed events remain eligible parents. During a test window the forecast history updates with earlier observed test events, while fitted parameters and spatial nuisance estimates remain frozen. This is sequential conditional forecasting, not a single forecast of an entire future catalogue at the training cutoff.

The factorial crosses three temporal forms (fixed-scale ML, common free-scale ML, Omori), two immigrant-background representations (cell and KDE-enriched), and two offspring-location representations (continuous and cell-lift). All share fixed magnitude productivity parameter r = 0.7 and the same training-only conditional-bin magnitude-rate estimate. No new magnitude-productivity search is included.

For time t and projected location z, the conditional intensity has the form

$$\lambda(t,z) = e^{a+dx_{\mathrm{trend}}(t)} b_{\mathrm{bg}}(z) + n \sum_{j:t_j<t} \kappa_j k_\theta(t-t_j) s_\sigma(z \mid z_j), \tag{10.6}$$

$$x_{\mathrm{trend}}(t) = 2t/24394 - 1. \tag{10.7}$$

Here κ_j is the shared, training-calibrated magnitude-productivity weight of parent j. Both spatial densities integrate to one over the fixed nine-cell domain. Consequently the integrated regional rate is the background amplitude plus the normalized temporal offspring contribution. The joint marked log score includes point-event log intensity, the full space-time compensator and the common magnitude density. It decomposes exactly into timing, conditional zone, conditional cell, within-cell area density and magnitude terms. Higher score is better. All location scores use the same projected-area measure in square kilometres; a joint gain is not a percentage improvement in accuracy.

Continuous offspring density is proportional to a fixed training-derived reference density times the radial factor (1 + squared distance / sigma squared) to power -2, normalized separately for each actual parent. The lift comparator uses target-cell-centre weights and spreads each resulting cell mass uniformly inside that cell; it also retains actual parent coordinates. The radial factor is isotropic in the declared equal-area projection, but reference weighting and domain truncation make the full density generally anisotropic and piecewise smooth.

Where the fixed nine-cell reference density changes across a cell boundary, multiplication by that density can give the offspring density a step discontinuity even though the radial factor is smooth. Normalization preserves its unit mass but does not remove those artificial grid-boundary jumps. This limits any interpretation as a smooth physical stress or transport field.

The KDE-enriched background mixes 95% training kernel density with a fixed 5% cell-density floor. Each kernel is separately normalized on the finite domain. Bandwidth selection uses spatial leave-one-out scores, removing exactly the scored row from both density and hierarchical counts. After selection the full-training plug-in density, including self-kernels, is frozen. It estimates observed event geography, not an identified immigrant distribution; in-sample fitting therefore includes nuisance reuse.

Smoothed past seismicity is a forecast construction with an established comparative record (Helmstetter et al., 2006; Werner et al., 2011); it need not be an estimate of an exogenous immigrant field. Hard window declustering (Gardner and Knopoff, 1974) and interaction-

based linkage (Reasenberg, 1985) impose different event assignments. Probability-weighted reconstruction (Zhuang et al., 2002; Marsan and Lengliné, 2008) represents uncertainty in those assignments and accounts for indirect triggering. Thus a better geographic density and recovery of physical immigrant geography are separate questions. The factorial below tests the former; the later benchmark in Section 11.5.2 estimates a training-weighted background before its matched attenuation comparison.

This undeclustered plug-in background uses all training events, including events generated by triggering. It therefore mixes background and clustered seismicity; leave-one-out bandwidth selection does not turn it into an identified immigrant density. Stochastic declustering instead estimates background and triggering contributions jointly or probabilistically (Zhuang et al., 2002). The retained self-kernels can increase in-sample density at training coordinates, a nuisance-fitting limitation rather than test-location inclusion.

Fixed ML uses the pinned normalized 195-mode mixture at one day. Common ML estimates its scale over 0.001-20 days; Omori estimates c over 0.0001-20 days and p over 1.01-2.5. Temporal kernels have unit mass over all positive lags, with exact finite-window exposure rather than window renormalization. Spatial scale sigma is selected once per fold by the fixed-ML/cell-background/continuous baseline from 5, 15, 50, 150, 500 and 1500 km, then shared across all twelve primary procedures. Free parameters are refitted separately within each procedure. These are conditional shared-scale comparisons, not independent spatial optimization for every model. The frozen protocol specifies bounds, starts and selection rules.

10.9 Temporal improvement is positive but conditional

The table below reports continuous-location comparisons. Differences are candidate minus reference, aggregated over the same 612 targets; nats/event is total gain divided by 612. These contrasts are descriptive; no confidence intervals are supplied.

Background and temporal contrast	Joint nats	Nats/event	Timing nats	Joint-positive windows
Cell: common ML - fixed ML	-1.25	-0.00204249	+1.4546	2/3
Cell: Omori - fixed ML	+67.9659	+0.111055	+23.4194	3/3
Cell: Omori - common ML	+69.2159	+0.113098	+21.9648	3/3
KDE-enriched: common ML - fixed ML	+5.61686	+0.00917788	+9.6524	2/3
KDE-enriched: Omori - fixed ML	+15.4742	+0.0252847	+10.3682	3/3
KDE-enriched: Omori - common ML	+9.85737	+0.0161068	+0.715779	1/3

With the KDE-enriched background, Omori's joint gains over fixed ML are +0.687236, +0.845910 and +13.941085 nats; timing gains are +3.042197, +2.293691 and +5.032289. The timing improvement therefore survives the stronger background in each window, although most joint gain occurs in the third.

The comparison against common-scale ML is less consistent: joint differences are -1.091786, -5.559124 and +16.508279 nats. Moreover, under KDE-enriched background and cell-lift geog-

raphy, Omori loses 10.186077 joint nats and 0.970870 timing nats to fixed ML. The evidence favours Omori over fixed ML within the continuous representation, not universal superiority of one kernel.

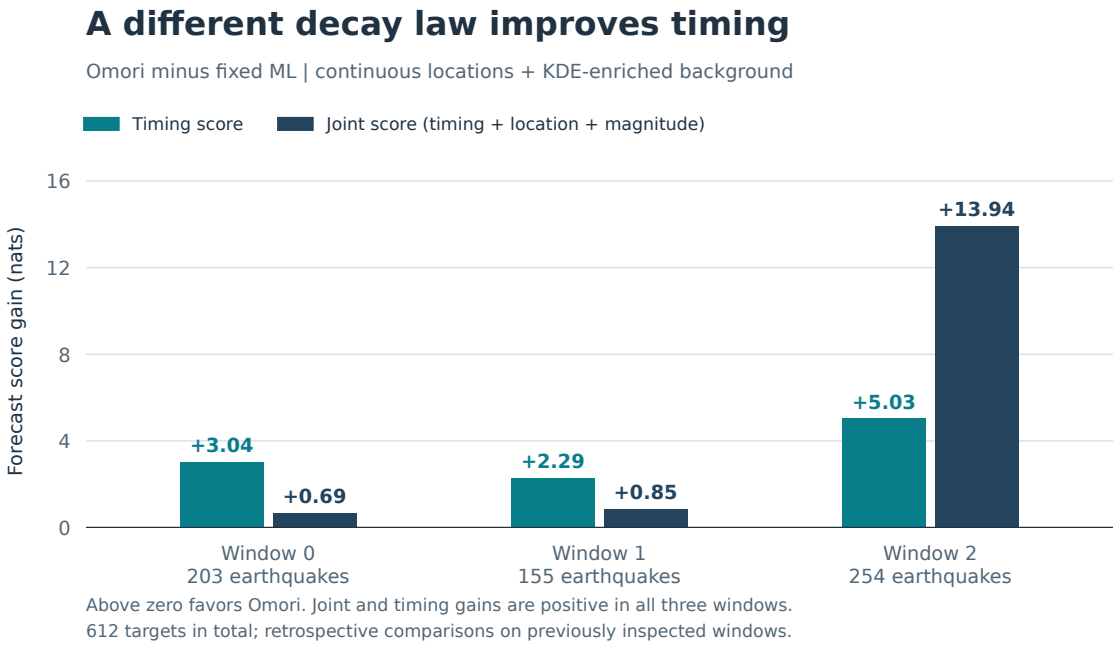


Figure 10.1: Temporal comparisons across the three historical evaluation windows.

10.10 The location improvement survives stronger background geography

Every continuous-minus-lift joint contrast wins all three windows under both backgrounds. The table below separates the spatial refinement from the effect of improving background geography. These contrasts are descriptive; no confidence intervals are supplied.

Temporal family	Continuous - lift, cell background (nats)	Continuous - lift, KDE-enriched background (nats)	KDE continuous–lift joint (nats/event)	KDE continuous–lift timing (nats)
Fixed ML	+1453.29	+1424.12	+2.327	+16.5456
Common ML	+1445.7	+1423.3	+2.32565	+15.1151
Omori	+1507.21	+1449.79	+2.36893	+27.8846

The stronger background reduces the continuous-location advantage by 22.397419-57.420970 nats. It explains some of that advantage, but does not remove it within the declared models.

Location dominates the remaining improvement. For Omori with KDE-enriched background, the +1449.785291 joint nats comprise +27.884617 timing, +14.810114 conditional-zone, +18.815997 conditional-cell and +1388.274563 within-cell area-density nats. Magnitude differences are zero. The timing aggregate is positive despite two losing windows (+35.932611, -2.308009, -5.739984); this spatial result must not be presented as uniform timing success.

Changing background geography itself improves continuous-model joint scores by 85.740055, 92.606919 and 33.248405 nats for fixed ML, common ML and Omori. For Omori that background change loses 1.757274 timing nats. At fixed common parameters, normalized spatial changes leave total regional intensity unchanged; their fitted timing effects arise through refitting.

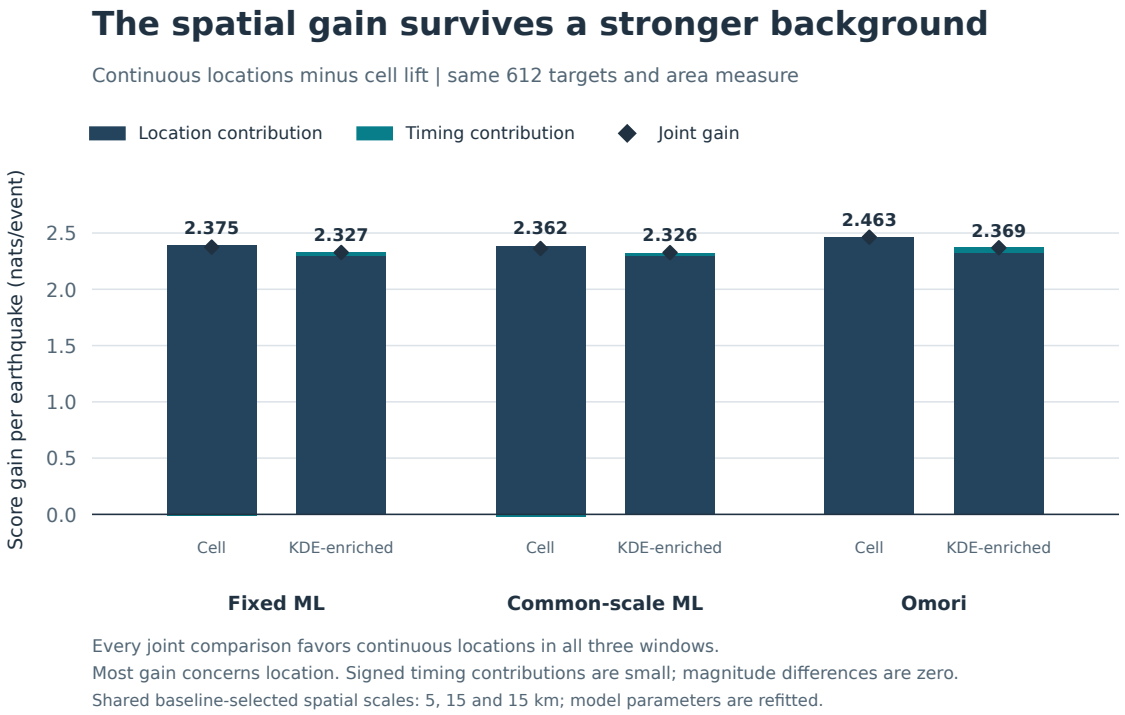


Figure 10.2: Continuous-location improvements under both background representations, with location and timing distinguished.

10.11 Combined procedure and scale sensitivity

Against the current matched fixed-ML/cell-background/lift comparator, Omori with continuous locations gains 1521.255407 joint nats using cell background, or 1554.503812 nats (2.540039 per event) using KDE-enriched background. The latter has +11.530195 timing nats and +1542.973617 net conditional-location nats; magnitude contributes zero. Joint gains are positive in all three windows, but timing changes are -2.961931, +4.108633 and +10.383493. These are direct fitted contrasts, not sums of earlier separate experiments. The previous sealed lift procedure selected its own sigma and is not the present shared-sigma comparator.

Training selected sigma values of 5, 15 and 15 km; KDE bandwidth was 15 km in all three windows. The first sigma lies at the primary grid's lower endpoint. Bounded sensitivity therefore retained separate fixed-ML/continuous fits at 1 and 2 km under both backgrounds. All twelve lost joint score against their primary reference, by 96.988003-308.063524 nats. Some improved timing while losing more location density. These sensitivity fits neither replace the primary models nor constitute an expanded-grid factorial.

The catalogue has 910 distinct coordinate pairs among 913 events, with three repeated pairs. Serialized decimal places do not establish location accuracy; these results identify neither a physical interaction distance nor a universal optimal bandwidth.

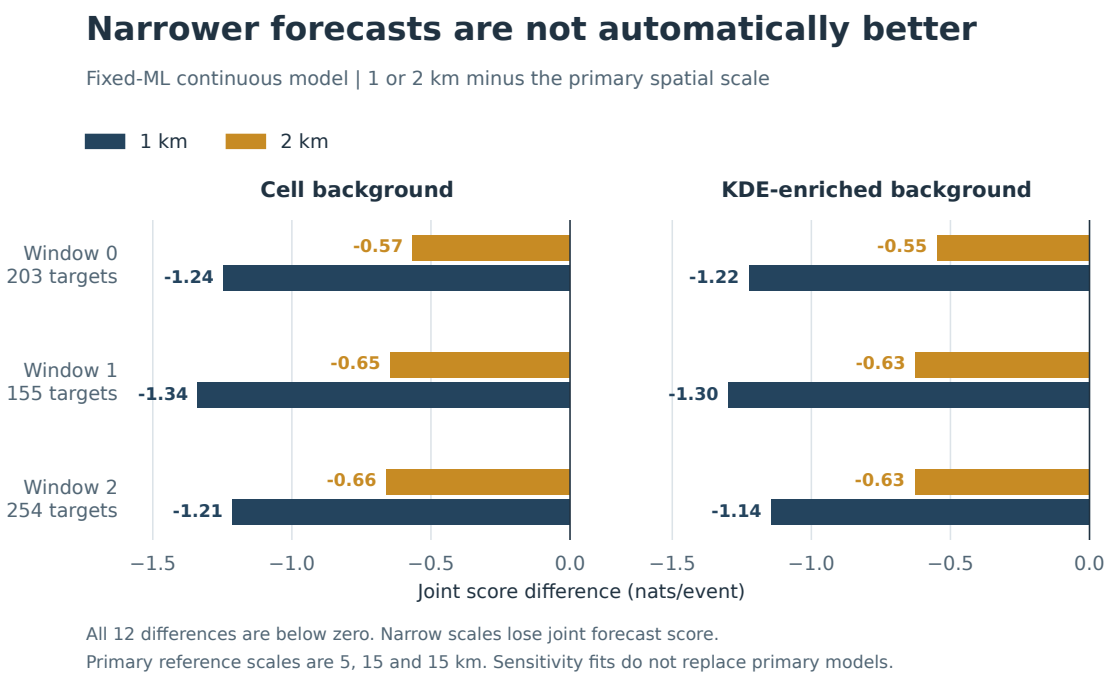


Figure 10.3: Joint-score losses from the declared 1 and 2 km sensitivity procedures.

10.12 Validation, scope and implications

All 63 observed procedures and 504 prescribed starts resolved, retaining 579 candidates and 75 exact floors. Including two fixed-seed engineering histories, totals are 87 procedures, 696 starts and solver returns, 803 candidates and 107 floors. Analytic gradients, finite differences, normalization, exact nesting and quadrature checks preceded observed fitting. An independent computational audit reconciled saved scores to within 3.28e-11 nats, below the unchanged 1e-4 tolerance. Near-optimal forecast spread passed the declared gate. Neither multiple starts nor that gate certifies a global optimum.

The strongest supported conclusion is a positive retrospective forecasting ablation: finer offspring-location representation retains a large location-score advantage under a stronger background, while Omori adds a smaller timing improvement against fixed ML in the continuous model. Background and temporal comparisons also contain losses that prevent a blanket preference claim. Constant-Poisson floors are numerical witnesses, not an optimized no-trigger comparator, so this experiment does not establish that physical triggering is present or required. It does not test inhibition.

These retrospective ablations reuse the same historical windows and overlapping training histories. Their observation-law and conditional-uncertainty interpretation follows Section 6.1. The subsequent continuous local-inhibition comparison is reported in Chapter 13; its matched excitation reference determines the incremental attenuation result. All primary arm-by-window absolute scores are retained in Appendix B.8.5; its small-scale sensitivity table retains every bounded sensitivity fit.

10.13 Reproducible geometry and fitting contract

Let A_c be projected cell area and N_c, N_z the training cell and zone counts. The hierarchical reference obeys

$$\pi_z = N_z/N, \qquad q(c \mid z) = \frac{N_c + 1/2}{N_z + 3/2}, \qquad b_{\mathrm{ref}}(z) = \frac{\pi_{z(c)} q(c \mid z(c))}{A_c} \quad (z \in c). \tag{10.8}$$

Writing $D_h(v) = (1 + \|v\|^2/h^2)^{-2}$, define $F_j(h) = \int_D D_h(v - z_j)\,\mathrm{d}v$. The actual smooth immigrant density and the parent-normalized offspring law are

$$b_{\mathrm{bg}}(z) = 0.05 b_{\mathrm{ref}}(z) + \frac{0.95}{N} \sum_{j=1}^{N} \frac{D_h(z - z_j)}{F_j(h)}, \tag{10.9}$$

$$s_\sigma(z \mid z_j) = \frac{b_{\mathrm{ref}}(z) D_\sigma(z - z_j)}{\int_D b_{\mathrm{ref}}(v) D_\sigma(v - z_j)\,\mathrm{d}v}, \tag{10.10}$$

$$W_{cj} = \frac{\pi_c D_\sigma(v_c - z_j)}{\sum_l \pi_l D_\sigma(v_l - z_j)}, \qquad s_{\mathrm{lift}}(z \mid z_j) = W_{cj}/A_c \quad (z \in c). \tag{10.11}$$

Here v_c is the projected rectangle centre and $\pi_c = \pi_{z(c)} q(c \mid z(c))$. Parents retain actual coordinates in both laws; only the lift's target weights use centres. The same KDE-enriched immigrant density is retained even in a lifted offspring arm; its cell-density floor can retain boundary jumps. Leave-one-out selection removes the scored row's identity from both the kernel mixture and hierarchical counts, retains other observations with duplicate coordinates, and normalizes every retained kernel separately. Exact bandwidth ties select the smaller bandwidth. Full-training plug-in nuisance density, including self contributions, is frozen after selection.

The equal-area coordinates are $x = R \cos\phi_0 (\lambda - \lambda_0)$ and $y = R(\sin\phi - \sin\phi_0)/\cos\phi_0$, with angular arguments in radians. Euclidean radial isotropy is specific to this projection. Multiplication by piecewise reference geography and finite-domain truncation generally destroys isotropy and translation invariance of the full density. Positive rectangle integrals are checked without clipping invalid values. For any parent, finite-window temporal exposure uses $S(L) - S(U)$ with $L = \max(t_{\mathrm{start}} - t_j, 0)$ and $U = \max(t_{\mathrm{end}} - t_j, 0)$; temporal kernels are never renormalized to the evaluation window.

All main fits use the same bounded training-likelihood selection rule. Fixed ML starts span the prescribed amplitude offset, trend and branching schedule. Common-scale ML and Omori initialize shared nuisance coordinates from their same-background/same-geography fixed-ML fit; this is a declared procedure dependency, not held-out selection. The constant-Poisson floor is an admissible numerical witness. Common-scale ML additionally retains the exact fixed-ML floor; Omori and ML are not nested within these finite parameter bounds. Background and continuous/lift families are not declared nested. The complete scalar numerical protocol, prescribed starts, bounds and engineering-control coordinates are retained in Appendix B.8.6.

Chapter 11

External Comparisons, Competitive ETAS Benchmarks, Robustness, and Sequence Concentration

11.1 Study overview

The unweighted inverse-power benchmark (EP) has a higher joint score than the background-weighted procedure (OC) in every primary California and Japan test window, with an episode reversal in the separate California backtest. OC improves California timing against fixed-scale Mittag-Leffler continuous (FC) by 0.119969 nats per target and location against the coarse Omori cell-lift (OL) by 2.140710, over 315 targets. These are different references, and the location gain primarily measures recovery of within-cell density. Japan retains that representation contrast but reverses the OC–FC timing direction in every window. Conditional calibration discrepancies in both OC and EP make these rankings comparisons among conditionally misspecified procedures, rather than validation of a calibrated reference. Section 6.1 states the common scope.

The 2010 El Mayor–Cucapah episode is the EP reversal in the five-episode backtest (Section 11.12). The 2019 Ridgecrest concentration concerns positive OC contrasts against other references and a retrospective active-time union, not an independent replication of that episode ranking. The episode-origin and calendar-year/active-union denominators below remain separate. In particular, concentration in a large sequence does not establish transferable performance across independently sampled sequences.

11.2 Question, status and relation to the preceding work

Established space–time clustering and forecasting work provides the competitive context (Console and Murru, 2001; Helmstetter et al., 2006; Werner et al., 2011). Forecast comparison and consistency testing ask different questions (Rhoades et al., 2011); residual and simulation diagnostics assess features that a relative log score can leave wrong (Bray and Schoenberg, 2013).

The question is how far the specific forecasting improvements reported in Chapter 10 travel when the region, comparator and diagnostic question change. There are four distinct questions: does a matched ablation retain its direction elsewhere; does the complete procedure beat serious alternatives; do its conditional probabilities resemble the recorded history; and where do the realised score differences occur? A positive answer to one does not answer the others.

Chapter 10 found a large location-score advantage for continuous offspring geography over a cell-lift representation in New Zealand, and a smaller timing advantage for Omori decay over fixed-scale Mittag-Leffler decay within the continuous representation. Its 612 historical targets are context for the programme, not an additional denominator for any table below.

The present chapter consolidates subsequent completed studies. Chapter 13 separately reports the local-inhibition extension and its negative observed result. The original manuscript and Addenda I–III remain historical source authorities; the present chapter integrates the completed studies into this unified monograph.

The historical comparisons retain their original fitted states. Later selected-state likelihood replays checked their scores without repeating optimizer searches; the magnitude-scaled benchmark in Section 11.5.2 is a separately identified fitting execution. The repository provenance map distinguishes those evidence types.

11.3 Models, forecast object and score

11.3.1 Arm names and what each comparison changes

The abbreviations are local procedure names, not claims of new model families. All excitation arms share the declared training-only magnitude recipe with productivity exponent $r = 0.7$ and score the magnitude mark once.

Arm	Temporal law	Offspring geography	Role
FC	Fixed-scale Mittag-Leffler	Continuous OC geography	Primary timing reference
FL	Fixed-scale Mittag-Leffler	Cell-lift	Spatial/context reference
MC	Free common-scale Mittag-Leffler	Continuous OC geography	Temporal context
OC	Omori	Continuous OC geography	Frozen target procedure
OL	Omori	Cell-lift	Primary location reference
EP	Omori ETAS adaptation	Unweighted inverse-power radial law	Competitive reference
N0	No offspring excitation	Smooth background	Refitted no-trigger reference
LI	Omori plus immigrant attenuation	Continuous OC geography	Inhibition extension in Chapter 13

The continuous OC offspring density multiplies a radial factor by a fixed training-derived reference density and normalizes over the finite domain separately for each parent. A cell-lift uses target-cell-centre weights and spreads each cell's offspring mass uniformly inside it; actual parent coordinates remain in the calculation. Thus OC-OL is a particular within-domain representation comparison. Both retain smooth immigrant geography. EP instead uses a separately fitted, pure inverse-power offspring kernel, normalized with respect to projected area without OC's reference-density multiplier. OC-EP compares whole procedures with different spatial laws; it is not an isolated temporal-kernel experiment.

The cell-lift and national-routing arms are coarse representation controls. Their losses establish the cost of those discretizations for the declared score, rather than competitive superiority over an equally resolved alternative. The separately fitted EP arm supplies the stronger spatial comparison. Keeping the testing population, forecast object and reference explicit

follows the forecast-evaluation distinction emphasized in CSEP methodology (Schorlemmer et al., 2007; Zechar et al., 2010).

N0 fixes branching at zero and refits its own background time-rate parameters on training data. It is neither an OC fit with its trigger switched off afterwards nor a constant-Poisson numerical floor. LI attenuates only the immigrant component of OC; its full definition and exact zero-inhibition nesting are given in Chapter 13.

11.3.2 Sequential conditional forecasting

At each origin, parameters and spatial nuisance choices use training data only and remain fixed during the test window. Every strictly earlier admissible observed event can enter the subsequent history, including earlier events in that test window. This scores a sequential conditional forecast. It does not supply the whole future catalogue at the origin, and it is not an issue-frozen nonhomogeneous Poisson forecast of the complete future path.

The finite domain, parent eligibility, exposure and projected-area measure are shared within each contrast. For an event intensity $\lambda(t, z)$, its regional ground intensity $R(t)$, full space-time exposure E, and common magnitude-mark score M, define

$$P = \sum_i \log \lambda(t_i, z_i), \qquad Q = \sum_i \log R(t_i), \tag{11.1}$$

$$E = \int_I \int_D \lambda(t, z)\, \mathrm{d}z\, \mathrm{d}t, \qquad C = -E, \tag{11.2}$$

$$S_{\text{time}} = Q + C, \qquad S_{\text{loc}} = P - Q, \tag{11.3}$$

$$S_{\text{joint}} = P + C + M = S_{\text{time}} + S_{\text{loc}} + M. \tag{11.4}$$

Higher is better. In a difference between procedures, the signed exposure contribution is minus the difference in E. Quiet periods therefore contribute even when they contain no targets. Common mark differences cancel; a positive location difference is not a statement about magnitude forecasting. Nats are natural-log score units, not percentage accuracy, correlations, causal effects or an estimated percentage of predictable earthquakes. Pooled nats per target divide the sum of window differences by the sum of target counts, never the unweighted average of window means.

This event-logsum and compensator construction is a logarithmic point-process score (the negative of an entropy-loss convention); its differences express log-score gain for the declared observation law (Daley and Vere-Jones, 2004).

These symbols are the narrative convention throughout Chapters 11–13. The source-field dictionary maps legacy labels without changing their values: LI's raw pooled key E denotes a point-logsum difference, while sequence timing_compensator denotes a signed difference in nonnegative integrated exposure whose score contribution has the opposite sign. Neither raw name overrides the definitions above.

Primary timing means OC-FC timing; primary location means OC-OL conditional location. These rows use different references and cannot be added to manufacture a combined gain. A direct OC-FL joint contrast, where reported, is a separate comparison.

11.4 California: successful matched transfer with uneven contributions

11.4.1 Population and chronological windows

The repaired California source contains 1,272 canonical unique preferred SCEDC/SCSN records meeting reported M $\geq$ 4.0, latitude 32-37 degrees, longitude -121 to -114 degrees and depth -5 to 40 km. Of these, 106 records in 1981-1985 provide warmup parents. Scored training begins in 1986; expanding training prefixes contain 851, 930 and 1,097 targets. The three half-open UTC test windows contain 79, 167 and 69 targets. Their total is 315 unique held-out targets. Reappearance of this denominator in the benchmark, calibration, concentration and inhibition studies reflects reuse of those same targets, not new samples.

11.4.2 Primary results

The California scores below use the original fitted states. Paired calendar-block intervals for the OC–EP joint and location contrasts are reported below using the method in Section 8.4.1.

Window (targets)	OC-FC timing nats	Timing nats/target	OC-OL location nats	Location nats/target
2011-2016 (79)	+0.099084	+0.001254	+91.803255	+1.162067
2016-2021 (167)	+32.722878	+0.195945	+465.829993	+2.789401
2021-2026 (69)	+4.968377	+0.072005	+116.690274	+1.691163
Pooled (315)	+37.790339	+0.119969	+674.323522	+2.140710

Both primary directions are positive in all three windows. Timing is nearly tied in the early window; the middle window supplies about 87% of the total timing difference. The timing pair also has a negative conditional-location difference in the middle window. Positive primary directions therefore do not imply uniformly improved components or a constant effect size.

Of the +674.323522 conditional-location nats, +653.602228 come from within-cell density. This is consistent with the distinction between a continuous density and uniform within-cell offspring mass. It does not establish a similarly large gain in allocation among broad zones or cells, nor general accuracy of individual event locations. Training selected both background bandwidth and inherited offspring scale at 5 km, the lower grid endpoint in every California fold. This is a boundary of the declared grid, not an identified physical interaction distance or proof of coordinate accuracy.

Thus about 96.93% of this location contrast comes from within-cell density. The analogous higher-threshold share is about 95.48%. These large shares are consequences of the particular uniform-within-cell reference; they are not percentages of correctly predicted locations or evidence of a physical interaction distance.

For context, OC-MC gains +21.177884 joint nats (+0.067231 per target), FC-FL gains +671.244340 location nats (+2.130934), and the direct OC-FL comparison gains +747.800021 joint nats (+2.373968). Each wins all three windows. Their reference arms and score components differ; these results are not additive.

The California accepted report supplies source selection, repair chronology and complete comparison components.

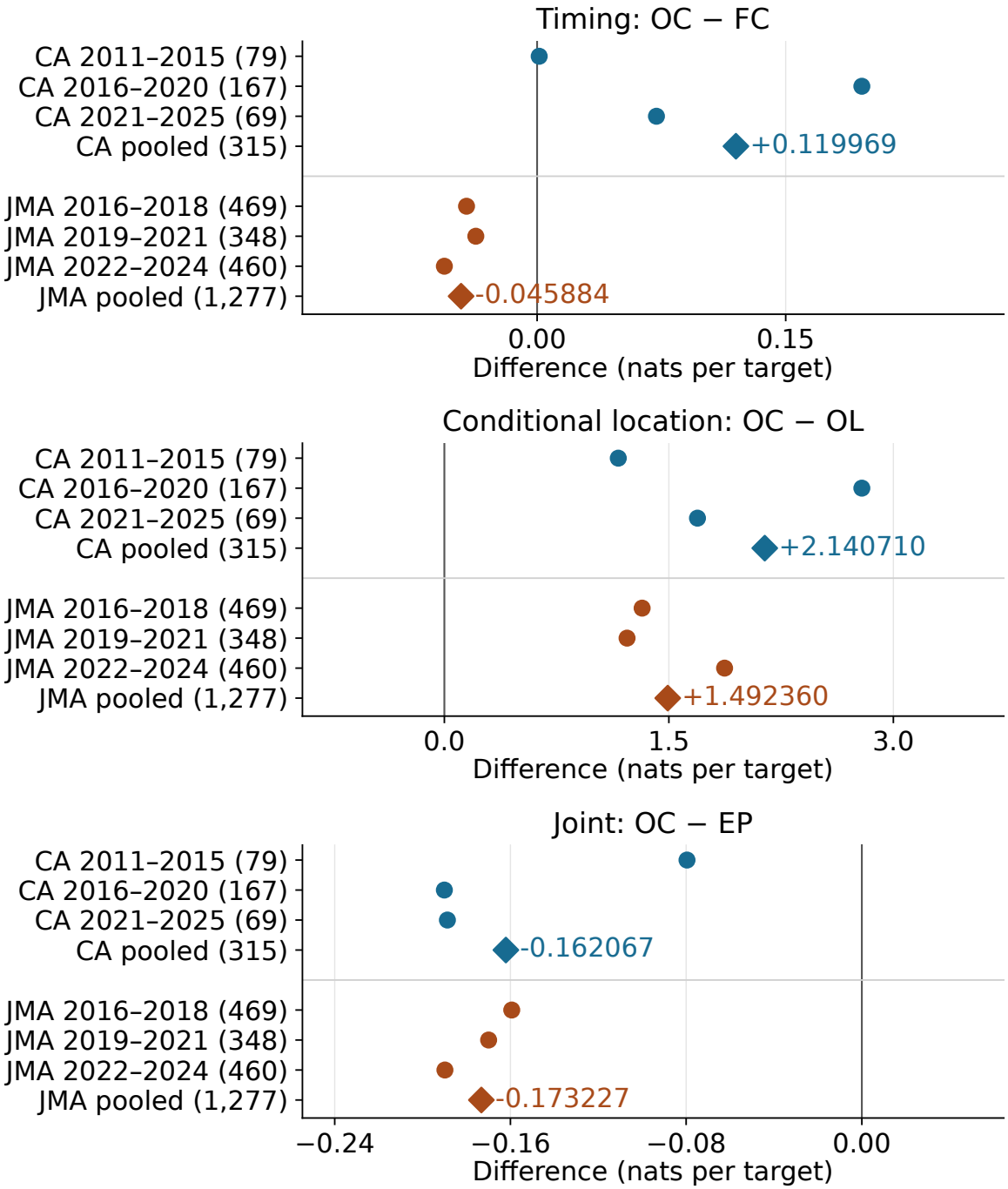


Figure 11.1: Matched timing and location comparisons in California and Japan. Realised proper-score contrasts from the original California M $\geq$ 4 cohort (315 targets) and the JMA reported-primary M $\geq$ 4.5 complete-case cohort (1,277 records). Each panel has its own reference and horizontal scale. Timing compares OC with FC; conditional location compares OC with OL; joint compares OC with EP. Pooled diamonds divide summed window nats by the regional total. California uses UTC years and JMA source-JST years. Different-reference columns and geographies are not added. Related retrospective windows have no event-iid uncertainty intervals.

11.5 Stronger benchmarks change the conclusion

OC exceeds the refitted N0 procedure by +1694.942967 joint nats over 315 targets, but EP exceeds OC in each primary California test window. The same 315 targets, common mark law and frozen OC states are used. EP was independently fitted using training data; it was not selected as a replacement downstream forecast after inspecting these scores. The table gives pooled point-score decompositions; paired calendar-block intervals follow in Table 11.4.

OC minus reference, pooled 315	Joint nats	Timing nats	Location nats	Joint nats/target
N0	+1694.94	+894.846	+800.097	+5.38077
EP	-51.0512	+0.33625	-51.3875	-0.162067

OC-N0 joint, timing and location differences are positive in all three windows. This supplies a serious no-trigger predictive comparison that a numerical Poisson floor could not supply. Its middle window contributes about 77.49% of the joint gain. The comparison is between complete procedures with shared trained geography; it does not establish causal triggering or identify immigrant labels.

OC-EP joint differences are -6.288819, -31.742512 and -13.019873 nats. Conditional location loses in each window. The pooled timing difference is only +0.001067 nats per target, with a small middle-window loss; EP's joint advantage is mainly spatial. Therefore the finding that OC improves on OL cannot be promoted into superiority over the strongest evaluated spatial benchmark.

The decomposition also matters. Against EP, the point-event difference is -54.591677 nats, partly offset by a +3.540472 signed exposure contribution. Against N0, these are +1612.690857 and +82.252111. A point-only comparison would omit quiet-period exposure. The common magnitude law contributes equally to every arm, so these joint-score contrasts measure space–time occurrence forecasting; all mark differences are zero.

The competitive benchmark report is the numeric authority for this section. The LI replay in Chapter 13 is separately sourced and differs at floating-point scale; it does not replace these archived benchmark values.

11.5.1 Calendar-block uncertainty of the spatial comparison

Using the paired block procedure in Section 8.4.1, the California OC–EP comparison separates a weak first-fold contrast from the larger subsequent deficits. The intervals below concern the joint score.

Fold	Contrast (nats)	Paired 95% interval	Nominal sign-flip p
F0	-6.289	$[-21.790, 3.296]$	0.57656
F1	-31.743	$[-77.229, -3.991]$	0.02558
F2	-13.020	$[-22.823, -4.861]$	0.00366
Pooled	-51.051	$[-99.930, -17.264]$	0.00018

Table 11.4: California OC minus EP: 90-day paired calendar-block uncertainty. The pooled bootstrap resamples within each fold.

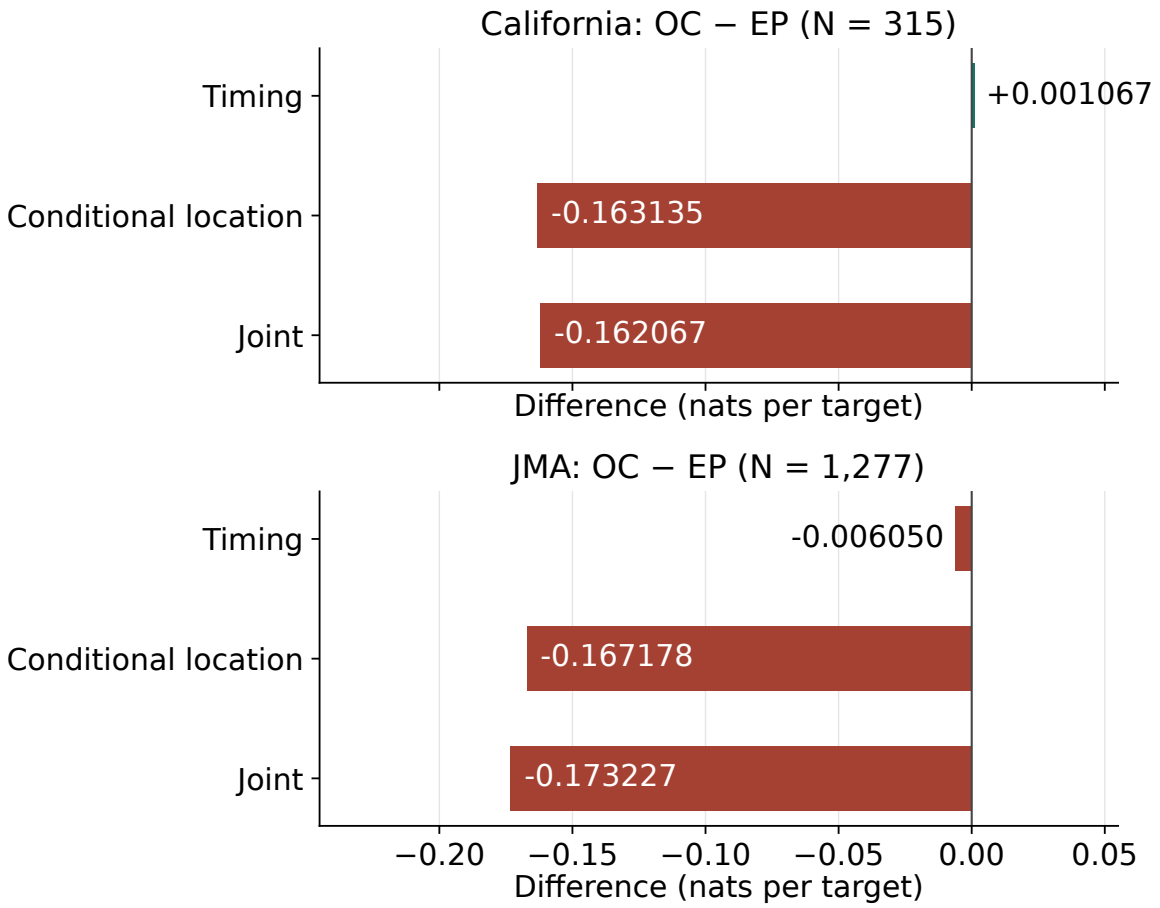


Figure 11.2: Timing and location components of OC relative to EP in California and Japan. Pooled OC-minus-EP timing, conditional-location and joint differences, normalized separately by 315 California targets and 1,277 JMA records. Joint equals timing plus conditional location because mark differences are zero. California has a tiny timing edge; JMA has a timing loss. The larger location deficit accounts for most joint loss in both regions. EP is a separately fitted complete procedure; relative superiority establishes neither EP calibration nor an isolated temporal-kernel effect.

The pooled joint deficit is -51.051 nats; its 180- and 365-day intervals are $[-96.894, -16.999]$ and $[-93.188, -17.257]$, with sign-flip values $p = 0.00130$ and $p = 0.00726$. The location component separately totals -51.387 nats, with a 90-day interval $[-98.567, -18.423]$ and $p = 0.00006$. Thus the roughly 51-nat joint loss is also visible after preserving within-block event dependence.

11.5.2 Magnitude-scaled ETAS and rupture-scaled attenuation

We fit an unweighted California excitation benchmark with a magnitude-scaled spatial kernel

$$s_j(z) = \frac{q-1}{\pi\sigma_j^2}\left(1 + \frac{\|z - z_j\|^2}{\sigma_j^2}\right)^{-q}, \qquad \sigma_j = \sigma_0 e^{\alpha_s(M_j - M_0)}.$$

This density is normalized on the plane; its mass inside the observation rectangle enters the compensator, retaining regional offspring leakage. Both this procedure and archived EP define intensities in $\text{day}^{-1}\,\text{km}^{-2}$ on the same observation domain $[T_0, T_1] \times R$. Both score events against that common measure and integrate their compensators over that domain. EP normalizes each parent's spatial density over R; here the in-region mass $\int_R s_j(z)\,\mathrm{d}z \le 1$ enters the triggering exposure. Allowing offspring outside R therefore changes the fitted forecast but does not invalidate a proper-score comparison of the two recorded-region intensities. It does prevent attributing their contrast solely to magnitude scaling. Training immigrant probabilities weight five successive background KDE updates, following the probability-weighting principle of stochastic declustering (Zhuang et al., 2002). The resulting density, with its inherited 5 km bandwidth and a 5% uniform floor, is shared by the two final fitting arms. Training event densities omit their own KDE component.

Both arms jointly estimate $(a, d, n, c, p, \sigma_0, \alpha_s, q)$. The attenuation arm also estimates $(\eta, \ell_0, \psi, \gamma)$ in

$$\lambda(t, z) = \{\mu(t) b_{\text{bg}}(z) + nH(t, z)\} \exp\left\{-\eta \sum_{t_j < t} \frac{e^{\gamma(M_j - M_0) - (t - t_j)/\psi}}{[1 + \|z - z_j\|^2/\ell_j^2]^2}\right\}, \qquad \ell_j = \ell_0 10^{0.5(M_j - M_0)}.$$

The fixed shadow-radius exponent is a statistical design choice motivated by empirical magnitude–rupture relations (Wells and Coppersmith, 1994). It supplies a magnitude-scaled isotropic comparison without identifying a physical Coulomb stress field. The uninhibited arm is retained as an exact nested candidate. At each final quadrature resolution, the training search includes the fitted null and three prescribed strictly positive attenuation starts, in addition to nuisance warm starts. The attenuation deficit uses a frozen scrambled-Sobol importance proposal during each training optimization; independent larger proposals verify the training ordering and the held-out numerical precision. The reported integration error measures numerical variability across independent scrambles. The saved field `training_power`= 14 denotes Sobol quadrature resolution, 2^{14} integration points, not statistical detection power.

All three baseline fits place σ_0 at the 2 km lower search bound. The combined magnitude-scaled, probability-weighted-background, joint-fitting and spatial-normalization procedure improves the held-out joint score over the earlier EP procedure by 14.58, 30.69, and 24.28 nats in F0, F1, and F2, respectively (69.55 nats in total). The comparison uses identical event and parent histories, fertility weights, and common mark scores. The gain confounds all four procedural changes and rests on a boundary spatial-scale estimate; it does not identify which change caused the improvement. Adding the rupture-scaled attenuation selects $\widehat{\eta} = 0$ in all three folds. The matched scored states are identical, so all three attenuation contrasts are exactly zero by nesting.

Fold	N_{train}	N_{test}	$\widehat{\sigma}_0$ (km)	$\widehat{\alpha}_s$	$\widehat{q}$	$\widehat{\eta}$	Test contrast (nats)
F0	851	79	2.000	0.617	2.440	0.000000	+0.0000
F1	930	167	2.000	0.623	2.470	0.000000	+0.0000
F2	1097	69	2.000	0.721	2.903	0.000000	+0.0000

Table 11.5: Magnitude-scaled total attenuation minus the matched magnitude-scaled ETAS baseline. The common mark term cancels in each contrast. Spatial scales, magnitude exponents and tail shapes shown are from the uninhibited fit. Each zero contrast follows exactly from selecting the nested training boundary $\hat{\eta} = 0$; it is not an interior-power test for physical inhibition.

Component ablation and the spatial-scale boundary

The 69.553-nat EP+ gain is not primarily a normalization effect. The balanced factorial gives Shapley contributions of 29.22 nats for training declustering, 30.03 nats for magnitude scaling, 0.064 nats for full-plane versus rectangle normalization, and 10.241 nats for joint temporal–spatial fitting (Table 11.7). The rectangle-normalized analogue A7 reaches 69.502 nats, so common-space normalization changes the total by 0.050 nats.

The original lower bound of 2.000 km is binding in every accepted fit. Releasing it to 0.050 km selects interior scales of 0.823, 0.839 and 0.821 km under full-plane normalization, with $\widehat{\alpha}_s$ near eight tenths (Table 11.8). The three-fold test gain rises to 77.056 nats when the background is redeclustered under the relaxed bound, and to 78.584 nats when the accepted EP+ weights are held fixed. The fixed-σ_0 profile curves in Table 11.9 confirm an interior training minimum rather than a boundary one; the earlier boundary was therefore a search-box artefact, not a likelihood maximum.

Arm	Norm.	KDE	α_s	F0	F1	F2	Total
A0	rect	unw	fixed	0.0000	0.0000	0.0000	0.0000
A1	rect	unw	fixed	5.0835	4.5498	0.6075	10.2408
A2	plane	unw	fixed	5.1056	4.6368	0.5877	10.3302
A3	rect	decl	fixed	12.6340	4.1432	21.2576	38.0347
A4	plane	decl	fixed	12.5908	4.2517	21.2477	38.0902
A5	rect	unw	free	5.6526	30.4042	2.7849	38.8417
A6	plane	unw	free	5.6457	30.4627	2.7808	38.8892
A7	rect	decl	free	14.6067	30.6127	24.2831	69.5025
A8	plane	decl	free	14.5795	30.6910	24.2824	69.5529

Table 11.6: Component ablation of the magnitude-scaled ETAS procedure, in held-out nats relative to archived EP. Positive values favour the new procedure. A0 is the archived EP score; A1–A8 form a balanced $2 \times 2 \times 2$ factorial over rectangle versus full-plane normalization, unweighted versus five-step training-declustered KDE background, and fixed versus free magnitude scaling, all with joint temporal–spatial fitting except A0. Columns abbreviate normalization (rect/plane), KDE (unw/decl) and magnitude scaling (fixed/free). Each arm uses the same histories, fertility weights and common mark measure.

California attenuation profile and upper bounds

At the S8 moderate scale pair, the training profile places the one-sided ninety-five per cent upper bound on η at 3.72×10^{-5}, 3.75×10^{-5} and 2.93×10^{-5} in F0, F1 and F2 after re-optimizing (a, d, n) at each fixed attenuation. The corresponding held-out contrast at the S8 parameter is -73.76, -10119.53 and -25.75 nats. The fixed-backbone grid in Table 11.10 shows that the upper bound and test contrast depend strongly on the spatial and temporal scales; the S8 pair is not a universal detection threshold. The training bounds lie far below

Component	Shapley (nats)	Sequential (nats)
Joint fitting (A0→A1)	—	10.241
Full-plane normalization	0.064	0.089
Training declustering	29.223	27.760
Magnitude scaling	30.026	31.463
Total (A8−A0)	69.553	69.553

Table 11.7: Additive decomposition of the 69.553-nat A8−A0 gain. Shapley values average every order of the three procedural factors and sum exactly with the joint-fitting increment; the sequential column follows A0→A1→A2→A4→A8.

Fold/weights	$\widehat{\sigma}_0$ (km)	$\widehat{\alpha}_s$	$\widehat{q}$	Test gain (nats)
F0	0.823	0.770	1.689	14.046
F1	0.839	0.773	1.710	38.365
F2	0.821	0.826	1.794	24.645
F0 (frozen KDE)	0.828	0.770	1.694	14.861
F1 (frozen KDE)	0.845	0.774	1.717	38.705
F2 (frozen KDE)	0.827	0.827	1.802	25.018

Table 11.8: Relaxed lower bound $\sigma_0 \geq 0.05$ km, full-plane normalization. The first three rows rerun five-step training declustering under the relaxed bound; the last three hold the accepted EP+ background weights fixed. Every fit remains training-only and every test score uses the common mark measure.

Fold	0.05	0.10	0.20	0.30	0.50	0.75	1.00	1.50	2.00	3.00	5.00	10.00
F0	8326.09	8267.59	8210.51	8181.35	8154.44	8144.84	8145.97	8160.09	8179.06	8217.18	8297.16	8457.99
F1	9403.82	9342.01	9281.27	9250.01	9220.89	9210.08	9210.73	9224.61	9243.62	9282.05	9365.19	9538.05
F2	10578.53	10502.93	10427.43	10388.37	10352.46	10339.83	10341.31	10358.93	10381.28	10432.94	10544.53	10765.56

Table 11.9: Training negative log-likelihood at fixed σ_0 (km), with $(a, d, n, c, p, \alpha_s, q)$ re-optimized at each point, full-plane normalization and accepted EP+ background weights. Lower values are better.

the S8 generating value, so under the EP+ kernel and the California training windows the zero boundary is strongly selected rather than a flat-direction artefact. The large negative held-out contrasts at that value are model-specific consequences of the same amplitude, not estimates of a physical stress-shadow strength.

Fold	ℓ/ψ (km, d)	Slope at zero	η_{upper}	$\Delta\ell_{\text{test}}(\eta_{\text{S8}})$
F0	10/2	-11450.1244	4.93×10^{-5}	-66.7915
F0	15/5	-15914.8956	3.63×10^{-5}	-74.6861
F0	20/5	-16690.5223	3.48×10^{-5}	-75.4810
F0	50/30	-24432.0608	2.17×10^{-5}	-91.5085
F1	10/2	-11745.3094	4.91×10^{-5}	-8421.9884
F1	15/5	-16299.7610	3.62×10^{-5}	-10126.6168
F1	20/5	-17086.5338	3.46×10^{-5}	-10476.3676
F1	50/30	-25195.7265	2.16×10^{-5}	-13728.4733
F2	10/2	-15174.0559	3.59×10^{-5}	-23.1668
F2	15/5	-20120.4700	2.71×10^{-5}	-28.4518
F2	20/5	-20995.3535	2.59×10^{-5}	-29.0287
F2	50/30	-29096.6446	1.68×10^{-5}	-43.7236

Table 11.10: California EP+ attenuation slices at the uninhibited training MLE. The slope is the one-sided training score derivative at zero; η_{upper} is where the training profile falls by the one-sided threshold; the last column is the held-out contrast at the S8 moderate parameter. Positive last-column values favour attenuation.

For this combined excitation improvement, the paired 90-day block calculation in Section 8.4.1 gives a 95% interval of $[32.239, 123.719]$ nats (nominal sign-flip $p = 0.00002$). The 180- and 365-day intervals are $[32.153, 120.312]$ and $[26.589, 122.164]$ nats, with nominal sign-flip values $p = 0.00010$ and $p = 0.00452$, respectively. The small reference probability and wide interval are compatible summaries of directional extremeness and score-magnitude variability. Adjacent blocks are not independent sequence replications, and independent sign reversal does not establish exchangeability under clustered, history-dependent seismicity. Read the multi-scale intervals and fold directions first; these values do not supply calibrated long-memory significance or fitted-parameter uncertainty.

11.6 Conditional adequacy is a separate question

The frozen OC conditional simulation and its complete coordinate results are developed in Chapter 12. Relative scores here do not supply an adequacy result for either OC or EP.

11.7 Robustness and concentration answer narrower questions

11.7.1 A higher threshold retains directions under a changed observation law

The base annual accounting retains every event's original history and all exposure, including years with zero targets. It refits no base parameter. A separately admitted $M \geq 4.5$ sensitivity rebuilds the population, eligible histories, training estimates, mark law and fitted procedures. Its 391-record population includes 33 warmup parents; training counts are 271, 287 and 340, and test counts 16, 53 and 18, totaling 87. Those 87 are a higher-threshold subset of the California history, not 87 additional independent records to add to 315.

The sensitivity applies the threshold to the provider's exact preferred-magnitude token before converting magnitude excess to binary64. Its relative mark-bin cutoff remains 0.5, corresponding to M5.0. Domain, depth/type/time rules, strict earlier-parent eligibility and $r = 0.7$ are retained. Beta, hierarchical reference masses, KDE bandwidth and model parameters are re-estimated from each training prefix. Six fixed-ML continuous spatial selectors, a lifted prerequisite, common-scale ML and the two Omori procedures remain in the roster; the selected fixed-ML arm is an alias of its winning selector, not an extra fit. The first fold served as the resource/correctness pilot, and its accepted files were reused without score-based continuation or replacement. The 87 targets fall below the inherited descriptive reference of 100 in total; 16 and 18 also fall below 20 per window. Those pragmatic references are not power guarantees.

The higher-threshold scores below retain the original fitted states and have likewise been checked by the later selected-state CPU replay and table reconciliation. These contrasts are descriptive; no confidence intervals are supplied.

Primary comparison and component	M ≥ 4.0 nats/target (315)	M ≥ 4.5 nats/target (87)	M ≥ 4.5 total nats	Positive windows
OC-FC timing	+0.119969	+0.152292	+13.2494	3/3
OC-OL location	+2.14071	+2.12387	+184.776	3/3
OC-MC joint context	+0.0672314	+0.0775184	+6.7441	2/3

The first higher-threshold OC-MC joint difference is -0.626348 nats, and its OC-FC timing gain is only +0.077968 total nats. The two smaller test windows contain fewer than 20 targets. The threshold columns describe distinct complete procedures and target/parent laws; their difference is not a paired performance improvement or a threshold selection result. EP was not refitted or evaluated in this sensitivity, and no new calibration was performed.

Against OL, the higher-threshold location decomposition is +5.881781 zone, +2.472120 conditional-cell and +176.422568 within-cell nats. Its OC–FC timing decomposition is +14.762895 event-ground nats minus 1.513472 integrated-exposure nats; OC–MC joint gain comprises +5.992897 timing and +0.751200 location nats. Each decomposition retains its own reference.

This threshold sensitivity changes recorded-data eligibility, rather than adding an observation-error likelihood. Completeness and magnitude-scale uncertainty remain distinct issues (Woessner and Wiemer, 2005; Mignan, 2012); the common observation-law scope in Section 6.1 applies.

11.7.2 Annual gains are concentrated

In the base analysis, calendar 2019 has 116 targets and contributes +29.280597 OC-FC timing nats, +392.773429 OC-OL location nats and +7.979400 OC-MC joint nats. At M ≥ 4.5, 37 targets in that year contribute +10.977080, +112.007721 and +5.288450 respectively. These are separate contrast/component columns. Negative annual contributions remain in the accepted tables.

The base 2019 signed shares are 77.48%, 58.247% and 37.678% of their respective pooled contrasts, including negative years in each denominator. At the higher threshold, ten of fifteen annual timing contributions are negative. Zero-target years 2015 and 2017 retain their

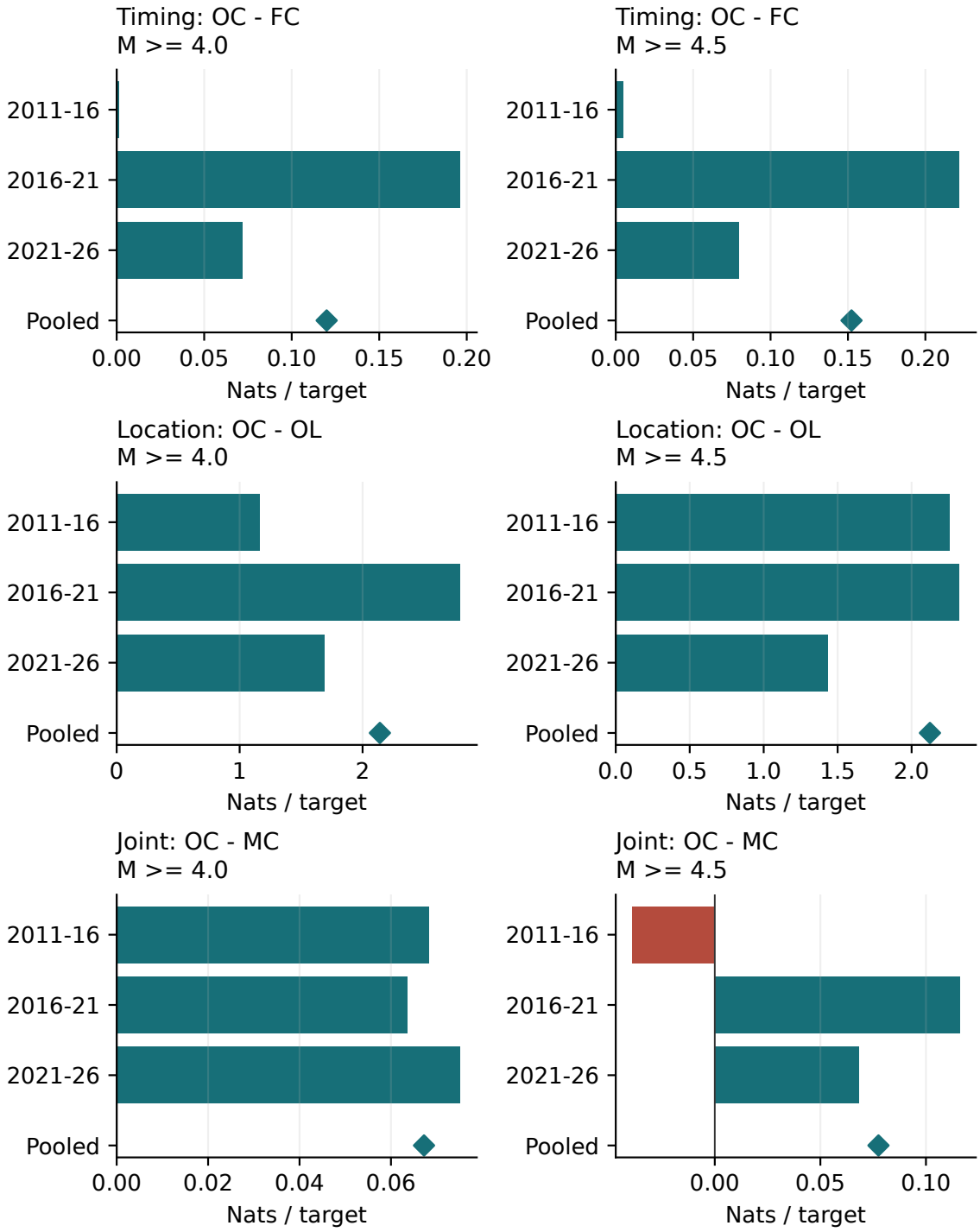


Figure 11.3: Fixed-window and event-weighted pooled primary score contrasts per target under M $\geq$ 4.0 and M $\geq$ 4.5. Bars retain the three windows and pooled diamonds divide summed nats by 315 or 87 targets. Each component uses its named reference; the negative early OC–MC joint contrast is retained. Cross-threshold differences are descriptive, not paired performance gains.

exposure terms; their per-target means are undefined, not zero. Appendix B.9.1 retains every annual row and both thresholds.

Annual partitions preserve subsequent parent history. They do not estimate a counterfactual catalogue without a sequence. Their purpose is to locate realised score contributions, not to remove inconvenient years or create independent replications. The robustness decision report records the full design and absent sensitivity laws.

11.7.3 One eligible active episode, with the complete complement retained

The finer sequence partition was chosen after annual inspection, including 2019. The frozen metadata rule marks regional time active when an admissible $M \geq 6$ anchor at u satisfies $u < t \leq u + 30$ days. Overlapping anchor windows merge globally before fold clipping; eligible prehistory is retained. This is a regional time partition, not a causal or spatial aftershock classification. The initial anchor belongs to the complement unless an earlier anchor already activates it.

Only one eligible episode intersects the scored test windows: a Ridgecrest-associated union lasting 31.406991204 days with 110 targets. The other 205 targets and 5,447.593008796 exposure days remain the complete complement. Thirteen anchors and ten global episodes occur in the admitted chronology, but nine episodes lie in pre-test history. They are not nine additional scored replications.

The active period contributes +30.058155064 of the +37.790339259 pooled OC–FC timing nats, approximately 79.54%. With only one scored episode, the archive cannot estimate reproducibility of that concentration across sequences. Integer UTC membership and global interval merging are fixed before fold clipping. One of the two test anchor targets is active and the other is in the complement; event-only anchor scores are not part of the public aggregate. The active and complement totals below are descriptive score partitions, without confidence intervals.

Pooled category	Targets	OC-FC timing nats	OC-OL location nats	OC-MC joint nats
Active period	110	+30.0582	+375.945	+9.87815
Complete complement	205	+7.73218	+298.379	+11.2997
Full original pool	315	+37.7903	+674.324	+21.1779

The six-target calendar-2019 complement is the non-active portion of the 116-target UTC calendar year 2019 within primary Fold 1, distinct from the 127-target causal-origin block and its 16-target complement in Section 11.12. This calendar-year complement has -0.777558 timing nats and -1.898745 OC-MC joint nats, although its location contribution is positive. The annual timing and MC-context gains therefore arise in the active period. Across the full fifteen test years, all three contextual totals remain positive in the complement; concentration does not mean every contextual gain is confined to that episode.

The stronger reference still wins both partitions. OC-EP joint score is -18.372230 nats in the active period and -32.678975 in the complete complement. Its active timing difference is -0.806683 and location difference -17.565547. Complement timing is +1.142932, outweighed by -33.821907 location nats. All original parents and full-domain exposure remain in every score; event-only scoring would answer a different question. The sequence audit report retains empty active rows, exact exposure accounting and the full partition.

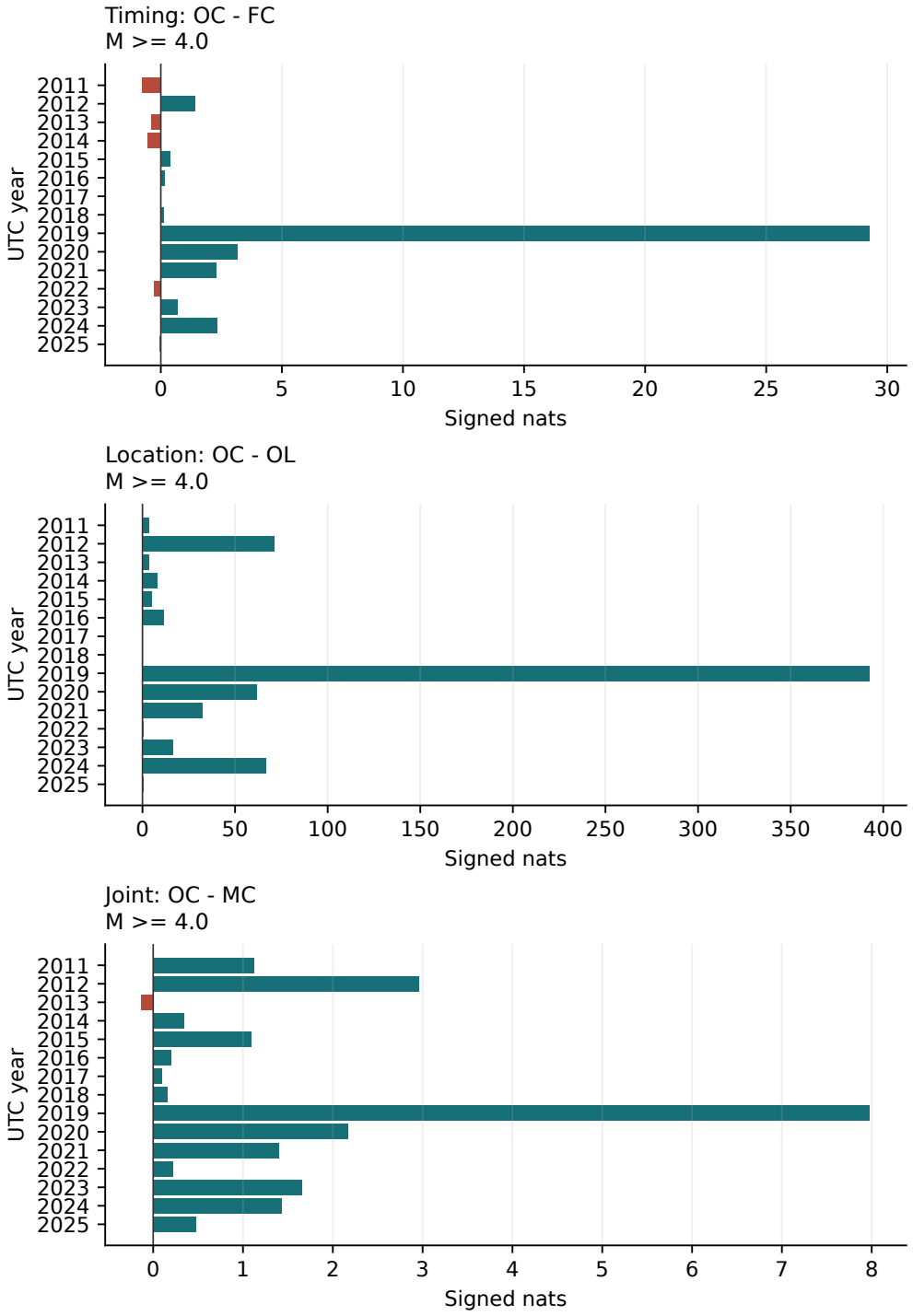


Figure 11.4: Annual signed score contributions for all fifteen California M $\geq$ 4.0 test years under the three distinct named comparisons. The panels use their own references; negative years remain. These are contributions to realised scores, not independent annual replications.

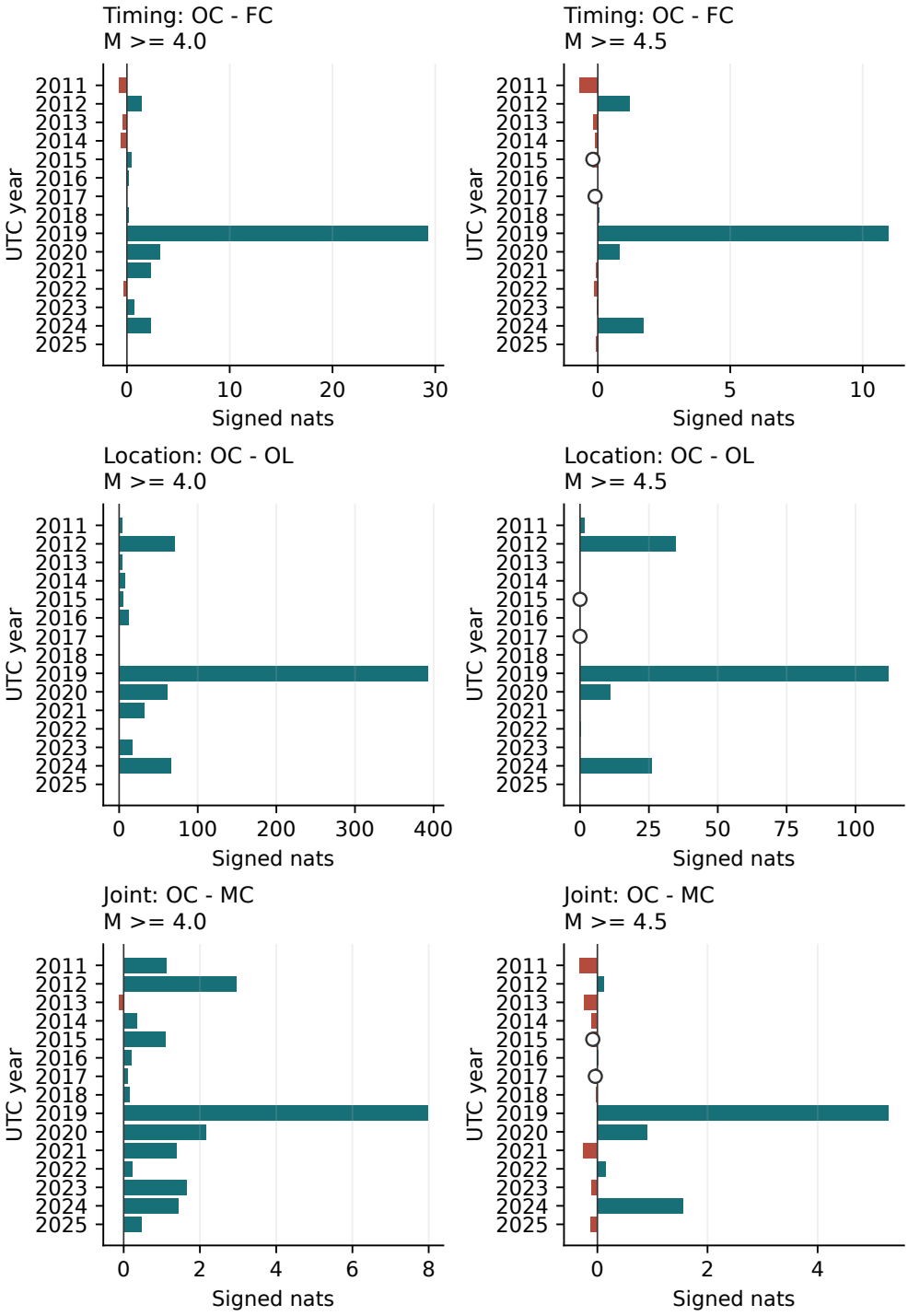


Figure 11.5: Annual signed score contributions under M $\geq$ 4.0 and M $\geq$ 4.5. All fifteen years remain; hollow circles identify zero-target years whose exposure contributions are retained. Component panels have distinct references. The nested populations and refitted procedures do not define a paired threshold gain.

The complete partition tables in Appendix B.9.3 retain empty active periods, full-domain exposures and absolute arm scores. Calendar-2019 and pooled rows repeat portions of the original folds for reporting and must not be added to them. The sequence partition changes neither the original fifteen selected states nor their eligible parent histories.

11.8 Japan: spatial direction transfers, combined descriptive transfer rule fails

11.8.1 The reported-record target

The JMA analysis processes 19 local annual archives for source-JST years 2006-2024. Its 4,019,138 source records yield 4,602 usable study records. The target requires agency J, natural subsidiary 1, determination K/k/A, allowed primary magnitude types J/D/V/d/v, observed numeric primary reported $M \geq 4.5$, depth 0-60 km and the fixed projected region. Primary and secondary magnitudes remain separate. This is an observed primary complete-case reported-record cohort, not homogeneous Mw, inferred latent magnitudes or proof of physical completeness.

Inspection identified 41,278 initially unresolved target observations: 41,277 with missing/undetermined primary information alone and one with an additional depth issue. An explicit, inspection-informed restriction requires numeric observed primary magnitude and an allowed primary type. It was documented and independently reviewed before fitting. Those records receive no assigned latent magnitude and are not asserted to be physically small. No complete target row was added, removed or modified by that clarification. One auxiliary depth-ambiguous subthreshold record remains unusable for diagnostics and has zero target, parent or beta-observation effect.

The 496 records in 2006-2010 supply warmup only. The following training prefixes and disjoint test windows use exact source-JST boundaries.

Origin	Scored training years	Training records	Test years	Test records
2016	2011-2015	2,829	2016-2018	469
2019	2011-2018	3,298	2019-2021	348
2022	2011-2021	3,646	2022-2024	460

The total scored denominator is 1,277. These are revised local records; catalogue inspection and observation adjudication preclude a pristine as-issued holdout claim. The JMA processing report gives the source restrictions and exact accounting.

11.8.2 Descriptive transfer rule and competitive context

The combined descriptive transfer rule requires both primary contrasts to exceed +0.0006 pooled total nats and each to be positive in at least two of three folds after audit pass. That threshold allows the declared numerical score tolerance; it is not a statistical significance boundary. The EP joint comparison is separate competitive context.

The 0.0006 threshold allows 0.0001 nat for each of two arm scores across three folds. It is a numerical comparison rule, not evidence that the three windows or their events are independent tests.

The JMA scores below retain the original fitted states and have been recomputed using the recovered cohort and reconstructed numerical features. These are descriptive contrasts, without confidence intervals.

Comparison and component	Total nats	Nats/record (1,277)	Direction in windows	Interpretation
OC-FC timing	-58.5935	-0.0458837	Negative 3/3	Timing direction reverses
OC-OL location	+1905.74	+1.49236	Positive 3/3	Spatial direction transfers
OC-EP joint	-221.211	-0.173227	Negative 3/3	Strong reference still wins
OC-MC joint context	-49.1946	-0.0385236	Negative 3/3	No universal Omori preference
FC-FL location context	+1847.64	+1.44686	Positive 3/3	Spatial result also under FC

The combined descriptive transfer rule is **not met**. OC-FC timing losses are approximately -19.9765, -12.8718 and -25.7452 nats. OC-OL location gains are approximately +619.711, +424.797 and +861.235. A large location total cannot compensate for a timing requirement with the wrong sign, and primary contrasts with different references must not be added.

EP exceeds OC jointly in all three windows. Conditional location accounts for 213.485907 of the 221.211492-nat deficit. OC also exceeds N0 on pooled joint score, but the strong-reference failure remains. These findings support a useful continuous-versus-lifted representation contrast while qualifying temporal transfer and overall competitive performance.

All background bandwidth selectors choose the lower 5 km endpoint, while FC spatial-scale selectors choose the interior 15 km value. The background trend slope reaches its lower bound for 7, 6 and 2 of seven models in the three folds. Results apply to the frozen grids and bounds; these choices do not identify physical length scales or measurement precision. The JMA forecast report retains full components, annual accounting and the combined descriptive transfer rule.

Appendix B.9.4 retains all seven pooled contrasts and their three window differences, including OC–FL and OC–N0. The nine annual source-JST blocks retain the same counts and all canonical score/exposure components.

11.9 Consolidated implications and limits

The completed evidence separates five conclusions. First, continuous offspring geography repeatedly improves conditional location relative to the declared cell-lift procedure, including California's higher-threshold sensitivity and the JMA reported-record cohort. Second, Omori's timing advantage over fixed-scale ML is conditional: positive in the original New Zealand/California comparisons, but negative in every JMA window. Third, excitation improves relative scores over a serious refitted no-trigger baseline, while EP beats OC on joint score in the primary California and Japan folds, mainly through location. Fourth, OC's conditional diagnostics disclose material discrepancies, so relative gains are not adequate-probability certification. Fifth, California gains are uneven and concentrated, with only one eligible scored sequence period and substantial full-complement contributions.

The historical archives have been inspected, processed and revised; later studies adaptively reuse outcomes and related chronological windows. Frozen training/test calculations within

a study do not erase that programme-level reuse. Overlapping training prefixes, earthquake dependence and small numbers of windows preclude event-iid confidence intervals and broad independent-replication claims. No Wilks inference, causal triggering claim, recovered immigrant law, physical stress range or absolute calibration result follows. Magnitude-type mixtures, reported-coordinate observation laws, finite-domain parent omissions, plug-in background estimates and grid/boundary choices remain material.

11.10 Numerical custody and review scope

Numerical agreement supports the reported calculations; it does not independently replicate earthquake science. The California source repair accounted for 30 procedures and 240 starts. The separate benchmark audited all six new fit procedures and reconciled scores within 8.19e-11 nats. The higher-threshold sensitivity accounted for 30 procedures and 240 starts, reconciling within 1.13e-11 nats. The sequence audit retained all 315 targets and passed 7,128 component checks with maximum discrepancy 4.55e-13 nats. These are different validation rosters and are not independent research samples to add together.

JMA fitting resolved 36 procedures, 288 starts, 42 floors and 330 slots. Maximum independently reconstructed score disagreement was 1.82e-10 nats against a 1e-4-nat tolerance; maximum stored near-optimal spread was 3.75e-5 nats per record against a strict 0.01 gate. Multistart completion and near-optimal-set stability are not global optimization certificates or uncertainty intervals.

The public reproduction bundle now reconstructs the California and Japan cohorts from the original provider records, including source-row identities, exclusion accounting, exact clocks and chronological partitions. The Japanese bundle also supplies recovered fitted states and the explicit projected-region predicate. The current evidence index retains the source pins, unchanged display scalars, source-specific field dictionary and claim bindings. The readable scientific protocol subset supplies the LI law, clocks and fitting contract used in Chapter 13. The review guide separately identifies the executed cohort reconstructions and the completed selected-state score replays. Neither operation reruns the original optimizer searches. Rebuildable event caches are outputs of the numerical pipeline, rather than required private inputs.

The detailed numerical record is organized once in Appendix B.9. It contains the annual, higher-threshold, sequence-partition and JMA backup tables, with their source-specific denominators and exposure signs; the standalone decision-report narratives are not repeated here.

11.11 Completed four-arm spatial comparison

The comparison contains three California folds with 315 targets and three New Zealand folds with 612 targets. Four procedures are evaluated in each fold: arm 1 uses the reference-density-weighted rational offspring kernel and inherited KDE background; arm 2 removes reference-density weighting; arm 3 adds identity-based leave-one-out background fitting; arm 4 uses the unweighted EP offspring shape with the inherited background procedure. The temporal and mark parameters remain training-selected and fixed. Background level, trend and reproduction coefficient are refitted under the declared arm-specific normalization. California uses the archived training-selected EP spatial scale and shape. The New Zealand OC archives did not contain EP spatial parameters, so a separate training-only prerequisite estimated these missing nuisance parameters before the four-arm comparison. Those values were then frozen:

each final arm optimizes only background level, trend and reproduction coefficient. This prerequisite is an explicit addition to the original execution specification, not a claim that the entire preparation fitted only three parameters.

Table 11.15: Held-out joint log-score contrasts in nats. Positive values favour the first named arm. Arm labels follow the text. These contrasts are descriptive; no confidence intervals are supplied.

Region	Fold	Targets	$2-1$	$3-2$	$4-2$	$4-1$
CA	0	79	-0.2962	1.4515	6.3629	6.0668
CA	1	167	1.0722	3.1070	29.9884	31.0606
CA	2	69	0.2776	0.7288	12.6035	12.8811
NZ	0	203	0.3421	2.1881	17.9074	18.2495
NZ	1	155	33.2980	-0.3822	-0.0583	33.2397
NZ	2	254	-1.2518	-3.2235	0.0346	-1.2172

Table 11.16: Regional totals for EP minus the OC baseline, in nats. The common mark score cancels. These descriptive totals have no confidence intervals.

Region	Timing	Location	Joint
CA	-0.8205	50.8289	50.0085
NZ	2.1452	48.1269	50.2721

The contrasts are descriptive comparisons of fitted procedures, not additive causal contributions. In particular, arm 4 minus arm 3 changes the offspring model and the background fitting procedure; it is not a pure tail effect. Arm 4 minus arm 2 is the direct EP versus unweighted-rational comparison with the same background procedure. The identity $(4-1) = (2-1) + (3-2) + (4-3)$ is exact bookkeeping, whereas $(2-1) + (3-2) + (4-2)$ is not the total. Refitting the shared nuisance coefficients further prevents a mechanical causal attribution to a single feature.

All finite retained-state and near-optimal forecast checks are supplied in `reproduction/regional/original_completion_20261101.zip`, with planned denominators retained. The closure verification checks the exact frozen spatial parameters, common temporal and mark parameters, three-coordinate final fits, and training-only winner selection across all six folds. It verifies original and portable case byte identities separately and requires exact parsed-content equality; serialization differences are not treated as changed scientific inputs. This is retrospective verification of the saved contract, not a new fit or a fresh validation cohort. The acceptance plan combines a predeclared independent raw geometry/score sample for California fold zero with saved custody, start coverage, selection and score-decomposition checks across all folds. It does not claim independent raw replay of every fit or New Zealand shape prerequisite. New Zealand results retain the archive-reuse disclosure; they are not a fresh untouched holdout.

11.12 Causal-window California backtests and JMA trend ablation

The California and JMA values in this section retain the original fitted states and audit records. The later replay recomputes the declared selected-state likelihoods and reconciles

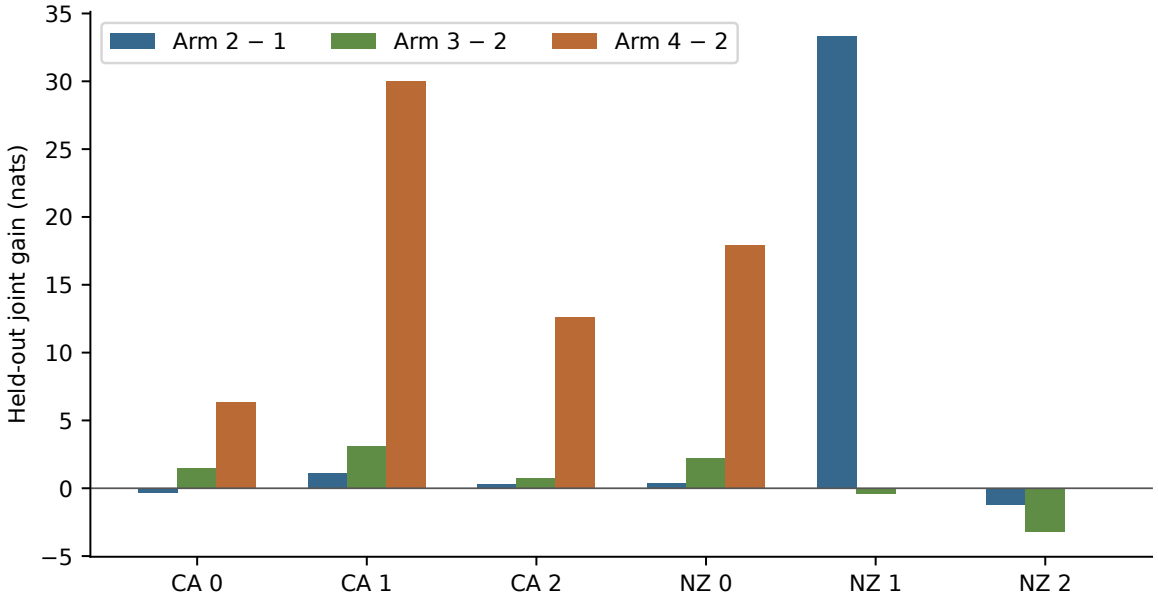


Figure 11.6: Saved held-out joint contrasts across all six folds. Each fold has its own refitted nuisance coefficients; these differences are not a causal mechanism decomposition. The original saved custody checks and independent sampled raw validation passed.

the full, active and complement comparisons. California episode geometry uses the archived projected coordinates, with the live-platform projection comparison retained separately. The original fits were not rerun, and their original audit coverage remains distinct.

The completed California map contains five eligible episodes, in 1994, 1999, 2004, 2010 and 2019. The two 1992 episodes are excluded by the predeclared training-count or zone-adequacy rule, not silently replaced by a more favourable sequence. Training expands from 1986 up to, but not including, each episode origin; the parent history retains the 1981 warmup. Each scored block ends at the earlier of the next eligible origin, one year after its origin, or the coverage cutoff. Parameters, spatial scale and nuisance inputs are selected only on the preceding training prefix.

The $M \geq 6$ episode origins were identified retrospectively from the catalogue. The roster is retrospective and exploratory. Overlapping training histories and a common archive mean that the five episodes are not independent replications. The active partition is the union of declared episode time intervals, and its complement fills the rest of the scored block; it does not causally label individual earthquakes as aftershocks.

Table 11.17: Held-out joint scores (nats) on the five eligible California blocks. Higher is better. These descriptive scores have no confidence intervals.

Year	Targets	FC	MC	OC	EP
1994	64	-527.0462	-525.5013	-526.7452	-521.5603
1999	75	-551.4448	-545.6841	-540.1365	-521.8412
2004	30	-349.2570	-347.5086	-347.8333	-341.1560
2010	125	-1044.6385	-1055.3761	-1050.2614	-1059.1042
2019	127	-696.3825	-679.3514	-670.2133	-647.9267

Table 11.18: OC minus FC joint contrasts, partitioned without changing fitted parameters. Full equals active plus complement, subject to the saved numerical tolerance. These contrasts are descriptive; no confidence intervals are supplied.

Year	Active N	Complement N	Full	Active	Complement
1994	49	15	0.3010	-0.0672	0.3682
1999	61	14	11.3083	11.3970	-0.0887
2004	7	23	1.4237	-0.1897	1.6134
2010	75	50	-5.6229	-6.0147	0.3918
2019	111	16	26.1693	26.3067	-0.1374

EP achieves the highest held-out joint score in the 1994, 1999, 2004 and 2019 blocks, but loses in the 2010 block. That block contains 125 targets: 75 active and 50 in the complement.

EP's joint score of -1059.1042 is below OC (-1050.2614), FC (-1044.6385) and MC (-1055.3761). OC also trails FC: its joint contrast is -5.6229 nats, comprising -6.0147 active nats and 0.3918 complement nats. EP's joint advantage in the three primary chronological folds is therefore not uniform across every retrospective episode block.

The 2019 episode block contains 127 targets (111 active and 16 in the complement). It starts at the $M \geq 6$ episode origin and runs for one year, so its denominator differs from the UTC calendar-2019 slice in Section 11.7.3.

For JMA, the free-trend states are the preserved training-selected parameters, rescored without gratuitous refitting. The stationary alternative fixes $d = 0$ and refits the prescribed starts and floors under the same training nuisance inputs. Both trends use identical target populations, the reported magnitude-lattice counting measure and all 1,277 targets across the three folds.

Table 11.19: JMA contrasts (nats) under free and stationary background trend. Positive values favour OC. These contrasts are descriptive; no confidence intervals are supplied.

Fold	Trend	Targets	OC–FC timing	OC–EP timing	OC–EP joint
0	free	469	-19.9765	-1.1943	-74.7591
0	$d = 0$	469	-20.0603	-5.8012	-76.4264
1	free	348	-12.8718	-1.4445	-59.1334
1	$d = 0$	348	-8.1665	-4.1719	-61.1146
2	free	460	-25.7452	-5.0868	-87.3190
2	$d = 0$	460	-9.6847	-3.6114	-84.6859
All	free	1277	-58.5935	-7.7256	-221.2115
All	$d = 0$	1277	-37.9115	-13.5846	-222.2269

These ablation contrasts diagnose sensitivity to the fitted trend within the declared procedures. They do not isolate a physical mechanism or establish independent cross-region validation.

Fixing the background trend at zero reduces the pooled OC–FC timing deficit but does not eliminate it. The OC–EP joint deficit also persists, so a lower-bound trend estimate alone does not explain the Japanese reversal.

New Zealand remains an adaptively reused archive in this manuscript. S06 supplies no fresh New Zealand cohort and does not remove the adaptive-reuse disclosure or the need for genuinely prospective validation.

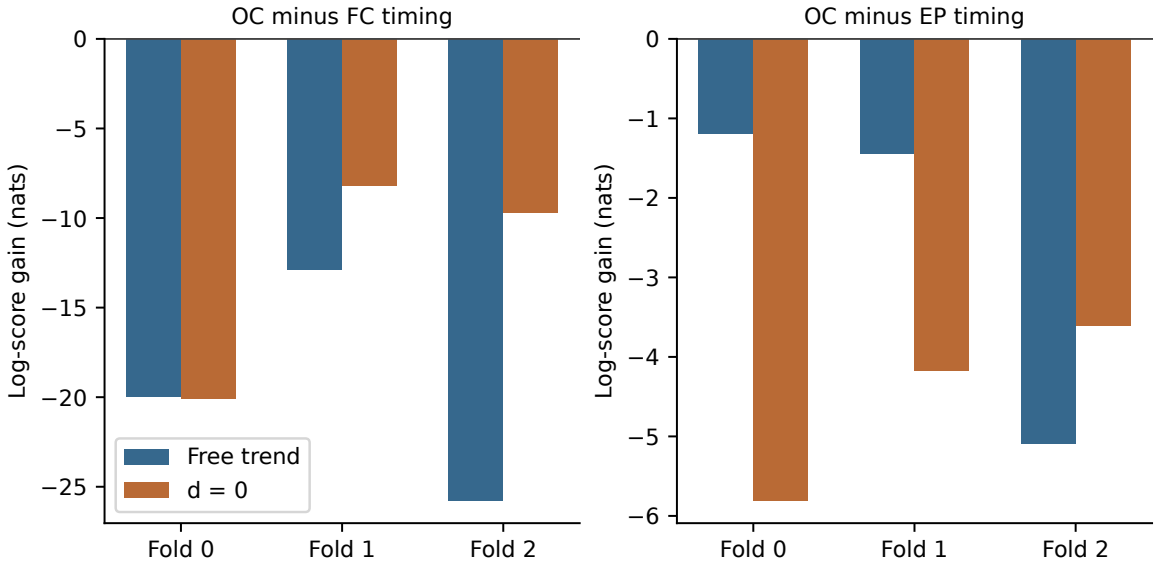


Figure 11.7: Saved JMA timing contrasts under free and stationary background trends. Each bar uses the same fold population; no new likelihood calculation or simulation was performed for the plot.

Chapter 12

Conditional Simulation Diagnostics of Frozen Forecasts

12.1 Conditional diagnostics: useful discrimination without overall adequacy

Forecast testing distinguishes comparison of competing predictive scores from consistency of observations with a specified forecast law (Schorlemmer et al., 2007; Rhoades et al., 2011). A model can outscore a comparator and still misrepresent counts, dependence or spatial allocation. Bray and Schoenberg (2013) discuss residual diagnostics that address such structure beyond an aggregate likelihood. Here complete branching futures supply a reference for the frozen fitted state; the conditioning and uncertainty interpretation are specified in Section 6.1.

12.1.1 Frozen target and diagnostic reference

The completed conditional check targets original frozen OC alone. For each of its three origins, 999 complete branching-future paths were generated conditional on that origin's fixed fitted state, nuisance choices and original prehistory. Descendants can generate further descendants. All 2,997 paths completed; failed and unknown counts are zero. They are dependent-event paths under the model, not 2,997 independent earthquake samples. The observed-path exposure is a realised predictable compensator; it is not a Poisson future-count mean.

Each origin has 84 recorded coordinates, covering count, time gaps and dependence, spatial transforms and allocation, scores and subblocks. The 252 entries are not 252 independent diagnostic tests: immigrant exposure is a deterministic bookkeeping coordinate at each fixed origin, and total-exposure and trigger-exposure ranks coincide because their difference is that fixed constant. Those entries are retained for accounting, explicitly flagged in Appendix B.10, and add no independent evidence of adequacy. The spatial transforms draw on the probability-integral-transform tradition (Rosenblatt, 1952; Schoenberg, 2003). A coordinate's insertion interval is

$$\left[\frac{1+\#\{S^{(r)}<S^{\mathrm{obs}}\}}{1000},\quad \frac{1+\#\{S^{(r)}\le S^{\mathrm{obs}}\}}{1000}\right]. \tag{12.1}$$

The resolution is 0.001. Boundary ranks are finite reference comparisons, not zero probability or exact population tail probabilities. Ties can create wide intervals. This descriptive collection has no asserted omnibus significance test, and its coordinates are not separate unadjusted discoveries. Empirical reference intervals describe simulated catalogues, not confidence intervals for fitted parameters or scientific validity.

12.1.2 Discrepant and ordinary-looking results coexist

Origin	Recorded count	Supplied simulated count interval	Count insertion interval	Selected diagnostic evidence
Early	79	121-453	[0.001, 0.001]	Gap KS, serial statistic and spatial KS exceed every reference
Middle	167	90-398	[0.704, 0.712]	Ordinary total hides yearly concentration and dependence
Late	69	69-279	[0.021, 0.028]	Marginal gap/spatial KS ordinary; dependence and variance discrepant

Early count is below all 999 simulated counts. Its exposure and trigger exposure are also below all references. Its gap KS is 0.222245 and spatial Rosenblatt KS is 0.167335, each above every reference. Yet the time event-logsum insertion is 0.583, the censored terminal tail 0.111 and within-cell absolute maximum 0.187. The model does not fail every coordinate in the same way.

Middle total count looks ordinary, but the five yearly counts are 12, 4, 8, 116 and 27. Its subblock balance maximum has insertion 0.998; the gap serial statistic exceeds all references. Its conditional-location score insertion is 0.364 and within-cell density score 0.312, while spatial KS, cross dependence and allocation show discrepancies. A satisfactory-looking total or score coordinate does not settle concentration or dependence.

Late gap KS has insertion 0.676 and spatial KS 0.810, yet serial dependence has insertion 0.998, the spatial cross statistic 0.992, and dimensionless location variance exceeds every reference. Across origins, cell 4 observed counts 15, 7 and 8 are below conditional allocations 23.493717, 16.132286 and 14.770085. These allocations sum conditional probabilities at observed event times; they are not unconditional event-count means. Empty-cell ties require particular care: one zero count has insertion [0.001, 0.563], which cannot be summarized by its lower endpoint.

Raw score totals also depend on target count and density measure. Early total location and joint scores exceed all simulated totals while its count is below all simulated counts; this does not validate its probabilities. Immigrant exposure ties identically across paths within a fixed state by construction. Neither that deterministic tie nor count-dependent mark totals validates the underlying recipe.

12.1.3 Diagnostic scope

OC shows material conditional diagnostic discrepancies in all three origins: early count and several dependence/spatial coordinates, middle concentration and dependence, and late dependence and location variance. The fixed-state forecast therefore lacks demonstrated conditional adequacy. That original coordinate-wise conclusion was descriptive rather than an omnibus rejection. EP was absent from that historical OC-only simulation experiment. The subsequent comparative conditional evaluation in Section 12.3 includes EP and the declared within-origin omnibus statistics. Simulation-based forecast testing separates these questions (Schorlemmer et al., 2007; Zechar et al., 2010). The reference conditions on the fitted state as defined in Section 6.1. Later training histories include earlier target periods; separate

per-origin simulations do not define a coherent cross-origin joint null. Cross-window patterns remain descriptive. Heavy-tailed productivity has infinite second moment under the fixed mark recipe, and normal/CLT count inference is not used. A censored terminal tail is survival information, not an additional completed gap.

The conditional diagnostic report and current evidence index retain the coordinate-level source and audit scope. All paths received identity/count/mark/exposure checks; independent computational replay of all 84 coordinates covered nine paths, three prescribed indices per origin, not all 2,997 complete vectors. The replay is a computational validation of the retained diagnostic vectors.

12.2 Complete diagnostic record and subsequent evaluation

Appendix B, Section B.10, retains the complete coordinate tables and origin-specific rank figures, their reference definitions, and the independent computational audit. The compact summary above and that full record describe the same frozen OC experiment.

The subsequent score-concentration and catalogue-robustness evaluation retained both discrepant and ordinary-looking diagnostics. Its completed results appear in Chapter 11; the frozen OC calibration target was not replaced by that follow-on.

12.3 Completed conditional forecast calibration

Each model and California origin has 999 conditional simulated futures and its observed vector. The following tests condition on the fitted state and past history; they do not integrate parameter-estimation uncertainty.

Table 12.2: Conditional count, original minimum-rank and continuous-coordinate pooled tests. All p-values use their declared inclusive-tail conventions.

Model	Fold	Conditional N	Min-rank	Quadratic	Fisher-type
OC	0	0.002	0.070	0.004	0.001
OC	1	0.594	0.075	0.003	0.004
OC	2	0.056	0.069	0.007	0.012
EP	0	0.002	0.070	0.004	0.002
EP	1	0.698	0.075	0.005	0.006
EP	2	0.036	0.069	0.009	0.008

The conditional count summary retains lower and upper inclusive empirical tails and reports twice the smaller tail, capped at one. The original minimum-rank test is retained with its exclusions and tie behaviour; it is not replaced by the continuous-coordinate statistics.

The pooled quadratic statistic uses eight declared coordinates: total exposure, time-event log sum, gap KS, gap serial correlation, spatial Rosenblatt KS maximum, absolute spatial cross-correlation, and dimensionless location-log-density mean and variance. Standardization and the ridge-regularized covariance are computed symmetrically on the observed vector plus all simulated vectors. The Fisher-type statistic is the preserved $-\sum_j \log \widetilde{p}_j$, not an independently calibrated chi-square reference or an assumption of independent coordinates. Both statistics use the inclusive upper rank in the same symmetric pool.

Table 12.3: Retained-coordinate and tie diagnostics. Equal counts include the observed vector and every exactly tied pooled statistic.

Model	Fold	Min-rank coordinates	Quadratic equal	Fisher equal
OC	0	82	1	1
OC	1	82	1	1
OC	2	82	1	1
EP	0	82	1	1
EP	1	82	1	1
EP	2	82	1	1

Continuous does not automatically mean tie-free: coordinate duplicates and equality with the observed statistic are retained in the public aggregate and interpreted conservatively. This is the declared conditional calibration and pooled omnibus scope. No CSEP L-, M- or S-test results are inferred, and a non-rejection is not proof that the model or physical inhibition mechanism is correct. The independent reconstruction coverage is reported separately below; saved arithmetic checks and raw replay remain distinct forms of evidence.

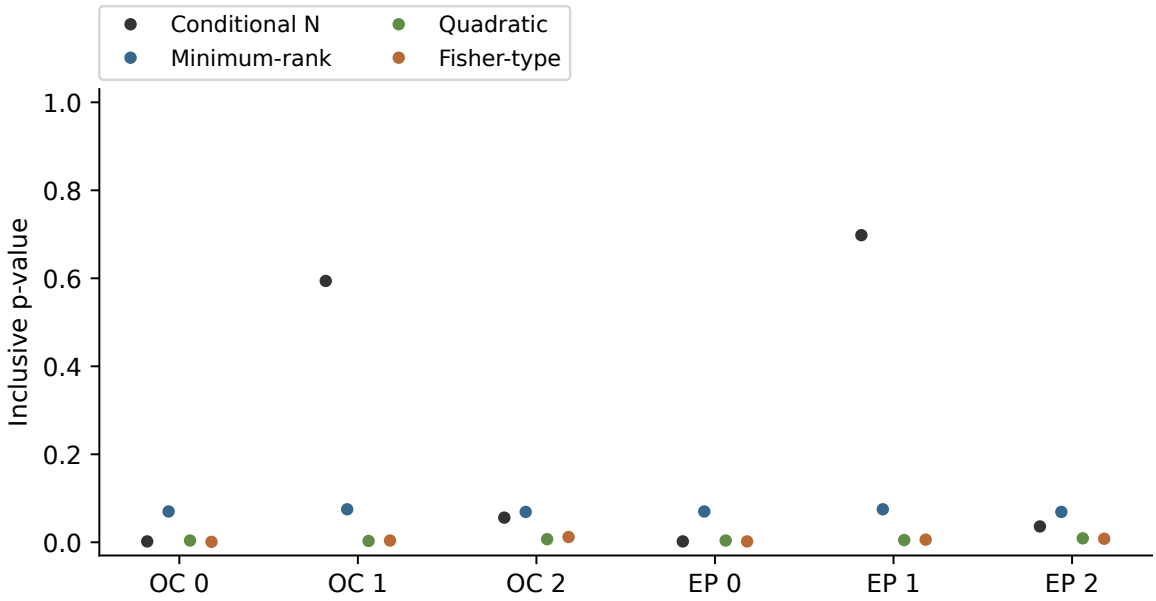


Figure 12.1: Reported inclusive conditional/pooled p-values.

Model	Fold	Completed futures	Planned futures	Status
OC	0	999	999	resolved
OC	1	999	999	resolved
OC	2	999	999	resolved
EP	0	999	999	resolved
EP	1	999	999	resolved
EP	2	999	999	resolved

Both continuous-statistic omnibus procedures reject at the declared five-percent level for each model and origin, while the retained minimum-rank procedure does not. Thus the EP

score advantage is not a conditional-adequacy certificate. These are separate within-origin tests, not a joint cross-origin or model-selection-adjusted rejection. The historical OC full-vector audit remains separate.

The original completed independent closure reconstructed count, marks, immigrant and triggering exposure, temporal event logs, and spatial point, location and joint scores directly from all 5,994 saved OC and EP futures (999 per model and origin). It also reconstructed all 84 diagnostic coordinates and all 11 event-detail columns on EP indices 0, 249, 499, 749 and 998 in each of the three origins, giving 15 predeclared full-vector replays. Count identities are exact; finite numerical discrepancies satisfy the declared score, diagnostic and spatial-CDF tolerances. This extends the earlier nine-path native sample and zero-full-vector scope; it does not assert full spatial-PIT replay on all futures.

The independent reconstruction uses incomplete-beta line integrals and a separately implemented positive-polar rectangle-integration kernel. The scoring route and source identities are documented in `reproduction/regional/README.md`.

Independent closure quantity	Value
Independent raw point-score paths	5994
EP paths with all 84 diagnostics	15
Maximum point-score error (nats)	1.17143e-09
Maximum full-vector score error (nats)	9.09495e-12
Maximum other diagnostic error	8.18545e-12
Maximum spatial-CDF error	6.66134e-15
Maximum event-log error (nats)	1.10134e-13
Maximum gap error	1.15463e-14

Chapter 13

Continuous Immigrant Local Inhibition and Synthetic Control Regimes

13.1 Study overview

Continuous immigrant-only local attenuation selects the zero-strength training boundary in all three California test windows, both with unit marks and with magnitude scaling. Nesting then makes its forecast identical to the matched background-weighted Omori reference on the same 315 targets. This boundary selection is not an interior-power test of physical inhibition. Raw magnitude-scaled total attenuation gives negative differences smaller than two nats per fold, a near tie on this score scale. The dependent-block interval summaries at the end of this chapter should be read under Section 8.4.1, rather than as calibrated tests of long-memory seismicity. The magnitude-scaled ETAS extension in Section 11.5.2 also selects the nested zero-strength boundary.

The synthetic controls distinguish an uninhibited reference fixed at generating parameters from a model allowed to fit the temporal backbone. All 80 joint-fit histories and 400 outcomes are complete. Broad/long immigrant attenuation has mean -10.37 nats (sample MCSE 8.56), median $+2.26$, 10/20 wins and 16.9 mean test events. Broad/long total attenuation has mean -1.26 nats (MCSE 2.63), median -1.26, 8/20 wins and 13.25 mean test events. The tables report every regime with event counts and sample MCSEs, including floor returns. The fitted-intercept and branching-ratio controls show why fixed-floor gains do not isolate attenuation. At zero strength, range and memory disappear from the law.

The intermediate S8 bank rejects 27.2% of proposals and retains 94.7 test events on average. Its raw mean is $+0.294$ nats (MCSE 2.777; 13/20 wins), while its training-BIC-gated sensitivity has mean $+2.495$ nats (MCSE 1.444; 9/20 wins and 12/20 active fits). Section 13.10.2 reports the boundary and training-resolution diagnostics. These finite-bank summaries do not establish uniform detection power.

Together the controls span almost unrealized narrow/short attenuation, strong count depletion in the broad/long banks, and a moderate regime with noisy per-history contrasts. None supplies an automatically well-powered physical absence test. Sample MCSEs describe variation across independently generated histories; they do not make events within one earthquake sequence independent or justify Gaussian inference under heavy-tailed generating laws. The common measurement and inferential scope is consolidated in Section 6.1.

13.2 Key question and review status

Does spatially local, temporary attenuation improve retrospective prediction beyond the matched OC procedure, and does the resulting procedure compete with EP? The question follows the positive continuous-location ablation in Chapter 10 and the more qualified external,

benchmark and diagnostic findings in Chapter 11. It requires an actual likelihood with full exposure, a frozen fitting rule, matched references, retained negative controls and complete accounting. A term described as inhibition is not evidence that a physical stress-depletion mechanism has been identified.

The initial and expanded controls retain their declared generating laws and comparator definitions. The joint-backbone comparisons and the moderately thinned bank evaluate fitted uninhibited explanations on the same histories as their inhibited counterparts.

13.3 Recorded-event design and comparator definitions

13.3.1 Same California history, not a new sample

The observation law selects preferred California records with reported $M \geq 4.0$, latitude 32-37 degrees, longitude -121 to -114 degrees and depth -5 to 40 km. The fixed source contains 1,272 unique events: 106 warmup parents in 1981-1985 and 1,166 evaluated records in 1986-2026. The latter comprise 851 initial training targets plus 315 subsequent test targets. Warmup is already included in 1,272; the counts must not be added a second time.

Fold	Test window, UTC	Training targets	Test targets	Exposure days
F0	2011-2016	851	79	1,826
F1	2016-2021	930	167	1,827
F2	2021-2026	1,097	69	1,826
Pooled test	2011-2026	Overlapping prefixes	315	5,479

These are the same 315 targets in the California replication, benchmark, conditional diagnostic and concentration studies. Their repeated evaluation is not independent replication. The archive and earlier annual/sequence outcomes had already been inspected. Parameters and nuisance choices use training data, remain frozen in test, and histories update sequentially after each observed timestamp. Only strictly earlier eligible events enter excitation and inhibition; timestamp ties are processed as a batch, with no within-batch self- or mutual influence.

13.3.2 The references answer different questions

OC means Omori temporal triggering with the finite-domain continuous, reference-density-weighted offspring law and smooth immigrant background. EP means a separately fitted Omori ETAS adaptation whose inverse-power radial offspring density is normalized directly against projected area without OC's reference weighting. EP is an excitation model, not a Poisson model. LI-OC is the matched extension comparison. LI-EP compares complete competing procedures with different spatial excitation laws and does not isolate inhibition alone.

For completeness, the other abbreviations in the programme are FC (fixed-scale Mittag-Leffler continuous), FL (its cell-lift counterpart), OL (Omori cell-lift), MC (free common-scale Mittag-Leffler continuous) and N0 (a separately refitted smooth no-trigger procedure). Their primary contrasts are detailed in Chapter 11. A constant-Poisson candidate here is a numerical optimization floor, distinct from N0 and EP.

13.4 The continuous immigrant-inhibition model

13.4.1 Intensity and attenuation field

For time t and recorded projected location z in the fixed finite domain D, the model is

$$\lambda_{\mathrm{LI}}(t,z) = e^{a+dx_{\mathrm{trend}}(t)} b_{\mathrm{bg}}(z) e^{-\eta G(t,z)} + nH_{\mathrm{OC}}(t,z), \tag{13.1}$$

$$x_{\mathrm{trend}}(t) = 2t/T_0 - 1, \qquad T_0 = 16436 \text{ days}, \tag{13.2}$$

$$G(t,z) = \sum_{j:t_j<t} e^{-(t-t_j)/\psi} \left(1 + \frac{\|z-z_j\|^2}{\ell^2}\right)^{-2}. \tag{13.3}$$

The observed clock is days since 1981-01-01 00:00:00 UTC.

Here a and d govern immigrant amplitude/trend; b_{bg} is the inherited training-only background density; n scales excitation H_{OC}; η is nonnegative inhibition strength; ℓ is its spatial range in kilometres; and ψ is its memory in days. The observed clock uses the frozen timestamp convention and strict earlier-integer-timestamp parent rule, including tied batches. Synthetic controls have their own day-zero origin and normalization, specified in Section 13.7.1. The stress mark multiplier is fixed to one: every eligible earlier recorded event contributes to G without magnitude weighting. No latent immigrant/offspring classification is required. Inhibition acts only on the immigrant term. It does not suppress H_{OC}, and it is not spatially renormalized. It therefore changes both local density and total immigrant rate. Renormalizing the inhibited density would define a different model.

The unit stress mark is a substantive restriction: at the same lag and separation, an eligible M4 event and an eligible M7.1 event contribute equally to the attenuation field. Their excitation productivity can still differ through κ_j. This model therefore does not represent magnitude-scaled stress release, a Coulomb stress tensor, or suppression of the full triggered sequence. It tests immigrant attenuation under one restricted statistical law.

More explicitly, the inherited excitation is a sum over all eligible strictly earlier parents:

$$H_{\mathrm{OC}}(t,z) = \sum_{j:t_j<t} \kappa_j k_O(t-t_j) s_\sigma(z \mid z_j), \tag{13.4}$$

$$k_O(u) = \frac{(p-1)c^{p-1}}{(u+c)^p}, \qquad u>0, \tag{13.5}$$

$$s_\sigma(z \mid z_j) = \frac{b_{\mathrm{ref}}(z)(1+\|z-z_j\|^2/\sigma^2)^{-2}}{Z_j}, \tag{13.6}$$

$$Z_j = \int_D b_{\mathrm{ref}}(v)(1+\|v-z_j\|^2/\sigma^2)^{-2}\,\mathrm{d}v. \tag{13.7}$$

Here κ_j is the shared training-calibrated magnitude-productivity weight, with fixed exponent $r = 0.7$; b_{ref} is the frozen training-derived reference geography. The background b_{bg} integrates to one on D. The temporal kernel integrates to one over all positive lags and each s_σ integrates to one on D; exposures use the actual finite time window. The training-only smooth background, parent eligibility and inherited excitation nuisance choices follow the matched OC procedure. LI introduces no new background bandwidth or excitation spatial-scale search. The stress radial factor in G itself is not a normalized spatial probability density; it is an attenuation field inside the exponential.

The temporal factor is the normalized Omori–Utsu law (Utsu et al., 1995); the reference-weighted spatial factor is a particular fitted forecasting choice rather than a physical stress law.

At $\eta = 0$, $\exp[-\eta G]$ equals one everywhere and LI exactly nests OC when shared coordinates agree. All ℓ and ψ choices then give the same law. Canonical $\ell = 30$ km and $\psi = 7$ days are display conventions at that boundary, not recovered scales. Even a positive procedure score difference at $\eta = 0$ could arise from refitting shared coordinates through another optimization path; it would not establish active attenuation. The separate approved selected-state diagnostics are therefore essential to the conclusion below.

13.4.2 Frozen fitting and selection contract

The fitted coordinates are $(a, d, n, \log c, p, \eta, \log \ell, \log \psi)$, where c and p parameterize the normalized Omori kernel. The physical bounds are fixed.

Coordinate	Bound	Coordinate	Bound
a	[-12, 0]	d	[-2, 2]
n	[0, 0.98]	c	[0.0001, 20] days
p	[1.01, 2.5]	η	[0, 4]
ℓ	[5, 150] km	ψ	[1/24, 365] days

Each main arm has eight prescribed starts. For LI fits only, shared a,d,n start from the accepted matched OC state. LI start 0 retains OC's c,p and $\eta = 0$; the other LI starts use the fixed c,p,η,ℓ,ψ schedule below. Synthetic OC control fits use a separate frozen initialization rule, not their own subsequently accepted fit: a0 = clip(log(0.6 max(Ntrain,1)/365), -11.5, -0.1), d0 = 0.2 and n0 = 0.35, with eight prescribed c,p rows in synthetic_common/OC_start_fill. The readable protocol subset gives both start schemes, floors, bounds and tolerances in full.

L-BFGS-B has maxiter 200, maxfun 20,000, maxls 200, ftol 1e-10 and gtol 2e-6. Bounded Powell is allowed only after failure of the declared first-method acceptance rule, from the same original start, with maxiter 400, maxfev 5,000, xtol 1e-6 and ftol 1e-10. These limits describe a bounded search, not a global optimization proof.

LI start: c (days), p	η	ℓ (km)	ψ (days)
0: accepted OC c,p	0	30	7
1: 0.0001, 1.1	0.1	5	1/24
2: 0.001, 1.3	0.3	15	0.25
3: 0.01, 1.1	0.7	50	7
4: 0.1, 1.3	1.5	150	30
5: 1, 1.7	3	5	365
6: 20, 2.5	4	150	1/24
7: 0.1, 1.01	0.5	30	365

Selection minimizes accepted training joint negative log likelihood. Exact ties prefer the OC floor, then the Poisson floor, then start and method ordinals. Held-out score does not select a candidate, method, computational epoch or seed. Candidates within 0.0004 training NLL units of the selected state receive the declared held-out spread check, requiring spread

strictly below 0.01 nats per target. This is an admission/sensitivity diagnostic over the stored near-optimal set; it is neither an uncertainty interval nor permission to select the best test forecast.

The protocol and case matrix supply the full start schedules, observation law and selection rule.

13.4.3 Proper scores and exposure

Write P for summed event log point intensity, Q for summed event log regional intensity, E for positive full-domain integrated intensity, C = -E for its signed score contribution and M for the common magnitude-mark score. Then

$$S_{\text{time}} = Q + C = Q - E, \qquad S_{\text{loc}} = P - Q, \tag{13.8}$$

$$S_{\text{joint}} = P + C + M = S_{\text{time}} + S_{\text{loc}} + M, \tag{13.9}$$

$$\Delta P = P_{\text{LI}} - P_R, \qquad \Delta C = E_R - E_{\text{LI}}, \tag{13.10}$$

$$\Delta S_{\text{joint}} = \Delta P + \Delta C + \Delta M. \tag{13.11}$$

All quiet-period and full-domain exposure is retained. Lower exposure can offset a worse event-intensity term; an event-only gain is insufficient. Within each arm, Q uses that arm's spatially integrated ground intensity under the fixed observed history, and all components use its same fitted state. Independently fitted arms can have different parameters. The shared conditional mark law is scored once and cancels in contrasts. Pooled scores sum all three window totals and divide by 315. These are nats of realised predictive log score, not percent accuracy, correlation, a physical effect size or evidence of calibration.

The source-field dictionary distinguishes these narrative symbols from legacy aggregate keys. In particular, LI's raw pooled E field means the point-event logsum difference, not positive exposure; its raw C is the signed exposure contribution. Sequence timing_compensator instead stores signed difference in nonnegative integrated exposure and enters the score negatively. Source keys and scalars remain unchanged.

13.5 Observed result: no selected inhibition and a numerical tie

All three selected LI fits have η **exactly zero**. The procedure selects no active attenuation term in these observed windows. Its joint contrasts against OC are numerical ties under the frozen aggregate sign tolerance of +/-0.0002 total nats (inclusive). The following score contrasts are descriptive, without confidence intervals.

Window	Targets	LI-OC joint nats/target	Selected η	Interpretation
F0	79	0	0	Numerical tie
F1	167	+1.128699e-12	0	Numerical tie
F2	69	0	0	Numerical tie
Pooled	315	+5.983898e-13	0 in 3/3	Numerical tie

The raw pooled difference is +1.88492776942439e-10 total nats. Its tiny positive floating-point sign is retained for arithmetic traceability and is not a forecast improvement. The frozen

aggregate sign tolerance concerns numerical comparison, not statistical significance or a fitter admission gate. Exact model nesting and finite arithmetic are separate issues: the selected law has no inhibition, while independently evaluated saved score arithmetic may differ at roundoff scale.

The zero-strength boundary leaves ℓ and ψ unidentified. Neither their canonical values nor bounds identify a physical inhibition footprint or duration. This result supports the narrower statement that the registered fitting procedure selected no active immigrant attenuation on this inspected recorded-coordinate history. It supplies no proof that earthquakes lack inhibition, no universal detection limit and no evidence of physical absence.

13.6 EP remains the stronger joint forecast

LI scores below EP in every observed window. Its pooled timing edge is tiny and mixed across windows; its larger conditional-location deficit accounts for the joint loss. All differences below are LI minus EP in nats per target. They are descriptive score contrasts, without confidence intervals.

Window (targets)	Timing	Conditional location	Joint	Mark difference
F0 (79)	+0.006244055	-0.085849360	-0.079605305	0
F1 (167)	-0.001101051	-0.188973874	-0.190074925	0
F2 (69)	+0.000389054	-0.189082869	-0.188693815	0
Pooled (315)	+0.00106746	-0.163135	-0.162067	0

The exact source-reported pooled LI-EP joint difference is -51.0512047628713 total nats, or -0.162067316707528 per target. Its timing and conditional-location totals are +0.336249525419333 and -51.3874542882905. These are predictive components, not causal shares or a decomposition of physical immigrant versus offspring generation.

The separately archived OC-EP benchmark in IV differs slightly at floating-point scale from this replay. Each is reported under its own accepted source identity; substituting one for the other would obscure provenance. Neither EP's better relative score nor LI's exact nesting supplies a probability-calibration result. The earlier conditional calibration concerned original OC only and found material discrepancies; Section 12.3 reports the later comparison including EP.

13.7 Every fixed-seed control, including the failures

13.7.1 Generating contract

Eight datasets, two seeds per regime, were fixed in advance. The null regimes have $\eta = 0$. The narrow/short law has positive generating strength $\eta = 0.5$, with $\ell = 5$ km and $\psi = 1/24$ day. The broad/long regime has $\eta = 2$, $\ell = 50$ km and $\psi = 30$ days. At the null, ℓ/ψ do not affect the law. The domain is 100 by 100 km; synthetic t is days since simulated history begins at day zero, with $x_{\text{trend}}(t) = 2t/760 - 1$. Warmup ends at day 30, training at day 395 and test at day 760. The observed 16,436-day normalization is not used for these controls. Known nuisance/background inputs are fixed equally for paired OC/LI fits. Chronological thinning uses accepted-event ancestry and fixed PCG64DXSM streams, with original caps/failures attached to their seeds. No failed or inconvenient seed is redrawn.

The controls test the registered fitting procedure on these laws. They do not mimic every catalogue feature, location-error process or earthquake mechanism. The public control table is complete.

Generating regime	Seed	Test targets	LI-OC joint nats	Nats/target
Uniform null	410001	116	-148.647608	-1.281445
Uniform null	410002	138	-4.930159	-0.035726
Smooth null	410101	128	-4.489917	-0.035077
Smooth null	410102	120	-7.638775	-0.063656
Narrow/short inhibition	410201	107	-1.297391	-0.012125
Narrow/short inhibition	410202	102	-25.357970	-0.248608
Broad/long inhibition	410301	23	+2.242679	+0.097508
Broad/long inhibition	410302	27	+6.771012	+0.250778

LI loses on six examples, including both histories from the positive-strength narrow/short law, and wins on the two broad/long examples. Neither narrow/short history rejected an immigrant proposal; under the recorded stream coupling, their event paths coincide with the same-seed uninhibited paths (Section 13.10). The first uniform-null loss is large and remains displayed. The broad/long examples contain only 23 and 27 test targets. Averaging these rows into a single success rate would discard the intentionally different laws and denominators.

These fixed examples do not estimate recovery probability. The losses remain compatible with finite-sample overfitting, weak information about the extra coordinates, or limitations of the bounded numerical search; two seeds per regime do not distinguish those explanations. In particular, a nested training floor at $\eta = 0$ protects training likelihood, not future log score. A large held-out loss therefore does not by itself prove optimizer failure. The LI initialization uses the accepted fitted OC state, not a guarantee that all starting coordinates equal the generating truth. The two broad/long wins do not establish general power, and the null losses do not identify a physical mechanism.

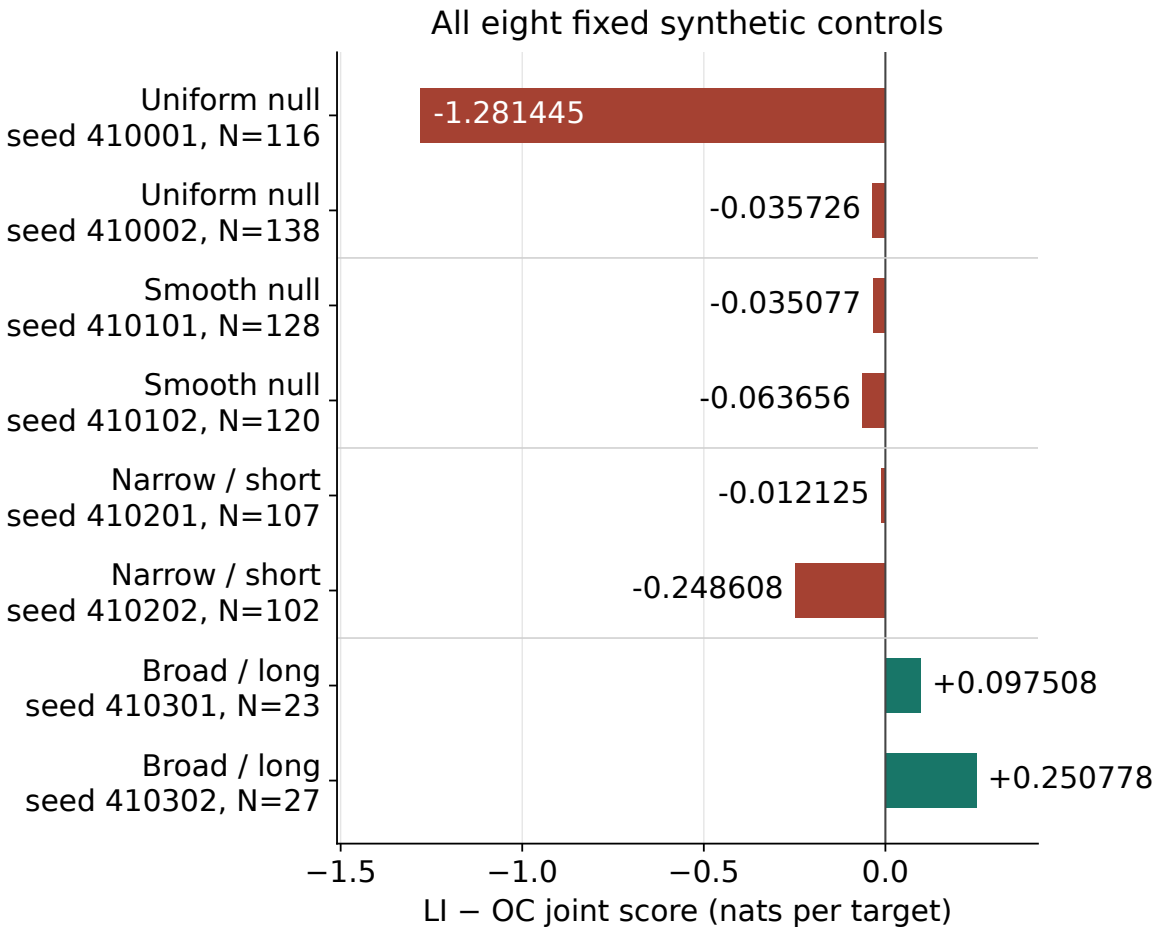


Figure 13.1: All eight prescribed LI-minus-OC fixed controls. The full linear scale retains the large uniform-null loss. Every seed and test count remains labelled. Positive means LI scored better; the rows are fixed examples, not estimated success probabilities.

13.7.2 What the controls do and do not show

Both broad/long wins show that the registered procedure can improve held-out score on those two fixed examples. The narrow/short law has positive strength, but neither retained history realizes immigrant thinning; their held-out losses do not measure failure to detect a realized thinning effect. No contrast here turns the observed zero-strength result into a physical absence test. Conversely, an active fitted term or better score would not identify a physical mechanism.

Two seeds per regime cannot estimate false-positive rate, power, parameter coverage or general recovery probability. Six losses do not prove absence of inhibition in the generating laws, and two wins do not establish reliable detection. The narrow/short scores do not locate a physical resolution threshold. The recorded-coordinate law has no admitted empirical location-error distribution; the engineering lower bound $\ell = 5$ km is a model/search constraint, not a measurement-accuracy assertion.

13.8 Identifiability and consolidated scientific implications

At $\eta = 0$ the likelihood does not depend on ℓ or ψ. Thus zero-strength selection does not estimate a zero range or duration: it removes those coordinates from the law. The null also lies at the boundary of a constrained parameter space. Exact nesting alone justifies neither ordinary Wilks inference nor a universal half-chi-square reference.

The finite multistart procedure is part of the estimand. Negative held-out differences are possible even for nested model families because training-defined selection and finite samples do not guarantee held-out superiority. The complete retained candidates and near-optimal checks reduce particular numerical concerns; they do not certify a global optimum or prove that no positive-inhibition state could ever perform better. The conclusion must stay about the registered procedure and this history.

Together with Chapter 11, the programme supports a qualified predictive conclusion. Continuous offspring geography improves location relative to the declared cell-lift procedure in the examined regional comparisons. Omori's timing advantage is conditional and reverses in Japan. OC and its zero-inhibition LI extension remain below EP jointly in the primary California folds; EP also wins in the primary Japan folds. OC and EP both show fixed-state conditional discrepancies in Section 12.3. California score gains are heterogeneous and concentrated in a single eligible scored sequence period while retaining positive contextual totals over the complete complement.

The new LI result adds a negative extension finding: no selected active inhibition and no predictive gain over matched OC in the observed windows. It does not overturn the earlier location ablation, reverse the serious-benchmark deficit, resolve calibration, recover physical labels or prove absence of a physical mechanism. The eight controls reinforce the need to distinguish procedural sensitivity from a physical absence claim.

All archives had been inspected and revised. Chronological test windows are related through overlapping training history and earthquake dependence, and the same California targets recur across studies. No pristine confirmation, broad independent sequence replication or causal conclusion follows.

13.9 Numerical validation

The unit-mark LI minus OC pooled difference is 1.884928×10^{-10} nats, a numerical tie. The selected training states pass independent score reconstruction and coarse/fine quadrature checks. All 19 training-defined near-optimal forecast spreads are below 0.01 nats per target; the largest is 0.00053761640079144. The aggregate checker passes 369 arithmetic and numerical checks. These checks assess the computed likelihoods and the bounded search, rather than provide uncertainty intervals for observed score contrasts. The paired block intervals for the observed total-intensity comparison are reported below.

13.10 Expanded fixed-backbone controls and magnitude-scaled inhibition

The original S04 completed campaign expands the fixed-seed control matrix and evaluates magnitude-scaled immigrant-only and total-intensity inhibition (LI+). The earlier eight examples above remain historical results under their original numerical procedure. The expanded results below use the same declared generating regimes and shared fixed background/excitation inputs; they are not full joint nuisance refits.

The original S04 histories with positive generating strength use the original unit-mark, immigrant-only LI law. Magnitude-scaled LI+ and total-intensity attenuation are alternative fitted procedures on those same histories, not their matched generating models. Their contrast therefore measures performance under that specified generator; it does not establish relative power when magnitude-scaled or total-intensity inhibition generates the data.

The original S04 expanded roster contains 80 distinct histories, 20 in each regime, with five paired procedure/gate combinations and 400 score rows. The stress field is

$$G_+(t,z) = \sum_{t_j<t} \exp\{\gamma Q_j - (t-t_j)/\psi\} \left(1 + \|z - z_j\|^2/\ell^2\right)^{-2}, \qquad \gamma = \tfrac{3}{4}\log 10.$$

Here Q_j is excess magnitude, so the reference excess is zero. Original LI uses the same field with unit magnitude weights. This bounded design retains a common shadow radius; it does not fit a magnitude-dependent rupture radius. With the shared fixed backbone written $B + nH$, the two LI+ intensities are

$$\lambda_{\text{imm}} = Be^{-\eta G_+} + nH, \qquad \lambda_{\text{total}} = (B + nH)e^{-\eta G_+}.$$

Score regularity of these laws. For the ideal continuous exponential marks used to define the LI+ controls, $\beta = 2$ and $r = 0.7$ give $\beta(1 + r) = 3.4 < 2\gamma$. Thus $\mathbb{E}[g_+^2/\kappa] = \infty$; the finite weighted-moment example in Chapter 5 does not cover this parameter setting. Proposition 5.2 supplies a separate null-law divergence argument under the stated fresh-mark and bounded-kernel assumptions. For total attenuation, $\partial_\eta \log \lambda_{\text{total}} = -G_+$, so the null expected bracket is $\mathbb{E}_0 \int (B + nH)G_+^2$, without the immigrant-only denominator damping. Its diagonal moment obstructions also fail here. At fixed positive generating strength, however, the exponential attenuation gives the finite-bracket bounds proved in Chapter 5. These are distinct statements about ideal null and positive generating laws, not a proof of the cause of particular held-out losses. Unit-mark immigrant LI has a different mark bound. The finite numerical histories, stopping rules and selected fitted procedures remain distinct from the ideal-law moment calculation.

The gate retains an active fit only when its training log-likelihood improvement exceeds $\frac{3}{2}\log N_{\text{train}}$.

The BIC-style gate is a training-only complexity penalty, not a Davies-calibrated test or a nominal-size guarantee. The gate can select the exact uninhibited floor. A held-out win and an active training fit are distinct outcomes. No seed was redrawn and no negative score was removed.

Regime	Procedure	N	Mean gain	MC SE	Median	Wins
Uniform null	LI	20	-2.9299	0.9259	-1.6435	3
Uniform null	LI+ immigrant gated	20	0.0000	0.0000	0.0000	0
Uniform null	LI+ immigrant	20	-2.9514	1.3451	-0.3835	4
Uniform null	LI+ total gated	20	0.0000	0.0000	0.0000	0
Uniform null	LI+ total	20	-74.6499	42.9783	-2.2181	1
Smooth null	LI	20	-3.4230	1.3263	-2.2716	4
Smooth null	LI+ immigrant gated	20	0.0000	0.0000	0.0000	0
Smooth null	LI+ immigrant	20	-3.6408	1.5156	-1.9787	4
Smooth null	LI+ total gated	20	0.0000	0.0000	0.0000	0
Smooth null	LI+ total	20	-448.5272	445.0217	-0.0604	3
Narrow/short	LI	20	-3.8713	1.3147	-2.3528	3
Narrow/short	LI+ immigrant gated	20	0.0000	0.0000	0.0000	0
Narrow/short	LI+ immigrant	20	-3.4720	1.6063	-1.1309	4
Narrow/short	LI+ total gated	20	0.0000	0.0000	0.0000	0
Narrow/short	LI+ total	20	-129.3344	103.6071	-0.6334	6
Broad/long	LI	20	33.8922	1.2083	33.4049	20
Broad/long	LI+ immigrant gated	20	28.3296	1.6682	26.6651	20
Broad/long	LI+ immigrant	20	28.3296	1.6682	26.6651	20
Broad/long	LI+ total gated	20	-5.4013	13.7572	3.9231	10
Broad/long	LI+ total	20	-5.3446	13.7585	4.4063	14

Mean and median gains are total held-out log-score differences from the shared uninhibited baseline; MC SE is the sample standard deviation divided by the square root of the number of independent seeded datasets within each regime. It is a descriptive finite-sample summary, not a calibrated confidence interval. The unbounded magnitude transforms and large retained losses do not by themselves establish finite score variance or a Gaussian approximation. The broad/long regime supports improved mean performance for original LI and immigrant-only LI+. Total-intensity attenuation has a negative mean even there, despite a positive median. The large negative means in other regimes are retained rather than trimmed.

In original S04, gating selects the uninhibited floor in both null regimes and in the positive-strength narrow/short regime. The latter bank has 0/2951 rejected immigrant proposals and 20/20 histories without realized thinning, including the initial seeds above; offspring are not thinned by this law. Under the recorded independent streams and unchanged runtime settings, each event path coincides with its same-seed uninhibited counterpart, although the generating laws differ. These rows describe fitting behavior on paths with no realized thinning, not failed detection of a realized effect or an estimated sensitivity–overfitting tradeoff. They do not establish nominal Type I error or general detection power.

The original S04 completed matrix combines accepted first-pass results with quadrature refinement and same-history finer-grid refits. Training-only selection preserves the prescribed starts, bounds and exact uninhibited floor. Finer-grid validation checks retained candidates, score decomposition and forecast spread. In the original audit, independent compiled CPU replays covered predeclared anchors; the remaining cases had per-case numerical checks and sampled-audit coverage, not individual independent replay. All aggregate tables and their

source identities are retained in the evidence bundle. The later complete original-control replay recomputes the retained selected and floor scores, checks the recorded training-gate arithmetic and reconciles the aggregate table, without refitting or generating replacement histories.

These results extend the synthetic evidence only. They do not replace the observed California comparison, for which the completed observed LI+ evaluation is reported in Section 13.11 below.

13.10.1 Matched magnitude-scaled generators (S04B)

S04B is a separate matched-generator extension, not a replacement for the unit-mark S04 controls above. Magnitude-scaled immigrant-only and total-intensity laws each generate narrow/short and broad/long positive controls. Only accepted events enter the strictly prior stress field and spawn descendants. Immigrant-only generation thins immigrant proposals; total-intensity generation also thins offspring proposals.

The completed extension contains 80 independent seeded histories and 400 paired procedure scores. Every regime–procedure row has 20 histories. The five procedures share each history and are not independent replications. The original null histories remain valid at zero inhibition; they are not regenerated or added to this positive-control denominator.

Generation uses the smooth background and the same day-based split as the original controls: warmup $[0, 30)$, training $[30, 395)$ and test $[395, 760)$, with horizon 760 days. For each generating law, narrow/short inhibition has $(\eta, \ell, \psi) = (0.5, 5\text{ km}, 1/24\text{ day})$ and broad/long inhibition has $(2.0, 50\text{ km}, 30\text{ days})$. The magnitude scaling and the two intensity definitions are those displayed above. Seeds are predeclared and retained without replacement: immigrant narrow/short 410401 through 410420, immigrant broad/long 410501 through 410520, total narrow/short 410601 through 410620, and total broad/long 410701 through 410720. Training uses all 8 prescribed starts plus the exact uninhibited floor at quadrature order 16; every retained finite state is checked at order 32 on training and test windows. The first seed in each regime is the predeclared independent CPU anchor. These settings and source identities are retained in the bundled frozen-method projection and protocol.

The realized generator counters are essential to interpretation. In the narrow/short immigrant regime, 19 of 20 histories reject no proposals; the entire bank rejects only one immigrant. In broad/long total generation, 98.2% of proposed offspring are rejected and 17 histories contain no accepted offspring. Table 13.12 gives the complete counters and phase counts. Narrow/short immigrant losses therefore do not demonstrate failed recovery of a strongly realized inhibitory signal.

All fits hold the shared background and excitation backbone fixed. Training-only selection and the complexity gate retain the exact uninhibited floor. Full saved-row arithmetic and numerical gates are audited; the original independent CPU objective and gradient replays cover 48 predeclared phase–model contexts from the anchor histories, rather than every history.

The recovered-input edition completed the full saved-state likelihood and score replay supplied in `reproduction/synthetic/`. It rebuilds quadrature from the raw matched histories and reproduces the selected/floor likelihoods, score contrasts and summary rows. This computation re-evaluates saved fits; it does not rerun their optimizer searches.

Under immigrant: narrow/short generation, the matched raw procedure has a negative mean gain and a negative median; its gated version has a negative mean. Under immigrant: broad/long generation, the matched raw procedure has a positive mean gain and a positive median; its gated version has a positive mean. Under total: narrow/short generation, the

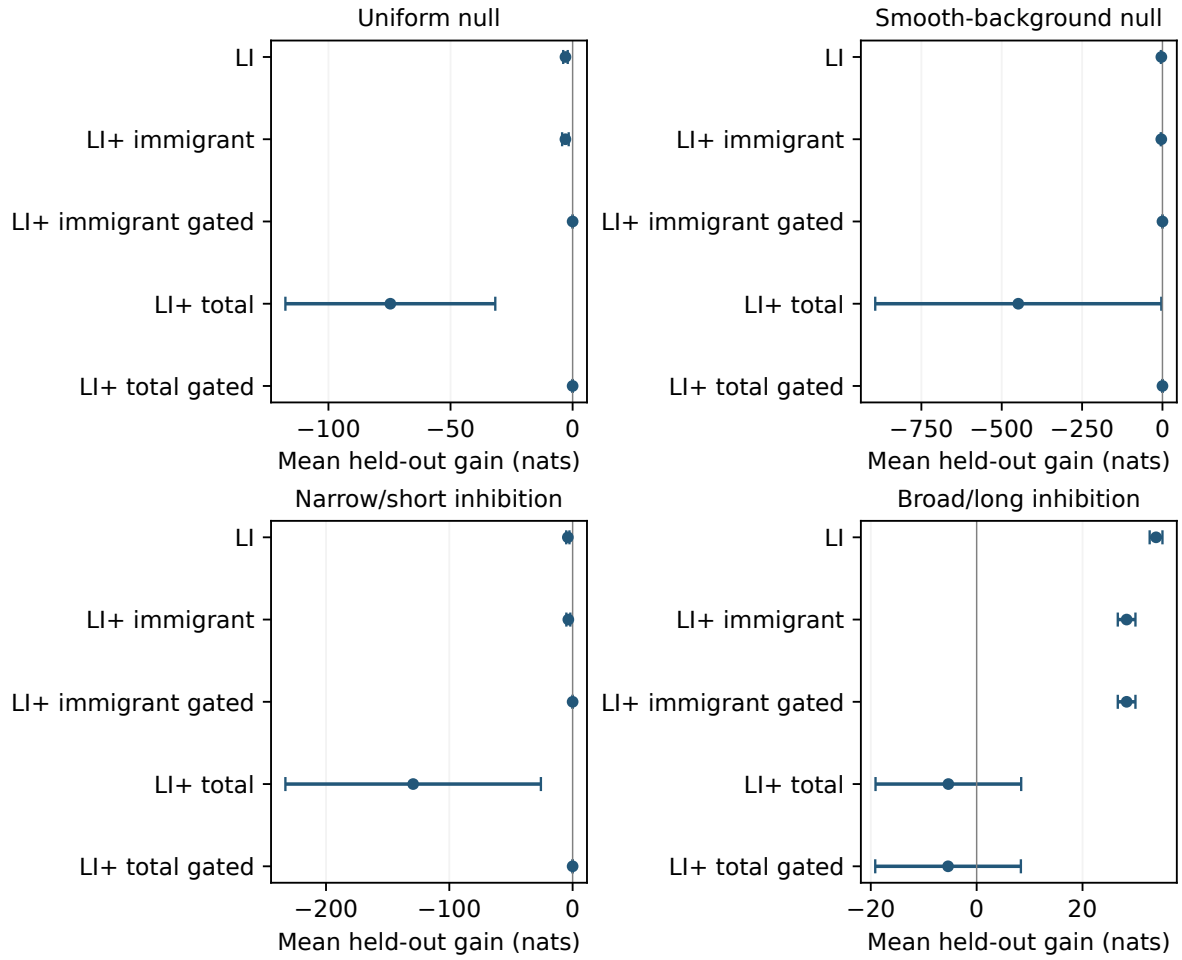


Figure 13.2: Original S04 unit-mark fixed-backbone controls. Points show mean held-out gain and bars show the descriptive sample standard deviation divided by the square root of the row denominator, labelled MC SE in the table. These bars are not calibrated confidence intervals. Panels retain their full linear ranges, including large losses; horizontal scales differ. Each plotted summary uses the full corresponding row denominator in the preceding table. The zero line is the shared uninhibited comparator.

matched raw procedure has a negative mean gain and a negative median; its gated version has a positive mean. Under total: broad/long generation, the matched raw procedure has a positive mean gain and a positive median; its gated version has a positive mean. Both matched raw procedures lose on average in their narrow/short settings. The positive gated mean under narrow/short total generation comes from a single winning active history and otherwise exact-floor ties; it does not demonstrate reliable recovery. Under narrow/short immigrant generation, the matched gate instead retains a single active loss and otherwise floor ties. In both broad/long generating settings, matched raw and gated procedures win on every history in this declared bank. These outcomes remain conditional on the fixed nuisance backbone and chosen regimes.

Mean and median gains below are total held-out log-score differences per simulated history against the fixed-backbone floor, not against a nuisance-refitted null and not per earthquake. MC SE is the within-regime sample standard deviation divided by the square root of the history denominator. It is descriptive: these heavy-tailed marked laws do not supply an automatic finite-variance or Gaussian guarantee. Wilson intervals describe bounded held-out-win indicators under the declared seeded simulation design; they are not intervals for the mean gain or calibrated detection power. A positive held-out gain, an active fit and an identified physical mechanism are different statements.

Table 13.8: S04B: Immigrant: narrow/short. Wins are strict positive gains; interval endpoints are probabilities.

Procedure	N	Mean	MC SE	Median	Wins	Wilson 95%	Active
LI	20	-3.9493	1.2142	-2.7555	2	[0.028, 0.301]	19
LI+ immigrant	20	-4.5674	1.6218	-2.0582	5	[0.112, 0.469]	20
LI+ immigrant gated	20	-1.3440	1.3440	0.0000	0	[0.000, 0.161]	1
LI+ total	20	-21.5766	9.8329	-3.1228	4	[0.081, 0.416]	19
LI+ total gated	20	-1.0043	1.0043	0.0000	0	[0.000, 0.161]	1

Table 13.9: S04B: Immigrant: broad/long. Wins are strict positive gains; interval endpoints are probabilities.

Procedure	N	Mean	MC SE	Median	Wins	Wilson 95%	Active
LI	20	44.3122	1.4559	43.3861	20	[0.839, 1.000]	20
LI+ immigrant	20	48.3868	0.9599	47.7879	20	[0.839, 1.000]	20
LI+ immigrant gated	20	48.3868	0.9599	47.7879	20	[0.839, 1.000]	20
LI+ total	20	19.4993	7.2001	35.3378	15	[0.531, 0.888]	19
LI+ total gated	20	19.3802	7.2159	35.3378	14	[0.481, 0.855]	18

Table 13.10: S04B: Total: narrow/short. Wins are strict positive gains; interval endpoints are probabilities.

Procedure	N	Mean	MC SE	Median	Wins	Wilson 95%	Active
LI	20	-1.7818	0.6357	-0.9047	5	[0.112, 0.469]	20
LI+ immigrant	20	-3.2420	1.2527	-0.5371	7	[0.181, 0.567]	20
LI+ immigrant gated	20	0.0000	0.0000	0.0000	0	[0.000, 0.161]	0
LI+ total	20	-14.5947	7.8248	-0.1064	10	[0.299, 0.701]	20
LI+ total gated	20	0.0686	0.0686	0.0000	1	[0.009, 0.236]	1

Table 13.11: S04B: Total: broad/long. Wins are strict positive gains; interval endpoints are probabilities.

Procedure	N	Mean	MC SE	Median	Wins	Wilson 95%	Active
LI	20	42.5503	1.7274	44.8818	20	[0.839, 1.000]	20
LI+ immigrant	20	49.2902	1.3502	48.5828	20	[0.839, 1.000]	20
LI+ immigrant gated	20	49.2902	1.3502	48.5828	20	[0.839, 1.000]	20
LI+ total	20	53.2332	2.1250	51.2906	20	[0.839, 1.000]	20
LI+ total gated	20	53.2332	2.1250	51.2906	20	[0.839, 1.000]	20

Active counts record off-floor selection ($\hat{\eta} > 0$), including the effect of the training-only gate. Only strength, spatial range and memory are fitted here; the shared background and excitation coordinates remain fixed. The historical compact audit exports score summaries and off-floor counts, while the current public package also supplies the fitted parameter vectors and solver traces in `reproduction/synthetic/s04b/fits/`. Zero gains remain ties, no negative gain is trimmed, and no seed is substituted. The following saved-record diagnostics distinguish realized inhibition and selected parameters from the generating-law labels.

Realized thinning and selected parameters

Table 13.12 counts actual rejected proposals over each complete generated history, including warmup. The narrow/short immigrant setting changes almost none of the realized event paths: nineteen histories have no rejected proposal and the remaining history rejects one immigrant. The separate random streams in the supplied generator imply that each zero-rejection immigrant-only path is identical to its same-seed uninhibited counterpart under the same runtime and other settings. This is a source-level coupling argument; no counterfactual generator replay is claimed. It does not make the two generating laws equal. The narrow/short total setting rejects offspring in every history and is a different control.

Table 13.12: S04B realized thinning. Rejected/proposed totals aggregate the whole histories in each regime. Zero rej. and Zero off. count histories with no rejected proposals and no accepted offspring, respectively. Mean counts use only the stated training and test windows. S4/S5 are narrow/short and broad/long immigrant attenuation; S6/S7 are the corresponding total-attenuation laws.

Regime	Imm. rejected	Off. rejected	Zero rej.	Zero off.	Train N	Test N
S4	1/2967	0/1120	19	0	86.65	109.40
S5	2518/3018	0/202	0	0	15.15	16.90
S6	1/3102	134/1335	0	0	95.80	111.55
S7	2525/3093	165/168	0	17	13.30	13.25

Broad/long thinning also changes the overall rate strongly. These are sparse-event controls against a floor whose nuisance backbone was fixed, not evidence that inhibition beats a trained uninhibited procedure. The separate background-intercept sensitivity below measures that comparator dependence.

Table 13.13 reports the selected raw fits before the complexity gate. The public package supplies every selected vector and solver trace. Boundary concentration is visible directly; an active fit is not parameter recovery. At zero strength, range and memory disappear from the likelihood, so their displayed values are arbitrary representatives of the same null intensity.

Table 13.13: S04B selected raw parameters: median above [minimum, maximum]. I_+ and T_+ denote LI+ immigrant and LI+ total. Each row uses 20 histories. B counts any box boundary, including zero strength; Z counts zero-strength fits. Boundary counts use the native coordinates $(\eta, \log \ell, \log \psi)$; exact equality and the recorded numerical tolerance give identical counts.

Regime	Model	$\hat{\eta}$	$\hat{\ell}$ (km)	$\hat{\psi}$ (days)	B	Z
		4.00	57.76	1.329		
S4	LI	[0.00, 4.00]	[5.00, 150.00]	[0.042, 365.000]	18	1
		2.27	19.15	1.645		
	I_+	[0.02, 4.00]	[5.00, 150.00]	[0.042, 365.000]	14	0
		0.16	5.62	0.363		
	T_+	[0.00, 0.89]	[5.00, 150.00]	[0.042, 365.000]	16	1
		2.71	67.19	46.791		
S5	LI	[0.78, 4.00]	[36.45, 150.00]	[30.108, 115.698]	5	0
		2.00	54.69	31.440		
	I_+	[0.56, 4.00]	[31.15, 150.00]	[19.086, 61.939]	4	0
		0.23	95.27	139.075		
	T_+	[0.00, 4.00]	[5.00, 150.00]	[7.000, 365.000]	12	1
		4.00	28.58	0.605		
S6	LI	[0.01, 4.00]	[5.00, 150.00]	[0.042, 365.000]	18	0
		2.43	49.82	0.571		
	I_+	[0.03, 4.00]	[5.00, 150.00]	[0.042, 24.086]	16	0
		0.23	5.28	0.111		
	T_+	[0.01, 2.70]	[5.00, 150.00]	[0.042, 34.715]	12	0
		2.98	59.83	42.448		
S7	LI	[0.86, 4.00]	[28.30, 150.00]	[14.873, 155.075]	9	0
		2.19	47.43	28.309		
	I_+	[0.51, 4.00]	[26.33, 99.70]	[11.353, 170.699]	5	0
		2.14	49.92	28.705		
	T_+	[0.49, 4.00]	[26.86, 117.12]	[11.164, 182.512]	4	0

The saved narrow/short immigrant example with unit-mark LI illustrates start dependence. A scale-matched starting point reaches zero strength with a positive one-sided negative-log-likelihood derivative in strength and zero shape derivatives; those signs satisfy the local constrained optimum conditions. Another prescribed start attains a lower training objective and is selected. Thus the null-face termination is not by itself a solver defect or a global-optimality certificate. The candidate also uses unit stress marks, so matching its scales to a magnitude-scaled generator does not make it the true generating law.

A training-fitted uninhibited background

The fixed comparator cannot lower its immigrant rate when broad/long inhibition reduces the event count. A separate sensitivity now fits its background intercept on each of the 80 existing training histories, holding the trend, excitation, spatial law, mark law and realized history fixed. The saved inhibited fits and original gates are unchanged. The saved procedures are also

compared with a training-only uninhibited refit of both the background rate and branching ratio.

Writing $b = e^a$, precomputed event features u_i, v_i, background time integral A, fixed triggering exposure H and common mark score M give

$$L(b) = \sum_i \log(bu_i + v_i) - bA - H + M.$$

This objective is concave in b. Endpoint derivatives or a bracketed derivative root select the global training optimum on the declared intercept box; that value is then frozen for test. The inherited box is $a \in [-12, 0]$. All 480 phase–model checks against the historical floor scores pass. The protocol, executable scalar fit, per-history components and independent output review are supplied under `experiments/s04b_refitted_floor/` and `validation/live_results/methodological_20261106/`.

The two-parameter comparator maximizes the jointly concave objective

$$L(b, n) = \sum_i \log(bu_i + nh_i) - bA - nE_{\text{trig}} + M, \qquad (b, n) \in [e^{-12}, 1] \times [0, 0.98].$$

For each n, a monotone score equation gives the optimal b; the monotone profile derivative then selects n, including its boundary optima. The box Karush–Kuhn–Tucker conditions and independent archived-score identities are checked for all 80 histories. All 20 S7 fits select $\hat{n} = 0$. The executable two-parameter control and per-history results are in `experiments/s04b_refitted_floor/run.py` and `validation/live_results/s04b_concave_20261111/`.

Table 13.14: S04B comparator sensitivity. Cells give mean held-out gain (MCSE), in nats, against the fixed floor, training-fitted intercept, and jointly fitted (a, n) comparator. MCSE is the sample standard deviation divided by the square root of the number of histories. The two win columns count strictly positive contrasts against their respective fitted comparators. Active counts describe the saved attenuation procedure; original gates are unchanged. A gate returning the original fixed floor can beat a refitted floor without selecting inhibition.

Regime	Saved procedure	Fixed floor	Fitted intercept	Fitted (a, n)	Wins a	Wins (a, n)	Active
S4: mean $N_{\text{train}} = 86.65$, $N_{\text{test}} = 109.40$							
S4	LI	-3.9493 (1.2142)	-3.2102 (1.0419)	-2.5062 (1.0019)	6/20	4/20	19/20
S4	LI+ immigrant	-4.5674 (1.6218)	-3.8283 (1.5001)	-3.1243 (1.3910)	5/20	5/20	20/20
S4	LI+ immigrant gated	-1.3440 (1.3440)	-0.6049 (1.1378)	0.0991 (1.0364)	13/20	14/20	1/20
S4	LI+ total	-21.5766 (9.8329)	-20.8374 (9.8741)	-20.1335 (9.5417)	5/20	5/20	19/20
S4	LI+ total gated	-1.0043 (1.0043)	-0.2651 (0.8096)	0.4389 (0.7296)	13/20	14/20	1/20
S5: mean $N_{\text{train}} = 15.15$, $N_{\text{test}} = 16.90$							
S5	LI	44.3122 (1.4559)	1.0885 (0.8462)	3.3455 (1.3055)	12/20	14/20	20/20
S5	LI+ immigrant	48.3868 (0.9599)	5.1631 (0.7036)	7.4200 (1.2945)	19/20	19/20	20/20
S5	LI+ immigrant gated	48.3868 (0.9599)	5.1631 (0.7036)	7.4200 (1.2945)	19/20	19/20	20/20
S5	LI+ total	19.4993 (7.2001)	-23.7244 (7.2583)	-21.4675 (6.5238)	4/20	5/20	19/20
S5	LI+ total gated	19.3802 (7.2159)	-23.8435 (7.2806)	-21.5865 (6.5500)	4/20	5/20	18/20
S6: mean $N_{\text{train}} = 95.80$, $N_{\text{test}} = 111.55$							
S6	LI	-1.7818 (0.6357)	-1.4757 (0.6454)	-0.8626 (0.6333)	7/20	8/20	20/20
S6	LI+ immigrant	-3.2420 (1.2527)	-2.9359 (1.2674)	-2.3228 (1.0470)	9/20	9/20	20/20
S6	LI+ immigrant gated	0.0000 (0.0000)	0.3061 (0.1270)	0.9192 (0.3695)	15/20	15/20	0/20
S6	LI+ total	-14.5947 (7.8248)	-14.2887 (7.8134)	-13.6756 (7.7435)	10/20	13/20	20/20
S6	LI+ total gated	0.0686 (0.0686)	0.3747 (0.1436)	0.9878 (0.3701)	15/20	15/20	1/20
S7: mean $N_{\text{train}} = 13.30$, $N_{\text{test}} = 13.25$							
S7	LI	42.5503 (1.7274)	-0.9462 (1.5646)	-5.0575 (2.0010)	10/20	8/20	20/20
S7	LI+ immigrant	49.2902 (1.3502)	5.7937 (0.9722)	1.6824 (0.8890)	18/20	15/20	20/20
S7	LI+ immigrant gated	49.2902 (1.3502)	5.7937 (0.9722)	1.6824 (0.8890)	18/20	15/20	20/20
S7	LI+ total	53.2332 (2.1250)	9.7367 (1.7197)	5.6255 (1.0844)	18/20	18/20	20/20
S7	LI+ total gated	53.2332 (2.1250)	9.7367 (1.7197)	5.6255 (1.0844)	18/20	18/20	20/20

Table 13.15: Held-out change from the fixed to the training-fitted uninhibited floor. Cells give mean (MCSE), using the same histories as Table 13.14; $C = -E$ is signed exposure and the mark contrast is zero.

Regime	ΔP	ΔC	Δ joint
S4: mean $N_{\text{train}} = 86.65$, $N_{\text{test}} = 109.40$			
S4	-5.6558 (2.1509)	4.9167 (2.0500)	-0.7392 (0.3407)
S5: mean $N_{\text{train}} = 15.15$, $N_{\text{test}} = 16.90$			
S5	-24.4963 (0.7762)	67.7200 (0.5160)	43.2237 (0.7837)
S6: mean $N_{\text{train}} = 95.80$, $N_{\text{test}} = 111.55$			
S6	0.5534 (2.1584)	-0.8595 (2.1831)	-0.3061 (0.1270)
S7: mean $N_{\text{train}} = 13.30$, $N_{\text{test}} = 13.25$			
S7	-21.8424 (1.2469)	65.3389 (0.7709)	43.4965 (0.7825)

Refitting the intercept reduces the broad/long immigrant mean gain from 48.3868 (0.9599) to 5.1631 (0.7036) nats and reverses the misspecified S5 total procedure from 19.4993 (7.2001) to -23.7244 (7.2583) nats. In S7, freeing (a, n) further reduces the total-procedure residual

from 9.7367 (1.7197) to 5.6255 (1.0844) nats; the fully joint temporal comparison in the next subsection reverses that residual to -1.2607 (2.6293) nats. Parentheses give MCSEs. The S4 gated procedures have 13/20 fitted-intercept wins but only 1/20 active fits. S6 immigrant and total gates each have 15/20 wins, with respectively 0/20 and 1/20 active fits. These wins therefore mostly compare two uninhibited floors and are not detections of inhibition.

13.10.2 Jointly fitted temporal backbone and inhibition

Both the uninhibited comparator and the inhibited alternatives use training data only, with matched parameter boxes and eight prescribed starts in each arm. The background and excitation coordinates (a, d, n, c, p) are fitted for the comparator and refitted jointly with the inhibition parameters for each alternative. The common spatial and mark laws remain fixed. The scientific specification is retained in `experiments/s04b_joint_nuisance/protocol.json`, with numerical provenance supplied separately in `docs/result_provenance.md`.

The training-selected fits are evaluated on held-out histories, including the exact uninhibited candidate in every alternative search. The comparison refits the temporal background and excitation in both arms; a selected positive attenuation must therefore improve on a fitted temporal explanation. Numerical execution and recovery records are retained in the companion repository, in `docs/result_provenance.md`.

In the diagnostic table below, boundary pairs report lower/upper counts for the named coordinate.

All 80 histories have completed training selections and held-out scores, giving 400 procedure outcomes.[1] Both arms jointly fit the temporal backbone (a, d, n, c, p); the common spatial and mark laws remain fixed. Eight starts are resolved in every arm. The exact fitted null is an eligible state in each inhibited search. Table rows report all twenty histories in each regime; positive near-zero gains remain strict wins.

A moderately thinned total-intensity control

The S8 generator uses $e^a = 0.25\,\mathrm{day}^{-1}$, $\eta = 0.4$, $\ell = 15\,\mathrm{km}$ and $\psi = 5$ days, with the preceding temporal-trigger and mark settings. The generating law is fixed before the twenty analysis histories are drawn. It rejects 27.21% of event proposals and retains 94.7 test events on average (range $[78, 112]$). Both arms jointly fit (a, d, n, c, p) on the 365-day training window; the total-attenuation arm also fits (η, ℓ, ψ). Its search includes the fitted null as an eligible state. The fitted parameters are frozen for the subsequent 365-day test window.

Raw selected fits reach at least one box boundary in 13/20 histories: $p = 2.5$ in 7, $n = 0.98$ in 6, $\eta = 4$ in 2 and $a = 0$ in 1, with overlapping counts. S8 applies its declared 0.01-nat training-score gate; 18/20 records fail the stricter inherited 0.0001-nat score and 0.0001 scaled-gradient criterion, with maximum scaled coarse/fine gradient difference 0.03324.

[1] The two recovery histories are S5 seed 410518 and S6 seed 410615; all other histories retain their saved completed fits.

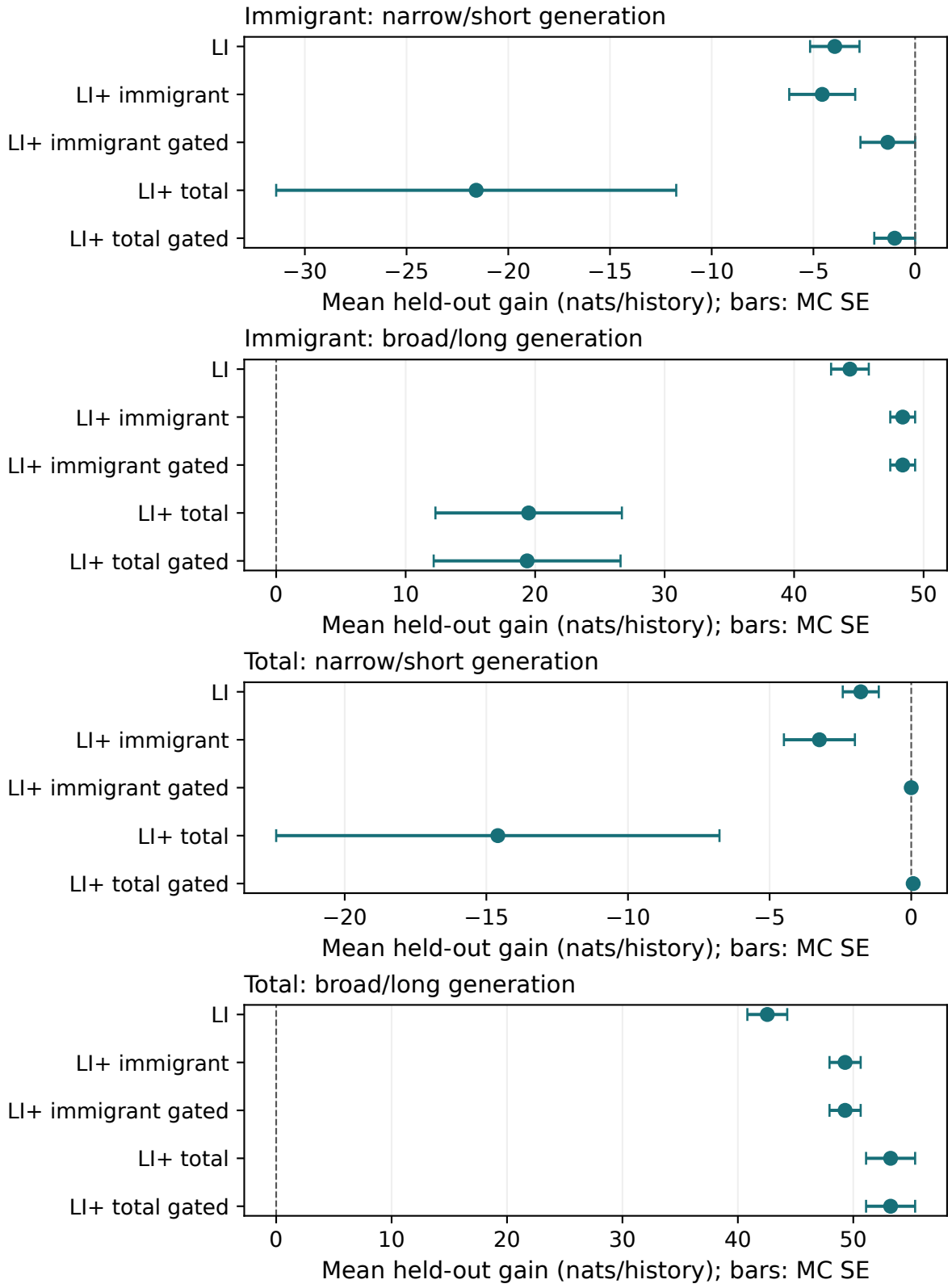


Figure 13.3: S04B matched-generator controls. Each panel keeps its own full linear range. Points show row means; bars show descriptive MC SE, not calibrated confidence intervals. Each procedure uses the same complete history denominator within its generating regime. The dashed zero line denotes the original fixed-backbone floor; regimes are not pooled. Table 13.14 gives the separate training-fitted-intercept comparison.

Table 13.16: Held-out joint-score gains against the jointly training-refitted uninhibited comparator, in nats. MCSE is the sample standard deviation divided by the square root of the number of histories. Wins are strictly positive gains; Wilson intervals quantify the win probability. Marked rows are the matched broad/long contrasts.

Regime	Procedure	Computed	Mean	MCSE	Median	Wins	Wilson interval
S4: mean $N_{\text{train}} = 86.65$, $N_{\text{test}} = 109.40$							
S4	LI (raw)	20/20	-14.8576	5.4708	-6.1519	3/20	[0.0524, 0.3604]
S4	LI+ immigrant (raw)	20/20	-9.7627	4.6732	-5.6971	6/20	[0.1455, 0.5190]
S4	LI+ immigrant (BIC)	20/20	0.0000	0.0000	0.0000	0/20	[0.0000, 0.1611]
S4	LI+ total (raw)	20/20	-41.9098	29.3814	-4.1493	4/20	[0.0807, 0.4160]
S4	LI+ total (BIC)	20/20	0.0000	0.0000	0.0000	0/20	[0.0000, 0.1611]
S5: mean $N_{\text{train}} = 15.15$, $N_{\text{test}} = 16.90$							
S5	LI (raw)	20/20	-4.3604	3.3375	-0.9636	8/20	[0.2188, 0.6134]
S5	LI+ immigrant (raw)†	20/20	-10.3701	8.5587	2.2590	10/20	[0.2993, 0.7007]
S5	LI+ immigrant (BIC)	20/20	-10.3701	8.5587	2.2590	10/20	[0.2993, 0.7007]
S5	LI+ total (raw)	20/20	-18.0129	7.4590	-3.4946	7/20	[0.1812, 0.5671]
S5	LI+ total (BIC)	20/20	-16.1057	7.4217	0.0000	1/20	[0.0089, 0.2361]
S6: mean $N_{\text{train}} = 95.80$, $N_{\text{test}} = 111.55$							
S6	LI (raw)	20/20	-12.7890	4.8712	-3.3593	3/20	[0.0524, 0.3604]
S6	LI+ immigrant (raw)	20/20	-15.7837	5.2080	-4.6434	2/20	[0.0279, 0.3010]
S6	LI+ immigrant (BIC)	20/20	0.0000	0.0000	0.0000	0/20	[0.0000, 0.1611]
S6	LI+ total (raw)	20/20	-30.4734	13.2726	-2.1916	6/20	[0.1455, 0.5190]
S6	LI+ total (BIC)	20/20	0.0828	0.0828	0.0000	1/20	[0.0089, 0.2361]
S7: mean $N_{\text{train}} = 13.30$, $N_{\text{test}} = 13.25$							
S7	LI (raw)	20/20	-30.5865	10.1921	-11.9125	5/20	[0.1119, 0.4687]
S7	LI+ immigrant (raw)	20/20	-1.8107	2.7358	-2.6127	8/20	[0.2188, 0.6134]
S7	LI+ immigrant (BIC)	20/20	-2.4696	2.6584	-2.6127	6/20	[0.1455, 0.5190]
S7	LI+ total (raw)†	20/20	-1.2607	2.6293	-1.2574	8/20	[0.2188, 0.6134]
S7	LI+ total (BIC)	20/20	-1.2435	2.6297	-1.2574	8/20	[0.2188, 0.6134]

Table 13.17: Search and numerical census for all twenty histories per regime. Boundary counts are lower/upper. Failed attempts count unsuccessful returns in the saved search records for the shared null and named alternative; raw/BIC rows reuse searches and must not be added. Near-zero is numerical, not statistical significance.

Regime	Procedure	Missing	a bounds	η bounds	Failed returns	Null gaps	Gate differences
S4	LI (raw)	0	0/4	0/6	0	0	0
S4	LI+ immigrant (raw)	0	0/1	0/6	1	0	0
S4	LI+ immigrant (BIC)	0	0/0	20/0	1	0	0
S4	LI+ total (raw)	0	0/0	0/0	0	0	0
S4	LI+ total (BIC)	0	0/0	20/0	0	0	0
S5	LI (raw)	0	0/10	0/7	0	0	0
S5	LI+ immigrant (raw)†	0	0/9	0/4	2	0	0
S5	LI+ immigrant (BIC)	0	0/9	0/4	2	0	0
S5	LI+ total (raw)	0	0/1	0/1	0	0	0
S5	LI+ total (BIC)	0	0/1	13/1	0	0	0
S6	LI (raw)	0	0/3	0/7	0	0	0
S6	LI+ immigrant (raw)	0	0/3	0/6	1	0	0
S6	LI+ immigrant (BIC)	0	0/0	20/0	1	0	0
S6	LI+ total (raw)	0	0/0	0/0	1	0	0
S6	LI+ total (BIC)	0	0/0	19/0	1	0	0
S7	LI (raw)	0	0/7	0/8	0	0	0
S7	LI+ immigrant (raw)	0	0/6	0/7	0	0	0
S7	LI+ immigrant (BIC)	0	0/5	2/7	0	0	0
S7	LI+ total (raw)†	0	0/4	0/7	0	0	0
S7	LI+ total (BIC)	0	0/4	1/7	0	0	0

Table 13.18: Floor and near-zero census for all twenty histories per regime. An alternative with zero inhibition and a better nuisance solution is a procedure and optimizer contrast, not evidence of inhibition.

Regime	Procedure	Off-floor	Floor returns	Near zero	Positive near zero	Robust positive	Robust negative
S4	LI (raw)	20	0	0	0	3	17
S4	LI+ immigrant (raw)	20	0	0	0	6	14
S4	LI+ immigrant (BIC)	0	20	20	0	0	0
S4	LI+ total (raw)	20	0	0	0	4	16
S4	LI+ total (BIC)	0	20	20	0	0	0
S5	LI (raw)	20	0	0	0	8	12
S5	LI+ immigrant (raw)†	20	0	0	0	10	10
S5	LI+ immigrant (BIC)	20	0	0	0	10	10
S5	LI+ total (raw)	20	0	0	0	7	13
S5	LI+ total (BIC)	7	13	13	0	1	6
S6	LI (raw)	20	0	0	0	3	17
S6	LI+ immigrant (raw)	20	0	0	0	2	18
S6	LI+ immigrant (BIC)	0	20	20	0	0	0
S6	LI+ total (raw)	20	0	0	0	6	14
S6	LI+ total (BIC)	1	19	19	0	1	0
S7	LI (raw)	20	0	0	0	5	15
S7	LI+ immigrant (raw)	20	0	0	0	8	12
S7	LI+ immigrant (BIC)	18	2	2	0	6	12
S7	LI+ total (raw)†	20	0	0	0	8	12
S7	LI+ total (BIC)	19	1	1	0	8	11

Table 13.19: Complete S8 held-out contrasts against the jointly fitted uninhibited comparator, in nats per history. Both rows use twenty histories with 94.7 test events on average. The later frozen-fit BIC sensitivity retains attenuation when the fine training gain exceeds $1.5 \log N_{\text{train}}$ and returns the exact fitted null otherwise. MCSE uses the sample standard deviation of all twenty gains, including floor-return zeros, divided by $\sqrt{20}$. Wilson intervals concern strict-positive win probability.

Selection	Mean	Median	MCSE	Wins	Active	Wilson interval
Raw	+0.2941	+2.5415	2.7773	13/20	20/20	[0.4329, 0.8188]
BIC	+2.4946	+0.0000	1.4436	9/20	12/20	[0.2582, 0.6579]

All twenty training fits select a positive attenuation amplitude. Individual test contrasts range from -38.2652 to +16.8335 nats, with 7 negative contrasts. The raw mean gain is smaller than its sample Monte Carlo standard error, and the win-probability interval includes 1/2. The training mesh is refined until selected score doubling agrees within 0.01 nat. Selected likelihoods are checked by doubling positive Gauss–Legendre quadrature; the largest held-out change is 0.000028 nats. The twenty histories and generating law are supplied in `reproduction/synthetic/s8_moderate_total/`; complete fits, CPU score checks and the aggregate are in `validation/live_results/s8_moderate_complete_20261111/`.

Oracle ceiling and estimation loss

An oracle that knows the generating attenuation parameters but not the nuisance backbone achieves a mean held-out gain of 18.11 nats over the jointly fitted null (median 12.86, MCSE 4.18). Against a fixed true-backbone null the mean is 16.43 nats, and against an intercept-refitted null it is 14.54 nats. The fitted procedure captures only 0.294 nats on average; its mean estimation loss relative to the oracle is -17.82 nats (median -8.10, MCSE 6.08). The bottleneck is parameter estimation, not the absence of a true generating effect.

Contrast	Mean	Median	SD	MCSE	Wins
True vs fixed true-backbone null	16.429	12.819	8.176	1.828	20
True vs refitted-intercept null	14.541	11.630	8.187	1.831	20
True vs jointly fitted null	18.112	12.865	18.700	4.182	19
Fitted inhibited minus true	-17.818	-8.097	27.204	6.083	2
Fitted inhibited minus fitted null	0.294	2.541	12.420	2.777	13

Table 13.20: S8 oracle ceiling and estimation loss over the twenty histories, in nats. The true law uses the S8 moderate attenuation parameter and the same fixed mark and spatial laws as the fitted procedure. The jointly fitted null is the actual comparator; the fixed and intercept-refitted nulls separate nuisance adaptation from attenuation. MCSE is the sample standard deviation divided by the square root of the history count.

13.11 Observed California fixed-backbone inhibition

The observed comparison uses the same three California folds and 315 targets. Background and excitation nuisance inputs are fixed and shared between each candidate and its exact uninhibited floor. The common mark term cancels from each held-out joint gain, reported as floor NLL minus selected NLL.

Original LI is retained unpenalized. Magnitude-scaled LI+ is evaluated with immigrant-only and total-intensity attenuation, each with raw training selection and with the declared training-only BIC-style gate. A gate may select the exact floor. A negative held-out gain is retained, while an unresolved numerical result remains unknown rather than being assigned zero.

The BIC-style gate is a complexity penalty, not a Davies-calibrated test or a guarantee of nominal size. A training-active fit, a selected floor and a held-out win are distinct outcomes. These observed folds have been inspected retrospectively.

Table 13.21: Held-out joint gains (nats) relative to the shared uninhibited floor. Displayed zeros select the nested $\hat{\eta} = 0$ training boundary: their algebraic forecast contrasts are exactly zero, rather than estimates of a zero physical effect. Paired block intervals for raw total-intensity attenuation are reported in Section 13.11.1.

Fold	Model	Gate	Status	Gain	Win	Active
0	LI	raw	resolved	0.0000	no	no
0	LI+ immigrant	raw	resolved	0.0000	no	no
0	LI+ immigrant	BIC	resolved	0.0000	no	no
0	LI+ total	raw	resolved	-0.2245	no	yes
0	LI+ total	BIC	resolved	0.0000	no	no
1	LI	raw	resolved	0.0000	no	no
1	LI+ immigrant	raw	resolved	0.0000	no	no
1	LI+ immigrant	BIC	resolved	0.0000	no	no
1	LI+ total	raw	resolved	-1.9529	no	yes
1	LI+ total	BIC	resolved	0.0000	no	no
2	LI	raw	resolved	0.0000	no	no
2	LI+ immigrant	raw	resolved	0.0000	no	no
2	LI+ immigrant	BIC	resolved	0.0000	no	no
2	LI+ total	raw	resolved	-0.2093	no	yes
2	LI+ total	BIC	resolved	0.0000	no	no

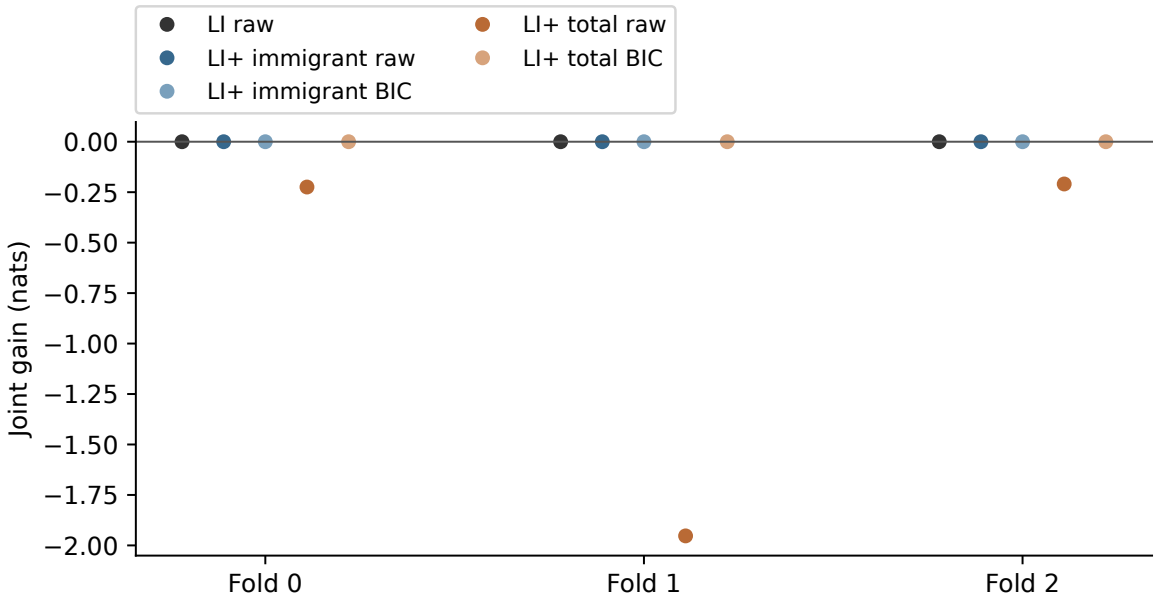


Figure 13.4: Held-out joint gains on the fixed three-fold population.

Fold	Training targets	Planned held-out targets
0	851	79
1	930	167
2	1097	69

All original LI and magnitude-scaled immigrant-only fits select their exact uninhibited floor. Raw total-intensity inhibition is active but loses held-out joint score in each fold. Its BIC-gated version selects the floor in every fold. These outcomes do not establish a predictive benefit from the revised inhibition mechanisms on this fixed observed population.

13.11.1 Uncertainty of the raw total-intensity contrasts

We apply the same paired calendar-block calculation to the raw total-intensity LI+ score minus its matched floor. The event sums and exposure integrals use identical calendar boundaries. The original log-Gauss event-gap map is retained; doubling its temporal order from 32 to 64 changes each block exposure by less than 10^{-11} nats on the original spatial grid.

Fold	Contrast (nats)	Paired 95% interval	Nominal sign-flip p
F0	-0.224	$[-0.787, 0.074]$	1.00000
F1	-1.953	$[-5.945, 0.143]$	0.47990
F2	-0.209	$[-0.458, -0.013]$	0.08712
Pooled	-2.387	$[-6.564, -0.033]$	0.07678

Table 13.22: California raw total-intensity LI+ minus its floor: paired 90-day calendar-block uncertainty.

The first two fold intervals include zero. The third excludes zero with 90- and 180-day blocks but includes it with annual blocks, $[-0.488, 0.020]$. The pooled percentile intervals are negative at all three block lengths: $[-6.564, -0.033]$, $[-6.426, -0.085]$, and $[-6.364, -0.090]$ for 90, 180, and 365 days. Corresponding pooled sign-flip values are 0.07678, 0.04560, and 0.06578. The pooled percentile intervals support a small negative contrast, while the sign-flip inference is sensitive to block length.

Part IV

Synthesis and Programme Boundaries

Chapter 14

Consolidated Discussion, Limits of Inference, and Conclusion

Across the completed comparisons, no statistically detectable held-out improvement from diagnostic attenuation is established against excitation baselines that fit the background and triggering nuisance parameters. Negative contrasts and nested zero-strength selections have different meanings: the latter return the excitation forecast algebraically and do not measure the power of an interior-effect test. Large spatial gains mainly correct coarse routing and uniform cell-lift representations. The unweighted inverse-power benchmark (EP) has higher scores than the background-weighted recipe (OC) in every California and Japan primary fold, with a reversal in the 2010 El Mayor–Cucapah episode backtest. These are comparisons of recorded-data forecast procedures under the common scope in Section 6.1.

Physical stress-shadow evidence is heterogeneous. Rate-and-state calculations can explain suppression and apparent exceptions (Harris and Simpson, 1998), while observations also motivate dynamic-triggering interpretations (Felzer and Brodsky, 2006) and expose difficulties resolving predicted decreases (Mallman and Zoback, 2007). The nonnegative radial attenuation fields tested here do not calculate those signed fault-stress histories. Proper logarithmic scoring concerns expected predictive reward on a specified observation measure (Gneiting and Raftery, 2007); a realized ranking does not identify a physical mechanism or establish conditional adequacy. Residual and simulation checks address that separate question (Bray and Schoenberg, 2013).

14.1 What the theory separates

The mathematical results delimit several distinct regimes. Finite positive-mode systems admit the stated first-moment stability conclusions under their link, mark and subcritical-triggering assumptions. Finite-horizon approximation does not establish an interchange of stationary and infinite-mode limits. Exact-memory counterexamples apply to specified no-feedback submodels, and the linear-response calculation concerns an auxiliary frozen-slope system. Full-law identifiability also requires assumptions stronger than those available in an observed finite catalogue. The preceding mathematical chapters state their assumptions and proofs. The new comparison in Section 8.7 directly fits an autonomous finite-mode subfamily satisfying the first-moment hypotheses. Its early and late folds select positive inhibition and lose held-out score; the middle fold selects $\eta = 0$ and is tied at displayed precision. This connects the stability model to an executed fitted comparison while keeping stability and predictive advantage distinct.

This separation matters for the empirical discussion. A physical model may contain loading or initial-stress memory while a finite catalogue lacks power to distinguish it from a restricted alternative. A diagnostic family may improve its training score because its likelihood geometry favours extra structure, without thereby identifying the full physical law. A small held-out loss may be numerically controlled yet limited as statistical evidence when folds overlap or were inspected. Conversely, a non-rejection is not evidence of equality. Failure to distinguish

two recipes leaves a set of explanations: the data may be insufficient, the candidate may be misspecified, the optimizer may be imperfect, or the tested effect may be small over the observed horizon. The controlled studies weaken some specific explanations, but do not select a unique physical one.

At $\eta = 0$, a separately varied stress kernel disappears from the diagnostic intensity when background and triggering are held fixed. It is therefore unidentified there. The analogous physical loading argument requires finite loading and initial-stress parameters: it cannot be carried into a singular small-feedback, large-loading limit without additional analysis. Non-rejection of a shared-kernel restriction at or near zero feedback consequently does not validate equality of two physical memories.

Synthetic diagnostics further clarify, but do not erase, the inferential gap. The historical comparison with nuisance quantities supplied produced 150 detections in 150 trials, whereas a fitted setting produced 57 in 150. These are different simple-versus-composite inferential tasks; their contrast is not an intrinsic information ceiling, and it is not proof that the remainder is avoidable algorithmic inefficiency. It offers no justification for fixing the real background merely to recover apparent power. Better optimization or a larger synthetic bank can improve numerical precision without changing the scientific question.

The transformed marks add another qualification. The magnitude excess has an exponential law, while the stress and productivity transforms have finite first and infinite second moments at the declared generating parameters. That does not prevent bounded-indicator Monte Carlo summaries, but it invalidates unqualified use of Fisher-information shortcuts based on second moments. Computational stability and statistical efficiency therefore have to be defended under the declared scoring rule, not inferred from formal differentiability alone.

14.2 What the temporal and mark extensions resolve

The magnitude-history and productivity studies retain negative matched results. A class-history law can change conditional mark probabilities and, through joint fitting, timing parameters without providing additional held-out information. Its effective-scale algebra concerns the fixed observed marked history; it does not establish equality of unconditional multistep future laws. Free productivity likewise changes the fitting problem and its finite-moment regime. Its failed continuation gate is part of the result, not an omitted successful analysis waiting to be inferred.

Exposure orthogonalization separates numerical geometry from the represented family. Better Gram conditioning is useful, but an interior optimum that maps feasibly into both coordinate boxes need not give a better predictive model. The negligible score change in Stage A is therefore consistent with the algebra. Its coarse Omori bridge must be distinguished from Extension O's continuously tuned temporal law. The latter improves excitation yet leaves matched and jointly tuned inhibition losing. The national temporal loss cannot be attributed solely to having fixed the original ML triggering kernel.

Common free ML time scale improves its own temporal comparison. Parent-class time scales add an inconsistent increment and fail the registered continuation rule, leaving observed association fitting unexecuted. This does not erase the earlier completed class-history experiment: the two branches have distinct references, parameters and gates. Every claim of nonexecution in this monograph refers to its specified branch.

14.3 What localization changes, and what it does not

Regional nonclosure gives a mathematical reason to test local structure: equal pooled depletion can conceal different sums of local rates. The completed progression in Chapter 10 supplies the empirical comparison that the earlier manuscript proposed. It moves from zone-local excitation to redistribution, distance routing, projected continuous density and a factorial with stronger immigrant geography. These changes improve particular location scores, but incremental inhibition remains negative, boundary-selected or tied. A reduced whole-model penalty after localization must not be confused with a positive inhibition-minus-matched-excitation contrast.

Continuous geography retains its advantage over the cell-lift representation under the richer background. Location dominates that benefit. Temporal and background contrasts still contain losses, including changes whose pooled joint sign differs from their timing sign and whose fold signs differ from their pooled sign. In these comparisons, normalization over the observation region leaves regional intensity unchanged at fixed shared parameters; timing changes after refitting do not imply that a normalized spatial density directly creates regional events. Nor can gains from separate historical stages be concatenated as if they were one fitted comparison.

The selected spatial scales describe a bounded fitting procedure, rather than identified physical interaction distances. Section 6.1 states the common recorded-coordinate and observation-law assumptions.

14.4 Transfer, competition and conditional adequacy are different questions

California retains the primary matched temporal and location directions, and the fitted target beats its no-trigger reference. EP nevertheless wins the joint comparison in every original primary window, chiefly through conditional location. The higher-magnitude sensitivity and annual/sequence accounting retain their separate denominators and do not convert retrospective inspection into untouched confirmation. The Ridgecrest active partition is a predetermined-in-form retrospective time union; it is not a causal assignment of individual aftershocks. These regional score conclusions retain the original fitted states and have now been checked by the declared selected-state score replays and table reconciliation; the original fitting searches and audit scopes remain separately identified.

The magnitude-scaled benchmark in Section 11.5.2 changes four ingredients together: magnitude-scaled offspring distance, training-declustered background weighting, joint temporal–spatial fitting, and full-plane normalization with regional leakage instead of EP's observation-rectangle normalization. The combined gain is 69.55 total held-out nats, and all three baseline spatial scales reach their 2.0 km lower search bound. Both laws have units $\mathrm{day}^{-1}\,\mathrm{km}^{-2}$ and are scored on the same time–region domain; the in-box mass of the full-plane kernel enters its compensator. This is a valid proper-score comparison, but it confounds all four changes and a boundary scale estimate. Selecting $\hat{\eta} = 0$ in every attenuation fit returns the matched baseline exactly. It does not establish zero physical inhibition or calibrated detection power.

The block analysis in Section 8.4.1 separates realized score signs from uncertainty under the chosen partitions. The second New Zealand fold's interval includes zero at 90, 180 and 365 days; the third includes zero at 90 days but excludes it at the two longer lengths. The 61 slowly decaying modes of the 195-mode New Zealand kernel and sequence clustering across

months and years make calendar blocks dependent, rather than independent exchangeable sequence replications. The small nominal sign-flip values are not calibrated long-memory significance levels. California's pooled OC–EP interval and the small LI+ total percentile intervals are negative, while sign-flip references vary with block length. Prioritize multi-scale interval sensitivity and fold consistency; neither the interval quantiles nor independent sign reversals eliminate the unmodeled dependence or parameter-estimation uncertainty.

Japan transfers the continuous-versus-lifted location direction but reverses the Omori-versus-fixed-ML timing direction in every scored window. The combined descriptive transfer rule therefore fails. EP again wins jointly. A statement that "the improvement transfers" without naming the component and reference would obscure a central negative finding. The explicit complete-primary selection rule also limits the population to which the JMA comparison applies; excluded unresolved records are not silently imputed.

The frozen OC conditional simulations ask another question. Material count, gap, spatial and subblock discrepancies survive the complete identity/count/mark/exposure audit. Tied diagnostic values require insertion intervals, particularly for empty cells; a single randomized rank would hide that ambiguity. The full raw replay has narrower path coverage than the aggregate audit. These diagnostics show material conditional inadequacy of OC across the three origins. Deterministic immigrant exposure is an accounting coordinate, not a random calibration test, and total/trigger exposure ranks are redundant. EP's log-score victory stands on its own; Section 12.3 shows fixed-state discrepancies for EP too (Schorlemmer et al., 2007; Zechar et al., 2010).

The completed matched magnitude-history study (Section 9.16) separates frequent small positive fitted gains from evidence calibrated against its empirical null bank. Matching productivity reduces the mean GLR relative to the misspecified candidate, but positive selection remains common. Exceedances of the same-bank empirical threshold do not establish independent Type I error control or a physical explanation.

The four-arm spatial comparison (Section 11.11) addresses a different contrast from the earlier coarse cell-lift checks. EP retains a regional joint advantage over OC in California and New Zealand, chiefly through location. Fold contrasts vary; the comparison neither establishes universal EP superiority nor causally decomposes tail shape and background effects.

The expanded California backtest (Section 11.12) has a distinct retrospective episode roster and variable procedure contrasts: EP does not win every episode. Its causal training prefixes do not make the inspected roster prospective or independent. In Japan, fixing the background trend at zero reduces the pooled OC–FC timing deficit but leaves every fold negative. The trend ablation therefore does not remove the reversal or rescue the combined descriptive transfer rule.

14.5 Continuous local inhibition and the zero-strength boundary

The LI extension attenuates only the immigrant contribution without spatial renormalization. The field uses unit stress marks and all eligible strictly earlier recorded parents. Every eligible event therefore contributes the same inhibition amplitude regardless of magnitude, while triggering is unattenuated. This is a restrictive diagnostic of immigrant attenuation, not a physical stress-shadow model or whole-intensity suppression. At zero strength it nests OC exactly, while range and memory disappear from the law.

All selected observed California LI states lie at that boundary. The tiny retained positive floating-point difference from OC is a numerical tie, not a predictive gain. The separately

archived OC–EP and LI–EP values retain distinct source identities even though their displayed rounded values agree. Range and memory display conventions at zero do not identify physical scales. EP's better score remains a competitive result rather than a calibration certificate.

Every fixed-seed regime matters. In the historical initial eight-dataset study, all four null controls lose held-out score, including a loss of 148.65 nats. This shows that nesting the null in training does not protect future performance: estimation variability and overfitting remain plausible. Held-out losses alone do not establish an optimization defect or estimate a false-positive rate. The original narrow/short histories realize no thinning despite positive generating strength; their losses do not test recovery of a realized thinning effect. Broad/long positive examples demonstrate success under those retained laws, with small test counts. The regimes have different laws and denominators; an aggregate success percentage would not estimate a meaningful common power. No inconvenient seed, cap or recovery history is replaced by a more favourable outcome.

The expanded matrix in Section 13.10 separates original LI, magnitude-scaled immigrant attenuation and total-intensity attenuation. Under the original unit-mark, immigrant-only generating law, magnitude-scaled immigrant-only inhibition has a positive mean gain in the broad/long regime, whereas total-intensity attenuation has a negative mean there. Both LI+ procedures are alternatives to that generating law, so this comparison does not measure their power under matched LI+ generation. The matrix reports descriptive Monte Carlo summaries by regime. Gating selects the floor in both null settings and in the narrow/short bank, where every path coincides with its same-seed uninhibited counterpart under the recorded stream coupling. That pathwise identity does not equate the generating laws or measure sensitivity to realized thinning.

The matched-generator extension in Section 13.10.1 shows why the uninhibited comparison must be fitted. The narrow/short immigrant bank rejects only one proposal across 20 histories. Broad/long immigrant attenuation rejects 83.4% of immigrant proposals but 78.2% of all proposals; broad/long total attenuation rejects 82.5% of all proposals, including 98.2% of offspring. Keeping these denominators explicit avoids overstating realized thinning. Intercept refitting changes the misspecified total-attenuation contrast on immigrant-thinned histories from $+19.50$ to -23.72 nats. On total-thinned histories, fitting the uninhibited intercept and branching ratio reduces the residual from $+9.74$ to $+5.63$ nats, with all 20 fitted branching ratios zero. Table 13.14 supplies the event counts and sample MCSEs. Gated wins in narrow/short settings mostly compare the old fixed floor with the refitted floor, rather than active attenuation.

Joint fitting in both arms gives a broad/long immigrant mean of -10.37 nats, median $+2.26$ and sample MCSE 8.56, with 10/20 wins and 16.9 mean test events. Broad/long total attenuation gives mean -1.26 nats, median -1.26 and MCSE 2.63, with 8/20 wins and 13.25 mean test events. These arithmetic signs do not establish well-powered negative conclusions: they admit a fitted temporal explanation for the count reduction while retaining considerable simulation-to-simulation variability.

The controls leave a gap between weak realized attenuation and strong evidence for recovery. The unit-mark narrow/short paths reject no immigrants, and the magnitude-scaled bank rejects one; the broad/long banks leave only 13.25–16.9 mean test events and nearly eliminate branching in the total-thinned regime. S8 instead rejects 27.2% of proposals and retains 94.7 mean test events, but its 20-history raw mean of $+0.294$ nats is small beside sample MCSE 2.777 (13/20 wins and 13/20 box-boundary fits). Applying the inherited training-BIC rule to the frozen fits gives a sensitivity mean of $+2.495$ nats, median 0 and MCSE 1.444, with 9/20 wins and 12/20 active fits. This gated endpoint was not declared in the executed S8 reporting protocol. The empirical MCSEs summarize this finite bank; they are not a

Gaussian mean-inference guarantee under the heavy-tailed generating laws. These regimes neither supply uniform detection power nor turn nested boundary selection into evidence of physical absence.

The completed magnitude-scaled observed comparison (Section 13.11) retains the same conclusion: immigrant-only LI+ selects no inhibition in every fold. Raw total-intensity attenuation selects positive inhibition but loses held-out joint score in every fold; its complexity-gated version returns to the exact uninhibited floor. This fixed-backbone comparison addresses the revised diagnostic procedures, not a general test of physical stress shadows.

The completed conditional comparison (Section 12.3) retains discrepancies for both OC and EP under the two continuous-statistic omnibus procedures at each origin. The retained minimum-rank procedure does not reject at the declared level. Better comparative log score and conditional adequacy are therefore distinct: EP wins important score comparisons without thereby passing these diagnostic checks. Fitted states are held fixed, and no joint cross-origin null or parameter-uncertainty integration is claimed.

14.6 Interpretation of the completed comparisons

The New Zealand programme adaptively reused the same 612 test events: earlier test scores informed later model design and parameter bounds. Freezing each stage did not erase this dependence (Dwork et al., 2015). Section 6.1 consolidates the consequences for recorded-data scope, dependent uncertainty, nested boundaries and retrospective reuse. Finite multistart searches and computational reconstructions support their stated numerical checks; they do not prove global optimality or a calibrated absence test.

The literature classifications in Appendix A.7 identify inherited triggering kernels, established stability machinery and restricted model-specific derivations. The contribution is the stated analysis and the executed procedure comparisons.

14.6.1 Minimum detectable effects and the power of the negative result

The regional score scales differ sharply. At 90-day blocks the two-sided 95 per cent minimum detectable effect is 9.24 nats in New Zealand, 48.30 nats in California and 15.98 nats in Japan. The 0.0077 nat-per-event New Zealand scale is much finer than the 0.0782 nat-per-event California scale because California's block scores are more dispersed. The completed EP+ gain of 69.553 nats is of the same order as one California 90-day MDE, while the S8 raw mean of 0.294 nats is far below its own MCSE. These calculations delimit what the negative and positive summaries can detect; they do not convert a non-significant contrast into evidence of physical absence.

14.7 Conclusion

Mathematical legitimacy, identification, estimation, competitive prediction and calibration are distinct achievements. The finite-mode and finite-history results provide a restricted foundation; fitted nuisance parameters, stronger benchmarks, calibration and transfer constrain its empirical reach. The tested inhibition additions do not establish predictive success on the observed catalogues. This does not imply physical absence. Positive ablations, negative contrasts, boundary selections and completed failed gates remain part of that assessment.

Region	Block (d)	SE_B (nats)	MDE_{95} (nats)	MDE_{95}/event	Required δ (%)
New Zealand	90	4.71	9.24	0.0151	17.4
New Zealand	180	5.42	10.62	0.0174	18.6
New Zealand	365	6.65	13.04	0.0213	20.6
California	90	24.64	48.30	0.1533	55.4
California	180	24.17	47.38	0.1504	54.8
California	365	26.96	52.84	0.1678	57.9
Japan	90	8.15	15.98	0.0125	15.8
Japan	180	8.15	15.98	0.0125	15.8
Japan	365	8.15	15.98	0.0125	15.8

Table 14.1: Minimum detectable effects at two-sided ninety-five per cent for regional total held-out scores. The final column is the fractional rate perturbation $\delta = \Delta\lambda/\lambda$ satisfying a quadratic small-perturbation score approximation. New Zealand and California use paired calendar-block scores; Japan uses the between-window standard error of three annual fold totals, repeated at each block label and not a calendar-block bootstrap. Contiguous dependent blocks are descriptive dependence-scale units, not independent replications.

Appendix A

Archival Source Map, Notation, Score Semantics, and Literature Classification

A.1 What the manuscript and its companion files reproduce

The internal project and addendum records are retained in `paper/supplements/project_source_records.bib`, as source-availability metadata rather than peer-reviewed bibliography entries. The exact consulted external preprint and its later journal publication remain distinct because the text cites version-specific equations.

The companion source builds this unified manuscript from retained LaTeX, native conversions and saved figures. The original compact-release availability statements below describe the historical baseline; the present delivery additionally embeds the sources, figures, claim ledger and authority copies needed for its editorial rebuild and reconciliation. The document build itself does not rerun historical scientific studies. Rebuilding a plot from a retained table is different from regenerating its histories, refitting its models and recalibrating its test. Each derived data file and figure records its source in the figure manifest. Illustrative mathematical figures are labelled as such; they are not synthetic catalogue outcomes.

The earlier 40-page manuscript is not redistributed in this public repository. The reconstructed New Zealand, California and Japan catalogue records are now supplied under `reproduction/`, together with the raw provider inputs, explicit selection rules and portable cohort-reconstruction commands. The catalogue providers, access locations, original snapshot hashes and terms are documented in `docs/data.md`. Reacquiring a live catalogue does not guarantee recovery of the historical snapshot. The earlier manuscript's frozen catalogue, original fitting code, complete optimization records and editable source remain unavailable.

The principal data snapshot was retrieved on 15 October 2026 from GeoNet's WFS interface. Its bundled selected CSV is `reproduction/nzca/expected/geonet_M5_shallow_1960_2026.csv`, with SHA256

```
feacccb7581d1c4c33f88874cd6b198c41639da6930a7fc35e46f910a594fb3e.
```

The saved metadata defines the exact geographic, depth, magnitude, event-type and clock rules. The 1960-origin observation window is 24,394 days, with 15 October 2026 excluded. The archived FDSN selection contains 914 events: all 913 primary IDs and one additional ID. These are two interfaces/snapshots from one provider, not independent physical catalogues.

A.2 Canonical authorities for the main results

Table A.1 gives the archive-relative entry points needed to inspect the main claims. Narrative summaries point to their underlying machine-readable outputs. A source report establishes

what was reported and retained; a hash establishes byte consistency. Neither alone certifies a physical mechanism or every historical execution.

The public availability register records the historical compact-release subset and references to the research archive. The historical source index is retained as metadata in historical source index; its old included paths describe release v4, not this repository. Current recovered inputs and executable replays are indexed separately in `reproduction/README.md`; current payload membership is recorded in `MANIFEST.json`. The earlier manuscript PDF remains outside this distribution.

Table A.1: Canonical source map. Display names identify archive-relative authorities; stable IDs map to the full paths in the source index. This is not a promise that every historical payload is embedded in a compact manuscript release.

Topic	Authority and scope
Finite-mode analysis	A_finite_mode_audit.md; accompanying `S21`: A_theorems.tex. State assumptions, construction, coupling and approximation scope.
Exact memory and response	B_memory_response.md; `S22`: B_propositions.tex. No-feedback obstructions and the auxiliary response are separate results.
Identification and score bounds	`S23`: C_identifiability.md; `S03`: restricted_score_bound.md; `S06`: power_response_corollary.md. Positive-feedback and fixed-rule assumptions apply locally to each proposition.
Amplitude profiling	`S12`: A_amplitude_proposition.md; `S13`: background_balance_corollary.md. Conditional negative-log-likelihood calculus, not a global optimizer theorem.
Reconstructed catalogue	`S24`: catalogue_metadata.json; `S25`: research_report.md. Archived new snapshot, selection and independent reconstruction.
Known-law comparison	`S09`: completed_evaluation.md; completed_results.json. All 2,149 scores accepted, including five validated imports; no new histories for this comparison.
Background comparison	`S11`: research_report.md; completion_summary.json. Accepted completion of the supplied-background and estimated-background procedures.
Optimizer robustness	`S14`: research_report.md. Ten selected difficult histories; the earlier preparation and failed strict branches are not substituted for completion.
Raw ablation	`S15`: research_report.md. Ten-history level/trend raw-statistic comparison, not calibrated power.
Level-only calibration	`S16`: research_report.md; `M09`: summary.json. All 2,149 fresh fits and accepted matched-null decisions.
Observed prediction	`S19`: research_report.md; `M18`: summary.json. Three folds, five declared families, 612 scored test events.
Heterogeneity and source sensitivity	`S20`: research_report.md; summary.json; A_source_comparison.json. Fixed-parameter temporal decomposition and matched source-value sensitivity.

A.3 Stable source supplement identifiers

The identifiers below retain the identity of historical documentary sources. Their exact original files are supplied in `reproduction/historical_fits/historical_fit_package.zip`; `reproduction/historical_fits/source_availability.json` maps each identifier to its archive member and hash. These documentary source identifiers are distinct from the later follow-up study identifiers. The mathematical arguments are given in Chapters 2–5. The public repository also retains `paper/supplements/explanatory_supplement.pdf`. Archive recovery supplies existing evidence and does not constitute a new scientific execution.

Table A.2: Stable documentary source keys, display filenames and public status. Full original archive-relative paths are retained in the source index for provenance.

Key	Archived document
S01	summary.json. *Recovered original*
S02	reference_score_geometry.md. *Recovered original*
S03	restricted_score_bound.md. *Recovered original*
S04	stable_process_irregular_score.md. *Recovered original*
S05	positive_eta_response.md. *Recovered original*
S06	power_response_corollary.md. *Recovered original*
S07	power_engine.py. *Recovered original*
S08	research_report.md. *Recovered original*
S09	completed_evaluation.md. *Recovered original*
S10	restricted_history.py. *Included*
S11	research_report.md. *Recovered original*
S12	A_amplitude_proposition.md. *Recovered original*
S13	background_balance_corollary.md. *Recovered original*
S14	research_report.md. *Recovered original*
S15	research_report.md. *Recovered original*
S16	research_report.md. *Recovered original*
S17	adequacy.py. *Included*
S18	observed_frozen.json. *Recovered original*
S19	research_report.md. *Recovered original*
S20	research_report.md. *Recovered original*
S21	A_theorems.tex. *Recovered original*
S22	B_propositions.tex. *Recovered original*
S23	C_identifiability.md. *Recovered original*
S24	catalogue_metadata.json. *Recovered original*
S25	research_report.md. *Recovered original*
S26	SUPERVISOR_REVIEW_ADDENDUM.txt. *Recovered original*
S27	research_handoff.md. *Recovered original*

A.4 Unified study-to-chapter map

The pinned library contains the baseline, Addenda I–V and the following completed follow-up strands. Their authoritative JSON remains in the named upstream runs; the ledger records exact files, fields and SHA-256 identities. The withdrawn F addendum is retained in the historical source archive; Addendum II in Chapter 9; Addendum III in Chapter 10; Addendum IV in Chapters 11–12; and Addendum V in Chapter 13. The section-level crosswalk accompanies this edition.

Table A.3: Complete follow-up roster and integrated destination. Dates are run identifiers, not independent evidence counts.

Pinned strand	Chapter
california_schema_repair_20261025T044000Z	11
competitive_benchmarks_20261025T065609Z	11
conditional_calibration_20261025T085151Z	12
kernel_and_continuous_space_20261024T183658Z	9 and 10
local_inhibition_gpu_resume_20261028T172954Z	13
magnitude_history_20261023T205716Z	9
magnitude_prediction_20261023T222049Z	9
multitype_temporal_decay_20261024T173017Z	9
orthogonal_stress_and_kernel_20261024T082952Z	9
productivity_slope_20261024T023501Z	9
score_robustness_20261025T102500Z	11
sequence_score_audit_20261025T193331Z	11
spatial_continuation_20261024T102859Z	10
spatial_local_inhibition_20261024T094130Z	10
temporal_spatial_background_20261024T211225Z	10
third_region_transfer_20261026T001254Z	11

A.5 Score-field custody

Section 2.6 defines the canonical $P, Q, E, C = -E, M$ notation. Historical temporal and LI raw `E` fields denote event-logsum differences; sequence `timing_compensator` denotes a signed difference in nonnegative exposure. The pinned field dictionary is preserved with its source identity. No archived key is renamed in the authority copy, and no sign is inferred from a key name alone. The source ledger distinguishes raw values, derived differences and displayed rounding.

A.6 Historical theoretical notation and scope

Table A.4: Core notation. Individual propositions specify any additional conditions and distinguish the physical and diagnostic uses.

Symbol	Meaning
T_i, M_i	Event time and fresh magnitude/mark; predictable intensity at T_i uses strictly earlier events.
Φ_s, h_e	Stress relaxation and triggering density. A shared restriction may relate them, but they are varied separately when testing identification.
X	Latent physical stress, including specified loading, initial state and event depletion.
$B(t)$	Independently specified diagnostic background; it may be nonstationary and is not the physical loading closure.
$G_s(t), H_e(t)$	Marked diagnostic depletion and normalized triggering features based on past history.
η	Nonnegative diagnostic inhibition coefficient; it is not automatically the original physical stress parameter.
n	Physical mean productivity $n = \mathbb{E}_F\kappa(M) < 1$; in the diagnostic convention $\mathbb{E}_F\kappa(Q) = 1$ and n is the coefficient outside H_e.
ℓ, F	In Chapter 5, log-likelihood and its negative; F elsewhere denotes the mark law and ℓ the auxiliary response slope. Local definitions fix these uses.
$A = e^a, p_i$	Background amplitude and fitted background mixture weight at event i. The weights are not identified causal parent labels.
$a_{\mathrm{rel}}, \beta_{\mathrm{rel}}$	Auxiliary-response coefficient $(1-n)\tau^{-\alpha}$ and exponent $1-\alpha$; a remains the background intercept.
γ	Feedback gain $m_1\ell$ in the auxiliary response; the stress-mark exponent is separately defined in the generating law.
α	Fractional relaxation index where used, with its permitted range stated in the relevant model.
β	Excess-mark rate in the exponential mark model; the Lyapunov weight in Theorem 3.1 is a separate local use. A plug-in mark-rate estimate is not automatically a joint marked-intensity MLE.

A.7 Focused literature comparisons

The contribution classifications distinguish inherited components, applications of established theory and model-specific derivations. Each mathematical result is stated and proved under its own hypotheses in the main chapters.

Table A.5: Contribution classifications and primary-source context.

Result	Classification and source comparison
Mittag–Leffler triggering	**Inherited component / application**. Davis et al. (2024a), equations (11)–(12), give the derivative-related triggering kernel and marked productivity.

Result	Classification and source comparison
Shared depletion and triggering	**Model specialization; no priority claim**. Petrillo et al. (2023), equations (10)–(11), uses an additive permanent correction; the present restriction multiplies a positive background by relaxing depletion and sets $h = -\Phi'$. An equation difference does not establish firstness.
Theorem 3.1	**Application of weak Harris theory**. The model-specific construction, drift and coupling verify the hypotheses of Hairer et al. (2011), Theorem 4.8 and Corollary 4.11.
Proposition 3.2	**Self-contained moment derivation**. A recent-jump truncation argument identifies the divergent second moments while retaining first-moment stability; no priority claim.
Propositions 3.3–3.4	**Coupling derivations / scope correction**. Quantitative half-line control and qualitative local truncation are distinct. The correction does not establish an invariant infinite-mode limit.
Proposition 3.5	**Compensated shot-noise specialization**. The displayed characteristic-function and moment arguments apply to the stated no-feedback submodel; no general nonlinear or novelty claim.
Proposition 3.6	**Self-contained linear specialization**. The explicit resolvent uses established Stieltjes/complete-Bernstein relaxation tools; see Meerschaert and Toaldo (2019), Section 2.1. Its nonlinear interpretation remains restricted.
Zero-feedback nonidentification	**Direct observation-law consequence**. At fixed finite loading and initial stress, stress memory vanishes from the diagnostic law at zero feedback. The singular divergent-loading closure is separate.
Positive-feedback identification	**Restricted nonlinear derivation; no priority claim**. The known-transform, empty-history, unbounded-mark experiment differs from the linear representation setting of Davis et al. (2026b), Theorem 3. It does not establish finite-catalogue estimation.
Visibility and score bounds	**Self-contained finite-horizon derivations**. Matched nonnegative histories and the stated weighted moment conditions are essential. No general information, boundary asymptotics or universal detection threshold follows.
Remark 4.5	**Conditional pooling algebra**. Equal total depletion need not give equal summed local rates. No empirical attribution to pooling follows.
Conditional amplitude profile	**Finite-history calculus / convexity application**. The conditional profile, its balance and curvature do not establish a global optimum over trend and inhibition.
Fixed-rule positive-inhibition response	**Likelihood differentiation derivation**. The fixed complete-data rule and positive-inhibition envelope differ from the fitted diagnostic setting in Kling and Vetter (2024). No monotonicity or boundary calibration is claimed.

Appendix B

Master Numerical Custody, Audit Rosters, and Parameter Ledger

B.1 Machine-readable custody and rebuild

The delivery's `validation/MASTER_CLAIM_LEDGER.json` binds empirical numbers and table entries to the retained authorities. Each entry identifies a file hash, a JSON pointer or explicit documentary locator, the displayed value and its derivation, with comparator, component and denominator where applicable. `validation/numeric_reconciliation_receipt.json` reports actual comparisons and unresolved limitations. Hash agreement establishes identity; it is not a theorem or a reconstruction of the generating computation.

The historical source maps below remain useful for original calculations that were not distributed in the compact public release. The unified source is self-contained for LaTeX compilation. The separate recovered-input bundle distributes earthquake records under provider terms and supplies executable cohort and numerical-replay entry points; it does not recover the original missing manuscript. The document build invokes Tectonic at the pinned version and the production validator checks references, citations, figures, log warnings and page edges, then renders every page for inspection. It imports no scientific fitting or simulation entry point.

B.2 Parameter and procedure custody

The original origin-specific predictive parameters are retained in Chapter 8 and in `paper/supplements/predictive_parameters.json`. The design chapter records the finite spectral recipe and historical generator. Addendum II's reconciled parameter/procedure tables and the A/B/O summaries bind the temporal extensions; the spatial summaries bind routing, density and factorial choices. The external/LI protocol subset supplies nuisance freezing, bounds, starts, fallback rules, exact floors and tie selection. Chapter 13 retains the full LI start schedule and all controls. A training-selected parameter, a numerical floor and a boundary display convention are recorded separately.

The main numerical rosters must remain local to their studies. Counts of procedures, prescribed starts, solver returns, retained candidates, exact floors and audited scores are not interchangeable. Raw-history score reconstruction is also distinct from checking exported aggregate arithmetic. The III–V audit answers those distinctions explicitly; a full aggregate calibration audit does not mean every simulated diagnostic was independently replayed from raw paths.

B.3 Completed evidence

The initial calibration retained 1,999 cases for each of two restrictions, or 3,998 rows, under its declared recovery policy. The later paired study completed 150 fresh pairs, or 300 full-fit cases.

The known-law and background comparisons reused specified saved histories and retained their reference/query identities. Fresh fitting of a saved history is not a fresh independent catalogue replication. The smaller planning cohorts were not pooled with the later cohort to improve its reported precision.

The observed five-family comparison completed all 477 planned records, 411 mandatory optimization start lineages and 30 selected numerical controls.

The final fixed-parameter audit completed 90 evaluations, 45 quadrature-order pairs, 18 temporal additivity controls and 913 source matches. That fixed-parameter audit did not introduce a spatial model, new fitting, new simulation or another catalogue download. Later spatial and external studies are separately integrated in Part III. Spatial and mark group tables partition event terms; their absence of a local compensator is a limit of the object computed, not a missing integral that can be supplied by arbitrarily allocating the national exposure.

B.4 Numerical control is not a statistical or physical certificate

The likelihood uses predictable pre-event features, the full observation window and its quiet intervals. Numerical controls compare the declared finite-mixture/clock representation and quadrature refinements at selected points. An empirical envelope or agreement at two orders is not a proven integration-error bound. Histories with extreme marks or closely spaced clocks remain in the procedure through declared numerical policies; unresolved outcomes cannot be silently converted into non-rejections.

Likewise, a solver's small gradient, successful exit or repeated winner is not a global optimality theorem. The ten-history robustness study materially reduces the appeal of merely increasing optimizer limits on that selected set, without validating all historical fits. The mathematical convexity of the conditional amplitude objective is narrower than joint convexity of the remaining profile. The manuscript retains these distinctions when interpreting the numerical evidence.

B.5 Computational and AI assistance

The completed summaries were accepted within the research archive's internal review. This manuscript does not present model-assisted agreement as independent academic refereeing. Human collaborators are Drefetr and magna-nz, identified by their reported GitHub handles. AI systems used as tools, as reported in the project history, include Claude Opus 5.5, DeepSeek 4.1 Flash, Gemini 4.0 Argon, GPT 6.0 Astra and GPT 6.1 Sol; these systems are not listed as authors. The bulk of the project work was performed by AI agents: Claude Opus 5.5 and GPT 6.0 Astra developed the initial outline and specification; GPT 6.1 Sol coordinated an agent swarm comprising GPT 6.1 Sol and DeepSeek 4.1 Flash subagents; GPT 6.0 Astra and Gemini 4.0 Argon provided review. These roles describe the development history reported by the collaborators, rather than independently verified model identities for every historical call. The release-v4 finalisation used native Codex agents without external DeepSeek calls. Reviewing retained sources and arithmetic is distinct from rerunning the historical studies.

B.6 Reading the detailed audit records

The following records supply the complete per-window, parameter, protocol and diagnostic tables supporting Chapters 9, 10 and 12. Their original comparison signs, denominators, failed gates and numerical representations are retained. Each chapter links to the relevant record; the main text retains its primary comparisons and interpretation.

B.7 Temporal and mark-extension audit tables

B.7.1 Implementation and validation

The new numerical core is standalone C++17 with a base-R interface. The preserved Python estimator uses its original CUDA/FP64 primary route and declared failure-only recovery. Core special-function support is explicitly limited to temporal shape 0.5 or 1; unsupported shapes fail. No general contour implementation was silently substituted.

Validation evidence includes:

- Original released ml.m executed under GNU Octave 11.3.0 after one private copyright-preserving dispatch-byte change; all 78 checked shape/scale/time cases passed. Maximum absolute survival/ML discrepancy was `2.33e-15`. This is a kernel reference bridge, not a full author-fit reproduction.
- Independent SciPy special-function/spectral-quadrature and high-precision Decimal checks on 84 kernel cases plus the saved clock fixtures. These mathematical audits used Python; R was tested separately as an end-to-end interface.
- Fixed-seed waiting-law checks on 120,000 variates across six shape/scale combinations, with declared CDF and 18 Laplace-transform checks.
- R 4.6.1 configuration/simulation/history/scoring round trip and malformed-input rejection checks. An initial unsupported-shape test-harness mutation failed to change its input; the failed receipt and corrected targeted rerun are retained, with no implementation change.
- Unchanged comparison smoke and six excluded timing cases; original starts, $\eta = 0$ candidate, quiet exposure, positive gaps, witnesses and `1e-4` numerical gates retained.
- Full saved-artifact source/input/seed/configuration/clock/checkpoint audit, followed by independent recomputation of all 120 outcome rows, six frequency cells and four paired contrasts.

All 360 mandatory optimizer records resolved through the original primary CUDA tier; no solver recovery was needed in the pilot. Local convergence remains distinct from a global optimum certificate. No event-cap or precision-gate failure occurred. The generator rejects excessive relative parent-child wait rounding above `1e-8` without redrawing, and retains incomplete constructions as ineligible for fitting. Finite two-double arithmetic is not arbitrary-precision path recovery.

B.7.2 Computational resources and preservation

The pilot used 977.383 seconds (16.29 minutes). Mean per-catalogue comparison wall time including process startup was 16.23 seconds; maximum was 29.84 seconds. New C++ generation across all 60 invocations used 2.207 measured wall seconds. These measurements are not a cross-language performance comparison.

The 20 forests contained 15,290 distinct event times, repeated across scenarios; per-forest counts ranged from 668 to 847. The run retained a single numerical-job slot and one numerical

thread per job. Prior numerical reservation wall times were conservatively charged before allocating the remaining 6,638-second pilot ceiling. The pilot finished well within it.

A read-only sampler observed 47 of 60 comparison processes. Maximum observed process-reported peak working set was 892.1 MiB. PyTorch reported at most 146.9 MiB of allocated tensor memory; this excludes driver/context/desktop memory. The recorded 16 GiB target was not an enforced OS memory limit, and unsampled host peaks remain unknown.

Source and numerical reviews required corrections before acceptance; the retained audit records distinguish those corrections from independent external peer review.

Original executable baseline: 3e76a174cf4f58ce601c6c9640c4d0c11168813f. Accepted local source commit: ae934318c1acf914da341f09072b1e59b0b1cc9c. Source-only patch application was checked against the original baseline. Original checkout, sealed v4 outputs, historical aggregates and historical F counts remain unchanged. No experiments were merged into those historical denominators.

B.7.3 Reproducibility artifacts

- `pilot_protocol_frozen.json` and `receipts/pilot_freeze.json`: original immutable scientific plan and hash.
- `results/pilot/plan.json`, `registry.jsonl`, per-case histories, configurations, raw comparisons and `outcomes.json`: every planned identity and outcome.
- `results/pilot_analysis/summary.json` and `method_outcomes.csv`: strict saved-output analysis.
- `methods/pilot_saved_artifact_audit.json` and `result_audit/statistical_recomputation.json`: independent saved-output and arithmetic checks.
- `reports/methods_and_limits.md`: full construction, estimand and interpretation details.
- `sources/source_manifest.json` and `reproduction_gaps.json`: source pins, versions, licences and exact-reproduction gaps.
- `reports/claim_register.json` and the separate final claim-review receipt: claim scope and acceptance.
- delivery/magnitude_history_implementation.patch: source-only implementation patch; full bundle includes synthetic outputs and review evidence.

The exact published supplement remains unavailable, the external reference project's released Japan CSV is empty and unshipped historical banks cannot be reconstructed from summaries. The fixed author preprint is separately pinned. A private original-source reference copy, article archives, worker prompts/provider logs and the unaccepted general draft remain excluded. This project's recovered regional catalogue records and selected simulation banks are separately supplied under `reproduction/`. Absolute frozen paths retain their historical identities; relocating or rebuilding requires an explicit source mapping, not silent weakening of historical guards.

B.7.4 Evidence map

- `pilot_protocol_frozen.json`: estimand, scenarios, sample plan, comparison contract, resources, analysis and exclusions.
- `sources/model_mapping.md`, `methods/construction_review.md`, `sources/ds16_cross_review.json`: source equations and independently reviewed finite-window construction.
- `sources/source_manifest.json`, `sources/reproduction_gaps.json`: exact source identities, versions, licences and missing assets.

- `sources/octave_reference/source_copy_identity.json`, `sources/octave_reference/comparison_report.json`, `sources/octave_reference/terminal_receipt.json`: private original-source Octave bridge.
- `sources/methods_independent_controls_cross_review.json`, `methods/independent_controls_receipt.json`: independent numerical controls and their scope.
- `methods/final_pilot_protocol_review.json`, `methods/freeze_confirmation.json`: pre-pilot protocol review and identity confirmation.

This methods note preceded the terminal pilot. The completed results, negative paired contrasts and independent saved-output audit are retained in Chapter 9 and in the numerical custody ledger; the prospective interpretation gate in the original note is recorded as historical.

B.7.5 Complete magnitude-extension fold accounting

The aggregate findings in Chapter 9 are accompanied by the full fold contrasts below, including both sensitivity settings and every declared comparator. The marked class-history losses remain visible separately from timing. For productivity, the common fitted magnitude law cancels; tiny last-digit timing/joint differences reflect saved floating-point summation. The tables are native transcriptions of the hash-bound aggregate packet, which retains original producer selectors and hashes.

Class-history fold contrasts

Comparison keys are local to this table.

H1 `E1-E0`
H2 `E1-I0`
H3 `E0-I0`
H4 `E1-E1_h0`
H5 `E1_h0-E0`

Table B.1: Class-history fold contrasts; primary. Total nats unless stated otherwise. Positive favors the candidate.

Fold	Key	Targets	Timing	ΔM	Joint	Joint/event
0	H1	203	-5.003978	0.359055	-4.644923	-0.022881
1	H1	155	4.911943	-2.885308	2.026635	0.013075
2	H1	254	-1.800546	-0.894770	-2.695315	-0.010611
0	H2	203	11.984757	0.359055	12.343812	0.060807
1	H2	155	4.939445	-2.885308	2.054137	0.013252
2	H2	254	-0.630368	-0.894770	-1.525137	-0.006004
0	H3	203	16.988735	0.000000	16.988735	0.083688
1	H3	155	0.027502	0.000000	0.027502	0.000177
2	H3	254	1.170178	0.000000	1.170178	0.004607
0	H4	203	0.204715	0.359055	0.563770	0.002777
1	H4	155	-0.094372	-2.885308	-2.979680	-0.019224
2	H4	254	0.037575	-0.894770	-0.857195	-0.003375
0	H5	203	-5.208693	-0.000000	-5.208693	-0.025659
1	H5	155	5.006315	0.000000	5.006315	0.032299
2	H5	254	-1.838120	-0.000000	-1.838120	-0.007237

Table B.2: Class-history fold contrasts; constant_background. Total nats unless stated otherwise. Positive favors the candidate.

Fold	Key	Targets	Timing	ΔM	Joint	Joint/event
0	H1	203	-6.371564	0.356263	-6.015301	-0.029632
1	H1	155	4.868998	-2.891175	1.977823	0.012760
2	H1	254	-2.104297	-0.893801	-2.998098	-0.011804
0	H2	203	3.639388	0.356263	3.995651	0.019683
1	H2	155	4.868992	-2.891175	1.977817	0.012760
2	H2	254	0.921431	-0.893801	0.027630	0.000109
0	H3	203	10.010952	0.000000	10.010952	0.049315
1	H3	155	-0.000006	0.000000	-0.000006	-0.000000
2	H3	254	3.025728	0.000000	3.025728	0.011912

Table B.3: Class-history fold contrasts; post1980_training. Total nats unless stated otherwise. Positive favors the candidate.

Fold	Key	Targets	Timing	ΔM	Joint	Joint/event
0	H1	203	-8.029804	-0.274203	-8.304007	-0.040906
1	H1	155	6.138077	-1.393217	4.744860	0.030612
2	H1	254	-1.869401	0.106705	-1.762696	-0.006940
0	H2	203	10.598707	-0.274203	10.324504	0.050860
1	H2	155	7.007711	-1.393217	5.614494	0.036223
2	H2	254	-1.755667	0.106705	-1.648962	-0.006492
0	H3	203	18.628512	0.000000	18.628512	0.091766
1	H3	155	0.869634	0.000000	0.869634	0.005611
2	H3	254	0.113734	0.000000	0.113734	0.000448

Productivity fold contrasts

Comparison keys are local to this table.

R1 `E0_free-E0_fixed`
R2 `I0_free-I0_fixed`
R3 `E0_free-I0_free`
R4 `E0_fixed-I0_fixed`

Table B.4: Productivity fold contrasts; primary. Total nats unless stated otherwise. Positive favors the candidate.

Fold	Key	Targets	Timing	ΔM	Joint	Joint/event
0	R1	203	-7.120621	0.000000	-7.120621	-0.035077
1	R1	155	8.166602	0.000000	8.166602	0.052688
2	R1	254	-6.420486	0.000000	-6.420486	-0.025278
0	R2	203	-6.917234	0.000000	-6.917234	-0.034075
1	R2	155	8.279067	0.000000	8.279067	0.053413
2	R2	254	-6.934000	0.000000	-6.934000	-0.027299
0	R3	203	16.785391	0.000000	16.785391	0.082687
1	R3	155	-0.084963	0.000000	-0.084963	-0.000548
2	R3	254	1.683695	0.000000	1.683695	0.006629

Fold	Key	Targets	Timing	ΔM	Joint	Joint/event
0	R4	203	16.988779	0.000000	16.988779	0.083689
1	R4	155	0.027501	0.000000	0.027501	0.000177
2	R4	254	1.170180	0.000000	1.170180	0.004607

Table B.5: Productivity fold contrasts; constant_background. Total nats unless stated otherwise. Positive favors the candidate.

Fold	Key	Targets	Timing	ΔM	Joint	Joint/event
0	R1	203	-7.770158	0.000000	-7.770158	-0.038277
1	R1	155	8.043488	0.000000	8.043488	0.051893
2	R1	254	-6.666621	0.000000	-6.666621	-0.026247
0	R2	203	-6.546142	0.000000	-6.546142	-0.032247
1	R2	155	8.043490	0.000000	8.043490	0.051893
2	R2	254	-6.814730	0.000000	-6.814730	-0.026830
0	R3	203	8.786939	0.000000	8.786939	0.043285
1	R3	155	0.000000	0.000000	0.000000	0.000000
2	R3	254	3.173846	0.000000	3.173846	0.012495
0	R4	203	10.010954	0.000000	10.010954	0.049315
1	R4	155	0.000001	0.000000	0.000001	0.000000
2	R4	254	3.025736	0.000000	3.025736	0.011912

Table B.6: Productivity fold contrasts; post1980_training. Total nats unless stated otherwise. Positive favors the candidate.

Fold	Key	Targets	Timing	ΔM	Joint	Joint/event
0	R1	203	-27.910113	0.000000	-27.910113	-0.137488
1	R1	155	-3.166988	0.000000	-3.166988	-0.020432
2	R1	254	-6.211334	0.000000	-6.211334	-0.024454
0	R2	203	-24.583234	0.000000	-24.583234	-0.121100
1	R2	155	-2.012969	0.000000	-2.012969	-0.012987
2	R2	254	-6.492439	0.000000	-6.492439	-0.025561
0	R3	203	15.301635	0.000000	15.301635	0.075378
1	R3	155	-0.284368	0.000000	-0.284368	-0.001835
2	R3	254	0.394837	0.000000	0.394837	0.001554
0	R4	203	18.628514	0.000000	18.628514	0.091766
1	R4	155	0.869650	0.000000	0.869650	0.005611
2	R4	254	0.113732	0.000000	0.113732	0.000448

B.7.6 Stage A complete fold score accounting

Comparison keys are local to this table.

A1 `I0_raw_ml-E0_ml`
A2 `I0_orth_ml-E0_ml`
A3 `I0_orth_ml-I0_raw_ml`
A4 `E0_omori-E0_ml`
A5 `I0_orth_omori-E0_omori`

A6 `I0_orth_omori-I0_orth_ml`
A7 `(I0_orth_omori-E0_omori)-(I0_orth_ml-E0_ml)`

Table B.7: Stage A complete fold score accounting; primary. Total nats unless stated otherwise. Positive favors the candidate.

Fold	Key	Targets	ΔQ	ΔC	Timing	ΔM	Joint
0	A1	203	-33.904743	16.916008	-16.988735	0.000000	-16.988735
0	A2	203	-33.904812	16.916043	-16.988769	0.000000	-16.988769
0	A3	203	-0.000069	0.000035	-0.000034	0.000000	-0.000034
0	A4	203	-1.475060	-1.735606	-3.210666	0.000000	-3.210666
0	A5	203	-33.783069	17.379036	-16.404033	0.000000	-16.404033
0	A6	203	-1.353318	-1.272613	-2.625930	0.000000	-2.625930
1	A1	155	-2.032892	2.005390	-0.027502	0.000000	-0.027502
1	A2	155	-2.032888	2.005386	-0.027503	0.000000	-0.027503
1	A3	155	0.000004	-0.000005	-0.000001	0.000000	-0.000001
1	A4	155	-33.909489	-2.045907	-35.955396	0.000000	-35.955396
1	A5	155	-2.989593	3.117053	0.127460	0.000000	0.127460
1	A6	155	-34.866194	-0.934240	-35.800434	0.000000	-35.800434
2	A1	254	-4.917340	3.747162	-1.170178	0.000000	-1.170178
2	A2	254	-4.917343	3.747154	-1.170189	0.000000	-1.170189
2	A3	254	-0.000003	-0.000008	-0.000011	0.000000	-0.000011
2	A4	254	-2.632651	3.627421	0.994770	0.000000	0.994770
2	A5	254	-0.000001	0.000002	0.000000	0.000000	0.000000
2	A6	254	2.284690	-0.119731	2.164959	0.000000	2.164959
0	A7	203	0.121742	0.462994	0.584736	0.000000	0.584736
1	A7	155	-0.956705	1.111667	0.154962	0.000000	0.154962
2	A7	254	4.917341	-3.747152	1.170189	0.000000	1.170189

Table B.8: Stage A complete fold score accounting; constant_background. Total nats unless stated otherwise. Positive favors the candidate.

Fold	Key	Targets	ΔQ	ΔC	Timing	ΔM	Joint
0	A1	203	-28.768977	18.758026	-10.010952	0.000000	-10.010952
0	A2	203	-30.329816	18.501711	-11.828105	0.000000	-11.828105
0	A3	203	-1.560839	-0.256314	-1.817153	0.000000	-1.817153
0	A4	203	-0.260758	-2.899086	-3.159844	0.000000	-3.159844
0	A5	203	-30.816545	19.934773	-10.881773	0.000000	-10.881773
0	A6	203	-0.747487	-1.466025	-2.213512	0.000000	-2.213512
1	A1	155	-0.000014	0.000021	0.000006	0.000000	0.000006
1	A2	155	-1.365149	0.838195	-0.526954	0.000000	-0.526954
1	A3	155	-1.365134	0.838174	-0.526960	0.000000	-0.526960
1	A4	155	-31.177940	-5.497944	-36.675884	0.000000	-36.675884
1	A5	155	-1.430681	0.895357	-0.535324	0.000000	-0.535324
1	A6	155	-31.243473	-5.440782	-36.684255	0.000000	-36.684255
2	A1	254	-9.491283	6.465555	-3.025728	0.000000	-3.025728
2	A2	254	-3.390062	2.893287	-0.496775	0.000000	-0.496775
2	A3	254	6.101221	-3.572267	2.528954	0.000000	2.528954
2	A4	254	-16.161294	12.563620	-3.597674	0.000000	-3.597674

Fold	Key	Targets	ΔQ	ΔC	Timing	ΔM	Joint
2	A5	254	0.000000	0.000000	0.000000	0.000000	0.000000
2	A6	254	-12.771232	9.670332	-3.100900	0.000000	-3.100900
0	A7	203	-0.486729	1.433061	0.946333	0.000000	0.946333
1	A7	155	-0.065533	0.057162	-0.008370	0.000000	-0.008370
2	A7	254	3.390062	-2.893287	0.496775	0.000000	0.496775

Table B.9: Stage A complete fold score accounting; post1980_training. Total nats unless stated otherwise. Positive favors the candidate.

Fold	Key	Targets	ΔQ	ΔC	Timing	ΔM	Joint
0	A1	203	-39.900996	21.272485	-18.628512	0.000000	-18.628512
0	A2	203	-39.375561	20.264481	-19.111080	0.000000	-19.111080
0	A3	203	0.525435	-1.008004	-0.482568	0.000000	-0.482568
0	A4	203	-2.317077	-0.828399	-3.145476	0.000000	-3.145476
0	A5	203	-0.000151	0.000117	-0.000033	0.000000	-0.000033
0	A6	203	37.058333	-21.092763	15.965570	0.000000	15.965570
1	A1	155	-1.170377	0.300743	-0.869634	0.000000	-0.869634
1	A2	155	-1.170359	0.300710	-0.869649	0.000000	-0.869649
1	A3	155	0.000018	-0.000033	-0.000015	0.000000	-0.000015
1	A4	155	-16.618485	3.450748	-13.167737	0.000000	-13.167737
1	A5	155	0.000000	0.000000	0.000000	0.000000	0.000000
1	A6	155	-15.448125	3.150038	-12.298088	0.000000	-12.298088
2	A1	254	-2.498407	2.384669	-0.113739	0.000000	-0.113739
2	A2	254	-2.498407	2.384669	-0.113739	0.000000	-0.113739
2	A3	254	0.000000	0.000000	0.000000	0.000000	0.000000
2	A4	254	1.932529	-0.625115	1.307414	0.000000	1.307414
2	A5	254	0.000000	0.000000	0.000000	0.000000	0.000000
2	A6	254	4.430937	-3.009784	1.421152	0.000000	1.421152
0	A7	203	39.375411	-20.264364	19.111047	0.000000	19.111047
1	A7	155	1.170359	-0.300710	0.869649	0.000000	0.869649
2	A7	254	2.498407	-2.384669	0.113739	0.000000	0.113739

B.7.7 Stage A aggregate sensitivity accounting

The comparison keys are defined with the corresponding primary aggregate table in Chapter 9.

Table B.10: Stage A complete aggregate score accounting; constant_background. Total nats unless stated otherwise. Positive favors the candidate. The same 612 targets enter each aggregate; positive-fold counts are out of three.

Key	ΔQ	ΔC	Timing	ΔM	Joint	Joint/event
AG1	-38.260275	25.223601	-13.036674	0.000000	-13.036674	-0.021302
AG2	-35.085027	22.233193	-12.851834	0.000000	-12.851834	-0.021000
AG3	3.175247	-2.990408	0.184840	0.000000	0.184840	0.000302
AG4	-47.599992	4.166589	-43.433403	0.000000	-43.433403	-0.070970
AG5	-32.247227	20.830130	-11.417097	0.000000	-11.417097	-0.018655
AG6	-44.762191	2.763526	-41.998666	0.000000	-41.998666	-0.068625

Key	ΔQ	ΔC	Timing	ΔM	Joint	Joint/event
AG7	2.837801	-1.403064	1.434737	0.000000	1.434737	0.002344

Table B.11: Stage A complete aggregate score accounting; post1980_training. Total nats unless stated otherwise. Positive favors the candidate. The same 612 targets enter each aggregate; positive-fold counts are out of three.

Key	ΔQ	ΔC	Timing	ΔM	Joint	Joint/event
AG1	-43.569781	23.957897	-19.611884	0.000000	-19.611884	-0.032046
AG2	-43.044328	22.949860	-20.094467	0.000000	-20.094467	-0.032834
AG3	0.525453	-1.008036	-0.482583	0.000000	-0.482583	-0.000789
AG4	-17.003033	1.997234	-15.005799	0.000000	-15.005799	-0.024519
AG5	-0.000151	0.000117	-0.000033	0.000000	-0.000033	-0.000000
AG6	26.041144	-20.952509	5.088635	0.000000	5.088635	0.008315
AG7	43.044177	-22.949743	20.094434	0.000000	20.094434	0.032834

B.7.8 Stage A training exposure geometry

Table B.12: Training exposure geometry for every primary and sensitivity case. These are Gram condition numbers, not likelihood information or sampling covariance.

Variant	Fold	γ_0	γ_1	Raw	Residual	Scaled raw	Scaled residual
Primary	0	1.473592	1.296771	137.804559	19.281470	124.155962	13.928203
Primary	1	1.753030	1.721009	101.246757	9.224213	58.817721	5.290263
Primary	2	1.819329	1.856766	108.420677	5.169906	44.708612	2.318729
Constant	0	1.473592	1.296771	137.804559	19.281470	124.155962	13.928203
Constant	1	1.753030	1.721009	101.246757	9.224213	58.817721	5.290263
Constant	2	1.819329	1.856766	108.420677	5.169906	44.708612	2.318729
Post-1980	0	1.457034	1.111932	258.340548	80.797916	97.008850	13.928203
Post-1980	1	1.771206	2.148674	178.062897	24.544460	36.377157	1.639521
Post-1980	2	1.765340	2.202126	154.072404	12.233693	42.055743	1.964848

B.7.9 Stage B complete fold score accounting

Comparison keys are local to this table.

B1 `E0_tau_free-E0_tau1`
B2 `E1_tau_class-E0_tau_free`
B3 `E1_tau_class-E0_tau1`

Table B.13: Stage B complete fold score accounting; primary. Total nats unless stated otherwise. Positive favors the candidate.

Fold	Key	Targets	ΔQ	ΔC	Timing	ΔM	Joint
0	B1	203	-0.133101	0.210113	0.077011	0.000000	0.077011
0	B2	203	-0.575950	0.107987	-0.467963	-0.000000	-0.467963
0	B3	203	-0.709052	0.318100	-0.390952	0.000000	-0.390952
1	B1	155	2.226571	0.827196	3.053767	0.000000	3.053767

Fold	Key	Targets	ΔQ	ΔC	Timing	ΔM	Joint
1	B2	155	4.745611	-1.885141	2.860470	0.000000	2.860470
1	B3	155	6.972182	-1.057945	5.914237	0.000000	5.914237
2	B1	254	2.312010	1.940885	4.252896	0.000000	4.252896
2	B2	254	1.128317	-2.512991	-1.384674	0.000000	-1.384674
2	B3	254	3.440327	-0.572105	2.868222	0.000000	2.868222

B.7.10 Stage B selected temporal scales

Table B.14: All selected Stage B temporal scales; all selected coordinates are interior.

Fold	Model	τ_0 days	τ_1 days	Training NLL
0	E0_tau1	1.000000	1.000000	1035.271005
0	E0_tau_free	0.931796	0.931796	1035.254019
0	E1_tau_class	0.863412	0.987068	1035.241963
1	E0_tau1	1.000000	1.000000	1319.780120
1	E0_tau_free	0.826384	0.826384	1319.503187
1	E1_tau_class	1.770186	0.539459	1317.635532
2	E0_tau1	1.000000	1.000000	1476.713690
2	E0_tau_free	0.402085	0.402085	1468.072664
2	E1_tau_class	1.352711	0.228245	1462.824072

B.7.11 Extension O complete fold score accounting

Comparison keys are local to this table.

O1 `E_omori-E_ml`
O2 `Iorth_matched-E_omori`
O3 `Iraw_matched-E_omori`
O4 `Iorth_joint-E_omori`
O5 `Iorth_joint-Iorth_matched`
O6 `Iorth_matched-Iraw_matched`

Table B.15: Extension O complete fold score accounting; constant_background. Total nats unless stated otherwise. Positive favors the candidate.

Fold	Key	Targets	ΔQ	ΔC	Timing	ΔM	Joint
0	O1	203	6.977360	-4.247158	2.730202	0.000000	2.730202
0	O2	203	-27.485414	19.107314	-8.378100	0.000000	-8.378100
0	O3	203	-30.527464	21.203330	-9.324134	0.000000	-9.324134
0	O4	203	-22.464138	15.555084	-6.909054	0.000000	-6.909054
0	O5	203	5.021276	-3.552230	1.469046	0.000000	1.469046
0	O6	203	3.042050	-2.096017	0.946033	0.000000	0.946033
1	O1	155	7.896044	-5.795535	2.100509	0.000000	2.100509
1	O2	155	-1.021226	0.635369	-0.385856	0.000000	-0.385856
1	O3	155	-1.643197	2.278392	0.635194	0.000000	0.635194
1	O4	155	0.364194	-1.788729	-1.424535	0.000000	-1.424535
1	O5	155	1.385419	-2.424098	-1.038679	0.000000	-1.038679

Fold	Key	Targets	ΔQ	ΔC	Timing	ΔM	Joint
1	O6	155	0.621972	-1.643022	-1.021050	0.000000	-1.021050
2	O1	254	16.526438	-7.064119	9.462319	0.000000	9.462319
2	O2	254	-1.357974	1.325013	-0.032962	0.000000	-0.032962
2	O3	254	-18.392939	12.835486	-5.557453	0.000000	-5.557453
2	O4	254	-1.120117	1.172210	0.052093	0.000000	0.052093
2	O5	254	0.237858	-0.152803	0.085055	0.000000	0.085055
2	O6	254	17.034964	-11.510473	5.524491	0.000000	5.524491

Table B.16: Extension O complete fold score accounting; post1980_training. Total nats unless stated otherwise. Positive favors the candidate.

Fold	Key	Targets	ΔQ	ΔC	Timing	ΔM	Joint
0	O1	203	5.580265	-2.102392	3.477873	0.000000	3.477873
0	O2	203	-38.642680	21.044140	-17.598539	0.000000	-17.598539
0	O3	203	-39.351608	22.384136	-16.967472	0.000000	-16.967472
0	O4	203	-28.578025	16.319335	-12.258690	0.000000	-12.258690
0	O5	203	10.064655	-4.724805	5.339850	0.000000	5.339850
0	O6	203	0.708928	-1.339996	-0.631067	0.000000	-0.631067
1	O1	155	1.191057	-0.573982	0.617074	0.000000	0.617074
1	O2	155	-1.307756	0.313343	-0.994412	0.000000	-0.994412
1	O3	155	-1.307753	0.313339	-0.994414	0.000000	-0.994414
1	O4	155	-0.828427	-0.296337	-1.124764	0.000000	-1.124764
1	O5	155	0.479328	-0.609680	-0.130352	0.000000	-0.130352
1	O6	155	-0.000003	0.000005	0.000002	0.000000	0.000002
2	O1	254	6.073799	-0.065525	6.008274	0.000000	6.008274
2	O2	254	-1.945172	1.769425	-0.175747	0.000000	-0.175747
2	O3	254	-1.945042	1.769314	-0.175728	0.000000	-0.175728
2	O4	254	-2.554853	2.087877	-0.466976	0.000000	-0.466976
2	O5	254	-0.609681	0.318452	-0.291229	0.000000	-0.291229
2	O6	254	-0.000130	0.000111	-0.000019	0.000000	-0.000019

Table B.17: Extension O complete fold score accounting; primary. Total nats unless stated otherwise. Positive favors the candidate.

Fold	Key	Targets	ΔQ	ΔC	Timing	ΔM	Joint
0	O1	203	5.049423	-2.601323	2.448100	0.000000	2.448100
0	O2	203	-28.233396	15.616474	-12.616922	0.000000	-12.616922
0	O3	203	-27.722639	15.467188	-12.255451	0.000000	-12.255451
0	O4	203	-17.687488	8.965567	-8.721921	0.000000	-8.721921
0	O5	203	10.545908	-6.650906	3.895001	0.000000	3.895001
0	O6	203	-0.510757	0.149286	-0.361471	0.000000	-0.361471
1	O1	155	3.993136	-0.808859	3.184277	0.000000	3.184277
1	O2	155	-2.408001	2.583159	0.175158	0.000000	0.175158
1	O3	155	-2.407986	2.583140	0.175155	0.000000	0.175155
1	O4	155	-1.878676	1.525240	-0.353436	0.000000	-0.353436
1	O5	155	0.529325	-1.057919	-0.528594	0.000000	-0.528594

Fold	Key	Targets	ΔQ	ΔC	Timing	ΔM	Joint
1	O6	155	-0.000015	0.000019	0.000003	0.000000	0.000003
2	O1	254	8.199586	-1.626460	6.573125	0.000000	6.573125
2	O2	254	-6.360917	4.665508	-1.695409	0.000000	-1.695409
2	O3	254	-6.360917	4.665508	-1.695409	0.000000	-1.695409
2	O4	254	-7.613588	4.885708	-2.727880	0.000000	-2.727880
2	O5	254	-1.252671	0.220200	-1.032470	0.000000	-1.032470
2	O6	254	-0.000000	0.000000	-0.000000	0.000000	-0.000000

B.7.12 Extension O aggregate sensitivity accounting

The comparison keys are defined with the corresponding primary aggregate table in Chapter 9.

Table B.18: Extension O complete aggregate score accounting; constant_background. Total nats unless stated otherwise. Positive favors the candidate. The same 612 targets enter each aggregate; positive-fold counts count joint gains exceeding 10^{-4} nats out of three. Positive values at or below that threshold are treated as numerical ties.

Key	ΔQ	ΔC	Timing	ΔM	Joint	Joint/event	Positive folds
OG1	31.399842	-17.106812	14.293030	0.000000	14.293030	0.023355	3
OG2	-29.864614	21.067696	-8.796918	0.000000	-8.796918	-0.014374	0
OG3	-50.563600	36.317208	-14.246392	0.000000	-14.246392	-0.023278	1
OG4	-23.220061	14.938565	-8.281496	0.000000	-8.281496	-0.013532	1
OG5	6.644553	-6.129131	0.515422	0.000000	0.515422	0.000842	2
OG6	20.698986	-15.249512	5.449474	0.000000	5.449474	0.008904	2

Table B.19: Extension O complete aggregate score accounting; post1980_training. Total nats unless stated otherwise. Positive favors the candidate. The same 612 targets enter each aggregate; positive-fold counts count joint gains exceeding 10^{-4} nats out of three. Positive values at or below that threshold are treated as numerical ties.

Key	ΔQ	ΔC	Timing	ΔM	Joint	Joint/event	Positive folds
OG1	12.845121	-2.741900	10.103221	0.000000	10.103221	0.016509	3
OG2	-41.895607	23.126909	-18.768698	0.000000	-18.768698	-0.030668	0
OG3	-42.604403	24.466788	-18.137615	0.000000	-18.137615	-0.029637	0
OG4	-31.961305	18.110875	-13.850430	0.000000	-13.850430	-0.022631	0
OG5	9.934302	-5.016033	4.918269	0.000000	4.918269	0.008036	1
OG6	0.708796	-1.339880	-0.631084	0.000000	-0.631084	-0.001031	0

B.7.13 Extension O selected continuous kernels

Table B.20: All excitation-selected continuous Omori kernels shared by matched inhibitory comparisons.

Variant	Fold	c days	p
constant_background	0	0.016103	1.165455
constant_background	1	0.019728	1.191021

Variant	Fold	c days	p
constant_background	2	0.007315	1.166448
post1980_training	0	0.028709	1.276663
post1980_training	1	0.027837	1.265051
post1980_training	2	0.007187	1.180165
primary	0	0.014609	1.146121
primary	1	0.020002	1.194027
primary	2	0.006777	1.152028

B.7.14 Engineering normalization, forecast, profile and map witnesses

Table B.21: All class-history engineering normalization witnesses. Offspring outside the window are retained separately.

History	Events	Proposed offspring	Outside	Radius	$\mathbf{1}^\top Kp$
E0_00	1582	622	13	0.350	0.350000
E0_01	1425	448	4	0.350	0.350000
E0_02	1408	433	2	0.350	0.350000
E1_active_00	1202	189	0	0.350	0.151019
E1_active_01	1162	142	2	0.350	0.151019
E1_active_02	1202	216	2	0.350	0.151019

Table B.22: Every saved engineering score witness from the magnitude-extension packet; nats, candidate minus reference. Histories and target counts are specific to each witness.

History	Contrast	Targets	Timing	ΔM	Joint
E0_00	E1-E0	612	-0.574423	-3.729507	-4.303930
E0_01	E1-E0	405	0.441037	-0.178469	0.262568
E0_02	E1-E0	372	0.407162	-0.574953	-0.167790
E1_active_00	E1-E0	339	0.369123	7.331815	7.700938
E1_active_01	E1-E0	386	0.755100	24.400509	25.155610
E1_active_02	E1-E0	363	0.669956	15.162250	15.832205
E0_r070_00	E0_free-E0_fixed	67	0.114099	0.000000	0.114099
E0_r070_01	E0_free-E0_fixed	93	0.085151	0.000000	0.085151
E0_r035_00	E0_free-E0_fixed	82	4.994451	0.000000	4.994451
E0_r035_01	E0_free-E0_fixed	87	11.847512	0.000000	11.847512

Table B.23: All primary sparse productivity support sets. None denotes no sampled grid node within two nats, not an empty confidence interval.

Case	Model	Selected r	Sampled support
fold_0	E0_free	0.471908	0.35, 0.50
fold_0	I0_free	0.472469	0.35, 0.50
fold_1	E0_free	0.605853	none
fold_1	I0_free	0.603756	none
fold_2	E0_free	0.610446	none

Case	Model	Selected r	Sampled support
fold_2	I0_free	0.609725	none

Table B.24: Every Extension O selected raw/residual map. Feasibility includes the destination fixed-coordinate restriction; coefficients are not clipped.

Case	Direction	Mapped intercept	Mapped trend	Feasible
constant / 0	raw to residual	-4.728423	-0.651341	False
constant / 0	residual to raw	-2.031941	2.146959	False
constant / 1	raw to residual	-4.354684	-0.063093	False
constant / 1	residual to raw	-3.681607	0.671691	False
constant / 2	raw to residual	-4.398560	-0.197474	False
constant / 2	residual to raw	-4.246945	0.093713	False
post-1980 / 0	raw to residual	-4.598880	0.106504	True
post-1980 / 0	residual to raw	-1.223168	3.311541	False
post-1980 / 1	raw to residual	-4.337005	0.815822	True
post-1980 / 1	residual to raw	-3.587742	1.724763	True
post-1980 / 2	raw to residual	-4.405585	-0.131728	True
post-1980 / 2	residual to raw	-4.181072	0.148335	True
primary / 0	raw to residual	-4.855097	-0.623668	True
primary / 0	residual to raw	-1.753179	2.114439	False
primary / 1	raw to residual	-4.367342	-0.014533	True
primary / 1	residual to raw	-3.688696	0.651718	True
primary / 2	raw to residual	-4.443303	-0.239259	True
primary / 2	residual to raw	-4.089207	0.122123	True

B.7.15 Stage A selected coarse-grid kernels

Table B.25: All Stage A excitation-selected coarse Omori kernels. The saved field named tau is the Omori scale c.

Case	c days	p
primary / 0	0.100000	1.300000
primary / 1	0.100000	1.300000
primary / 2	0.100000	1.700000
constant / 0	0.100000	1.300000
constant / 1	0.100000	1.300000
constant / 2	0.100000	1.700000
post-1980 / 0	0.100000	1.700000
post-1980 / 1	0.100000	1.700000
post-1980 / 2	0.100000	1.700000

B.8 New Zealand spatial audit tables

In the tables, $I_{\mathrm{d,g}}$ denotes distance excitation with global stress (the saved identifier `I_distance_globalG`). A stacked mantissa and power of ten form one scientific-notation value; their full displayed precision is retained.

B.8.1 C1 fold and zone accounting

Per-fold comparisons

Fold	Contrast	Events	Event	ΔC	Regional timing	Conditional zone	Joint
0	$E0_{local}$-$E0_{global}$	203	103.29	1.81424	-2.08324	107.188	105.105
0	$I0_{local}$-$I0_{global}$	203	134.194	-12.3089	14.0738	107.812	121.885
0	$I0_{local}$-$E0_{local}$	203	-3.00076	2.7929	-0.831697	0.623835	-0.207862
0	$I0_{global}$-$E0_{global}$	203	-33.9047	16.916	-16.9887	-1.13687×10^{-13}	-16.9887
1	$E0_{local}$-$E0_{global}$	155	113.7	5.30083	1.79756	117.203	119.001
1	$I0_{local}$-$I0_{global}$	155	116.119	-0.282139	0.579009	115.258	115.837
1	$I0_{local}$-$E0_{local}$	155	0.385632	-3.57753	-1.24604	-1.94585	-3.1919
1	$I0_{global}$-$E0_{global}$	155	-2.03292	2.00544	-0.0274878	1.13687×10^{-13}	-0.0274878
2	$E0_{local}$-$E0_{global}$	254	121.606	4.07498	2.11386	123.567	125.681
2	$I0_{local}$-$I0_{global}$	254	126.523	0.327849	3.28405	123.567	126.851
2	$I0_{local}$-$E0_{local}$	254	-4.43908×10^{-6}	1.47191×10^{-6}	-2.6669×10^{-6}	-3.00271×10^{-7}	-2.96717×10^{-6}
2	$I0_{global}$-$E0_{global}$	254	-4.91733	3.74714	-1.17019	-5.11591×10^{-13}	-1.17019
0	($I0_{local}$-$E0_{local}$)-($I0_{global}$-$E0_{global}$)	203	30.904	-14.1231	16.157	0.623835	16.7809
1	($I0_{local}$-$E0_{local}$)-($I0_{global}$-$E0_{global}$)	155	2.41856	-5.58296	-1.21855	-1.94585	-3.16441
2	($I0_{local}$-$E0_{local}$)-($I0_{global}$-$E0_{global}$)	254	4.91732	-3.74714	1.17018	-3.00271×10^{-7}	1.17018

Exact zone decomposition

Fold	Contrast	Zone	Events	Event	ΔC	Sum
0	$E0_{local}$-$E0_{global}$	south	43	24.4051	2.40447	26.8096
0	$E0_{local}$-$E0_{global}$	central	37	-2.32422	17.1847	14.8605
0	$E0_{local}$-$E0_{global}$	north	123	81.2094	-17.775	63.4344
0	$I0_{local}$-$I0_{global}$	south	43	37.0734	-1.66038	35.413
0	$I0_{local}$-$I0_{global}$	central	37	9.42696	10.7779	20.2049
0	$I0_{local}$-$I0_{global}$	north	123	87.694	-21.4264	66.2676

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Fold	Contrast	Zone	Events	Event	ΔC	Sum
0	$I0_{local}$-$E0_{local}$	south	43	0.532678	-0.243298	0.289381
0	$I0_{local}$-$E0_{local}$	central	37	-0.366395	0.224714	-0.141682
0	$I0_{local}$-$E0_{local}$	north	123	-3.16705	2.81148	-0.355561
0	$I0_{global}$-$E0_{global}$	south	43	-12.1355	3.82156	-8.31398
0	$I0_{global}$-$E0_{global}$	central	37	-12.1176	6.63152	-5.48605
0	$I0_{global}$-$E0_{global}$	north	123	-9.65163	6.46293	-3.18871
1	$E0_{local}$-$E0_{global}$	south	115	130.41	-109.594	20.8165
1	$E0_{local}$-$E0_{global}$	central	23	-12.0792	40.2793	28.2001
1	$E0_{local}$-$E0_{global}$	north	17	-4.63109	74.6154	69.9844
1	$I0_{local}$-$I0_{global}$	south	115	128.878	-108.752	20.1265
1	$I0_{local}$-$I0_{global}$	central	23	-9.14046	36.0405	26.9
1	$I0_{local}$-$I0_{global}$	north	17	-3.61925	72.4293	68.8101
1	$I0_{local}$-$E0_{local}$	south	115	-2.77278	1.28366	-1.48912
1	$I0_{local}$-$E0_{local}$	central	23	2.62957	-3.62209	-0.992518
1	$I0_{local}$-$E0_{local}$	north	17	0.528839	-1.2391	-0.710261
1	$I0_{global}$-$E0_{global}$	south	115	-1.24078	0.441674	-0.799105
1	$I0_{global}$-$E0_{global}$	central	23	-0.309142	0.616751	0.307609
1	$I0_{global}$-$E0_{global}$	north	17	-0.483004	0.947012	0.464008
2	$E0_{local}$-$E0_{global}$	south	66	-1.36256	33.6904	32.3278
2	$E0_{local}$-$E0_{global}$	central	129	108.706	-54.8	53.906
2	$E0_{local}$-$E0_{global}$	north	59	14.2625	25.1847	39.4472
2	$I0_{local}$-$I0_{global}$	south	66	0.601783	32.4053	33.0071
2	$I0_{local}$-$I0_{global}$	central	129	110.286	-55.8122	54.4742
2	$I0_{local}$-$I0_{global}$	north	59	15.6351	23.7347	39.3698
2	$I0_{local}$-$E0_{local}$	south	66	-2.68952×10^{-6}	1.02233×10^{-6}	-1.66719×10^{-6}
2	$I0_{local}$-$E0_{local}$	central	129	-6.95975×10^{-7}	-4.98334×10^{-7}	-1.19431×10^{-6}
2	$I0_{local}$-$E0_{local}$	north	59	-1.05358×10^{-6}	9.47913×10^{-7}	-1.05669×10^{-7}
2	$I0_{global}$-$E0_{global}$	south	66	-1.96435	1.28506	-0.679292
2	$I0_{global}$-$E0_{global}$	central	129	-1.58037	1.01213	-0.568246
2	$I0_{global}$-$E0_{global}$	north	59	-1.3726	1.44995	0.077351
0	($I0_{local}$-$E0_{local}$)-($I0_{global}$-$E0_{global}$)	south	43	12.6682	-4.06485	8.60336
0	($I0_{local}$-$E0_{local}$)-($I0_{global}$-$E0_{global}$)	central	37	11.7512	-6.40681	5.34437
0	($I0_{local}$-$E0_{local}$)-($I0_{global}$-$E0_{global}$)	north	123	6.48459	-3.65144	2.83315
1	($I0_{local}$-$E0_{local}$)-($I0_{global}$-$E0_{global}$)	south	115	-1.532	0.841991	-0.690012

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Fold	Contrast	Zone	Events	Event	ΔC	Sum
1	($I0_{local}$-$E0_{local}$)-($I0_{global}$-$E0_{global}$)	central	23	2.93872	-4.23884	-1.30013
1	($I0_{local}$-$E0_{local}$)-($I0_{global}$-$E0_{global}$)	north	17	1.01184	-2.18611	-1.17427
2	($I0_{local}$-$E0_{local}$)-($I0_{global}$-$E0_{global}$)	south	66	1.96435	-1.28506	0.679291
2	($I0_{local}$-$E0_{local}$)-($I0_{global}$-$E0_{global}$)	central	129	1.58037	-1.01213	0.568245
2	($I0_{local}$-$E0_{local}$)-($I0_{global}$-$E0_{global}$)	north	59	1.3726	-1.44995	-0.0773511

B.8.2 W window and target-region accounting

Per-window event and compensator decomposition

Fold	Contrast	Events	Event	ΔC	Timing	Zone	Cell given zone	Joint
0	E_{wfree}-E_{w0}	203	2.74011	-0.446709	0.865688	1.42771	0	2.2934
0	E_{wfree}-E_{w1}	203	106.03	1.36751	-1.21754	108.615	0	107.398
0	$Ilocal_{wfree}$-E_{wfree}	203	-2.00297	2.09027	-0.704286	0.791583	0	0.0872966
0	$Iglobal_{wfree}$-E_{wfree}	203	-24.9079	13.7159	-10.0284	-1.16355	0	-11.1919
0	$Ilocal_{w0}$-E_{w0}	203	-3.00086	2.79298	-0.831722	0.623842	0	-0.20788
0	$Iglobal_{w0}$-E_{w0}	203	-2.63908	1.63403	-0.528321	-0.476723	0	-1.00504
0	$Ilocal_{w1}$-E_{w1}	203	-4.09687×10^{-5}	2.90728×10^{-5}	-1.18959×10^{-5}	1.13687×10^{-13}	0	-1.18959×10^{-5}
0	$Iglobal_{w1}$-E_{w1}	203	-33.9047	16.916	-16.9887	2.55795×10^{-13}	0	-16.9887
0	$Ilocal_{wfree}$-$Iglobal_{wfree}$	203	22.9049	-11.6256	9.32411	1.95513	0	11.2792
1	E_{wfree}-E_{w0}	155	9.62251	-1.4458	-0.561462	8.73818	0	8.17672
1	E_{wfree}-E_{w1}	155	123.323	3.85493	1.23607	125.942	0	127.178
1	$Ilocal_{wfree}$-E_{wfree}	155	0.561647	-3.79426	-1.29001	-1.9426	0	-3.23262
1	$Iglobal_{wfree}$-E_{wfree}	155	-0.646655	1.28603	0.0912276	0.548152	0	0.639379
1	$Ilocal_{w0}$-E_{w0}	155	0.385602	-3.57749	-1.24603	-1.94586	0	-3.19188
1	$Iglobal_{w0}$-E_{w0}	155	-3.47829×10^{-5}	5.21426×10^{-5}	1.5626×10^{-5}	1.73366×10^{-6}	0	1.73597×10^{-5}
1	$Ilocal_{w1}$-E_{w1}	155	0	0	0	0	0	0
1	$Iglobal_{w1}$-E_{w1}	155	-2.03289	2.00539	-0.0275043	-1.42109×10^{-13}	0	-0.0275043

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Fold	Contrast	Events	Event	ΔC	Timing	Zone	Cell given zone	Joint
1	$\mathrm{Ilocal_{wfree}}$-$\mathrm{Iglobal_{wfree}}$	155	1.2083	-5.0803	-1.38124	-2.49075	0	-3.872
2	$\mathrm{E_{wfree}}$-$\mathrm{E_{w0}}$	254	0.282971	-0.199715	-3.54035	3.6236	0	0.0832556
2	$\mathrm{E_{wfree}}$-$\mathrm{E_{w1}}$	254	121.889	3.87531	-1.4265	127.191	0	125.764
2	$\mathrm{Ilocal_{wfree}}$-$\mathrm{E_{wfree}}$	254	-1.02061	0.528631	-0.298122	-0.193855	0	-0.491977
2	$\mathrm{Iglobal_{wfree}}$-$\mathrm{E_{wfree}}$	254	-7.33546	5.0003	-1.65631	-0.678847	0	-2.33516
2	$\mathrm{Ilocal_{w0}}$-$\mathrm{E_{w0}}$	254	1.92136×10^{-5}	-1.83895×10^{-5}	7.43356×10^{-7}	8.07663×10^{-8}	0	8.24122×10^{-7}
2	$\mathrm{Iglobal_{w0}}$-$\mathrm{E_{w0}}$	254	1.92136×10^{-5}	-1.83895×10^{-5}	7.43356×10^{-7}	8.07663×10^{-8}	0	8.24122×10^{-7}
2	$\mathrm{Ilocal_{w1}}$-$\mathrm{E_{w1}}$	254	-2.23743×10^{-5}	2.54481×10^{-6}	-1.98295×10^{-5}	-2.84217×10^{-13}	0	-1.98295×10^{-5}
2	$\mathrm{Iglobal_{w1}}$-$\mathrm{E_{w1}}$	254	-4.91733	3.74713	-1.1702	-2.84217×10^{-13}	0	-1.1702
2	$\mathrm{Ilocal_{wfree}}$-$\mathrm{Iglobal_{wfree}}$	254	6.31485	-4.47167	1.35819	0.484993	0	1.84318

Proper target-region exposures

Fold	Contrast	Target	Event	ΔC	Sum
0	$\mathrm{E_{wfree}}$-$\mathrm{E_{w0}}$	0	-0.807672	-0.131685	-0.939358
0	$\mathrm{E_{wfree}}$-$\mathrm{E_{w0}}$	1	2.56945	-0.429232	2.14022
0	$\mathrm{E_{wfree}}$-$\mathrm{E_{w0}}$	2	0.978328	0.114208	1.09254
0	$\mathrm{E_{wfree}}$-$\mathrm{E_{w1}}$	0	23.5975	2.27278	25.8703
0	$\mathrm{E_{wfree}}$-$\mathrm{E_{w1}}$	1	0.245243	16.7555	17.0007
0	$\mathrm{E_{wfree}}$-$\mathrm{E_{w1}}$	2	82.1877	-17.6608	64.527
0	$\mathrm{Ilocal_{wfree}}$-$\mathrm{E_{wfree}}$	0	0.422907	-0.17919	0.243717
0	$\mathrm{Ilocal_{wfree}}$-$\mathrm{E_{wfree}}$	1	-0.202542	0.165563	-0.0369782
0	$\mathrm{Ilocal_{wfree}}$-$\mathrm{E_{wfree}}$	2	-2.22334	2.10389	-0.119442
0	$\mathrm{Iglobal_{wfree}}$-$\mathrm{E_{wfree}}$	0	-8.61409	3.12587	-5.48823
0	$\mathrm{Iglobal_{wfree}}$-$\mathrm{E_{wfree}}$	1	-9.74315	5.60211	-4.14104
0	$\mathrm{Iglobal_{wfree}}$-$\mathrm{E_{wfree}}$	2	-6.55061	4.98793	-1.56268
0	$\mathrm{Ilocal_{w0}}$-$\mathrm{E_{w0}}$	0	0.532671	-0.243296	0.289375
0	$\mathrm{Ilocal_{w0}}$-$\mathrm{E_{w0}}$	1	-0.366428	0.224736	-0.141692
0	$\mathrm{Ilocal_{w0}}$-$\mathrm{E_{w0}}$	2	-3.1671	2.81154	-0.355562
0	$\mathrm{Iglobal_{w0}}$-$\mathrm{E_{w0}}$	0	-0.605392	0.373352	-0.23204
0	$\mathrm{Iglobal_{w0}}$-$\mathrm{E_{w0}}$	1	-1.34317	0.675289	-0.667881
0	$\mathrm{Iglobal_{w0}}$-$\mathrm{E_{w0}}$	2	-0.690514	0.58539	-0.105123
0	$\mathrm{Ilocal_{w1}}$-$\mathrm{E_{w1}}$	0	-2.32084×10^{-5}	6.56795×10^{-6}	-1.66405×10^{-5}
0	$\mathrm{Ilocal_{w1}}$-$\mathrm{E_{w1}}$	1	-1.97879×10^{-5}	1.13973×10^{-5}	-8.3906×10^{-6}
0	$\mathrm{Ilocal_{w1}}$-$\mathrm{E_{w1}}$	2	2.02764×10^{-6}	1.11076×10^{-5}	1.31352×10^{-5}
0	$\mathrm{Iglobal_{w1}}$-$\mathrm{E_{w1}}$	0	-12.1355	3.82156	-8.31397
0	$\mathrm{Iglobal_{w1}}$-$\mathrm{E_{w1}}$	1	-12.1176	6.63152	-5.48605
0	$\mathrm{Iglobal_{w1}}$-$\mathrm{E_{w1}}$	2	-9.65164	6.46292	-3.18871
0	$\mathrm{Ilocal_{wfree}}$-$\mathrm{Iglobal_{wfree}}$	0	9.037	-3.30506	5.73194

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Fold	Contrast	Target	Event	ΔC	Sum
0	$\text{Ilocal}_{\text{wfree}}$-$\text{Iglobal}_{\text{wfree}}$	1	9.54061	-5.43655	4.10406
0	$\text{Ilocal}_{\text{wfree}}$-$\text{Iglobal}_{\text{wfree}}$	2	4.32727	-2.88403	1.44324
1	E_{wfree}-E_{w0}	0	1.64758	1.76103	3.40861
1	E_{wfree}-E_{w0}	1	6.60185	-1.17024	5.43161
1	E_{wfree}-E_{w0}	2	1.37308	-2.03658	-0.663505
1	E_{wfree}-E_{w1}	0	132.058	-107.833	24.2251
1	E_{wfree}-E_{w1}	1	-5.47731	39.109	33.6317
1	E_{wfree}-E_{w1}	2	-3.258	72.5788	69.3208
1	$\text{Ilocal}_{\text{wfree}}$-$\text{E}_{\text{wfree}}$	0	-2.69947	1.32516	-1.37431
1	$\text{Ilocal}_{\text{wfree}}$-$\text{E}_{\text{wfree}}$	1	2.65283	-3.85229	-1.19946
1	$\text{Ilocal}_{\text{wfree}}$-$\text{E}_{\text{wfree}}$	2	0.60829	-1.26714	-0.65885
1	$\text{Iglobal}_{\text{wfree}}$-$\text{E}_{\text{wfree}}$	0	-0.188234	0.0366368	-0.151597
1	$\text{Iglobal}_{\text{wfree}}$-$\text{E}_{\text{wfree}}$	1	-0.136832	0.481554	0.344722
1	$\text{Iglobal}_{\text{wfree}}$-$\text{E}_{\text{wfree}}$	2	-0.321589	0.767843	0.446254
1	$\text{Ilocal}_{\text{w0}}$-$\text{E}_{\text{w0}}$	0	-2.7728	1.28367	-1.48912
1	$\text{Ilocal}_{\text{w0}}$-$\text{E}_{\text{w0}}$	1	2.62957	-3.62208	-0.992516
1	$\text{Ilocal}_{\text{w0}}$-$\text{E}_{\text{w0}}$	2	0.52883	-1.23908	-0.710248
1	$\text{Iglobal}_{\text{w0}}$-$\text{E}_{\text{w0}}$	0	-1.01594×10^{-5}	7.07895×10^{-6}	-3.08047×10^{-6}
1	$\text{Iglobal}_{\text{w0}}$-$\text{E}_{\text{w0}}$	1	-1.32397×10^{-5}	1.75729×10^{-5}	4.33326×10^{-6}
1	$\text{Iglobal}_{\text{w0}}$-$\text{E}_{\text{w0}}$	2	-1.13838×10^{-5}	2.74907×10^{-5}	1.61069×10^{-5}
1	$\text{Ilocal}_{\text{w1}}$-$\text{E}_{\text{w1}}$	0	0	0	0
1	$\text{Ilocal}_{\text{w1}}$-$\text{E}_{\text{w1}}$	1	0	0	0
1	$\text{Ilocal}_{\text{w1}}$-$\text{E}_{\text{w1}}$	2	0	0	0
1	$\text{Iglobal}_{\text{w1}}$-$\text{E}_{\text{w1}}$	0	-1.24076	0.441663	-0.799097
1	$\text{Iglobal}_{\text{w1}}$-$\text{E}_{\text{w1}}$	1	-0.309135	0.616737	0.307602
1	$\text{Iglobal}_{\text{w1}}$-$\text{E}_{\text{w1}}$	2	-0.482999	0.94699	0.463991
1	$\text{Ilocal}_{\text{wfree}}$-$\text{Iglobal}_{\text{wfree}}$	0	-2.51124	1.28853	-1.22271
1	$\text{Ilocal}_{\text{wfree}}$-$\text{Iglobal}_{\text{wfree}}$	1	2.78966	-4.33384	-1.54418
1	$\text{Ilocal}_{\text{wfree}}$-$\text{Iglobal}_{\text{wfree}}$	2	0.929879	-2.03498	-1.1051
2	E_{wfree}-E_{w0}	0	-0.215898	-0.637351	-0.853249
2	E_{wfree}-E_{w0}	1	-2.47187	0.930783	-1.54109
2	E_{wfree}-E_{w0}	2	2.97074	-0.493147	2.47759
2	E_{wfree}-E_{w1}	0	-1.57849	33.053	31.4745
2	E_{wfree}-E_{w1}	1	106.234	-53.8692	52.3649
2	E_{wfree}-E_{w1}	2	17.2332	24.6915	41.9248
2	$\text{Ilocal}_{\text{wfree}}$-$\text{E}_{\text{wfree}}$	0	-0.833143	0.399447	-0.433696
2	$\text{Ilocal}_{\text{wfree}}$-$\text{E}_{\text{wfree}}$	1	-0.262932	0.201889	-0.061043
2	$\text{Ilocal}_{\text{wfree}}$-$\text{E}_{\text{wfree}}$	2	0.0754677	-0.0727058	0.00276194
2	$\text{Iglobal}_{\text{wfree}}$-$\text{E}_{\text{wfree}}$	0	-3.47924	1.8192	-1.66004
2	$\text{Iglobal}_{\text{wfree}}$-$\text{E}_{\text{wfree}}$	1	-1.6334	1.16994	-0.463461
2	$\text{Iglobal}_{\text{wfree}}$-$\text{E}_{\text{wfree}}$	2	-2.22283	2.01117	-0.211657
2	$\text{Ilocal}_{\text{w0}}$-$\text{E}_{\text{w0}}$	0	5.45463×10^{-6}	-4.18448×10^{-6}	1.27015×10^{-6}
2	$\text{Ilocal}_{\text{w0}}$-$\text{E}_{\text{w0}}$	1	9.29142×10^{-6}	-8.64058×10^{-6}	6.50837×10^{-7}
2	$\text{Ilocal}_{\text{w0}}$-$\text{E}_{\text{w0}}$	2	4.46759×10^{-6}	-5.56446×10^{-6}	-1.09687×10^{-6}

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Fold	Contrast	Target	Event	ΔC	Sum
2	$\mathrm{Iglobal_{w0}}$-$\mathrm{E_{w0}}$	0	5.45463×10^{-6}	-4.18448×10^{-6}	1.27015×10^{-6}
2	$\mathrm{Iglobal_{w0}}$-$\mathrm{E_{w0}}$	1	9.29142×10^{-6}	-8.64058×10^{-6}	6.50837×10^{-7}
2	$\mathrm{Iglobal_{w0}}$-$\mathrm{E_{w0}}$	2	4.46759×10^{-6}	-5.56446×10^{-6}	-1.09687×10^{-6}
2	$\mathrm{Ilocal_{w1}}$-$\mathrm{E_{w1}}$	0	-1.51857×10^{-5}	8.72726×10^{-7}	-1.43129×10^{-5}
2	$\mathrm{Ilocal_{w1}}$-$\mathrm{E_{w1}}$	1	-2.74149×10^{-6}	6.87368×10^{-7}	-2.05412×10^{-6}
2	$\mathrm{Ilocal_{w1}}$-$\mathrm{E_{w1}}$	2	-4.44716×10^{-6}	9.84713×10^{-7}	-3.46245×10^{-6}
2	$\mathrm{Iglobal_{w1}}$-$\mathrm{E_{w1}}$	0	-1.96436	1.28506	-0.679301
2	$\mathrm{Iglobal_{w1}}$-$\mathrm{E_{w1}}$	1	-1.58037	1.01212	-0.568248
2	$\mathrm{Iglobal_{w1}}$-$\mathrm{E_{w1}}$	2	-1.3726	1.44995	0.077349
2	$\mathrm{Ilocal_{wfree}}$-$\mathrm{Iglobal_{wfree}}$	0	2.64609	-1.41975	1.22634
2	$\mathrm{Ilocal_{wfree}}$-$\mathrm{Iglobal_{wfree}}$	1	1.37046	-0.968047	0.402418
2	$\mathrm{Ilocal_{wfree}}$-$\mathrm{Iglobal_{wfree}}$	2	2.29829	-2.08387	0.214419

B.8.3 D window and target-cell accounting

Per-window event and compensator decomposition

Fold	Contrast	Events	Event	ΔC	Timing	Zone	Cell given zone	Joint
0	$\mathrm{E_{bridge}}$-$\mathrm{E_{global}}$	203	103.29	1.81423	-2.08323	107.188	2.27374×10^{-13}	105.105
0	$\mathrm{I_{bridge}}$-$\mathrm{I_{global}}$	203	134.194	-12.3088	14.0738	107.812	5.68434×10^{-14}	121.885
0	$\mathrm{I_{bridge}}$-$\mathrm{E_{bridge}}$	203	-3.00086	2.79298	-0.831722	0.623842	5.68434×10^{-14}	-0.20788
0	$\mathrm{E_{distance}}$-$\mathrm{E_{bridge}}$	203	3.45382	-0.00881186	0.67817	-5.19369	7.96052	3.44501
0	$\mathrm{E_{distance}}$-$\mathrm{E_{global}}$	203	106.744	1.80542	-1.40506	101.994	7.96052	108.55
0	$\mathrm{I_{distance}}$-$\mathrm{E_{distance}}$	203	1.89808	-1.44588	1.22564	-0.500322	-0.27312	0.4522
0	$\mathrm{I_{d,g}}$-$\mathrm{E_{distance}}$	203	-7.75799	5.42857	-2.45323	0.248731	-0.124921	-2.32942
0	$\mathrm{I_{distance}}$-$\mathrm{I_{d,g}}$	203	9.65607	-6.87445	3.67887	-0.749053	-0.148199	2.78162
1	$\mathrm{E_{bridge}}$-$\mathrm{E_{global}}$	155	113.7	5.30079	1.79755	117.203	7.10543×10^{-14}	119.001
1	$\mathrm{I_{bridge}}$-$\mathrm{I_{global}}$	155	116.119	-0.282137	0.57901	115.258	-4.26326×10^{-14}	115.837
1	$\mathrm{I_{bridge}}$-$\mathrm{E_{bridge}}$	155	0.385602	-3.57749	-1.24603	-1.94586	2.84217×10^{-14}	-3.19188
1	$\mathrm{E_{distance}}$-$\mathrm{E_{bridge}}$	155	34.2708	-1.00246	-0.43472	-2.64504	36.3481	33.2684
1	$\mathrm{E_{distance}}$-$\mathrm{E_{global}}$	155	147.971	4.29832	1.36283	114.558	36.3481	152.269
1	$\mathrm{I_{distance}}$-$\mathrm{E_{distance}}$	155	4.4095	-7.15026	-2.26963	-5.46472	4.99359	-2.74076

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Fold	Contrast	Events	Event	ΔC	Timing	Zone	Cell given zone	Joint
1	$I_{d,g}$-$E_{distance}$	155	-0.409991	0.72405	0.0964919	0.196743	0.0208243	0.31406
1	$I_{distance}$-$I_{d,g}$	155	4.81949	-7.87431	-2.36612	-5.66147	4.97277	-3.05482
2	E_{bridge}-E_{global}	254	121.606	4.07505	2.11386	123.567	-2.84217×10^{-13}	125.681
2	I_{bridge}-I_{global}	254	126.523	0.327865	3.28404	123.567	-2.27374×10^{-13}	126.851
2	I_{bridge}-E_{bridge}	254	1.91887×10^{-5}	-1.83711×10^{-5}	7.37475×10^{-7}	8.00863×10^{-8}	-5.68434×10^{-14}	8.17562×10^{-7}
2	$E_{distance}$-E_{bridge}	254	37.1782	0.952261	-3.50292	2.13441	39.4989	38.1304
2	$E_{distance}$-E_{global}	254	158.784	5.02731	-1.38906	125.701	39.4989	163.811
2	$I_{distance}$-$E_{distance}$	254	1.56192	-1.54844	0.527849	-0.144115	-0.370255	0.0134791
2	$I_{d,g}$-$E_{distance}$	254	-3.31077	2.33536	-0.496628	-0.362222	-0.116559	-0.97541
2	$I_{distance}$-$I_{d,g}$	254	4.87269	-3.8838	1.02448	0.218107	-0.253695	0.988889

Proper target-region exposures

Fold	Contrast	Target	Event	ΔC	Sum
0	E_{bridge}-E_{global}	0	22.2146	2.23149	24.4461
0	E_{bridge}-E_{global}	1	2.1905	0.155685	2.34618
0	E_{bridge}-E_{global}	2	0	0.0172983	0.0172983
0	E_{bridge}-E_{global}	3	0	0.0719026	0.0719026
0	E_{bridge}-E_{global}	4	6.10393	9.56305	15.667
0	E_{bridge}-E_{global}	5	-8.42815	7.54978	-0.87837
0	E_{bridge}-E_{global}	6	0	-0.0762874	-0.0762874
0	E_{bridge}-E_{global}	7	0	-0.0762874	-0.0762874
0	E_{bridge}-E_{global}	8	81.2094	-17.6224	63.587
0	I_{bridge}-I_{global}	0	33.4739	-1.54093	31.933
0	I_{bridge}-I_{global}	1	3.59941	-0.107507	3.49191
0	I_{bridge}-I_{global}	2	0	-0.0119452	-0.0119452
0	I_{bridge}-I_{global}	3	0	0.045096	0.045096
0	I_{bridge}-I_{global}	4	15.2089	5.99776	21.2066
0	I_{bridge}-I_{global}	5	-5.78193	4.73508	-1.04685
0	I_{bridge}-I_{global}	6	0	-0.0919586	-0.0919586
0	I_{bridge}-I_{global}	7	0	-0.0919586	-0.0919586
0	I_{bridge}-I_{global}	8	87.6939	-21.2424	66.4515
0	I_{bridge}-E_{bridge}	0	0.493526	-0.225793	0.267734
0	I_{bridge}-E_{bridge}	1	0.0391442	-0.015753	0.0233912
0	I_{bridge}-E_{bridge}	2	0	-0.00175033	-0.00175033
0	I_{bridge}-E_{bridge}	3	0	0.000940318	0.000940318
0	I_{bridge}-E_{bridge}	4	-0.273347	0.125062	-0.148285
0	I_{bridge}-E_{bridge}	5	-0.0930809	0.0987334	0.00565254
0	I_{bridge}-E_{bridge}	6	0	0.0120667	0.0120667
0	I_{bridge}-E_{bridge}	7	0	0.0120667	0.0120667
0	I_{bridge}-E_{bridge}	8	-3.1671	2.78741	-0.379696
0	$E_{distance}$-E_{bridge}	0	1.52369	-1.56967	-0.0459846
0	$E_{distance}$-E_{bridge}	1	-1.7412	0.755243	-0.985952
0	$E_{distance}$-E_{bridge}	2	0	0.110556	0.110556

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Fold	Contrast	Target	Event	ΔC	Sum
0	$E_{distance}$-E_{bridge}	3	0	0.0441556	0.0441556
0	$E_{distance}$-E_{bridge}	4	4.45311	-3.1711	1.28201
0	$E_{distance}$-E_{bridge}	5	5.39786	0.300324	5.69819
0	$E_{distance}$-E_{bridge}	6	0	0.209397	0.209397
0	$E_{distance}$-E_{bridge}	7	0	0.202572	0.202572
0	$E_{distance}$-E_{bridge}	8	-6.17964	3.10972	-3.06993
0	$E_{distance}$-E_{global}	0	23.7383	0.661813	24.4001
0	$E_{distance}$-E_{global}	1	0.449301	0.910928	1.36023
0	$E_{distance}$-E_{global}	2	0	0.127854	0.127854
0	$E_{distance}$-E_{global}	3	0	0.116058	0.116058
0	$E_{distance}$-E_{global}	4	10.557	6.39195	16.949
0	$E_{distance}$-E_{global}	5	-3.03028	7.8501	4.81982
0	$E_{distance}$-E_{global}	6	0	0.133109	0.133109
0	$E_{distance}$-E_{global}	7	0	0.126284	0.126284
0	$E_{distance}$-E_{global}	8	75.0297	-14.5127	60.5171
0	$I_{distance}$-$E_{distance}$	0	0.8838	-0.414348	0.469452
0	$I_{distance}$-$E_{distance}$	1	0.781575	-0.317988	0.463587
0	$I_{distance}$-$E_{distance}$	2	0	-0.0379157	-0.0379157
0	$I_{distance}$-$E_{distance}$	3	0	-0.0363839	-0.0363839
0	$I_{distance}$-$E_{distance}$	4	2.77134	-2.05343	0.717913
0	$I_{distance}$-$E_{distance}$	5	1.61461	-1.85018	-0.235565
0	$I_{distance}$-$E_{distance}$	6	0	-0.0388164	-0.0388164
0	$I_{distance}$-$E_{distance}$	7	0	-0.0368891	-0.0368891
0	$I_{distance}$-$E_{distance}$	8	-4.15325	3.34007	-0.813181
0	$I_{d,g}$-$E_{distance}$	0	-2.12819	1.14069	-0.987506
0	$I_{d,g}$-$E_{distance}$	1	-0.380981	0.0873014	-0.293679
0	$I_{d,g}$-$E_{distance}$	2	0	0.00993793	0.00993793
0	$I_{d,g}$-$E_{distance}$	3	0	0.00991384	0.00991384
0	$I_{d,g}$-$E_{distance}$	4	-2.43525	1.23782	-1.19744
0	$I_{d,g}$-$E_{distance}$	5	-0.603029	1.00225	0.399222
0	$I_{d,g}$-$E_{distance}$	6	0	0.0100631	0.0100631
0	$I_{d,g}$-$E_{distance}$	7	0	0.0100022	0.0100022
0	$I_{d,g}$-$E_{distance}$	8	-2.21053	1.9206	-0.289934
0	$I_{distance}$-$I_{d,g}$	0	3.01199	-1.55504	1.45696
0	$I_{distance}$-$I_{d,g}$	1	1.16256	-0.405289	0.757266
0	$I_{distance}$-$I_{d,g}$	2	0	-0.0478537	-0.0478537
0	$I_{distance}$-$I_{d,g}$	3	0	-0.0462978	-0.0462978
0	$I_{distance}$-$I_{d,g}$	4	5.20659	-3.29124	1.91535
0	$I_{distance}$-$I_{d,g}$	5	2.21764	-2.85243	-0.634786
0	$I_{distance}$-$I_{d,g}$	6	0	-0.0488796	-0.0488796
0	$I_{distance}$-$I_{d,g}$	7	0	-0.0468913	-0.0468913
0	$I_{distance}$-$I_{d,g}$	8	-1.94272	1.41947	-0.523247
1	E_{bridge}-E_{global}	0	110.126	-101.801	8.32533
1	E_{bridge}-E_{global}	1	20.2845	-7.30626	12.9783
1	E_{bridge}-E_{global}	2	0	-0.487084	-0.487084
1	E_{bridge}-E_{global}	3	0	0.128688	0.128688
1	E_{bridge}-E_{global}	4	-9.51127	24.322	14.8107
1	E_{bridge}-E_{global}	5	-2.5679	15.8286	13.2607
1	E_{bridge}-E_{global}	6	0	0.155773	0.155773
1	E_{bridge}-E_{global}	7	0	0.155773	0.155773
1	E_{bridge}-E_{global}	8	-4.63109	74.3039	69.6728
1	I_{bridge}-I_{global}	0	109.576	-101.018	8.55788
1	I_{bridge}-I_{global}	1	19.3021	-7.25013	12.0519
1	I_{bridge}-I_{global}	2	0	-0.483342	-0.483342
1	I_{bridge}-I_{global}	3	0	0.115145	0.115145
1	I_{bridge}-I_{global}	4	-8.77441	21.7624	12.988
1	I_{bridge}-I_{global}	5	-0.366054	14.1629	13.7968
1	I_{bridge}-I_{global}	6	0	0.151209	0.151209

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Fold	Contrast	Target	Event	ΔC	Sum
1	I_{bridge}-I_{global}	7	0	0.151209	0.151209
1	I_{bridge}-I_{global}	8	-3.61925	72.1269	68.5077
1	I_{bridge}-E_{bridge}	0	-1.16362	1.19239	0.0287677
1	I_{bridge}-E_{bridge}	1	-1.60917	0.0855783	-1.52359
1	I_{bridge}-E_{bridge}	2	0	0.00570522	0.00570522
1	I_{bridge}-E_{bridge}	3	0	-0.0115721	-0.0115721
1	I_{bridge}-E_{bridge}	4	0.669144	-2.18714	-1.51799
1	I_{bridge}-E_{bridge}	5	1.96042	-1.42337	0.537049
1	I_{bridge}-E_{bridge}	6	0	-0.0025868	-0.0025868
1	I_{bridge}-E_{bridge}	7	0	-0.0025868	-0.0025868
1	I_{bridge}-E_{bridge}	8	0.52883	-1.2339	-0.705074
1	$E_{distance}$-E_{bridge}	0	5.55823	10.7002	16.2585
1	$E_{distance}$-E_{bridge}	1	8.45207	7.72506	16.1771
1	$E_{distance}$-E_{bridge}	2	0	0.636456	0.636456
1	$E_{distance}$-E_{bridge}	3	0	0.0127908	0.0127908
1	$E_{distance}$-E_{bridge}	4	19.1858	-15.8708	3.31501
1	$E_{distance}$-E_{bridge}	5	1.36242	-1.14838	0.214035
1	$E_{distance}$-E_{bridge}	6	0	0.0124583	0.0124583
1	$E_{distance}$-E_{bridge}	7	0	0.0104879	0.0104879
1	$E_{distance}$-E_{bridge}	8	-0.287695	-3.0808	-3.3685
1	$E_{distance}$-E_{global}	0	115.684	-91.1003	24.5838
1	$E_{distance}$-E_{global}	1	28.7366	0.4188	29.1554
1	$E_{distance}$-E_{global}	2	0	0.149372	0.149372
1	$E_{distance}$-E_{global}	3	0	0.141479	0.141479
1	$E_{distance}$-E_{global}	4	9.67452	8.45122	18.1257
1	$E_{distance}$-E_{global}	5	-1.20548	14.6802	13.4747
1	$E_{distance}$-E_{global}	6	0	0.168232	0.168232
1	$E_{distance}$-E_{global}	7	0	0.166261	0.166261
1	$E_{distance}$-E_{global}	8	-4.91878	71.2231	66.3043
1	$I_{distance}$-$E_{distance}$	0	-4.06396	3.6889	-0.375064
1	$I_{distance}$-$E_{distance}$	1	2.61989	-0.628783	1.99111
1	$I_{distance}$-$E_{distance}$	2	0	-0.0468517	-0.0468517
1	$I_{distance}$-$E_{distance}$	3	0	-0.0439742	-0.0439742
1	$I_{distance}$-$E_{distance}$	4	0.965733	-5.09986	-4.13412
1	$I_{distance}$-$E_{distance}$	5	4.32996	-3.58338	0.74658
1	$I_{distance}$-$E_{distance}$	6	0	-0.0479659	-0.0479659
1	$I_{distance}$-$E_{distance}$	7	0	-0.0465053	-0.0465053
1	$I_{distance}$-$E_{distance}$	8	0.557883	-1.34185	-0.783963
1	$I_{d,g}$-$E_{distance}$	0	0.0401512	-0.0628927	-0.0227416
1	$I_{d,g}$-$E_{distance}$	1	-0.117479	0.0114469	-0.106032
1	$I_{d,g}$-$E_{distance}$	2	0	0.00104176	0.00104176
1	$I_{d,g}$-$E_{distance}$	3	0	0.00102631	0.00102631
1	$I_{d,g}$-$E_{distance}$	4	-0.00263744	0.152015	0.149378
1	$I_{d,g}$-$E_{distance}$	5	-0.0595624	0.119992	0.0604293
1	$I_{d,g}$-$E_{distance}$	6	0	0.00109003	0.00109003
1	$I_{d,g}$-$E_{distance}$	7	0	0.00108551	0.00108551
1	$I_{d,g}$-$E_{distance}$	8	-0.270463	0.499245	0.228782
1	$I_{distance}$-$I_{d,g}$	0	-4.10411	3.75179	-0.352322
1	$I_{distance}$-$I_{d,g}$	1	2.73737	-0.64023	2.09714
1	$I_{distance}$-$I_{d,g}$	2	0	-0.0478935	-0.0478935
1	$I_{distance}$-$I_{d,g}$	3	0	-0.0450005	-0.0450005
1	$I_{distance}$-$I_{d,g}$	4	0.96837	-5.25187	-4.2835
1	$I_{distance}$-$I_{d,g}$	5	4.38952	-3.70337	0.68615
1	$I_{distance}$-$I_{d,g}$	6	0	-0.0490559	-0.0490559
1	$I_{distance}$-$I_{d,g}$	7	0	-0.0475908	-0.0475908
1	$I_{distance}$-$I_{d,g}$	8	0.828346	-1.84109	-1.01274
2	E_{bridge}-E_{global}	0	-1.40194	28.6553	27.2534
2	E_{bridge}-E_{global}	1	0.0508683	4.961	5.01187

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Fold	Contrast	Target	Event	ΔC	Sum
2	E_{bridge}-E_{global}	2	-0.0115199	0.0740448	0.0625249
2	E_{bridge}-E_{global}	3	0	-0.152646	-0.152646
2	E_{bridge}-E_{global}	4	102.851	-31.9031	70.9478
2	E_{bridge}-E_{global}	5	5.85504	-22.7443	-16.8893
2	E_{bridge}-E_{global}	6	0	0.0490929	0.0490929
2	E_{bridge}-E_{global}	7	0	0.0490929	0.0490929
2	E_{bridge}-E_{global}	8	14.2625	25.0865	39.349
2	I_{bridge}-I_{global}	0	0.476653	27.5623	28.039
2	I_{bridge}-I_{global}	1	0.099788	4.77177	4.87156
2	I_{bridge}-I_{global}	2	0.0253261	0.0712205	0.0965466
2	I_{bridge}-I_{global}	3	0	-0.155466	-0.155466
2	I_{bridge}-I_{global}	4	103.676	-32.4923	71.1839
2	I_{bridge}-I_{global}	5	6.61009	-23.1644	-16.5543
2	I_{bridge}-I_{global}	6	0	0.0462665	0.0462665
2	I_{bridge}-I_{global}	7	0	0.0462665	0.0462665
2	I_{bridge}-I_{global}	8	15.6351	23.6422	39.2773
2	I_{bridge}-E_{bridge}	0	5.16499×10^{-6}	-3.55438×10^{-6}	1.6106×10^{-6}
2	I_{bridge}-E_{bridge}	1	1.86419×10^{-7}	-6.15359×10^{-7}	-4.28939×10^{-7}
2	I_{bridge}-E_{bridge}	2	9.33217×10^{-8}	-9.18446×10^{-9}	8.41372×10^{-8}
2	I_{bridge}-E_{bridge}	3	0	-2.4051×10^{-8}	-2.4051×10^{-8}
2	I_{bridge}-E_{bridge}	4	7.2178×10^{-6}	-5.02665×10^{-6}	2.19115×10^{-6}
2	I_{bridge}-E_{bridge}	5	2.06441×10^{-6}	-3.58359×10^{-6}	-1.51918×10^{-6}
2	I_{bridge}-E_{bridge}	6	0	-1.08341×10^{-8}	-1.08341×10^{-8}
2	I_{bridge}-E_{bridge}	7	0	-1.08341×10^{-8}	-1.08341×10^{-8}
2	I_{bridge}-E_{bridge}	8	4.46174×10^{-6}	-5.53622×10^{-6}	-1.07449×10^{-6}
2	$E_{distance}$-E_{bridge}	0	2.12054	-4.12809	-2.00755
2	$E_{distance}$-E_{bridge}	1	-0.276221	2.69055	2.41433
2	$E_{distance}$-E_{bridge}	2	-0.142578	0.056243	-0.0863347
2	$E_{distance}$-E_{bridge}	3	0	0.289847	0.289847
2	$E_{distance}$-E_{bridge}	4	37.3107	-28.3568	8.95392
2	$E_{distance}$-E_{bridge}	5	-4.49236	30.5329	26.0406
2	$E_{distance}$-E_{bridge}	6	0	0.0988367	0.0988367
2	$E_{distance}$-E_{bridge}	7	0	0.0944831	0.0944831
2	$E_{distance}$-E_{bridge}	8	2.65809	-0.325752	2.33234
2	$E_{distance}$-E_{global}	0	0.718607	24.5272	25.2459
2	$E_{distance}$-E_{global}	1	-0.225353	7.65155	7.4262
2	$E_{distance}$-E_{global}	2	-0.154098	0.130288	-0.0238098
2	$E_{distance}$-E_{global}	3	0	0.137201	0.137201
2	$E_{distance}$-E_{global}	4	140.162	-60.2598	79.9018
2	$E_{distance}$-E_{global}	5	1.36268	7.78862	9.1513
2	$E_{distance}$-E_{global}	6	0	0.14793	0.14793
2	$E_{distance}$-E_{global}	7	0	0.143576	0.143576
2	$E_{distance}$-E_{global}	8	16.9206	24.7607	41.6813
2	$I_{distance}$-$E_{distance}$	0	-1.53184	0.632313	-0.899524
2	$I_{distance}$-$E_{distance}$	1	0.165263	-0.266383	-0.101119
2	$I_{distance}$-$E_{distance}$	2	0.0866314	-0.00400427	0.0826271
2	$I_{distance}$-$E_{distance}$	3	0	-0.00395532	-0.00395532
2	$I_{distance}$-$E_{distance}$	4	1.10552	-0.78696	0.318564
2	$I_{distance}$-$E_{distance}$	5	1.17792	-0.478157	0.699764
2	$I_{distance}$-$E_{distance}$	6	0	-0.00397012	-0.00397012
2	$I_{distance}$-$E_{distance}$	7	0	-0.00397023	-0.00397023
2	$I_{distance}$-$E_{distance}$	8	0.558417	-0.633355	-0.0749378

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Fold	Contrast	Target	Event	ΔC	Sum
2	$I_{d,g}$-$E_{distance}$	0	-1.52164	0.731077	-0.790564
2	$I_{d,g}$-$E_{distance}$	1	-0.0211498	0.133934	0.112784
2	$I_{d,g}$-$E_{distance}$	2	-0.0197161	0.0020338	-0.0176823
2	$I_{d,g}$-$E_{distance}$	3	0	0.00204564	0.00204564
2	$I_{d,g}$-$E_{distance}$	4	-0.246055	0.235195	-0.01086
2	$I_{d,g}$-$E_{distance}$	5	-0.469565	0.277432	-0.192132
2	$I_{d,g}$-$E_{distance}$	6	0	0.00207327	0.00207327
2	$I_{d,g}$-$E_{distance}$	7	0	0.00206385	0.00206385
2	$I_{d,g}$-$E_{distance}$	8	-1.03264	0.949504	-0.0831372
2	$I_{distance}$-$I_{d,g}$	0	-0.0101955	-0.0987639	-0.108959
2	$I_{distance}$-$I_{d,g}$	1	0.186413	-0.400316	-0.213903
2	$I_{distance}$-$I_{d,g}$	2	0.106347	-0.00603807	0.100309
2	$I_{distance}$-$I_{d,g}$	3	0	-0.00600096	-0.00600096
2	$I_{distance}$-$I_{d,g}$	4	1.35158	-1.02215	0.329424
2	$I_{distance}$-$I_{d,g}$	5	1.64749	-0.755589	0.891897
2	$I_{distance}$-$I_{d,g}$	6	0	-0.00604339	-0.00604339
2	$I_{distance}$-$I_{d,g}$	7	0	-0.00603409	-0.00603409
2	$I_{distance}$-$I_{d,g}$	8	1.59106	-1.58286	0.00819931

B.8.4 Secondary common-space window arithmetic

Per-window arithmetic

Fold	D candidate	Events	Event	ΔC	Timing	Zone	Cell given zone	Joint
0	E_{global}	203	-106.03	-1.36753	1.21755	-108.615	1.42109×10^{-14}	-107.398
0	I_{global}	203	-139.935	15.5485	-15.7712	-108.615	1.42109×10^{-14}	-124.387
0	E_{bridge}	203	-2.74011	0.446709	-0.865688	-1.42771	1.42109×10^{-14}	-2.2934
0	I_{bridge}	203	-5.74097	3.23969	-1.69741	-0.803867	1.42109×10^{-14}	-2.50128
0	$E_{distance}$	203	0.713712	0.437897	-0.187518	-6.6214	7.96052	1.15161
0	$I_{distance}$	203	2.61179	-1.00798	1.03812	-7.12172	7.6874	1.60381
0	$I_{d,g}$	203	-7.04427	5.86647	-2.64074	-6.37267	7.8356	-1.17781
1	E_{global}	155	-123.323	-3.85499	-1.23609	-125.942	0	-127.178
1	I_{global}	155	-125.356	-1.84955	-1.26358	-125.942	0	-127.205
1	E_{bridge}	155	-9.62251	1.4458	0.561462	-8.73818	0	-8.17672
1	I_{bridge}	155	-9.23691	-2.13169	-0.684567	-10.684	0	-11.3686
1	$E_{distance}$	155	24.6483	0.443333	0.126742	-11.3832	36.3481	25.0916
1	$I_{distance}$	155	29.0578	-6.70693	-2.14289	-16.8479	41.3417	22.3509
1	$I_{d,g}$	155	24.2383	1.16738	0.223234	-11.1865	36.3689	25.4057
2	E_{global}	254	-121.889	-3.87533	1.42649	-127.191	0	-125.764
2	I_{global}	254	-126.806	-0.128169	0.256312	-127.191	-1.42109×10^{-14}	-126.934
2	E_{bridge}	254	-0.282971	0.199715	3.54035	-3.6236	0	-0.0832556
2	I_{bridge}	254	-0.282952	0.199697	3.54035	-3.6236	-1.42109×10^{-14}	-0.0832547
2	$E_{distance}$	254	36.8952	1.15198	0.037426	-1.48919	39.4989	38.0472
2	$I_{distance}$	254	38.4571	-0.396465	0.565275	-1.63331	39.1287	38.0607
2	$I_{d,g}$	254	33.5844	3.48733	-0.459202	-1.85141	39.3824	37.0718

Both stage-specific raw-row audits, source/input pins, raw event time/excess/category ordering, training proportions/smoothing, identical windows and shared beta were checked before arithmetic. Per-cell P+C contributions are retained as aggregates in summary.json; raw

catalogue rows, event arrays and caches are not exported. Original primary tables and fit/start denominators remain unchanged.

This comparison was planned after the W result and before opening D outcomes. It is secondary adaptive reporting and does not turn reused windows into independent validation or establish physical causality.

B.8.5 Complete factorial and bounded-scale score record

The model identifiers below encode temporal family, background (cell or smooth KDE-enriched) and offspring geography (cont or lift). Each fold's twelve rows use the same selected baseline spatial scale and test targets. Absolute log density scores depend on the declared square-kilometre measure; only matched differences carry the comparison interpretation. The common magnitude score is retained rather than absorbed into location. These are all primary arms, including the losing temporal/lift combinations.

Fold	Arm	*N*	Timing	Zone	Cell given zone	Within-cell density	Mark	Joint
F0	fixedML_cell_cont	203	-334.242	-109.786	-25.8562	-1968.98	34.2981	-2404.57
F0	commonML_cell_cont	203	-333.662	-109.884	-25.8391	-1968.77	34.2981	-2403.86
F0	Omori_cell_cont	203	-331.395	-110.326	-25.7421	-1964.35	34.2981	-2397.51
F0	fixedML_cell_lift	203	-331.846	-113.936	-27.8384	-2514	34.2981	-2953.33
F0	commonML_cell_lift	203	-331.499	-114.038	-27.8009	-2514	34.2981	-2953.04
F0	Omori_cell_lift	203	-329.13	-114.862	-27.7262	-2514	34.2981	-2951.42
F0	fixedML_smooth_cont	203	-337.85	-109.568	-25.6119	-1941.49	34.2981	-2380.23
F0	commonML_smooth_cont	203	-336.616	-109.995	-25.5726	-1940.56	34.2981	-2378.45
F0	Omori_smooth_cont	203	-334.808	-111.005	-25.5254	-1942.5	34.2981	-2379.54
F0	fixedML_smooth_lift	203	-362.798	-115.451	-27.6931	-2467.26	34.2981	-2938.9
F0	commonML_smooth_lift	203	-368.404	-118.704	-27.791	-2465.77	34.2981	-2946.37
F0	Omori_smooth_lift	203	-370.741	-121.533	-27.9344	-2465.09	34.2981	-2951
F1	fixedML_cell_cont	155	-158.445	-106.119	-35.2124	-1461.23	-3.84474	-1764.85
F1	commonML_cell_cont	155	-157.564	-106.198	-35.2265	-1461.37	-3.84474	-1764.2
F1	Omori_cell_cont	155	-150.352	-103.402	-37.9133	-1451.06	-3.84474	-1746.57
F1	fixedML_cell_lift	155	-158.719	-99.5906	-48.3143	-1907.4	-3.84474	-2217.87
F1	commonML_cell_lift	155	-154.521	-100.104	-48.3944	-1907.4	-3.84474	-2214.27
F1	Omori_cell_lift	155	-155.095	-97.8026	-49.8916	-1907.4	-3.84474	-2214.03
F1	fixedML_smooth_cont	155	-156.905	-106.779	-35.2247	-1444.76	-3.84474	-1747.51
F1	commonML_smooth_cont	155	-148.447	-107.784	-35.4844	-1445.55	-3.84474	-1741.11
F1	Omori_smooth_cont	155	-154.611	-107.477	-35.9637	-1444.77	-3.84474	-1746.67
F1	fixedML_smooth_lift	155	-160.171	-102.05	-48.8902	-1883.91	-3.84474	-2198.87
F1	commonML_smooth_lift	155	-142.14	-105.305	-49.7509	-1881.47	-3.84474	-2182.51
F1	Omori_smooth_lift	155	-152.303	-108.517	-50.5331	-1879.42	-3.84474	-2194.62

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Fold	Arm	N	Timing	Zone	Cell given zone	Within-cell density	Mark	Joint
F2	fixedML_cell_cont	254	-507.66	-172.693	-58.1854	-2676.78	-26.9106	-3442.23
F2	commonML_cell_cont	254	-507.667	-172.23	-57.8413	-2680.19	-26.9106	-3444.83
F2	Omori_cell_cont	254	-495.18	-169.315	-62.8372	-2645.34	-26.9106	-3399.59
F2	fixedML_cell_lift	254	-499.65	-170.042	-59.1399	-3137.99	-26.9106	-3893.74
F2	commonML_cell_lift	254	-497.357	-169.958	-59.0612	-3137.99	-26.9106	-3891.28
F2	Omori_cell_lift	254	-492.084	-167.66	-60.7759	-3137.99	-26.9106	-3885.42
F2	fixedML_smooth_cont	254	-494.298	-171.266	-58.4698	-2647.22	-26.9106	-3398.16
F2	commonML_smooth_cont	254	-494.337	-171.008	-58.0393	-2650.43	-26.9106	-3400.73
F2	Omori_smooth_cont	254	-489.266	-169.026	-58.8698	-2640.15	-26.9106	-3384.22
F2	fixedML_smooth_lift	254	-482.63	-169.703	-59.712	-3073.31	-26.9106	-3812.26
F2	commonML_smooth_lift	254	-483.972	-171.463	-60.3154	-3072.04	-26.9106	-3814.7
F2	Omori_smooth_lift	254	-483.526	-172.269	-60.7074	-3071.18	-26.9106	-3814.6

Every small-scale sensitivity fit

Each row changes only the declared fixed-ML continuous procedure's fixed spatial scale, under its named background, relative to that background's primary selected-scale procedure. These rows are not an expanded-grid twelve-arm factorial and do not select the primary forecast. Every joint difference is negative; occasional timing improvements are retained.

Fold	Primary reference	Sensitivity km	Primary km	Joint delta	Timing delta	Joint per target
F0	fixedML_cell_cont	1	5	-252.691	-4.91036	-1.24479
F0	fixedML_cell_cont	2	5	-114.734	-2.58132	-0.565193
F0	fixedML_smooth_cont	1	5	-248.566	-8.51238	-1.22447
F0	fixedML_smooth_cont	2	5	-111.301	-3.90217	-0.548279
F1	fixedML_cell_cont	1	15	-207.359	2.87971	-1.3378
F1	fixedML_cell_cont	2	15	-100.135	2.52249	-0.646035
F1	fixedML_smooth_cont	1	15	-201.315	0.647017	-1.2988
F1	fixedML_smooth_cont	2	15	-96.988	0.821518	-0.625729
F2	fixedML_cell_cont	1	15	-308.064	15.0179	-1.21285
F2	fixedML_cell_cont	2	15	-167.638	11.0334	-0.659992
F2	fixedML_smooth_cont	1	15	-290.547	8.74533	-1.14389
F2	fixedML_smooth_cont	2	15	-159.63	5.66373	-0.628465

B.8.6 Numerical protocol coordinates and controls

The factorial study fixes a nine-cell domain, a six-point primary spatial grid and eight starts per procedure. Background amplitude, trend and branching coordinates use $a \in [-12, 0]$, $d \in [-2, 2]$ and $n \in [0, 0.98]$. Common ML scale ranges from 0.001 to 20 days; Omori uses $c \in [0.0001, 20]$ days and $p \in [1.01, 2.5]$. The primary bandwidth and spatial grids are 5, 15, 50, 150, 500 and 1500 km. The background mixes 95% training KDE with a 5% cell-density floor.

Geometry uses latitude edges $-48.5, -43.5, -38.5, -33.5$ degrees, longitude edges $165, 170, 175, 180$ degrees, Earth radius 6371.0088 km and projection origin $(-41, 172.5)$ degrees. These are finite-domain numerical choices, not physical scale estimates.

The exact start offsets, optimizer settings, finite-difference steps, quadrature controls and both fixed-seed engineering laws are retained without alteration in the supplementary TeX fragment `paper/supplements/factorial_protocol_full.tex` and the bundled frozen protocol. Selected estimates above remain distinct from search limits and initial values. The configuration is a reproducibility contract, not another set of empirical replications.

B.9 External-comparison numerical backup

These tables retain the detailed saved displays supporting Chapter 11. A repeated subtotal or reporting partition is not an additional observation. Values, signs and source-specific rounding are preserved. The complete bundle supplies the underlying JSON; this appendix does not reproduce raw catalogue rows or refit any procedure.

B.9.1 California annual accounting

The original $M \geq 4.0$ population contains 315 targets. Annual boundaries partition events and exposure without resetting admissible history. Pooled means divide summed nats by 315 rather than averaging annual means. Signed 2019 shares include negative annual contributions in their denominators. The three columns use separate references and cannot be added.

Primary score and exact comparator (Omori_smooth_cont minus...)	Pooled gain, nats	Nats/target	2019 signed share
Timing: fixedML_ smooth_cont	37.790339	0.119969	77.48%
Conditional location: Omori_smooth_lift	674.323522	2.140710	58.247%
Joint: commonML_ smooth_cont	21.177884	0.067231	37.678%

UTC window	Targets	Timing vs fixedML, nats (per target)	Location vs Omori lift, nats (per target)	Joint vs commonML, nats (per target)
[2011,2016)	79	0.099084 (0.001254)	91.803255 (1.162067)	5.392213 (0.068256)
[2016,2021)	167	32.722878 (0.195945)	465.829993 (2.789401)	10.609008 (0.063527)
[2021,2026)	69	4.968377 (0.072005)	116.690274 (1.691163)	5.176663 (0.075024)

UTC year	Targets	Timing vs fixedML	Location vs Omori lift	Joint vs commonML
2011	14	-0.793154	3.79999	1.12678
2012	30	1.43786	71.4105	2.96013
2013	13	-0.389569	3.70658	-0.128884

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UTC year	Targets	Timing vs fixedML	Location vs Omori lift	Joint vs commonML
2014	12	-0.549005	7.99345	0.345606
2015	10	0.392957	4.89271	1.08859
2016	12	0.171398	11.7828	0.203358
2017	4	-0.0317705	-0.19626	0.0986852
2018	8	0.124995	-0.468682	0.16032
2019	116	29.2806	392.773	7.9794
2020	27	3.17766	61.9387	2.16724
2021	14	2.29243	32.6733	1.39997
2022	6	-0.277276	0.515857	0.224872
2023	11	0.685377	16.446	1.65334
2024	31	2.31922	66.5511	1.42642
2025	7	-0.0513726	0.503992	0.47207

The timing gain against fixedML decomposes into +40.898879 event-ground-log nats and -3.108539 exposure nats. The signed exposure contribution is **minus integrated intensity**, so a higher integrated intensity contributes negatively. The location gain against Omori lift consists of +17.206741 zone, +3.514553 conditional-cell and +653.602228 within-cell-density nats. The joint gain against commonML consists of +18.749060 timing and +2.428824 conditional-location nats, with zero mark contrast. These decompositions belong to their named comparator; they cannot be combined into a single decomposition across the three rows.

The source is `validation/authorities/bf4d50b9ecefc531_aggregates.json`. The reconstruction retains 60 arm rows, 45 contrast rows and 12 arm windows. Its 2,097 identity checks passed; the maximum independent discrepancy was $1.1368683772161603 \times 10^{-13}$ nats, below the 10^{-4} gate. The larger $3.9642412019702533 \times 10^{-7}$ comparison discrepancy concerned rounded historical totals, not a changed fitted state.

B.9.2 Higher-threshold roster, components and annual contributions

These displays concern the distinct 87-target M $\geq$ 4.5 population. The protocol retained every start, floor, candidate and finite return, including the first-fold resource pilot. Absolute arm totals include the common mark score once; it cancels in the contrasts. Zero-target years retain exposure and have undefined per-target means.

Planned item	Planned	Completed or audited
Fitted procedures	30	30
Prescribed optimizer starts	240	240
Exact nested floors	33	33
Accepted candidate slots	273	273
Finite solver returns	Retain every feasible return	240
Model-window statuses including prerequisite	15	15 complete
Retained primary-arm windows	12	12 complete
Annual arm blocks / contrast blocks	60 / 45	60 / 45
Test targets	16 / 53 / 18	87 total

Primary component and exact reference for OC minus reference	Pooled nats	Nats per target
Timing versus `fixedML_smooth_cont`	13.249423	0.152292
Conditional location versus `Omori_smooth_lift`	184.776469	2.123867
Joint versus `commonML_smooth_cont`	6.744097	0.077518

UTC window	Targets	Timing versus fixedML nats (per target)	Location versus Omori lift nats (per target)	Joint versus commonML nats (per target)
[2011,2016)	16	0.077968 (0.004873)	36.075692 (2.254731)	-0.626348 (-0.039147)
[2016,2021)	53	11.741293 (0.221534)	122.870533 (2.318312)	6.141657 (0.115880)
[2021,2026)	18	1.430162 (0.079453)	25.830245 (1.435014)	1.228788 (0.068266)

Primary arm	Timing nats	Conditional-location nats	Joint nats
`fixedML_smooth_cont`	-182.481890	-803.122764	-984.052721
`commonML_smooth_cont`	-175.225364	-803.571207	-977.244638
`Omori_smooth_cont`	-169.232467	-802.820007	-970.500541
`Omori_smooth_lift`	-183.351046	-987.596477	-1169.395589

UTC year	Targets	Timing versus fixedML	Location versus Omori lift	Joint versus commonML
2011	2	-0.69304	1.60649	-0.32573
2012	10	1.21515	34.7119	0.125733
2013	2	-0.151209	-0.171337	-0.237623
2014	2	-0.106284	-0.0714092	-0.104797
2015	0	-0.186651	0	-0.0839305
2016	2	-0.00588702	-0.106569	0.0171039
2017	0	-0.102727	0	-0.0373898
2018	3	0.0496577	-0.141071	-0.0308888
2019	37	10.9771	112.008	5.28845
2020	11	0.823168	11.1105	0.904382
2021	2	-0.0669458	-0.215809	-0.260059
2022	1	-0.125996	0.36667	0.151796
2023	2	-0.0280045	-0.134367	-0.101869
2024	12	1.72875	25.9029	1.56265
2025	1	-0.0776466	-0.0891843	-0.123725

Declared comparison	Original M4 nats (per 315 targets)	M4.5 nats (per 87 targets)
Timing OC versus fixedML	37.790339 (0.119969)	13.249423 (0.152292)
Location OC versus Omori lift	674.323522 (2.140710)	184.776469 (2.123867)
Joint OC versus commonML	21.177884 (0.067231)	6.744097 (0.077518)

The source is `validation/authorities/e93db40b7871acc7_approved_aggregates_v1.json`. The numerical audit covered 30 procedures, 240 starts, 33 floors, 273 candidates, 103 saved report artifacts and 84 caches; all 15 model-window statuses were complete. The maximum score discrepancy was $1.1226575225009583 \times 10^{-11}$ nats; annual reconstruction differed by at most $1.1368683772161603 \times 10^{-13}$. The full candidate, start, return, floor and training-selection tables are supplied in `reproduction/regional/original_primary_california_20261025.zip`, member `m45/private_reporting/candidate_tables_v1.json`. Its public copy replaces machine account roots with portable placeholders; the manifest retains the original hash 8ba55b42dae764c005e30bb3fe4200dbc7206d4a6f018aee2bc0760c5c50bb30 and verifies unchanged numerical fields. Historical compiler provenance remains unknown. None of these audit counts is a new empirical replication.

B.9.3 Complete sequence partition

Counts and durations are common to every arm and contrast. Calendar 2019, pooled rows and the episode row repeat portions of the original folds and must not be added to them. Empty active rows have zero exposure and totals, with undefined per-target means. The sole scored episode contains 110 targets over exactly 2,713,564,040,000 microseconds; exact integer membership is retained in the source.

Primary comparison: OC minus EP. All terms are nats; larger is better for OC. Timing = ground-event term - positive integrated intensity E. The exposure column is therefore -ΔE, not ΔE. Joint = timing + conditional location + inherited mark; the mark contrast is exactly zero in every row. The full regional compensator is retained independently of the target count. Active and complement rows and the calendar-2019 total use nine decimal places; other total rows and the repeated episode row use six significant digits. The accepted JSON retains full precision.

Reporting range / category	Targets	Anchor targets	Exposure days
2011-2016 / Active	0	0	0.000000000
2011-2016 / Complement	79	0	1826.000000000
2011-2016 / Total	79	0	1826
2016-2021 / Active	110	1	31.406991204
2016-2021 / Complement	57	1	1795.593008796
2016-2021 / Total	167	2	1827
2021-2026 / Active	0	0	0.000000000
2021-2026 / Complement	69	0	1826.000000000
2021-2026 / Total	69	0	1826

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Reporting range / category	Targets	Anchor targets	Exposure days
Calendar 2019 / Active	110	1	31.406991204
Calendar 2019 / Complement	6	1	333.593008796
Calendar 2019 / Total	116	2	365.000000000
Pooled 2011-2026 / Active	110	1	31.406991204
Pooled 2011-2026 / Complement	205	1	5447.593008796
Pooled 2011-2026 / Total	315	2	5479
2016-2021 / Episode 9 (Ridgecrest)	110	1	31.407

Range / category	Ground-event Δ	Exposure contribution -ΔE	Timing Δ	Location Δ	Joint Δ	Joint Δ/target
2011-2016 / Active	+0.000000000	+0.000000000	+0.000000000	+0.000000000	+0.000000000	null
2011-2016 / Complement	-1.064622768	+1.557903123	+0.493280355	-6.782099452	-6.288819097	-0.079605305
2011-2016 / Total	-1.06462	+1.5579	+0.49328	-6.7821	-6.28882	-0.0796053
2016-2021 / Active	-0.872592079	+0.065909253	-0.806682826	-17.565547269	-18.372230095	-0.167020274
2016-2021 / Complement	-0.679833552	+1.302640826	+0.622807275	-13.993089629	-13.370282354	-0.234566357
2016-2021 / Total	-1.55243	+1.36855	-0.183876	-31.5586	-31.7425	-0.190075
2021-2026 / Active	+0.000000000	+0.000000000	+0.000000000	+0.000000000	+0.000000000	null
2021-2026 / Complement	-0.587174150	+0.614018871	+0.026844721	-13.046717939	-13.019873217	-0.188693815
2021-2026 / Total	-0.587174	+0.614019	+0.0268447	-13.0467	-13.0199	-0.188694
Calendar 2019 / Active	-0.872592079	+0.065909253	-0.806682826	-17.565547269	-18.372230095	-0.167020274
Calendar 2019 / Complement	-0.071315734	+0.390295314	+0.318979580	-2.116996329	-1.798016750	-0.299669458
Calendar 2019 / Total	-0.943907813	+0.456204567	-0.487703246	-19.682543598	-20.170246844	-0.173881438
Pooled 2011-2026 / Active	-0.872592079	+0.065909253	-0.806682826	-17.565547269	-18.372230095	-0.167020274
Pooled 2011-2026 / Complement	-2.331630469	+3.474562820	+1.142932351	-33.821907020	-32.678974668	-0.159409633
Pooled 2011-2026 / Total	-3.20422	+3.54047	+0.33625	-51.3875	-51.0512	-0.162067
2016-2021 / Episode 9 (Ridgecrest)	-0.872592	+0.0659093	-0.806683	-17.5655	-18.3722	-0.16702

Three contextual comparisons. Each column has its own reference and primary component; the columns must never be added. Parentheses divide that row's summed nats by that row's target count. Pooled means use the summed original-fold scores divided by 315,

110 or 205 as appropriate; they do not average fold means. All sixteen reporting rows per contrast are retained, including the sole episode and both empty active rows.

Range / category	OC-fixedML timing: nats (per target)	OC-OL location: nats (per target)	OC-commonML joint: nats (per target)
2011-2016 / Active	+0.000000000 (null)	+0.000000000 (null)	+0.000000000 (null)
2011-2016 / Complement	+0.099084005 (+0.001254228)	+91.803254519 (+1.162066513)	+5.392213175 (+0.068255863)
2011-2016 / Total	+0.099084005 (+0.001254228)	+91.803254519 (+1.162066513)	+5.392213175 (+0.068255863)
2016-2021 / Active	+30.058155064 (+0.273255955)	+375.944700694 (+3.417679097)	+9.878145249 (+0.089801320)
2016-2021 / Complement	+2.664723302 (+0.046749532)	+89.885292282 (+1.576934952)	+0.730862724 (+0.012822153)
2016-2021 / Total	+32.722878366 (+0.195945379)	+465.829992976 (+2.789401156)	+10.609007973 (+0.063526994)
2021-2026 / Active	+0.000000000 (null)	+0.000000000 (null)	+0.000000000 (null)
2021-2026 / Complement	+4.968376888 (+0.072005462)	+116.690274452 (+1.691163398)	+5.176663249 (+0.075024105)
2021-2026 / Total	+4.968376888 (+0.072005462)	+116.690274452 (+1.691163398)	+5.176663249 (+0.075024105)
Calendar 2019 / Active	+30.058155064 (+0.273255955)	+375.944700694 (+3.417679097)	+9.878145249 (+0.089801320)
Calendar 2019 / Complement	-0.777558252 (-0.129593042)	+16.828728374 (+2.804788062)	-1.898745079 (-0.316457513)
Calendar 2019 / Total	+29.280596812 (+0.252418938)	+392.773429068 (+3.385977837)	+7.979400169 (+0.068787932)
Pooled 2011-2026 / Active	+30.058155064 (+0.273255955)	+375.944700694 (+3.417679097)	+9.878145249 (+0.089801320)
Pooled 2011-2026 / Complement	+7.732184195 (+0.037717972)	+298.378821253 (+1.455506445)	+11.299739148 (+0.055120679)
Pooled 2011-2026 / Total	+37.790339259 (+0.119969331)	+674.323521946 (+2.140709593)	+21.177884396 (+0.067231379)
2016-2021 / Episode 9 (Ridgecrest)	+30.058155064 (+0.273255955)	+375.944700694 (+3.417679097)	+9.878145249 (+0.089801320)

Absolute pooled arm accounting. The common target counts are 110 active, 205 complement and 315 total. These totals preserve the shared mark law once; contrast marks cancel. The closed aggregate also contains all fold, calendar-2019 and episode arm components, zone/cell/within-cell splits, exposures and per-target means.

Table B.47: Absolute pooled timing accounting: the ground-event sum and nonnegative exposure are retained separately.

Arm / category	Q	E	Timing
OC / Active	+451.279601588	+69.960547415	+381.319054173
OC / Complement	-447.826953842	+298.942177507	-746.769131348
OC / Total	+3.452647747	+368.902724922	-365.450077175
EP / Active	+452.152193667	+70.026456668	+382.125736999
EP / Complement	-445.495323373	+302.416740327	-747.912063700

Arm / category	Q	E	Timing
EP / Total	+6.656870295	+372.443196995	-365.786326700
fixedML / Active	+424.214370841	+72.953471732	+351.260899109
fixedML / Complement	-461.660601615	+292.840713928	-754.501315543
fixedML / Total	-37.446230774	+365.794185660	-403.240416434
OL / Active	+426.452950532	+54.629531543	+371.823418989
OL / Complement	-417.533182279	+353.640495229	-771.173677508
OL / Total	+8.919768253	+408.270026772	-399.350258519
commonML / Active	+440.199252278	+73.917986263	+366.281266016
commonML / Complement	-455.378304069	+295.102099395	-750.480403464
commonML / Total	-15.179051791	+369.020085657	-384.199137448

Table B.48: Absolute pooled location, mark and joint scores for the same categories and fitted states.

Arm / category	Location	M	Joint
OC / Active	-796.930791629	-3.899138947	-419.510876403
OC / Complement	-2069.601458524	+0.113243545	-2816.257346327
OC / Total	-2866.532250153	-3.785895402	-3235.768222730
EP / Active	-779.365244361	-3.899138947	-401.138646308
EP / Complement	-2035.779551504	+0.113243545	-2783.578371659
EP / Total	-2815.144795865	-3.785895402	-3184.717017967
fixedML / Active	-793.603858831	-3.899138947	-446.242098669
fixedML / Complement	-2074.107449411	+0.113243545	-2828.495521410
fixedML / Total	-2867.711308243	-3.785895402	-3274.737620079
OL / Active	-1172.875492323	-3.899138947	-804.951212281
OL / Complement	-2367.980279776	+0.113243545	-3139.040713739
OL / Total	-3540.855772099	-3.785895402	-3943.991926020
commonML / Active	-791.771148720	-3.899138947	-429.389021652
commonML / Complement	-2077.189925556	+0.113243545	-2827.557085475
commonML / Total	-2868.961074276	-3.785895402	-3256.946107126

The source is `validation/authorities/682a9c7ecf12bc8c_approved_aggregates_v2.json`. All 15 states, 80 arm rows and 64 contrast rows were retained. Independent reconstruction passed 7,128 component checks with maximum discrepancy $4.547473508864641 \times 10^{-13}$ nats against 10^{-4}. Production and independent controls passed 161 and 101 checks; exporter controls numbered 12, with 11 projection controls and 3,888 actual projection checks. A changed artifact pin was rejected before scientific launch and the original restored; no model, denominator or scientific state changed. This custody incident and the detailed execution telemetry remain in the protected source archive rather than being presented as scientific observations.

B.9.4 JMA complete comparison roster

Each arm scores 469, 348 and 460 records in its three windows, totalling 1,277. Rows below retain the two primary components and all competitive/contextual comparisons. The combined descriptive transfer rule is not demonstrated.

Comparison	Component	Pooled delta nats	Delta nats/target	Fold 0 delta	Fold 1 delta	Fold 2 delta	N
OC–FC	Timing	-58.5935	-0.0458837	-19.9765	-12.8718	-25.7452	1277
OC–OL	Location	1905.74	1.49236	619.711	424.797	861.235	1277
OC–MC	Joint	-49.1946	-0.0385236	-22.8544	-5.63338	-20.7069	1277
FC–FL	Location	1847.64	1.44686	589.61	411.737	846.29	1277
OC–FL	Joint	1777.98	1.39231	554.925	384.497	838.558	1277
OC–EP	Joint	-221.211	-0.173227	-74.7591	-59.1334	-87.319	1277
OC–N0	Joint	3262.47	2.55479	912.194	460.219	1890.05	1277

Annual rows use the nine registered source-JST calendar years. The full-window and annual score tables include canonical P, Q, nonnegative exposure E, signed contribution $C = -E$, immigrant and offspring exposures, mark/joint/timing/location components, observed target counts N, $N - E$, and conditional allocation residual summaries. These are saved-score aggregates; they are not event-level output.

The source is `validation/authorities/ad5c7b19cf47d0be_forecast_findings_v1.json`. The procedure and stability rosters contain 36 procedures, 288 starts, 42 floors and 330 slots. The largest independently reconstructed score difference was 1.81899×10^{-10} nats; the largest near-optimal spread was 3.74731×10^{-5} nats per record against the strict 0.01 gate. These saved checks do not certify global optimization or forecast calibration.

B.10 Conditional calibration: full diagnostic record

The complete reference shows material count, temporal and spatial discrepancies for the original frozen Omori continuous (OC) forecast, alongside some ordinary-looking diagnostics. It does not support a broad adequacy claim. This is a retrospective, exploratory, fixed-state conditional check of 315 already inspected targets: 79, 167 and 69 in [2011,2016), [2016,2021) and [2021,2026). No parameters or nuisance choices were refitted.

The target remains original OC, with its density-weighted rational offspring spatial kernel and smooth background. Thread01's separately fitted optimized smooth inverse-power ETAS comparator (EP) exceeded OC by 51.051204763 pooled joint nats. That relative predictive comparison did not replace the frozen target, and it does not establish EP's adequacy. These diagnostics do not calibrate EP.

B.10.1 Reference and interpretation

Each origin has 999 complete dependent branching-future paths, conditional on its own original prehistory and fitted/nuisance state. All 2,997 planned paths are complete; failed and unknown counts are both zero. Simulated events can generate descendants. Sequential observed scores use every strictly earlier admissible event. The issue-frozen NHPP is a different forecast object and has no empirical arm here. The observed-path exposure is a realised predictable compensator, not a Poisson future-count mean.

For each coordinate, the complete tables report the observed value y, inclusive counts $L = \#\{Y_b \le y\}$ and $H = \#\{Y_b \ge y\}$, ties, its insertion interval and the supplied empirical reference interval. With $B = 999$ complete paths, the insertion interval is

$$\left[\frac{1 + \#\{Y_b < y\}}{1000}, \frac{1 + \#\{Y_b \le y\}}{1000}\right].$$

The inclusive lower and upper tails are $(1 + L)/1000$ and $(1 + H)/1000$. Resolution is 0.001. A boundary rank is a finite comparison, not a zero probability or an exact population tail

probability. Ties can make insertion intervals wide. All 84 coordinates per origin form one descriptive collection; individual tails are not separate unadjusted discoveries, and no omnibus significance test is asserted. Reference intervals describe simulated catalogues, not confidence intervals for fitted parameters or model validity.

The coordinate `immigrant_exposure` is marked *fixed* in every table: its reference variance is zero by construction and it is not a stochastic adequacy diagnostic. Adding it to trigger exposure produces total exposure, so the latter two insertion intervals are duplicates within an origin. The full accounting vector is preserved, but its length must not be interpreted as the number of independent tests.

B.10.2 What the complete vector shows

Early origin: count 79 is below all 999 simulated counts (L=0,H=999,ties=0; insertion 0.001), with supplied count reference interval 121 to 453. Exposure 114.54932024184329 and trigger exposure 38.82182627332281 are also below all references. Count minus exposure is -35.54932024184329 (L=10,H=989; insertion 0.011). Gap KS 0.22224496322920195 and gap serial statistic 0.025311536051231393 both exceed every reference. Spatial Rosenblatt KS 0.16733517789142327 exceeds every reference; location dimensionless mean 2.4494637138726643 is below all references and variance 10.653917170486158 is above all. Yet time event-logsum insertion is 0.583, censored tail 0.111 and within-cell absolute maximum 0.187. The second year count 30 has insertion interval [0.370,0.409], with 39 ties. This is not uniformly poor on every coordinate.

Middle origin: total count 167 looks ordinary in its reference (L=711,H=296,ties=8; insertion [0.704,0.712], supplied count interval 90 to 398). Balance -4.373290651999895 has insertion 0.391. This total hides subblock concentration: yearly counts are 12,4,8,116,27; subblock balance maximum 0.6836875842626084 has L=997,H=2 and insertion 0.998. The fourth year's balance is +25.931179418686256 (insertion 0.998). Gap KS 0.15088891163993823 has H=8; serial 0.05049115130656217 exceeds all 999. Spatial KS 0.16932136137657972 has H=4, cross statistic 0.027677364202278305 exceeds all 999, and cell-mass residual maximum 1.278459418085198 has H=11. Conditional-location logscore insertion 0.364, within-cell density score 0.312, terminal tail 0.237 and within-cell absolute maximum 0.341 illustrate ordinary-looking components alongside those discrepancies.

Late origin: count 69 lies toward the low end (L=27,H=979,ties=7; insertion [0.021,0.028], supplied interval 69 to 279). Exposure 82.98011402768624 has insertion 0.012, while balance -13.980114027686241 has insertion 0.110. Gap KS 0.0881227829042357 has insertion 0.676, spatial KS 0.1129696501693383 has insertion 0.810 and terminal tail 1.586732938471152 has insertion 0.700. However serial statistic 0.02308692546525953 has H=2 (insertion 0.998), spatial cross statistic 0.021671079327220788 has H=8 (0.992), and dimensionless location variance 12.30394407102264 exceeds all references. Within-cell absolute maximum has insertion 0.499. A reasonable marginal shape statistic therefore does not settle dependence or concentration.

Across the descriptive cell tables, cell 4 counts are 15,7,8 against conditional expected allocations 23.493716546234523,16.132286198379887,14.770085109767768. These allocations sum conditional probabilities at the observed event times; they are not unconditional count means. Their count-minus-allocation values have insertion ranks 0.074,0.029,0.047. Other cells differ in direction: middle cell 7 count 124 has insertion 0.992, and late cell 3 count-minus-allocation +5.305104300093006 has insertion 0.991. Empty cells often tie extensively: early cell 8 count zero has 562 ties and insertion [0.001,0.563], so its lower endpoint alone is misleading.

Raw total log scores depend on the number of events and the score measure. For example early joint and location log scores exceed all reference totals while count is below all references; this is not evidence of adequate probabilities. Dimensionless location summaries, allocation residuals and spatial transforms provide complementary information. Immigrant exposure is deterministic within each fixed state: all 999 references tie at 75.72749396852048,65.12820992122843,51.19150695498898 respectively. Its [0.001,1.000] insertion interval is expected by construction, not an adequacy result. Mark totals also vary with count and do not independently validate the mark or productivity recipe.

B.10.3 Independent computational audit and limits

The independent campaign audit passed identity/hash/status/seed checks and all-path raw count, mark and temporal-exposure checks for 2,997 paths. The full 84-coordinate raw diagnostic replay was sampled: indices 0,499,998 in each origin, nine complete paths. Maximum sampled absolute score difference was 2.2737367544323206e-12 nats (gate 1e-4); maximum non-score difference 3.410605131648481e-13 and spatial CDF difference 6.217248937900877e-15. This does not mean all 2,997 paths received full-vector independent replay. Original observed reconstructions are separately bound by the approved report's execution and oracle receipt hashes. Numerical agreement checks implementation; it does not validate the forecast's scientific adequacy.

The fitted state, training history and nuisance choices are fixed; fitting, selection, measurement and catalogue-revision uncertainty are excluded. Later observed training histories include earlier target periods, so independent per-origin simulations do not define a coherent cross-origin joint null. Cross-window patterns and any pooled quantities remain descriptive. There is no event-iid confidence interval, Wilks inference, causal triggering conclusion, untouched holdout, general calibration or validation claim. Heavy-tailed productivity has infinite second moment under the fixed mark recipe; normal/CLT count arguments are not used. Terminal exposure is censored survival information, never an added completed gap. No observed rounding model, post-observation metric choice or seed extension was introduced.

B.10.4 Evidence and transition

The sole numerical reporting input is approved scalar report, SHA256 731a4a1481fc4dee9297a8cccba6012f1096803f5551171501f22bbd64233f93. Definitions are in frozen protocol, fixed methods and independent design. Audit evidence is campaign audit. The complete HTML table and the appendix cover all 252 coordinates. Rendering-only revised rank figures are early, middle, late; the JSON, rather than plotted glyphs, is authoritative.

The subsequent completed score-concentration and catalogue-robustness study retained these discrepant and ordinary-looking diagnostics. Its exact annual accounting preserved every event history and full exposure before the separately admitted higher-threshold sensitivity. Those completed results appear in Chapter 11; the frozen OC target was retained by that follow-on.

B.10.5 Complete coordinate appendix

Years are chronological, zero-based within each five-year window. Cells 0 to 8 are row-major latitude then longitude. Values are rounded for display; the approved JSON retains full precision. Every row has planned=complete=999 and failed/unknown=0. RI denotes the supplied empirical reference interval; it is not an inferential acceptance band.

california__fold__0

Coordinate	Observed	L	H	Ties	Insertion interval	RI
count	79	0	999	0	[0.001, 0.001]	[121, 453]
exposure	114.549	0	999	0	[0.001, 0.001]	[135.568, 486.759]
immigrant__exposure (fixed)	75.7275	999	999	999	[0.001, 1.000]	[75.7275, 75.7275]
trigger__exposure	38.8218	0	999	0	[0.001, 0.001]	[59.8403, 411.031]
count__minus__exposure	-35.5493	10	989	0	[0.011, 0.011]	[-27.6883, 30.0604]
time__event__logsum	-176.459	582	417	0	[0.583, 0.583]	[-286.513, 722.925]
timing__logscore	-291.008	832	167	0	[0.833, 0.833]	[-457.245, 177.529]
location__logscore	-816.442	999	0	0	[1.000, 1.000]	[-3711.06, -1114.36]
mark__logscore	6.45748	995	4	0	[0.996, 0.996]	[-99.9643, -0.728886]
joint__logscore	-1100.99	999	0	0	[1.000, 1.000]	[-3577.51, -1463.15]
zone__logscore	-74.7927	993	6	0	[0.994, 0.994]	[-165.841, -82.9881]
cell__given__zone__ logscore	-42.5394	989	10	0	[0.990, 0.990]	[-118.87, -46.1742]
within__cell__density__ logscore	-699.11	999	0	0	[1.000, 1.000]	[-3254.52, -969.257]
subblock__balance__max	0.461367	960	39	0	[0.961, 0.961]	[0.0994411, 0.481693]
gap__ks	0.222245	999	0	0	[1.000, 1.000]	[0.0301227, 0.11125]
gap__serial	0.0253115	999	0	0	[1.000, 1.000]	[-0.0125117, 0.012663]
censored__tail	0.363034	110	889	0	[0.111, 0.111]	[0.172443, 4.35742]
spatial__rosenblatt__ks__ max	0.167335	999	0	0	[1.000, 1.000]	[0.0369932, 0.120491]
spatial__rosenblatt__ cross__abs	0.0136967	979	20	0	[0.980, 0.980]	[0.000180014, 0.0134149]
cell__mass__residual__ max	0.944775	940	59	0	[0.941, 0.941]	[0.273419, 1.25152]
location__dimensionless__ log__mean	2.44946	0	999	0	[0.001, 0.001]	[3.22345, 5.12333]
location__dimensionless__ log__variance	10.6539	999	0	0	[1.000, 1.000]	[5.42532, 8.42119]
within__cell__ dimensionless__absmax	6.18894	186	813	0	[0.187, 0.187]	[6.05597, 6.93833]
within__cell__ dimensionless__sum	137.844	0	999	0	[0.001, 0.001]	[280.681, 1543.46]
year__count[0]	14	17	988	6	[0.012, 0.018]	[16, 137]
year__count[1]	30	408	630	39	[0.370, 0.409]	[14, 102]
year__count[2]	13	16	990	7	[0.010, 0.017]	[14, 121]
year__count[3]	12	19	986	6	[0.014, 0.020]	[14, 152]
year__count[4]	10	2	998	1	[0.002, 0.003]	[15, 137]
year__exposure[0]	22.3768	7	992	0	[0.008, 0.008]	[23.4745, 132.734]
year__exposure[1]	30.6867	382	617	0	[0.383, 0.383]	[22.4225, 104.079]
year__exposure[2]	20.8572	4	995	0	[0.005, 0.005]	[22.4249, 111.59]
year__exposure[3]	21.2066	10	989	0	[0.011, 0.011]	[22.4176, 150.938]
year__exposure[4]	19.4221	0	999	0	[0.001, 0.001]	[22.9658, 124.594]
year__count__minus__ exposure[0]	-8.3768	65	934	0	[0.066, 0.066]	[-10.4097, 14.7207]
year__count__minus__ exposure[1]	-0.686685	470	529	0	[0.471, 0.471]	[-10.1202, 14.0157]
year__count__minus__ exposure[2]	-7.85718	88	911	0	[0.089, 0.089]	[-10.827, 14.85]
year__count__minus__ exposure[3]	-9.2066	64	935	0	[0.065, 0.065]	[-11.3972, 14.9039]
year__count__minus__ exposure[4]	-9.42206	40	959	0	[0.041, 0.041]	[-10.3287, 17.6716]
cell__count[0]	1	898	288	187	[0.712, 0.899]	[0, 5]
cell__count[1]	5	325	763	89	[0.237, 0.326]	[1, 41]
cell__count[2]	39	169	852	22	[0.148, 0.170]	[27, 178]
cell__count[3]	2	260	854	115	[0.146, 0.261]	[0, 23]
cell__count[4]	15	0	999	0	[0.001, 0.001]	[35, 218]

Continued on next page

Coordinate	Observed	L	H	Ties	Insertion interval	RI
cell_count[5]	2	12	994	7	[0.006, 0.013]	[3, 42]
cell_count[6]	7	660	414	75	[0.586, 0.661]	[1, 31]
cell_count[7]	8	227	834	62	[0.166, 0.228]	[4, 56]
cell_count[8]	0	562	999	562	[0.001, 0.563]	[0, 5]
cell_conditional_ expected[0]	0.264795	22	977	0	[0.023, 0.023]	[0.266605, 2.60346]
cell_conditional_ expected[1]	3.39765	0	999	0	[0.001, 0.001]	[4.20337, 38.4866]
cell_conditional_ expected[2]	34.7782	49	950	0	[0.050, 0.050]	[32.7803, 180.086]
cell_conditional_ expected[3]	2.42206	0	999	0	[0.001, 0.001]	[2.77168, 21.7863]
cell_conditional_ expected[4]	23.4937	0	999	0	[0.001, 0.001]	[39.125, 218.36]
cell_conditional_ expected[5]	5.33786	0	999	0	[0.001, 0.001]	[7.36546, 39.439]
cell_conditional_ expected[6]	3.11359	40	959	0	[0.041, 0.041]	[3.00673, 27.2229]
cell_conditional_ expected[7]	5.78427	1	998	0	[0.002, 0.002]	[7.40276, 53.2884]
cell_conditional_ expected[8]	0.407903	0	999	0	[0.001, 0.001]	[0.483074, 4.0325]
cell_exposure[0]	0.558912	837	162	0	[0.838, 0.838]	[0.321129, 2.05508]
cell_exposure[1]	5.71935	149	850	0	[0.150, 0.150]	[4.99354, 37.8893]
cell_exposure[2]	43.1983	134	865	0	[0.135, 0.135]	[37.6058, 176.193]
cell_exposure[3]	4.01574	317	682	0	[0.318, 0.318]	[3.17289, 20.7453]
cell_exposure[4]	36.1404	0	999	0	[0.001, 0.001]	[45.597, 209.277]
cell_exposure[5]	7.84627	2	997	0	[0.003, 0.003]	[8.53853, 38.8116]
cell_exposure[6]	6.71515	666	333	0	[0.667, 0.667]	[3.56374, 27.1171]
cell_exposure[7]	9.77314	103	896	0	[0.104, 0.104]	[8.74061, 47.3811]
cell_exposure[8]	0.582052	0	999	0	[0.001, 0.001]	[0.582164, 3.19678]
cell_count_minus_ conditional_expected[0]	0.735205	918	81	0	[0.919, 0.919]	[-0.38687, 2.15471]
cell_count_minus_ conditional_expected[1]	1.60235	744	255	0	[0.745, 0.745]	[-4.11639, 5.82373]
cell_count_minus_ conditional_expected[2]	4.22184	808	191	0	[0.809, 0.809]	[-10.355, 9.62315]
cell_count_minus_ conditional_expected[3]	-0.422056	473	526	0	[0.474, 0.474]	[-3.21669, 4.60473]
cell_count_minus_ conditional_expected[4]	-8.49372	73	926	0	[0.074, 0.074]	[-11.0784, 10.7925]
cell_count_minus_ conditional_expected[5]	-3.33786	139	860	0	[0.140, 0.140]	[-5.78535, 6.84044]
cell_count_minus_ conditional_expected[6]	3.88641	940	59	0	[0.941, 0.941]	[-3.31407, 5.27876]
cell_count_minus_ conditional_expected[7]	2.21573	748	251	0	[0.749, 0.749]	[-5.47284, 6.7455]
cell_count_minus_ conditional_expected[8]	-0.407903	571	428	0	[0.572, 0.572]	[-0.680562, 2.20553]
cell_count_minus_ exposure[0]	0.441088	782	217	0	[0.783, 0.783]	[-0.369183, 2.33848]
cell_count_minus_ exposure[1]	-0.719352	442	557	0	[0.443, 0.443]	[-4.47976, 7.52299]
cell_count_minus_ exposure[2]	-4.19828	332	667	0	[0.333, 0.333]	[-14.4546, 18.9275]
cell_count_minus_ exposure[3]	-2.01574	198	801	0	[0.199, 0.199]	[-3.20664, 6.01319]
cell_count_minus_ exposure[4]	-21.1404	7	992	0	[0.008, 0.008]	[-15.7661, 19.7682]
cell_count_minus_ exposure[5]	-5.84627	32	967	0	[0.033, 0.033]	[-6.13766, 8.00333]
cell_count_minus_ exposure[6]	0.284852	614	385	0	[0.615, 0.615]	[-3.39293, 6.60789]

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Coordinate	Observed	L	H	Ties	Insertion interval	RI
cell_count_minus_ exposure[7]	-1.77314	367	632	0	[0.368, 0.368]	[-6.5223, 9.49155]
cell_count_minus_ exposure[8]	-0.582052	571	428	0	[0.572, 0.572]	[-0.60909, 2.45499]

california_fold_1

Coordinate	Observed	L	H	Ties	Insertion interval	RI
count	167	711	296	8	[0.704, 0.712]	[90, 398]
exposure	171.373	766	233	0	[0.767, 0.767]	[106.123, 397.718]
immigrant_exposure (fixed)	65.1282	999	999	999	[0.001, 1.000]	[65.1282, 65.1282]
trigger_exposure	106.245	766	233	0	[0.767, 0.767]	[40.9949, 332.589]
count_minus_exposure	-4.37329	390	609	0	[0.391, 0.391]	[-24.363, 26.3737]
time_event_logsum	326.504	963	36	0	[0.964, 0.964]	[-265.132, 583.383]
timing_logscore	155.131	971	28	0	[0.972, 0.972]	[-404.099, 178.164]
location_logscore	-1377.21	363	636	0	[0.364, 0.364]	[-3078.53, -854.24]
mark_logscore	-15.039	680	319	0	[0.681, 0.681]	[-81.6814, 1.52589]
joint_logscore	-1237.12	948	51	0	[0.949, 0.949]	[-3090.7, -1158.67]
zone_logscore	-54.3512	998	1	0	[0.999, 0.999]	[-141.063, -64.9197]
cell_given_zone_ logscore	-45.4093	852	147	0	[0.853, 0.853]	[-94.9976, -36.3447]
within_cell_density_ logscore	-1277.45	311	688	0	[0.312, 0.312]	[-2827.3, -745.781]
subblock_balance_max	0.683688	997	2	0	[0.998, 0.998]	[0.109334, 0.534276]
gap_ks	0.150889	991	8	0	[0.992, 0.992]	[0.0338237, 0.133599]
gap_serial	0.0504912	999	0	0	[1.000, 1.000]	[-0.0147719, 0.0133319]
censored_tail	0.616134	236	763	0	[0.237, 0.237]	[0.187709, 4.19745]
spatial_rosenblatt_ks_ max	0.169321	995	4	0	[0.996, 0.996]	[0.0419072, 0.13574]
spatial_rosenblatt_ cross_abs	0.0276774	999	0	0	[1.000, 1.000]	[0.00016408, 0.0154671]
cell_mass_residual_ max	1.27846	988	11	0	[0.989, 0.989]	[0.290536, 1.12216]
location_dimensionless_ log_mean	4.53739	906	93	0	[0.907, 0.907]	[3.02673, 5.1304]
location_dimensionless_ log_variance	8.76988	976	23	0	[0.977, 0.977]	[5.13867, 8.72968]
within_cell_ dimensionless_absmax	6.21235	340	659	0	[0.341, 0.341]	[6.00424, 6.92997]
within_cell_ dimensionless_sum	488.361	732	267	0	[0.733, 0.733]	[205.4, 1400.6]
year_count[0]	12	71	952	24	[0.048, 0.072]	[10, 90]
year_count[1]	4	1	999	1	[0.001, 0.002]	[11, 84]
year_count[2]	8	8	995	4	[0.005, 0.009]	[11, 101]
year_count[3]	116	985	14	0	[0.986, 0.986]	[11, 82]
year_count[4]	27	514	525	40	[0.475, 0.515]	[10, 78]
year_exposure[0]	18.2911	73	926	0	[0.074, 0.074]	[17.2083, 89.5413]
year_exposure[1]	14.8072	0	999	0	[0.001, 0.001]	[17.6359, 78.9419]
year_exposure[2]	17.9604	26	973	0	[0.027, 0.027]	[17.9169, 106.21]
year_exposure[3]	90.0688	980	19	0	[0.981, 0.981]	[18.2227, 78.0068]
year_exposure[4]	30.2459	643	356	0	[0.644, 0.644]	[18.1292, 75.4797]
year_count_minus_ exposure[0]	-6.29107	100	899	0	[0.101, 0.101]	[-9.38564, 12.2754]
year_count_minus_ exposure[1]	-10.8072	13	986	0	[0.014, 0.014]	[-9.30059, 11.8518]
year_count_minus_ exposure[2]	-9.96035	15	984	0	[0.016, 0.016]	[-9.09994, 12.8146]

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Coordinate	Observed	L	H	Ties	Insertion interval	RI
year_count_minus_ exposure[3]	25.9312	997	2	0	[0.998, 0.998]	[-8.60357, 13.2453]
year_count_minus_ exposure[4]	-3.24589	323	676	0	[0.324, 0.324]	[-9.7793, 11.4455]
cell_count[0]	0	717	999	717	[0.001, 0.718]	[0, 4]
cell_count[1]	5	452	634	87	[0.366, 0.453]	[1, 27]
cell_count[2]	17	13	988	2	[0.012, 0.014]	[19, 134]
cell_count[3]	5	732	343	76	[0.657, 0.733]	[0, 20]
cell_count[4]	7	0	999	0	[0.001, 0.001]	[25, 191]
cell_count[5]	3	80	969	50	[0.031, 0.081]	[2, 30]
cell_count[6]	4	454	626	81	[0.374, 0.455]	[0, 27]
cell_count[7]	124	991	8	0	[0.992, 0.992]	[2, 45]
cell_count[8]	2	947	129	77	[0.871, 0.948]	[0, 4]
cell_conditional_ expected[0]	0.196435	0	999	0	[0.001, 0.001]	[0.283822, 2.71701]
cell_conditional_ expected[1]	3.28105	8	991	0	[0.009, 0.009]	[3.47798, 24.3377]
cell_conditional_ expected[2]	20.5286	5	994	0	[0.006, 0.006]	[23.4844, 129.722]
cell_conditional_ expected[3]	1.64839	0	999	0	[0.001, 0.001]	[2.24293, 18.5057]
cell_conditional_ expected[4]	16.1323	0	999	0	[0.001, 0.001]	[30.1783, 188.077]
cell_conditional_ expected[5]	3.84289	0	999	0	[0.001, 0.001]	[5.56033, 25.2496]
cell_conditional_ expected[6]	1.9814	0	999	0	[0.001, 0.001]	[2.81796, 26.1705]
cell_conditional_ expected[7]	119.072	991	8	0	[0.992, 0.992]	[5.88285, 43.0544]
cell_conditional_ expected[8]	0.316679	1	998	0	[0.002, 0.002]	[0.375522, 3.23759]
cell_exposure[0]	0.35138	620	379	0	[0.621, 0.621]	[0.346286, 2.45038]
cell_exposure[1]	6.37401	553	446	0	[0.554, 0.554]	[4.15007, 23.3833]
cell_exposure[2]	26.8893	9	990	0	[0.010, 0.010]	[28.566, 135.655]
cell_exposure[3]	4.98962	768	231	0	[0.769, 0.769]	[2.61991, 17.8111]
cell_exposure[4]	28.8743	0	999	0	[0.001, 0.001]	[35.9872, 179.332]
cell_exposure[5]	6.65003	29	970	0	[0.030, 0.030]	[6.59611, 21.634]
cell_exposure[6]	4.61809	406	593	0	[0.407, 0.407]	[3.21846, 24.2814]
cell_exposure[7]	91.4855	988	11	0	[0.989, 0.989]	[7.1327, 42.7387]
cell_exposure[8]	1.14108	930	69	0	[0.931, 0.931]	[0.457852, 2.56708]
cell_count_minus_ conditional_expected[0]	-0.196435	724	275	0	[0.725, 0.725]	[-0.428208, 2.02942]
cell_count_minus_ conditional_expected[1]	1.71895	779	220	0	[0.780, 0.780]	[-3.38599, 5.56352]
cell_count_minus_ conditional_expected[2]	-3.52859	230	769	0	[0.231, 0.231]	[-8.6423, 9.3221]
cell_count_minus_ conditional_expected[3]	3.35161	942	57	0	[0.943, 0.943]	[-2.80122, 4.55026]
cell_count_minus_ conditional_expected[4]	-9.13229	28	971	0	[0.029, 0.029]	[-9.23256, 10.2033]
cell_count_minus_ conditional_expected[5]	-0.842886	410	589	0	[0.411, 0.411]	[-4.91408, 6.04513]
cell_count_minus_ conditional_expected[6]	2.0186	850	149	0	[0.851, 0.851]	[-3.22868, 4.4536]
cell_count_minus_ conditional_expected[7]	4.92771	951	48	0	[0.952, 0.952]	[-4.88563, 5.91796]
cell_count_minus_ conditional_expected[8]	1.68332	974	25	0	[0.975, 0.975]	[-0.546403, 1.68373]
cell_count_minus_ exposure[0]	-0.35138	102	897	0	[0.103, 0.103]	[-0.383332, 2.17322]
cell_count_minus_ exposure[1]	-1.37401	346	653	0	[0.347, 0.347]	[-3.72152, 7.35179]

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Coordinate	Observed	L	H	Ties	Insertion interval	RI
cell_count_minus_ exposure[2]	-9.88929	59	940	0	[0.060, 0.060]	[-12.9982, 15.0907]
cell_count_minus_ exposure[3]	0.0103764	604	395	0	[0.605, 0.605]	[-2.76441, 5.53505]
cell_count_minus_ exposure[4]	-21.8743	4	995	0	[0.005, 0.005]	[-13.5997, 17.6117]
cell_count_minus_ exposure[5]	-3.65003	109	890	0	[0.110, 0.110]	[-5.03823, 6.97647]
cell_count_minus_ exposure[6]	-0.618089	474	525	0	[0.475, 0.475]	[-3.22321, 7.21084]
cell_count_minus_ exposure[7]	32.5145	998	1	0	[0.999, 0.999]	[-5.52971, 8.40555]
cell_count_minus_ exposure[8]	0.858916	895	104	0	[0.896, 0.896]	[-0.474084, 1.85003]

california_fold_2

Coordinate	Observed	L	H	Ties	Insertion interval	RI
count	69	27	979	7	[0.021, 0.028]	[69, 279]
exposure	82.9801	11	988	0	[0.012, 0.012]	[84.6391, 273.416]
immigrant_exposure (fixed)	51.1915	999	999	999	[0.001, 1.000]	[51.1915, 51.1915]
trigger_exposure	31.7886	11	988	0	[0.012, 0.012]	[33.4476, 222.225]
count_minus_exposure	-13.9801	109	890	0	[0.110, 0.110]	[-21.5162, 25.7304]
time_event_logsum	-146.593	609	390	0	[0.610, 0.610]	[-232.883, 319.678]
timing_logscore	-229.573	782	217	0	[0.783, 0.783]	[-348.998, 48.5116]
location_logscore	-672.878	969	30	0	[0.970, 0.970]	[-2179.9, -658.585]
mark_logscore	4.79567	953	46	0	[0.954, 0.954]	[-55.1853, 7.87956]
joint_logscore	-897.655	975	24	0	[0.976, 0.976]	[-2266.97, -899.621]
zone_logscore	-55.0781	981	18	0	[0.982, 0.982]	[-120.766, -56.7314]
cell_given_zone_ logscore	-47.4487	375	624	0	[0.376, 0.376]	[-80.9155, -25.3484]
within_cell_density_ logscore	-570.351	974	25	0	[0.975, 0.975]	[-1993.36, -568.099]
subblock_balance_max	0.506742	912	87	0	[0.913, 0.913]	[0.13028, 0.597709]
gap_ks	0.0881228	675	324	0	[0.676, 0.676]	[0.0389069, 0.151529]
gap_serial	0.0230869	997	2	0	[0.998, 0.998]	[-0.0164741, 0.0158146]
censored_tail	1.58673	699	300	0	[0.700, 0.700]	[0.204758, 4.1602]
spatial_rosenblatt_ks_ max	0.11297	809	190	0	[0.810, 0.810]	[0.0465445, 0.155071]
spatial_rosenblatt_ cross_abs	0.0216711	991	8	0	[0.992, 0.992]	[0.000277444, 0.0182526]
cell_mass_residual_ max	0.9353	928	71	0	[0.929, 0.929]	[0.2906, 1.13691]
location_dimensionless_ log_mean	3.03231	37	962	0	[0.038, 0.038]	[2.97277, 5.07106]
location_dimensionless_ log_variance	12.3039	999	0	0	[1.000, 1.000]	[5.11568, 8.90204]
within_cell_ dimensionless_absmax	6.22259	498	501	0	[0.499, 0.499]	[5.93567, 6.92541]
within_cell_ dimensionless_sum	160.576	29	970	0	[0.030, 0.030]	[157.923, 976.089]
year_count[0]	14	188	851	40	[0.149, 0.189]	[8, 69]
year_count[1]	6	13	994	8	[0.006, 0.014]	[7, 77]
year_count[2]	11	112	913	26	[0.087, 0.113]	[7, 83]
year_count[3]	31	793	219	13	[0.781, 0.794]	[7, 75]
year_count[4]	7	28	983	12	[0.017, 0.029]	[7, 68]
year_exposure[0]	16.938	138	861	0	[0.139, 0.139]	[15.059, 59.8745]

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Coordinate	Observed	L	H	Ties	Insertion interval	RI
year_exposure[1]	13.1913	2	997	0	[0.003, 0.003]	[14.5469, 73.6176]
year_exposure[2]	15.3397	79	920	0	[0.080, 0.080]	[14.0186, 78.5319]
year_exposure[3]	23.0921	616	383	0	[0.617, 0.617]	[14.0072, 74.5098]
year_exposure[4]	14.419	44	955	0	[0.045, 0.045]	[13.6782, 64.7594]
year_count_minus_ exposure[0]	-2.93805	299	700	0	[0.300, 0.300]	[-7.67249, 12.2186]
year_count_minus_ exposure[1]	-7.19134	41	958	0	[0.042, 0.042]	[-7.9307, 11.0882]
year_count_minus_ exposure[2]	-4.33967	189	810	0	[0.190, 0.190]	[-8.34279, 12.6448]
year_count_minus_ exposure[3]	7.90791	932	67	0	[0.933, 0.933]	[-7.93494, 11.3002]
year_count_minus_ exposure[4]	-7.41897	34	965	0	[0.035, 0.035]	[-7.75287, 10.9252]
cell_count[0]	0	814	999	814	[0.001, 0.815]	[0, 2]
cell_count[1]	6	711	359	71	[0.641, 0.712]	[0, 27]
cell_count[2]	30	576	457	34	[0.543, 0.577]	[12, 87]
cell_count[3]	11	946	66	13	[0.934, 0.947]	[0, 16]
cell_count[4]	8	1	998	0	[0.002, 0.002]	[15, 103]
cell_count[5]	0	7	999	7	[0.001, 0.008]	[1, 21]
cell_count[6]	4	669	440	110	[0.560, 0.670]	[0, 22]
cell_count[7]	9	32	977	10	[0.023, 0.033]	[9, 107]
cell_count[8]	1	864	367	232	[0.633, 0.865]	[0, 4]
cell_conditional_ expected[0]	0.185909	45	954	0	[0.046, 0.046]	[0.179725, 1.28819]
cell_conditional_ expected[1]	2.61701	68	931	0	[0.069, 0.069]	[2.40165, 27.2712]
cell_conditional_ expected[2]	27.3997	493	506	0	[0.494, 0.494]	[15.2385, 88.3632]
cell_conditional_ expected[3]	5.6949	871	128	0	[0.872, 0.872]	[1.61982, 13.481]
cell_conditional_ expected[4]	14.7701	2	997	0	[0.003, 0.003]	[18.0943, 99.7113]
cell_conditional_ expected[5]	3.34589	11	988	0	[0.012, 0.012]	[3.5531, 17.7813]
cell_conditional_ expected[6]	2.04302	61	938	0	[0.062, 0.062]	[1.90358, 19.411]
cell_conditional_ expected[7]	12.5871	19	980	0	[0.020, 0.020]	[13.2619, 101.837]
cell_conditional_ expected[8]	0.356408	111	888	0	[0.112, 0.112]	[0.323762, 2.89666]
cell_exposure[0]	0.234267	533	466	0	[0.534, 0.534]	[0.233696, 1.18385]
cell_exposure[1]	5.1514	671	328	0	[0.672, 0.672]	[2.90187, 23.9996]
cell_exposure[2]	25.8568	386	613	0	[0.387, 0.387]	[19.0713, 87.7123]
cell_exposure[3]	6.82953	923	76	0	[0.924, 0.924]	[2.02464, 12.601]
cell_exposure[4]	20.3487	0	999	0	[0.001, 0.001]	[23.4101, 96.2069]
cell_exposure[5]	4.21047	0	999	0	[0.001, 0.001]	[4.43963, 16.6524]
cell_exposure[6]	3.42026	504	495	0	[0.505, 0.505]	[2.35546, 18.9387]
cell_exposure[7]	16.1866	22	977	0	[0.023, 0.023]	[16.3028, 98.9965]
cell_exposure[8]	0.742081	847	152	0	[0.848, 0.848]	[0.414219, 2.09758]
cell_count_minus_ conditional_expected[0]	-0.185909	771	228	0	[0.772, 0.772]	[-0.303696, 1.47732]
cell_count_minus_ conditional_expected[1]	3.38299	943	56	0	[0.944, 0.944]	[-3.05927, 4.33262]
cell_count_minus_ conditional_expected[2]	2.60032	753	246	0	[0.754, 0.754]	[-6.85874, 7.80752]
cell_count_minus_ conditional_expected[3]	5.3051	990	9	0	[0.991, 0.991]	[-2.29715, 4.0011]
cell_count_minus_ conditional_expected[4]	-6.77009	46	953	0	[0.047, 0.047]	[-7.93975, 8.92137]
cell_count_minus_ conditional_expected[5]	-3.34589	46	953	0	[0.047, 0.047]	[-3.71602, 5.01355]

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Coordinate	Observed	L	H	Ties	Insertion interval	RI
cell_count_minus_ conditional_expected[6]	1.95698	894	105	0	[0.895, 0.895]	[-2.4857, 3.84926]
cell_count_minus_ conditional_expected[7]	-3.5871	162	837	0	[0.163, 0.163]	[-6.95699, 7.65505]
cell_count_minus_ conditional_expected[8]	0.643592	873	126	0	[0.874, 0.874]	[-0.509632, 2.01469]
cell_count_minus_ exposure[0]	-0.234267	285	714	0	[0.286, 0.286]	[-0.277815, 1.46093]
cell_count_minus_ exposure[1]	0.848601	703	296	0	[0.704, 0.704]	[-2.98993, 5.32921]
cell_count_minus_ exposure[2]	4.14318	771	228	0	[0.772, 0.772]	[-9.05911, 12.7817]
cell_count_minus_ exposure[3]	4.17047	956	43	0	[0.957, 0.957]	[-2.12513, 4.97781]
cell_count_minus_ exposure[4]	-12.3487	9	990	0	[0.010, 0.010]	[-10.6048, 13.7031]
cell_count_minus_ exposure[5]	-4.21047	16	983	0	[0.017, 0.017]	[-3.74352, 6.04436]
cell_count_minus_ exposure[6]	0.579741	719	280	0	[0.720, 0.720]	[-2.40103, 5.9591]
cell_count_minus_ exposure[7]	-7.18655	60	939	0	[0.061, 0.061]	[-8.77738, 12.1147]
cell_count_minus_ exposure[8]	0.257919	683	316	0	[0.684, 0.684]	[-0.422793, 2.35711]

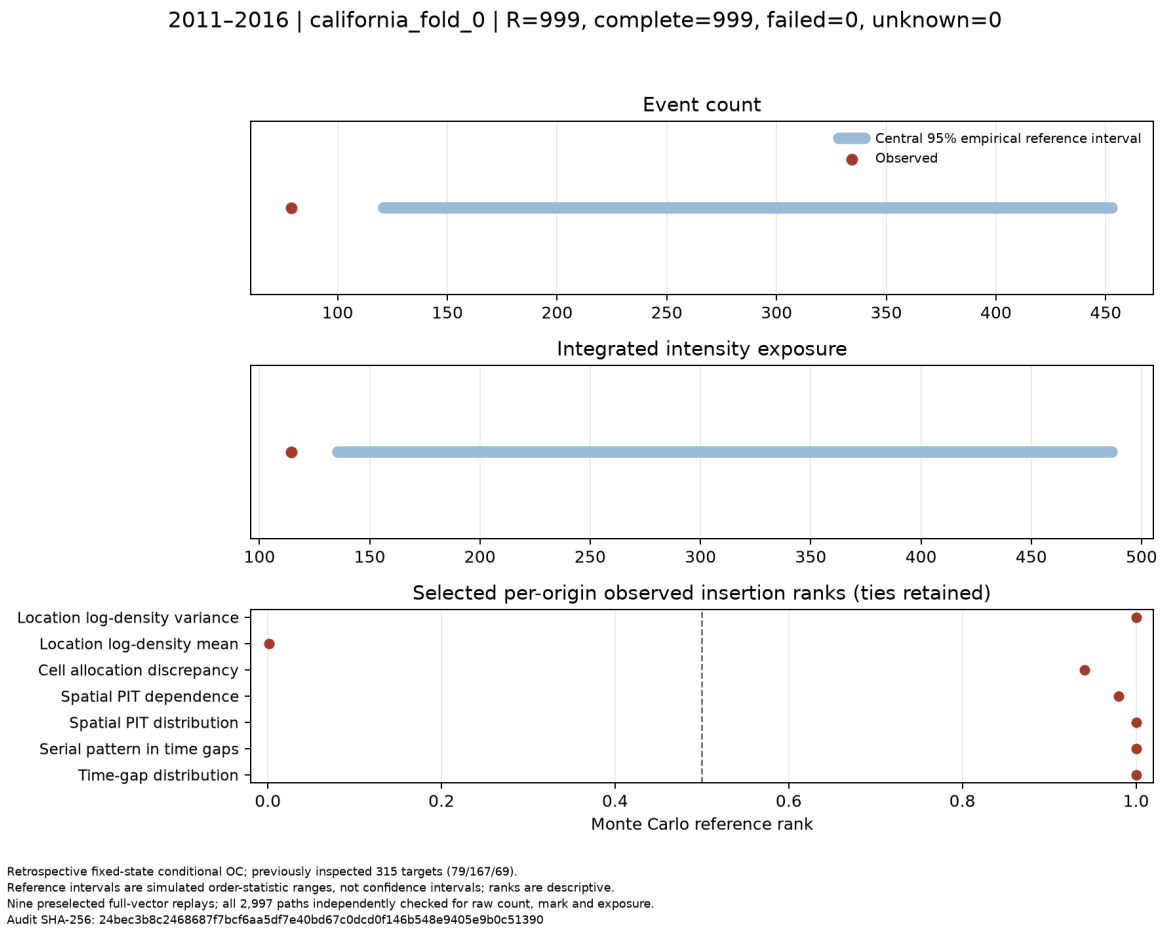


Figure B.1: Conditional diagnostic insertion intervals for frozen OC Fold 0 (2011–2016; 79 observed targets). Immigrant exposure is deterministic; its ties carry no adequacy evidence. Other ties and ordinary-looking coordinates coexist with discrepancies. These are descriptive reference comparisons.

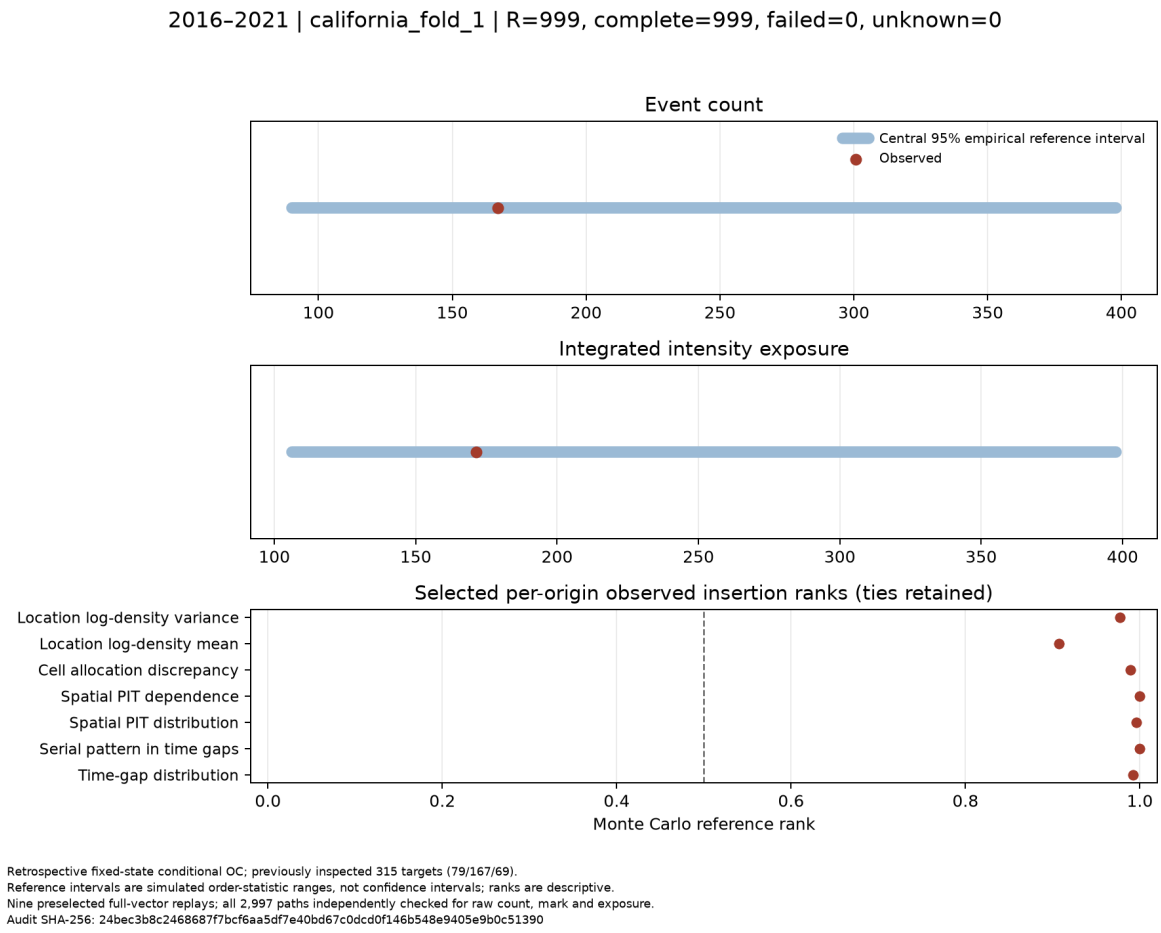


Figure B.2: Conditional diagnostic insertion intervals for frozen OC Fold 1 (2016–2021; 167 observed targets). Immigrant exposure is deterministic; its ties carry no adequacy evidence. Other ties and ordinary-looking coordinates coexist with discrepancies. These are descriptive reference comparisons.

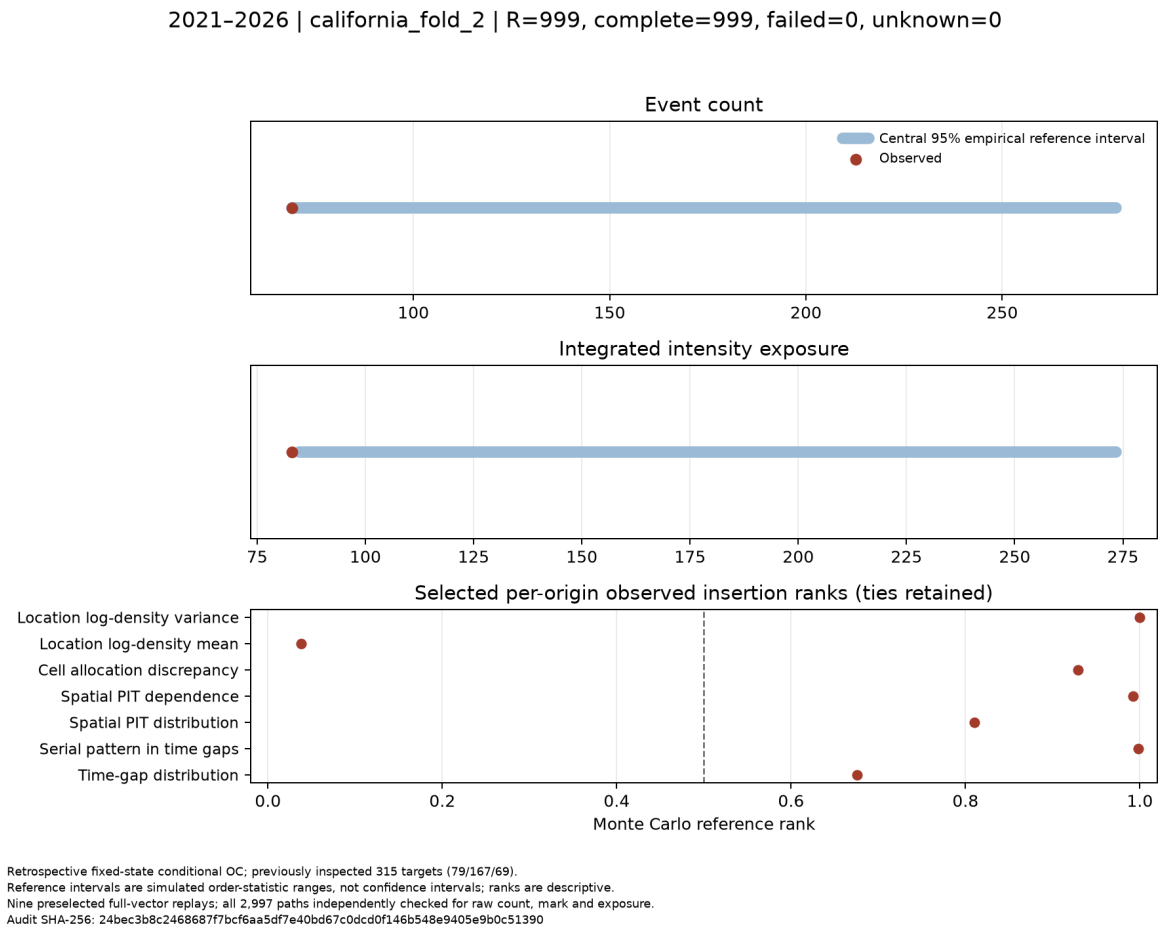


Figure B.3: Conditional diagnostic insertion intervals for frozen OC Fold 2 (2021–2026; 69 observed targets). Immigrant exposure is deterministic; its ties carry no adequacy evidence. Other ties and ordinary-looking coordinates coexist with discrepancies. These are descriptive reference comparisons.

Bibliography

Mark Bebbington and David Harte. The linked stress release model for spatio-temporal seismicity: formulations, procedures and applications. *Geophysical Journal International*, 154(3):925–946, 2003. doi: 10.1046/j.1365-246X.2003.02015.x.

Konstantin Borovkov and Mark S. Bebbington. A stochastic two-node stress transfer model reproducing Omori's law. *Pure and Applied Geophysics*, 160(8):1429–1445, 2003. doi: 10.1007/s00024-003-2354-8.

Andrew Bray and Frederic Paik Schoenberg. Assessment of point process models for earthquake forecasting. *Statistical Science*, 28(4):510–520, 2013. doi: 10.1214/13-STS440. URL https://arxiv.org/abs/1312.5934.

Pierre Brémaud and Laurent Massoulié. Stability of nonlinear Hawkes processes. *The Annals of Probability*, 24(3):1563–1588, 1996. doi: 10.1214/aop/1065725193.

Pierre Brémaud and Laurent Massoulié. Power spectra of general shot noises and Hawkes point processes with a random excitation. *Advances in Applied Probability*, 34(1):205–222, 2002. doi: 10.1239/aap/1019160957.

Rodolfo Console and Maura Murru. A simple and testable model for earthquake clustering. *Journal of Geophysical Research: Solid Earth*, 106(B5):8699–8711, 2001. doi: 10.1029/2000JB900269. URL https://agupubs.onlinelibrary.wiley.com/doi/abs/10.1029/2000JB900269.

Daryl J. Daley and David Vere-Jones. Scoring probability forecasts for point processes: the entropy score and information gain. *Journal of Applied Probability*, 41(A):297–312, 2004. doi: 10.1239/jap/1082552206.

Robert B. Davies. Hypothesis testing when a nuisance parameter is present only under the alternative. *Biometrika*, 64(2):247–254, 1977. doi: 10.1093/biomet/64.2.247.

Robert B. Davies. Hypothesis testing when a nuisance parameter is present only under the alternative. *Biometrika*, 74(1):33–43, 1987. doi: 10.1093/biomet/74.1.33.

Louis Davis, Boris Baeumer, and Ting Wang. A fractional Hawkes process model for earthquake aftershock sequences. *Journal of the Royal Statistical Society Series C: Applied Statistics*, 73(5):1185–1202, 2024a. doi: 10.1093/jrsssc/qlae031.

Louis Davis, Boris Baeumer, and Ting Wang. A multidimensional fractional Hawkes process for multiple earthquake mainshock aftershock sequences, 2024b. URL https://arxiv.org/abs/2404.01478v1. Exact author preprint v1, submitted 1 April 2024; consulted model definitions in Section 2. Related journal version: doi:10.1080/00031305.2025.2588128.

Louis Davis, Boris Baeumer, and Ting Wang. A multivariate fractional Hawkes process for multiple earthquake mainshock aftershock sequences. *The American Statistician*, 80(3):453–462, 2026a. doi: 10.1080/00031305.2025.2588128.

Louis Davis, Conor Kresin, Boris Baeumer, and Ting Wang. Multivariate representations of univariate marked Hawkes processes. *Scandinavian Journal of Statistics*, 53(1):140–174, 2026b. doi: 10.1111/sjos.70030.

davlo199. MDFHPCode: source snapshot. GitHub repository, 2024. URL https://github.com/davlo199/MDFHPCode/tree/435f317ca8843e991cf979ff5e6657491e4849ba. Source commit `435f317ca8843e991cf979ff5e6657491e4849ba` dated 1 April 2024 UTC; inspected in the preceding study on 23 October 2026. Repository-account author identifier retained. Repository copyright and licence notices remain separate from paper citations.

James H. Dieterich. A constitutive law for rate of earthquake production and its application to earthquake clustering. *Journal of Geophysical Research: Solid Earth*, 99(B2):2601–2618, 1994. doi: 10.1029/93JB02581.

Cynthia Dwork, Vitaly Feldman, Moritz Hardt, Toniann Pitassi, Omer Reingold, and Aaron Roth. The reusable holdout: Preserving validity in adaptive data analysis. *Science*, 349(6248):636–638,

2015. doi: 10.1126/science.aaa9375.

Karen R. Felzer and Emily E. Brodsky. Decay of aftershock density with distance indicates triggering by dynamic stress. *Nature*, 441(7094):735–738, 2006. doi: 10.1038/nature04799. URL https://pubs.usgs.gov/publication/70028503.

J. K. Gardner and Leon Knopoff. Is the sequence of earthquakes in Southern California, with aftershocks removed, Poissonian? *Bulletin of the Seismological Society of America*, 64(5):1363–1367, 1974. doi: 10.1785/BSSA0640051363. URL https://pubs.geoscienceworld.org/ssa/bssa/article-pdf/64/5/1363/5318629/bssa0640051363.pdf.

Roberto Garrappa. Numerical evaluation of two and three parameter Mittag-Leffler functions. *SIAM Journal on Numerical Analysis*, 53(3):1350–1369, 2015. doi: 10.1137/140971191.

Tilmann Gneiting and Adrian E Raftery. Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477):359–378, 2007. doi: 10.1198/016214506000001437.

GNS Science. GeoNet Aotearoa New Zealand Earthquake Catalogue. Continuously updated dataset, n.d. URL https://doi.org/10.21420/0S8P-TZ38. Dataset documentation accessed 24 October 2026. Scientific analyses use the separately identified frozen 913-event snapshot, not a new download.

Sebastian Hainzl. Rate-dependent incompleteness of earthquake catalogs. *Seismological Research Letters*, 87(2A):337–344, 2016. doi: 10.1785/0220150211.

Martin Hairer, Jonathan C. Mattingly, and Michael Scheutzow. Asymptotic coupling and a general form of Harris' theorem with applications to stochastic delay equations. *Probability Theory and Related Fields*, 149(1–2):223–259, 2011. doi: 10.1007/s00440-009-0250-6.

Charles R. Harris, K. Jarrod Millman, Stéfan J. van der Walt, et al. Array programming with NumPy. *Nature*, 585(7825):357–362, 2020. doi: 10.1038/s41586-020-2649-2.

Ruth A. Harris and Robert W. Simpson. Suppression of large earthquakes by stress shadows: A comparison of Coulomb and rate-and-state failure. *Journal of Geophysical Research: Solid Earth*, 103(B10):24439–24451, 1998. doi: 10.1029/98JB00793. URL https://pubs.usgs.gov/publication/70244158.

Alan G. Hawkes. Spectra of some self-exciting and mutually exciting point processes. *Biometrika*, 58(1):83–90, 1971. doi: 10.1093/biomet/58.1.83.

Agnès Helmstetter, Yan Y. Kagan, and David D. Jackson. Comparison of short-term and time-independent earthquake forecast models for southern California. *Bulletin of the Seismological Society of America*, 96(1):90–106, 2006. doi: 10.1785/0120050067. URL https://central.scec.org/node/3700.

John D. Hunter. Matplotlib: A 2D graphics environment. *Computing in Science & Engineering*, 9(3):90–95, 2007. doi: 10.1109/mcse.2007.55.

Valerie Isham and Mark Westcott. A self-correcting point process. *Stochastic Processes and their Applications*, 8(3):335–347, 1979. doi: 10.1016/0304-4149(79)90008-5.

Geoffrey C. P. King, Ross S. Stein, and Jian Lin. Static stress changes and the triggering of earthquakes. *Bulletin of the Seismological Society of America*, 84(3):935–953, 1994. doi: 10.1785/BSSA0840030935.

José C. F. Kling and Mathias Vetter. On goodness-of-fit testing for self-exciting point processes, 2024. URL https://arxiv.org/abs/2407.09130.

Francesco Mainardi. On some properties of the Mittag–Leffler function $E_\alpha(-t^\alpha)$, completely monotone for $t > 0$ with $0 < \alpha < 1$. *Discrete and Continuous Dynamical Systems - B*, 19(7):2267–2278, 2014. doi: 10.3934/dcdsb.2014.19.2267.

Ellen P. Mallman and Mark D. Zoback. Assessing elastic Coulomb stress transfer models using seismicity rates in southern California and southwestern Japan. *Journal of Geophysical Research: Solid Earth*, 112(B3):B03304, 2007. doi: 10.1029/2005JB004076. URL https://agupubs.onlinelibrary.wiley.com/doi/full/10.1029/2005JB004076.

David Marsan and Olivier Lengliné. Extending earthquakes' reach through cascading. *Science*, 319(5866):1076–1079, 2008. doi: 10.1126/science.1148783. URL https://pubmed.ncbi.nlm.nih.gov/18292339/.

Mark M. Meerschaert and Bruno Toaldo. Relaxation patterns and semi-Markov dynamics. *Stochastic Processes and their Applications*, 129(8):2850–2879, 2019. doi: 10.1016/j.spa.2018.08.004.

Arnaud Mignan. Functional shape of the earthquake frequency-magnitude distribution and completeness magnitude. *Journal of Geophysical Research: Solid Earth*, 117(B8):B08302, 2012. doi: 10.1029/2012JB009347. URL https://agupubs.onlinelibrary.wiley.com/doi/abs/10.1029/2012JB009347.

Yosihiko Ogata. Statistical models for earthquake occurrences and residual analysis for point processes. *Journal of the American Statistical Association*, 83(401):9–27, 1988. doi: 10.1080/01621459.1988.10478560.

Yosihiko Ogata. Space-time point-process models for earthquake occurrences. *Annals of the Institute of Statistical Mathematics*, 50(2):379–402, 1998. doi: 10.1023/A:1003403601725.

Tom Parsons. Global Omori law decay of triggered earthquakes: Large aftershocks outside the classical aftershock zone. *Journal of Geophysical Research: Solid Earth*, 107(B9):2199, 2002. doi: 10.1029/2001JB000646. URL https://agupubs.onlinelibrary.wiley.com/doi/full/10.1029/2001JB000646.

Giuseppe Petrillo, Jiancang Zhuang, and Eugenio Lippiello. Including stress relaxation in point-process model for seismic occurrence, 2023. URL https://arxiv.org/abs/2302.00827.

Paul Reasenberg. Second-order moment of central California seismicity, 1969–1982. *Journal of Geophysical Research: Solid Earth*, 90(B7):5479–5495, 1985. doi: 10.1029/JB090iB07p05479. URL https://agupubs.onlinelibrary.wiley.com/doi/abs/10.1029/JB090iB07p05479.

David A. Rhoades, Danijel Schorlemmer, Matthew C. Gerstenberger, Annemarie Christophersen, J. Douglas Zechar, and Masajiro Imoto. Efficient testing of earthquake forecasting models. *Acta Geophysica*, 59(4):728–747, 2011. doi: 10.2478/s11600-011-0013-5. URL https://link.springer.com/article/10.2478/s11600-011-0013-5.

Murray Rosenblatt. Remarks on a multivariate transformation. *The Annals of Mathematical Statistics*, 23(3):470–472, 1952. doi: 10.1214/aoms/1177729394.

Frederic Paik Schoenberg. Multidimensional residual analysis of point process models for earthquake occurrences. *Journal of the American Statistical Association*, 98(464):789–795, 2003. doi: 10.1198/016214503000000710.

Danijel Schorlemmer, Matthew C. Gerstenberger, Stefan Wiemer, David D. Jackson, and David A. Rhoades. Earthquake likelihood model testing. *Seismological Research Letters*, 78(1):17–29, 2007. doi: 10.1785/gssrl.78.1.17.

Steven G. Self and Kung-Yee Liang. Asymptotic properties of maximum likelihood estimators and likelihood ratio tests under nonstandard conditions. *Journal of the American Statistical Association*, 82(398):605–610, 1987. doi: 10.1080/01621459.1987.10478472.

Ross S. Stein. The role of stress transfer in earthquake occurrence. *Nature*, 402(6762):605–609, 1999. doi: 10.1038/45144.

Shinji Toda, Ross S. Stein, Keith Richards-Dinger, and Serkan B. Bozkurt. Forecasting the evolution of seismicity in southern California: Animations built on earthquake stress transfer. *Journal of Geophysical Research: Solid Earth*, 110(B5):B05S16, 2005. doi: 10.1029/2004JB003415.

Tokuji Utsu, Yosihiko Ogata, and Ritsuko S. Matsu'ura. The centenary of the Omori formula for a decay law of aftershock activity. *Journal of Physics of the Earth*, 43(1):1–33, 1995. doi: 10.4294/jpe1952.43.1.

David Vere-Jones. Earthquake prediction: a statistician's view. *Journal of Physics of the Earth*, 26(2):129–146, 1978. doi: 10.4294/jpe1952.26.129.

David Vere-Jones. On the variance properties of stress release models. *Australian Journal of Statistics*, 30A:123–135, 1988. doi: 10.1111/j.1467-842X.1988.tb00469.x.

Pauli Virtanen, Ralf Gommers, Travis E. Oliphant, et al. SciPy 1.0: fundamental algorithms for scientific computing in Python. *Nature Methods*, 17(3):261–272, 2020. doi: 10.1038/s41592-019-0686-2.

Donald L. Wells and Kevin J. Coppersmith. New empirical relationships among magnitude, rupture length, rupture width, rupture area, and surface displacement. *Bulletin of the Seismological Society of America*, 84(4):974–1002, 1994. doi: 10.1785/BSSA0840040974.

Maximilian J. Werner, Agnès Helmstetter, David D. Jackson, and Yan Y. Kagan. High-resolution long-term and short-term earthquake forecasts for California. *Bulletin of the Seismological Society of America*, 101(4):1630–1648, 2011. doi: 10.1785/0120090340. URL https://pubs.geoscienceworld.org/ssa/bssa/article-abstract/101/4/1630/349546/High-Resolution-Long-Term-and-Short-Term?redirectedFrom=fulltext.

Jochen Woessner and Stefan Wiemer. Assessing the quality of earthquake catalogues: Estimating the magnitude of completeness and its uncertainty. *Bulletin of the Seismological Society of America*, 95 (2):684–698, 2005. doi: 10.1785/0120040007. URL https://www.research-collection.ethz.ch/items/81dd05e3-3e12-4b42-921f-e15ed81f402d.

J. D. Zechar, M. C. Gerstenberger, and D. A. Rhoades. Likelihood-based tests for evaluating space-rate-magnitude earthquake forecasts. *Bulletin of the Seismological Society of America*, 100(3):1184–1195, 2010. doi: 10.1785/0120090192.

Xiao-Gu Zheng and David Vere-Jones. Application of stress release models to historical earthquakes from North China. *Pure and Applied Geophysics*, 135(4):559–576, 1991. doi: 10.1007/BF01772406.

Xiaogu Zheng and David Vere-Jones. Further applications of the stochastic stress release model to historical earthquake data. *Tectonophysics*, 229(1–2):101–121, 1994. doi: 10.1016/0040-1951(94)90007-8.

Jiancang Zhuang, Yosihiko Ogata, and David Vere-Jones. Stochastic declustering of space-time earthquake occurrences. *Journal of the American Statistical Association*, 97(458):369–380, 2002. doi: 10.1198/016214502760046925.